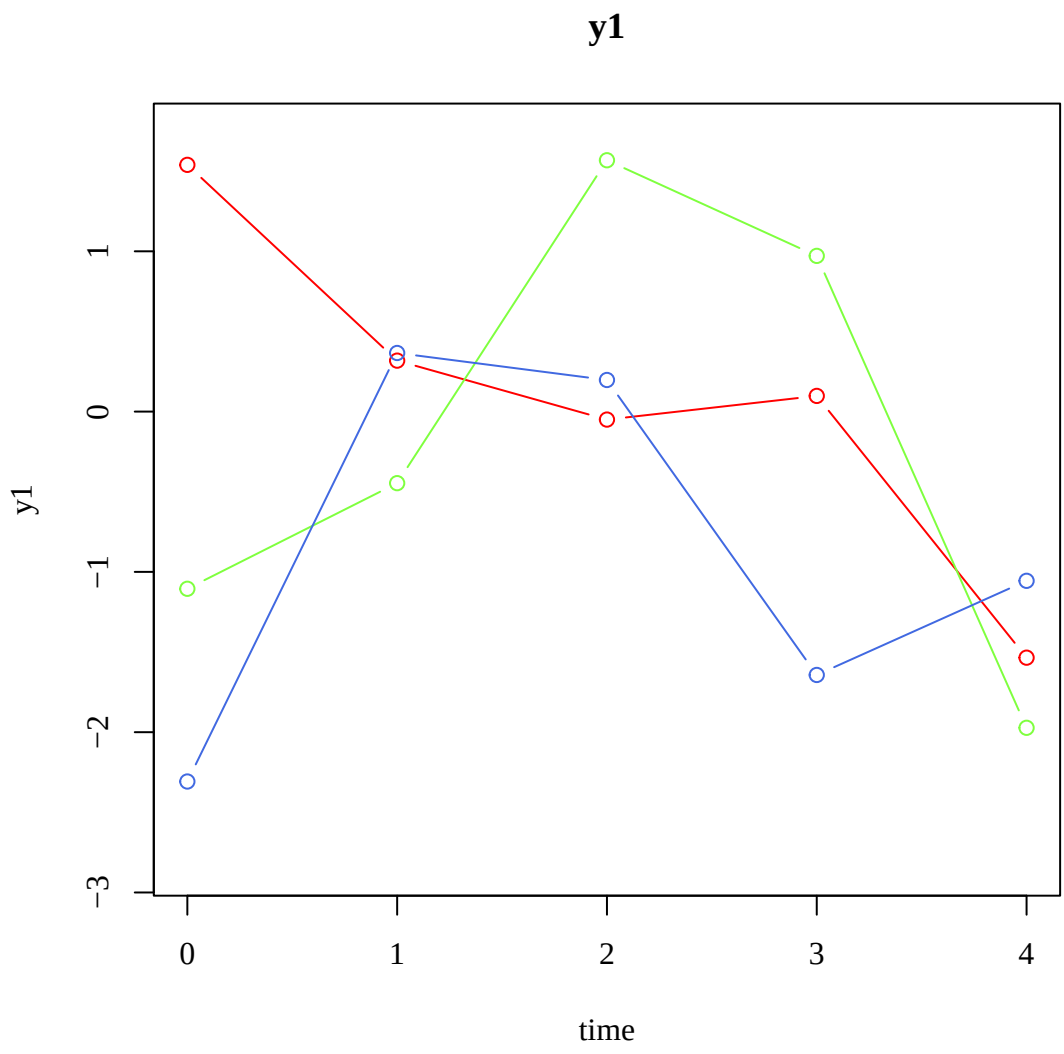
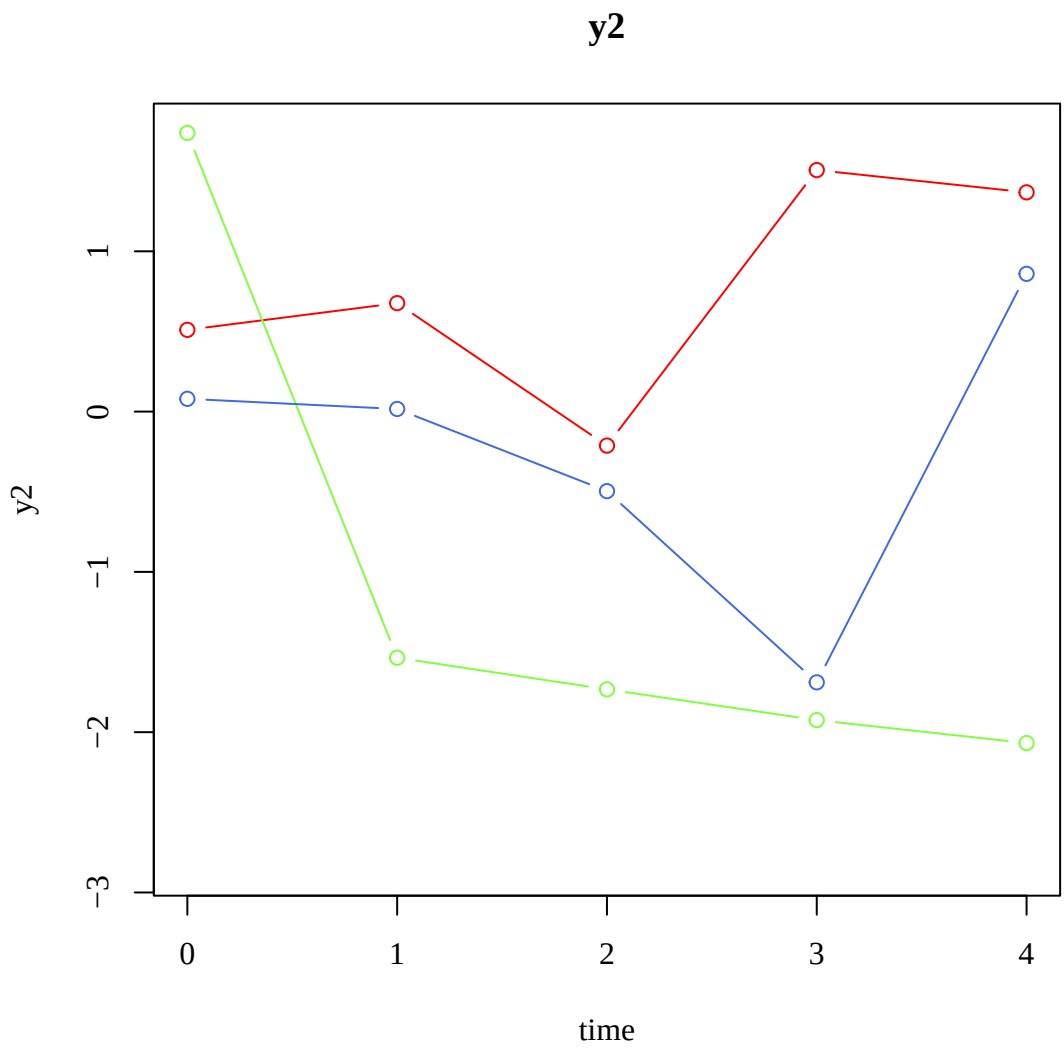
simStateSpace: Internal Tests

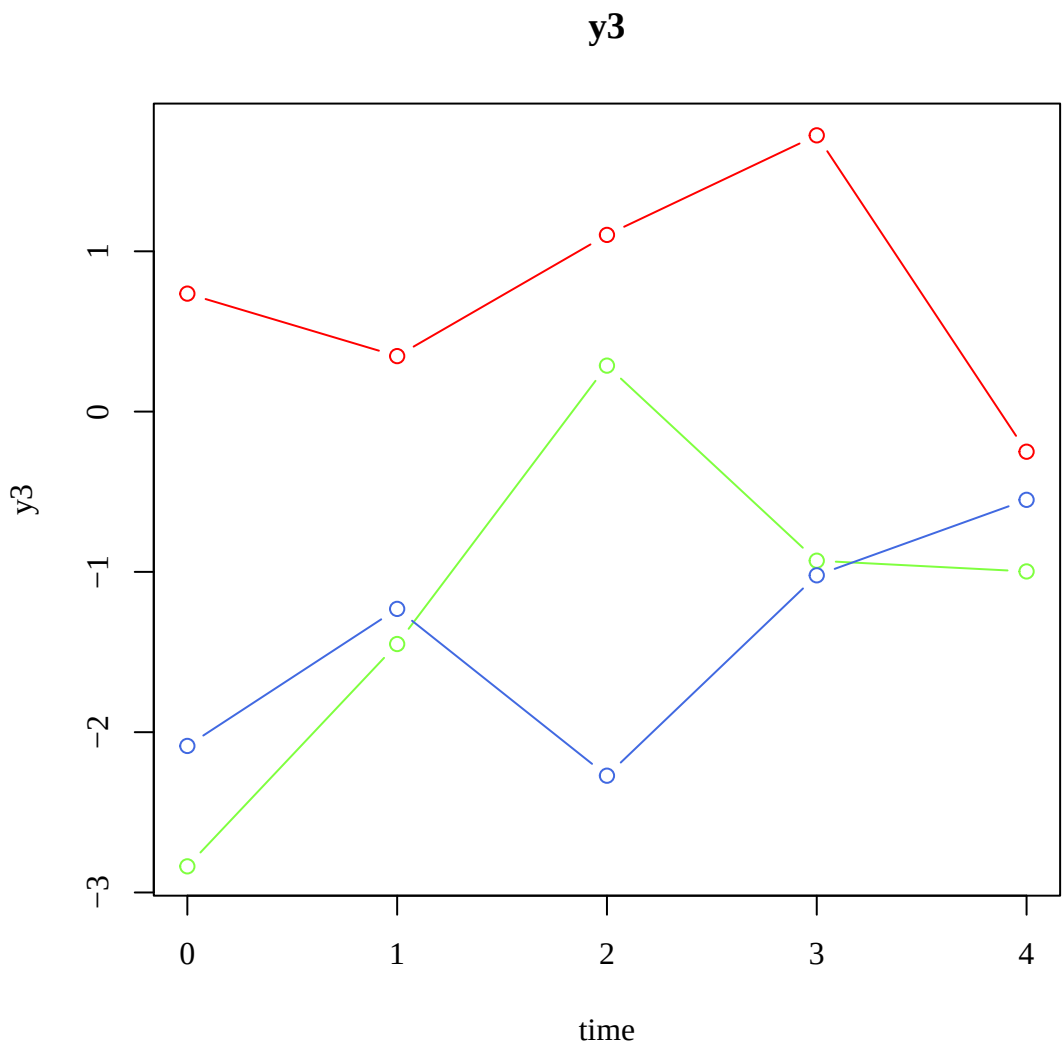
Ivan Jacob Agaloos Pesigan

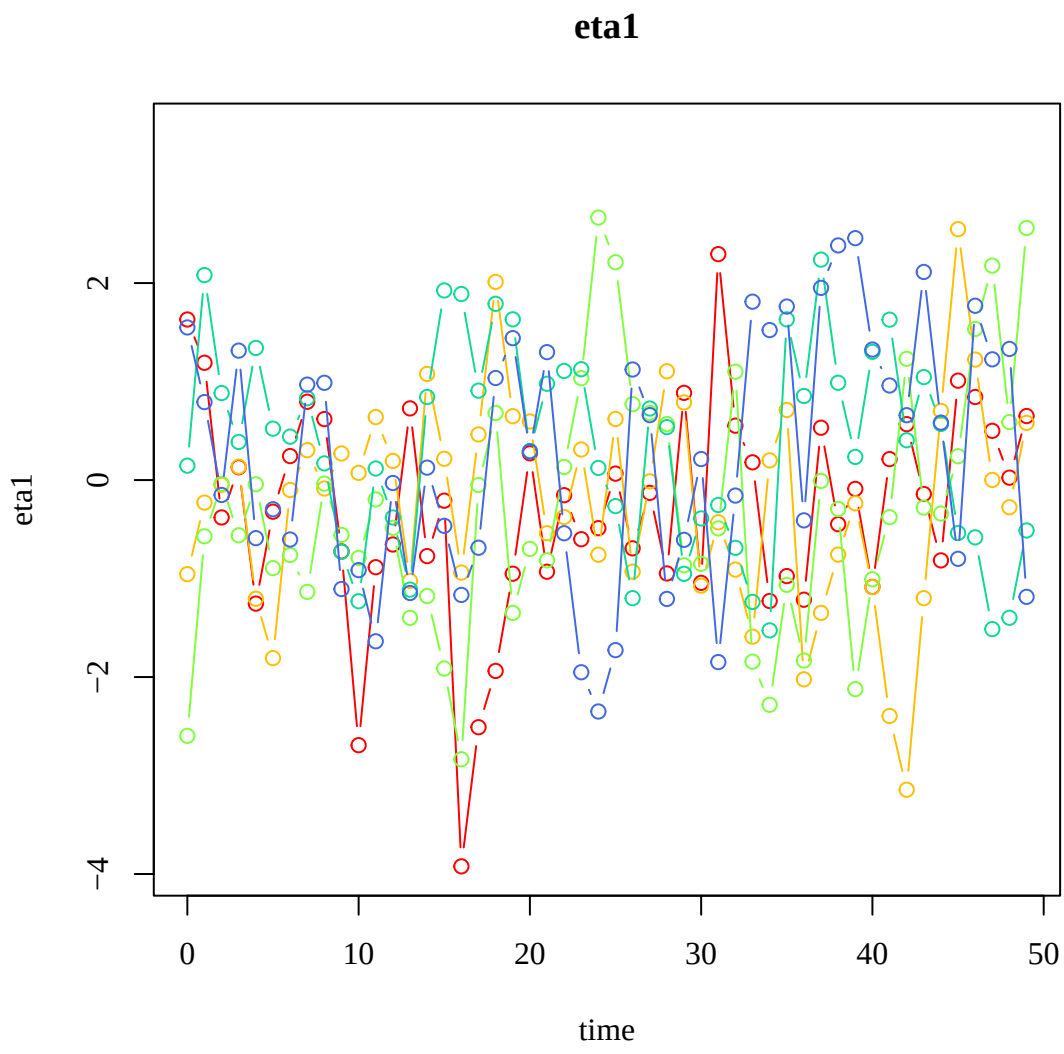
Tests

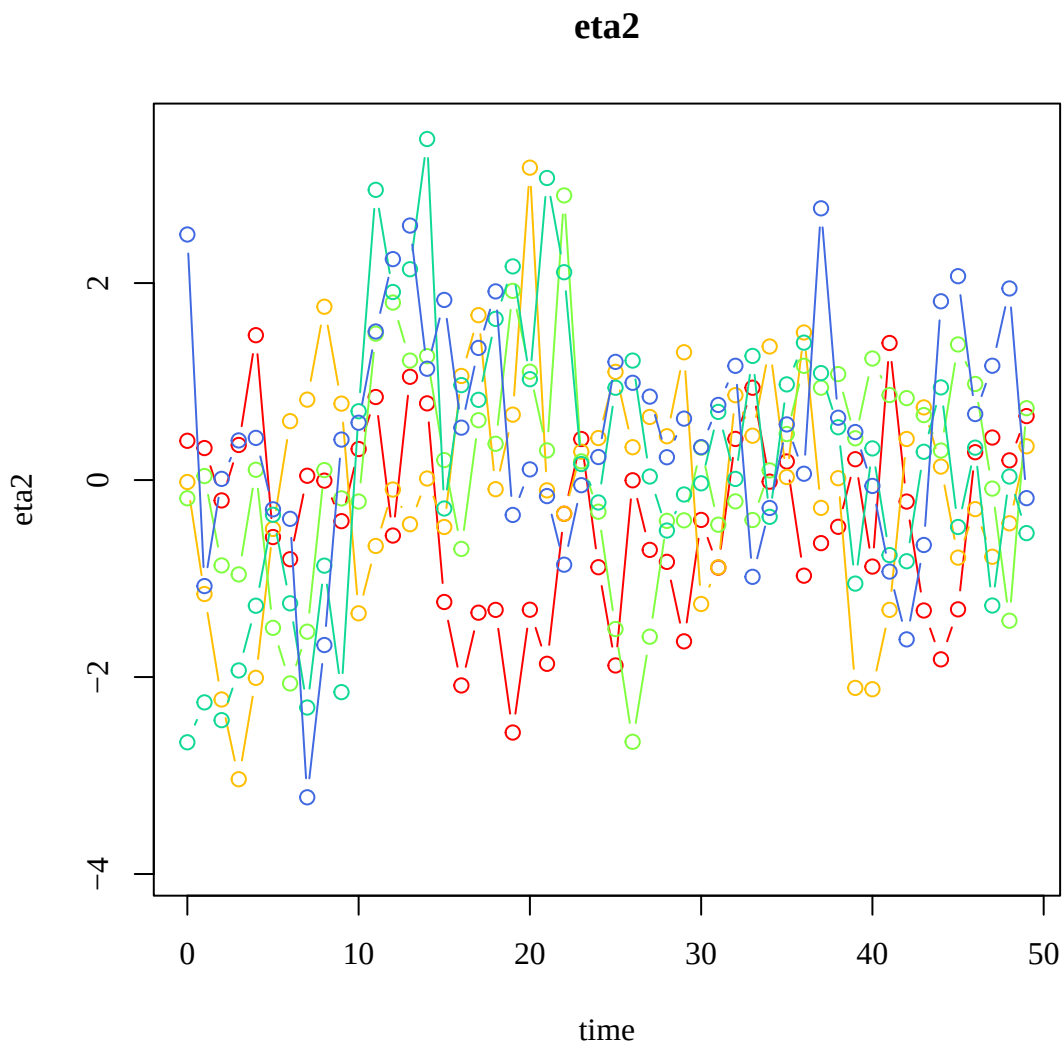
```
#> test-simStateSpace-ou-2-ssm
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-sim-ssm-fixed

#> Call:
#> simStateSpace::SimSSMFixed(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, nu = nu, lambda = lambda,
#>     theta = theta, type = 0, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMFixed`
#> to convert it to a data frame or a matrix.
```

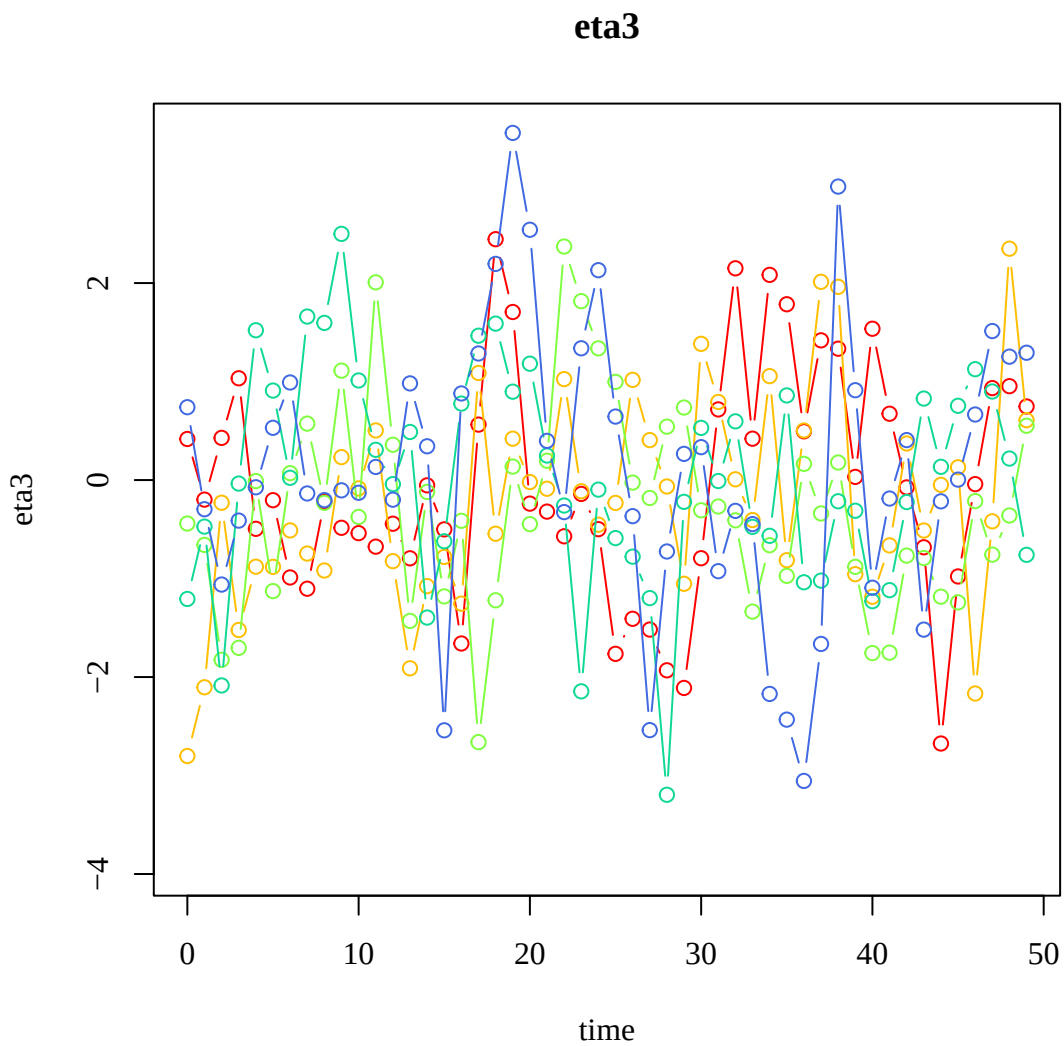

y1
y1
1
0
−1
−2
−3
0
1
2
3
4
time


y2
y2
1
0
−1
−2
−3
0
1
2
3
4
time

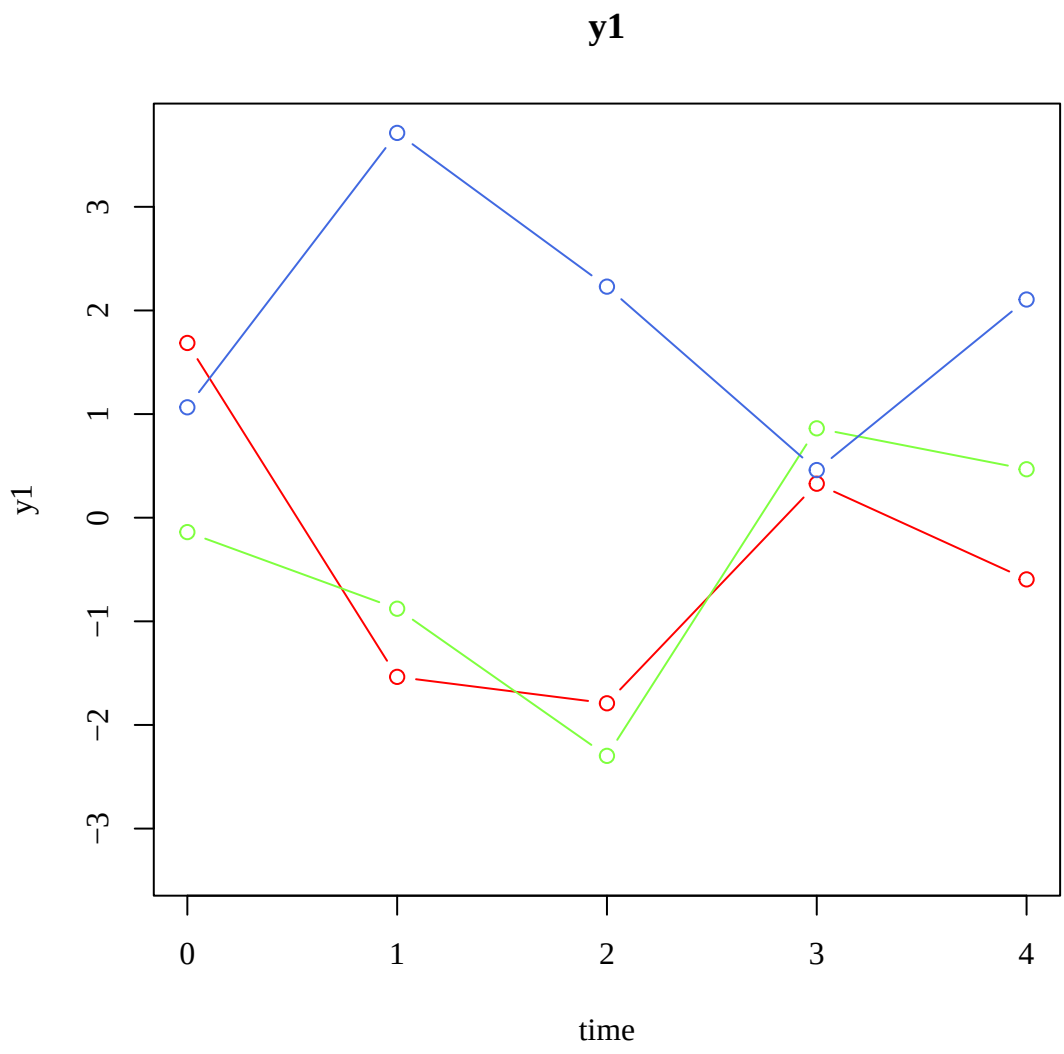

y3
y3
1
0
−1
−2
−3
0
1
2
3
4
time

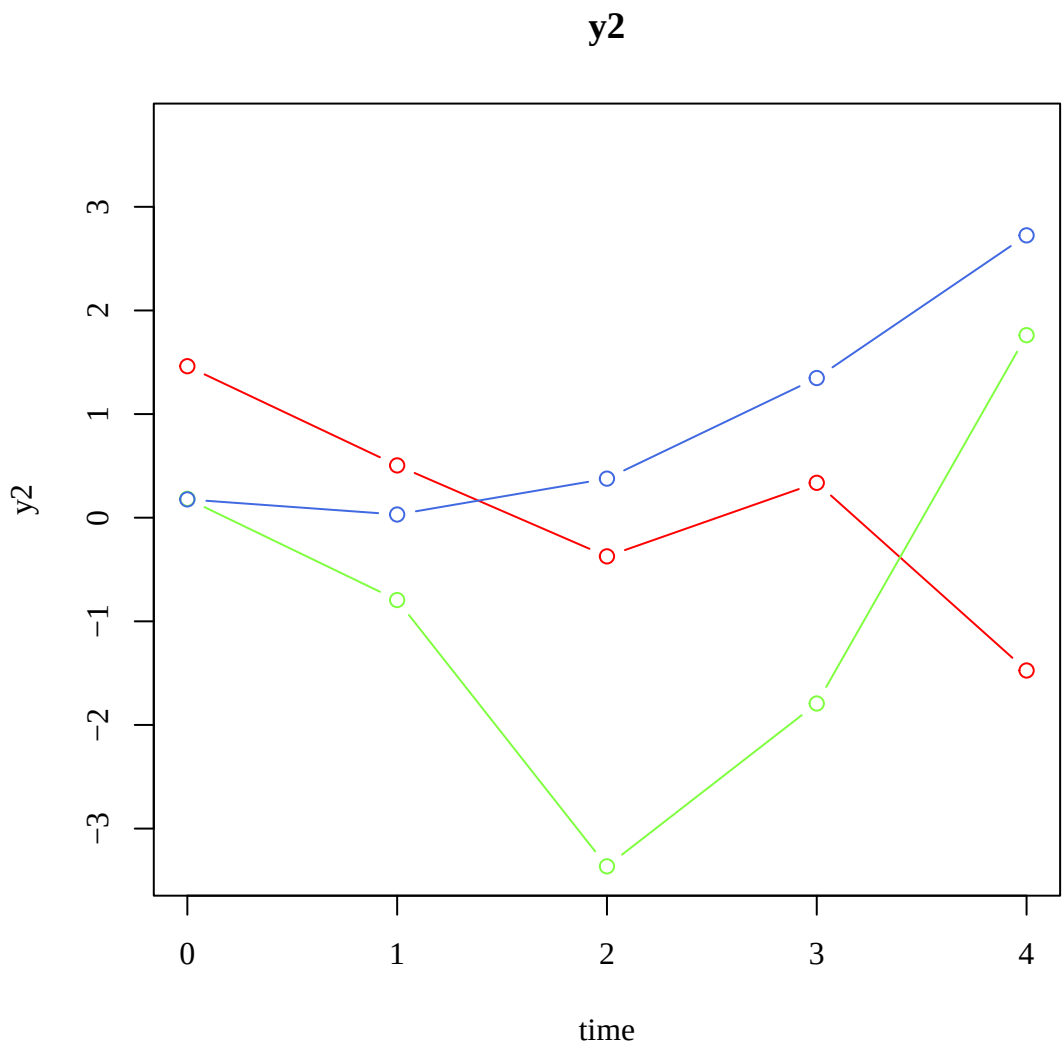

eta1
eta1
2
0
−2
−4
0
10
20
30
40
50
time

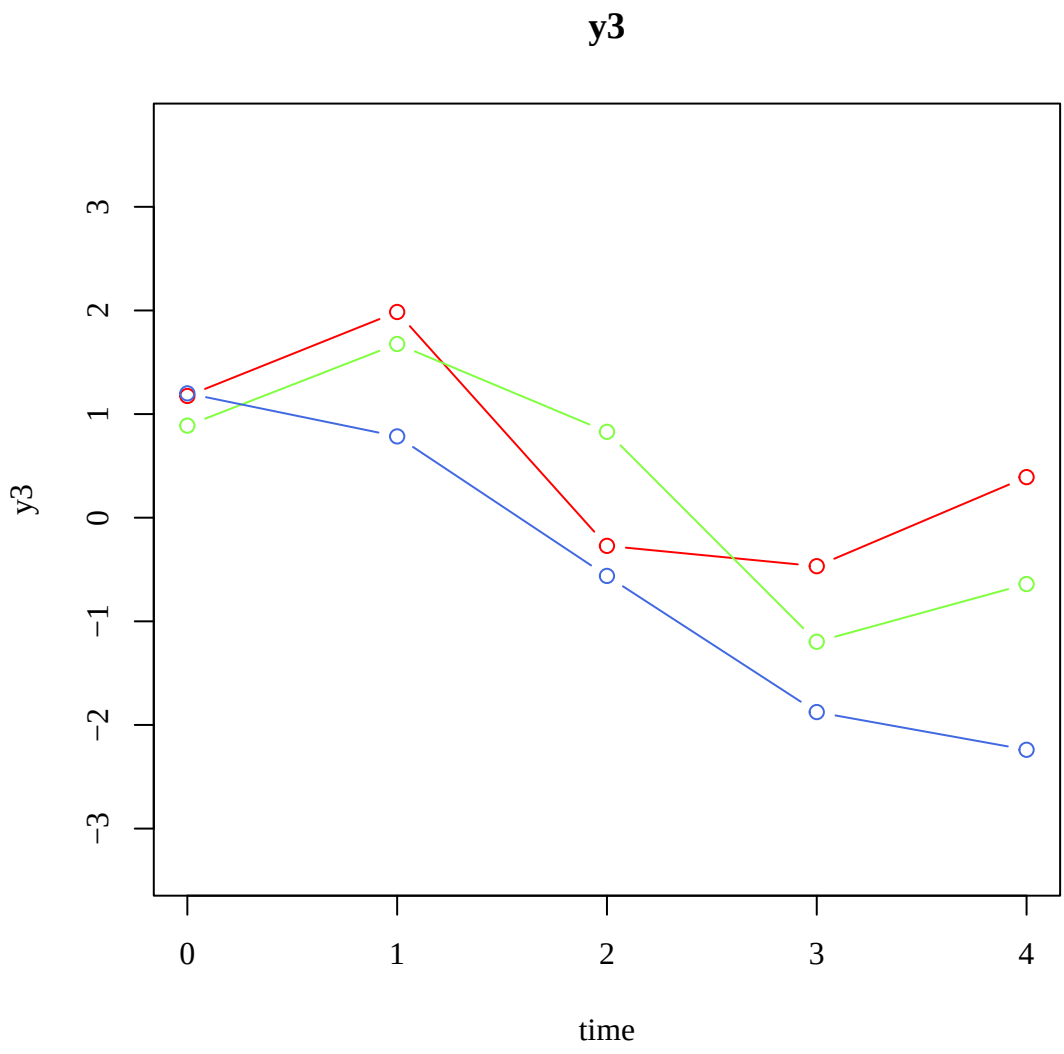

eta2
eta2
2
0
−2
−4
0
10
20
30
40
50
time

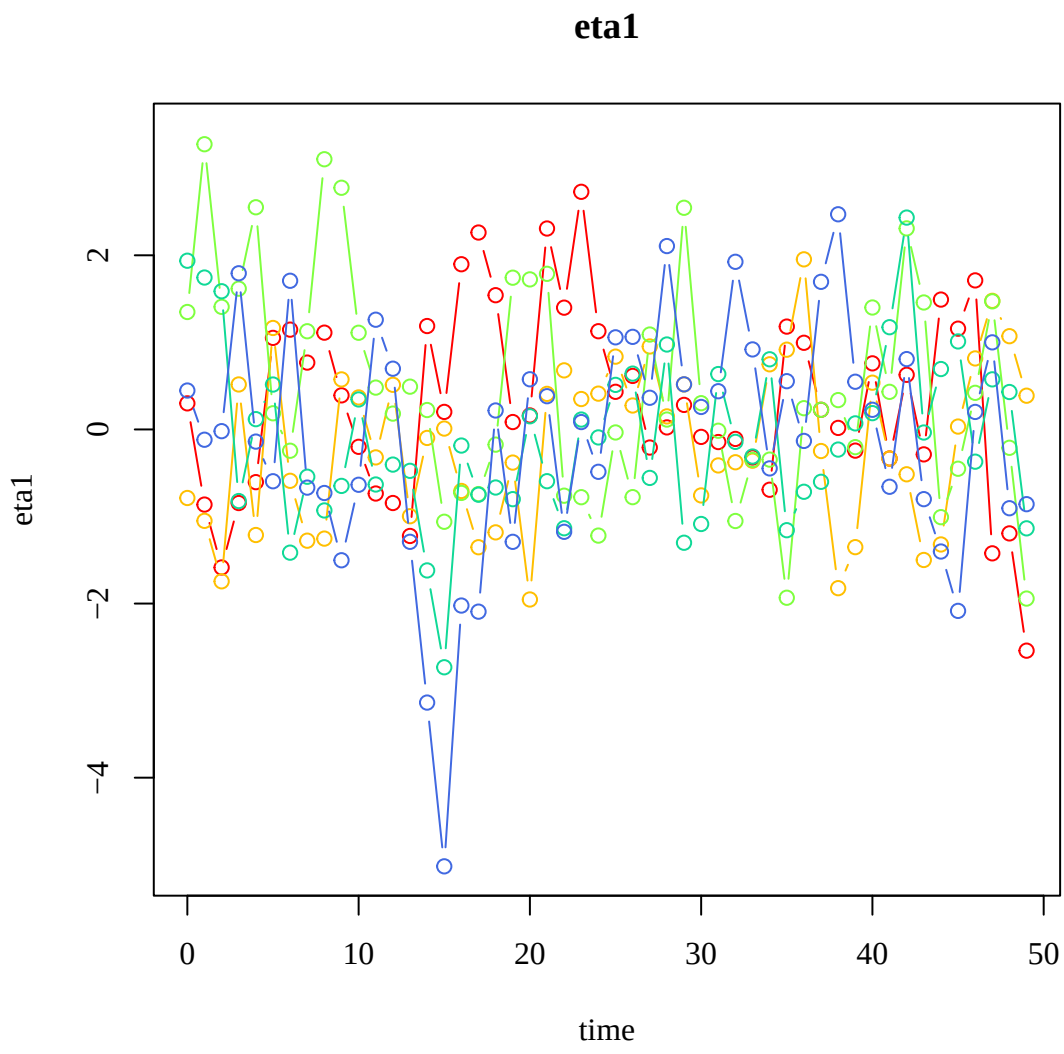


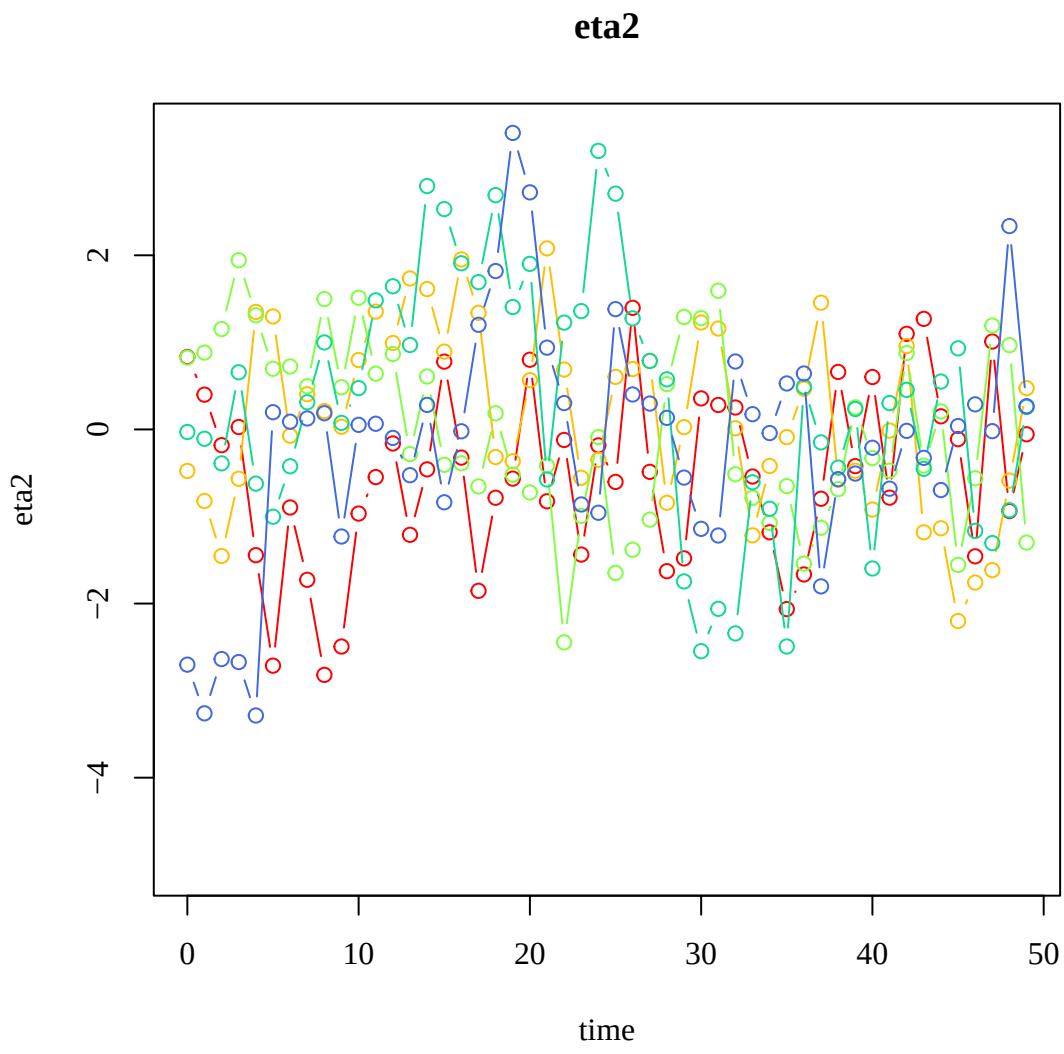

```
#> Call:
#> simStateSpace::SimSSMFixed(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, nu = nu, lambda = lambda,
#>     theta = theta, gamma_eta = gamma_eta, x = x, type = 1, time = time,
#>     burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMFixed`
#> to convert it to a data frame or a matrix.
```

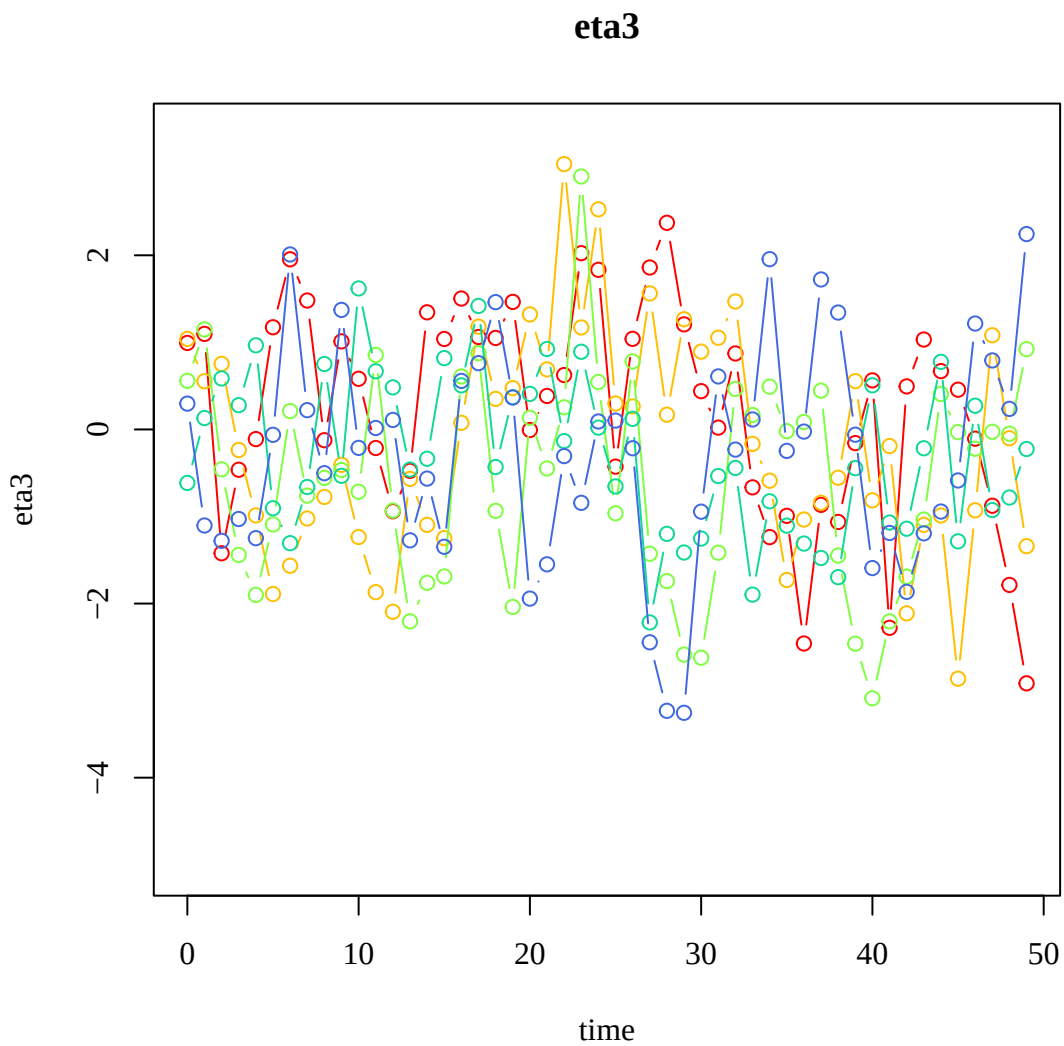

y1
y1
3
2
1
0
−1
−2
−3
0
1
2
3
4
time


y2
y2
3
2
1
0
−1
−2
−3
0
1
2
3
4
time

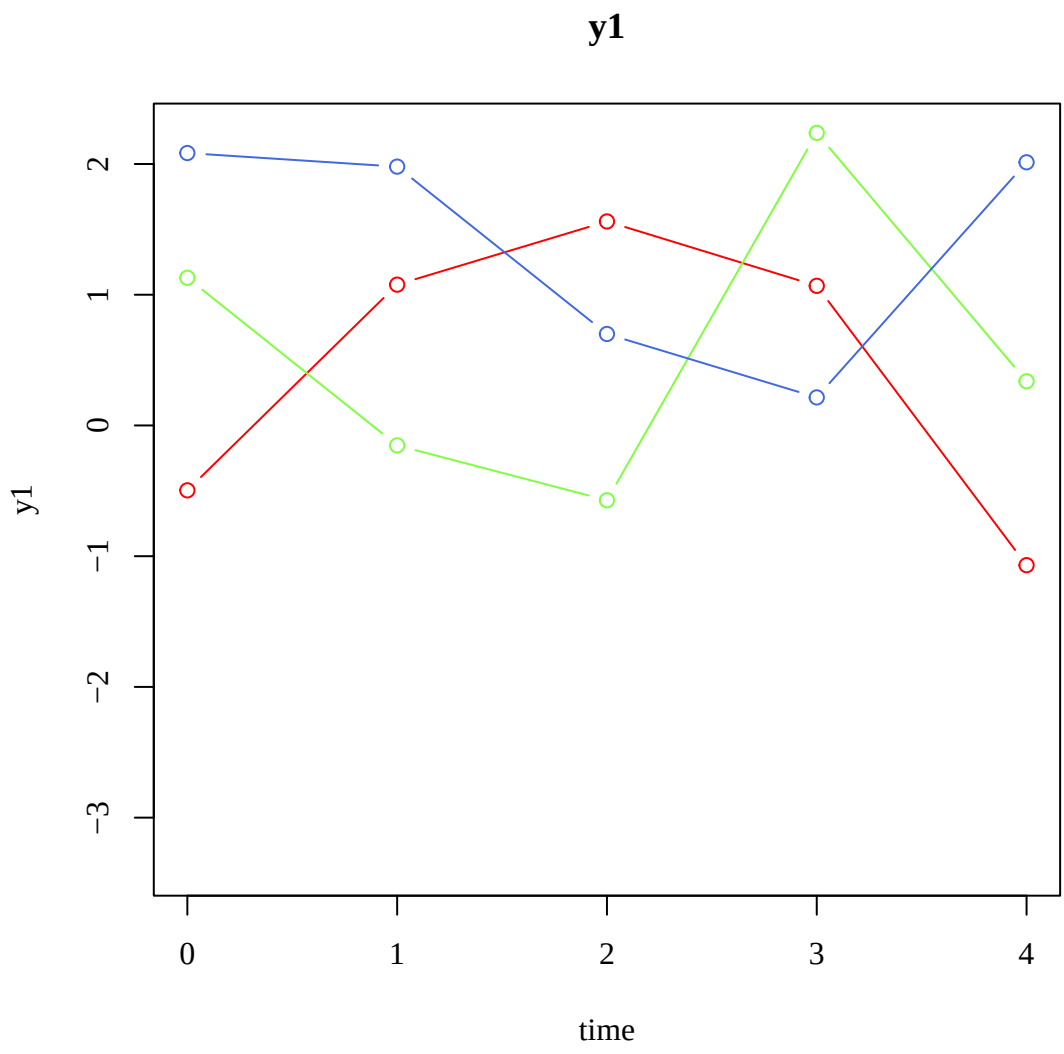

y3
y3
3
2
1
0
−1
−2
−3
0
1
2
3
4
time

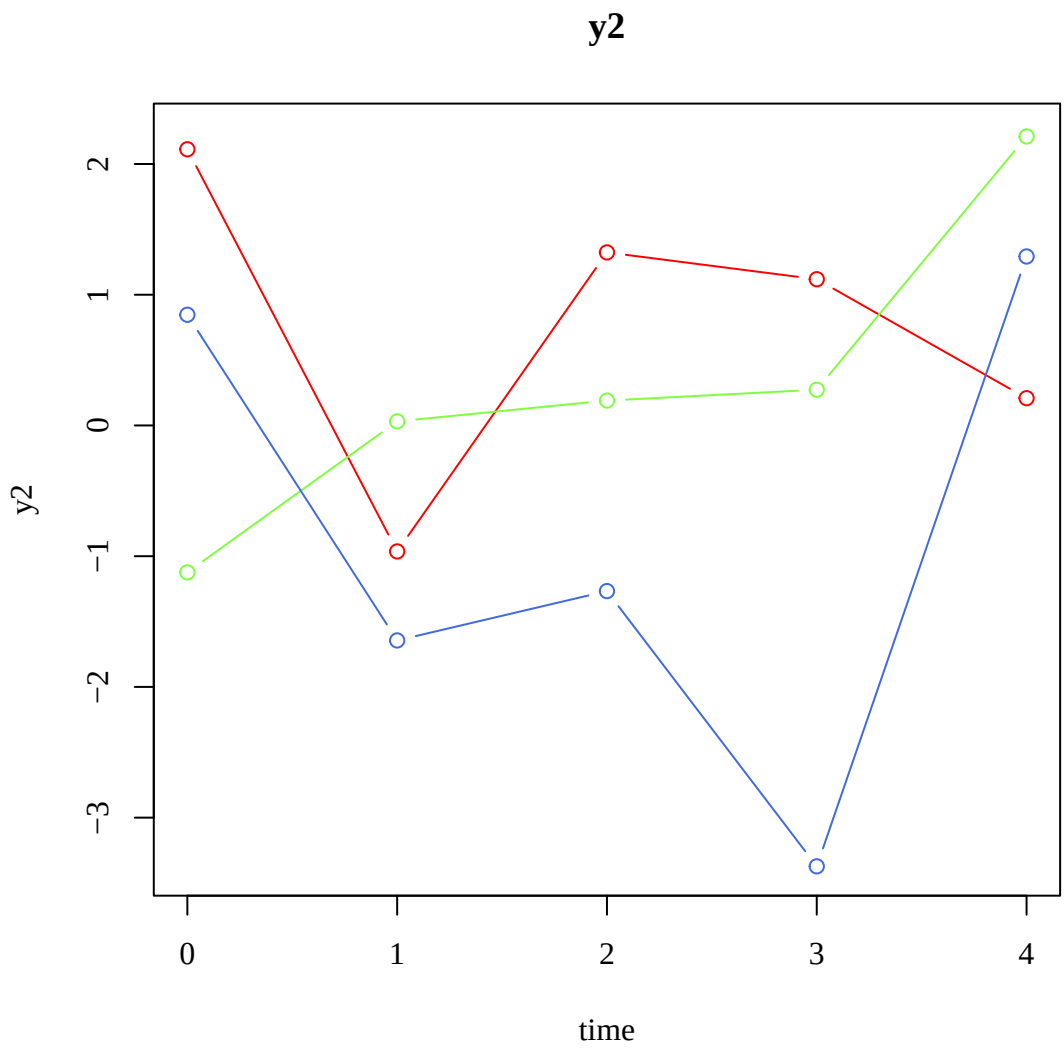

eta1
eta1
2
0
−2
−4
0
10
20
30
40
50
time

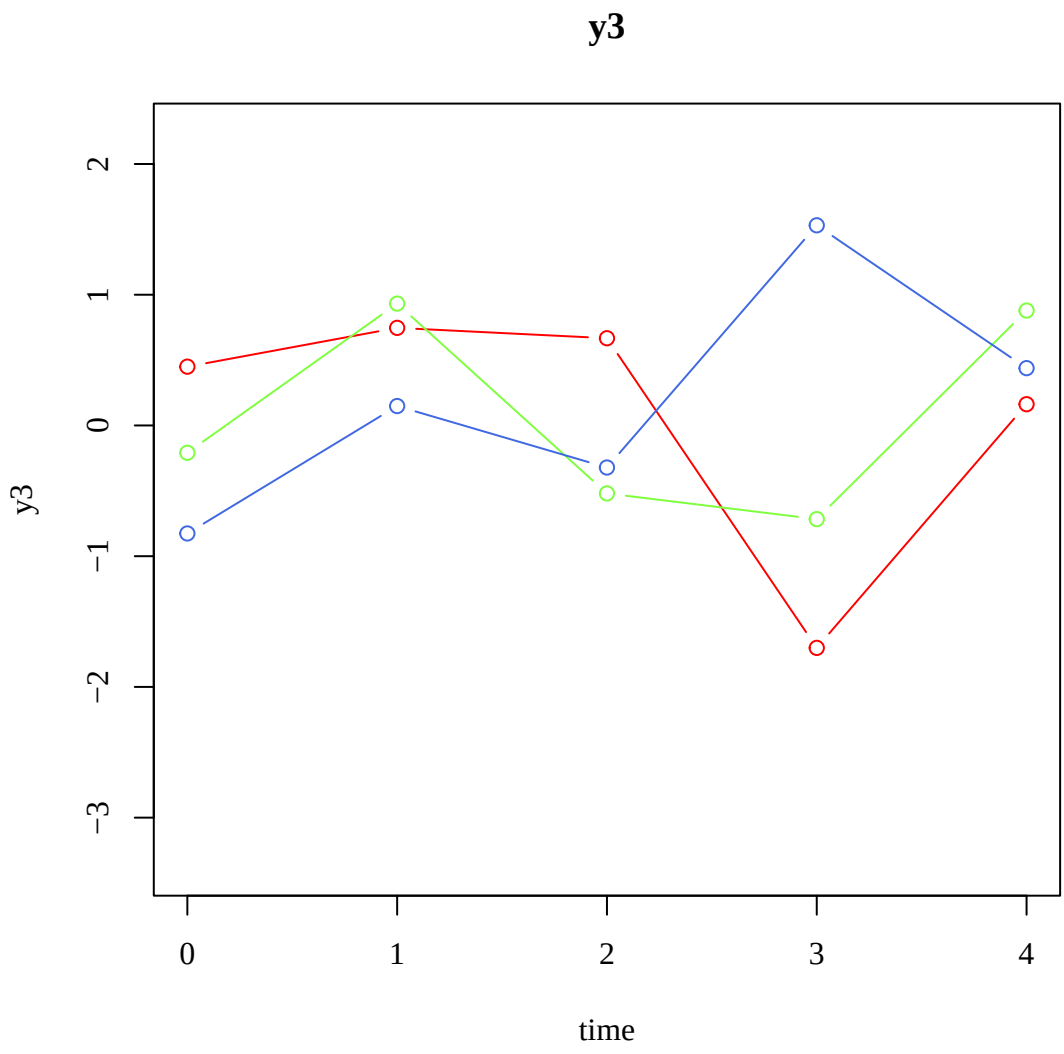

eta2
eta2
2
0
−2
−4
0
10
20
30
40
50
time

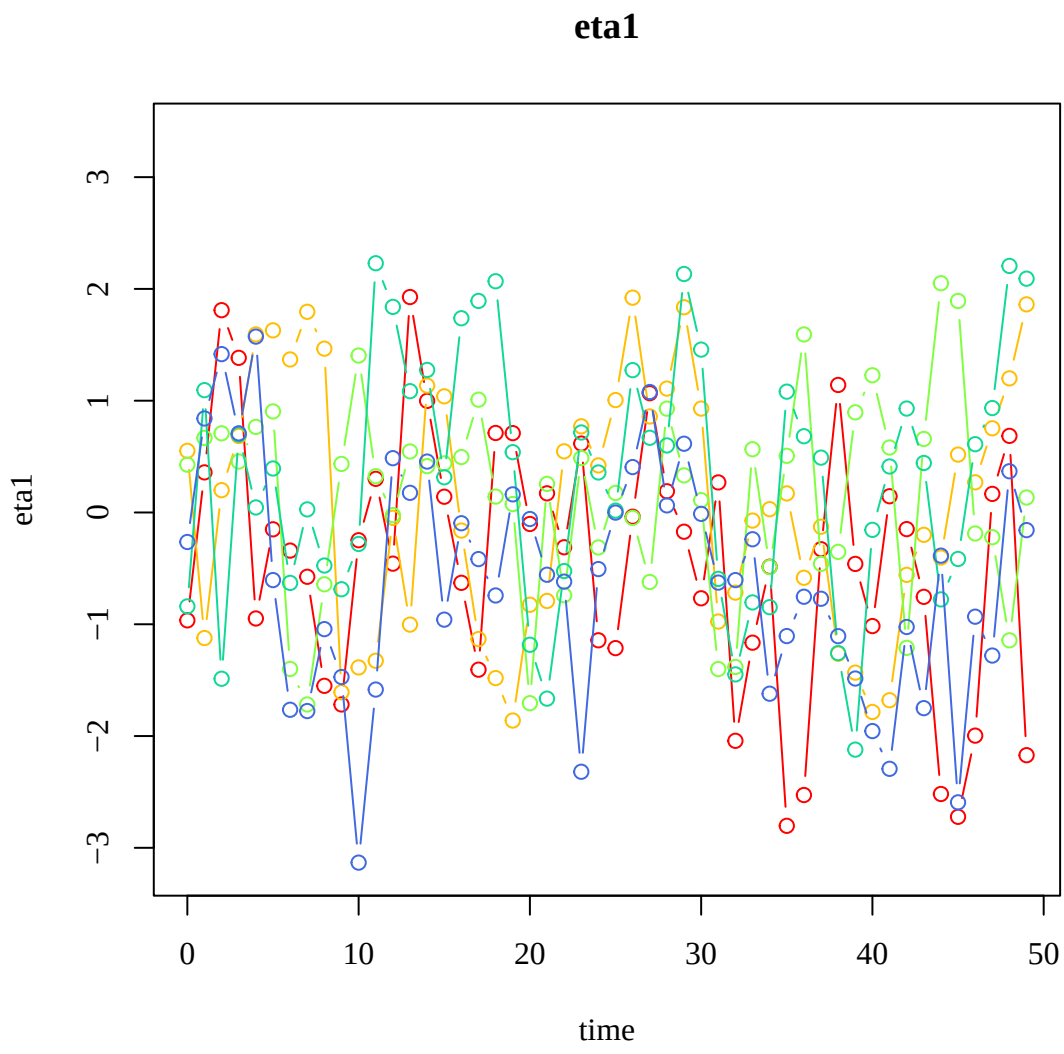



```
#> Call:
#> simStateSpace::SimSSMFixed(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, nu = nu, lambda = lambda,
#>     theta = theta, gamma_y = gamma_y, gamma_eta = gamma_eta,
#>     x = x, type = 2, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMFixed`
#> to convert it to a data frame or a matrix.
```


y1
y1
2
1
0
−1
−2
−3
0
1
2
3
4
time


y2
y2
2
1
0
−1
−2
−3
0
1
2
3
4
time


y3
y3
2
1
0
−1
−2
−3
0
1
2
3
4
time


eta1
eta1
3
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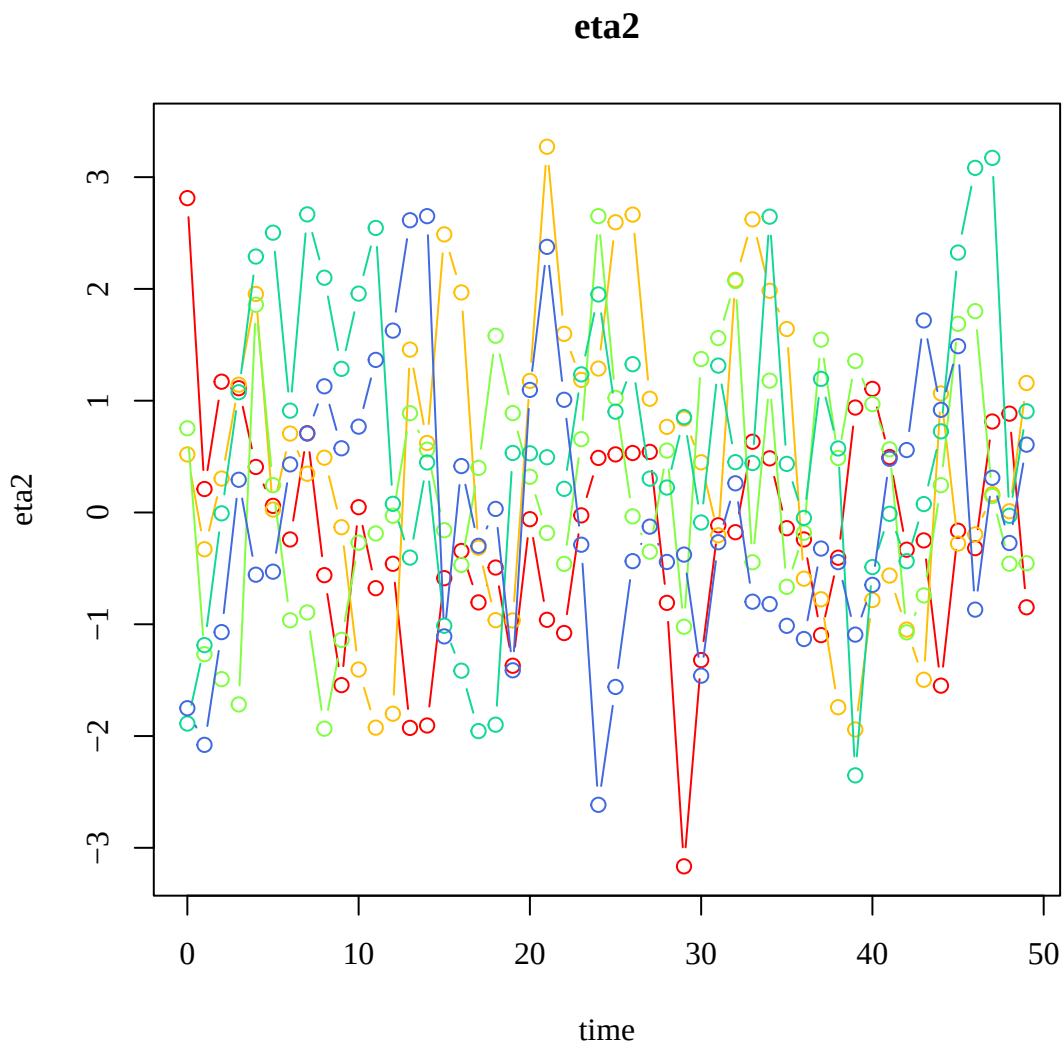
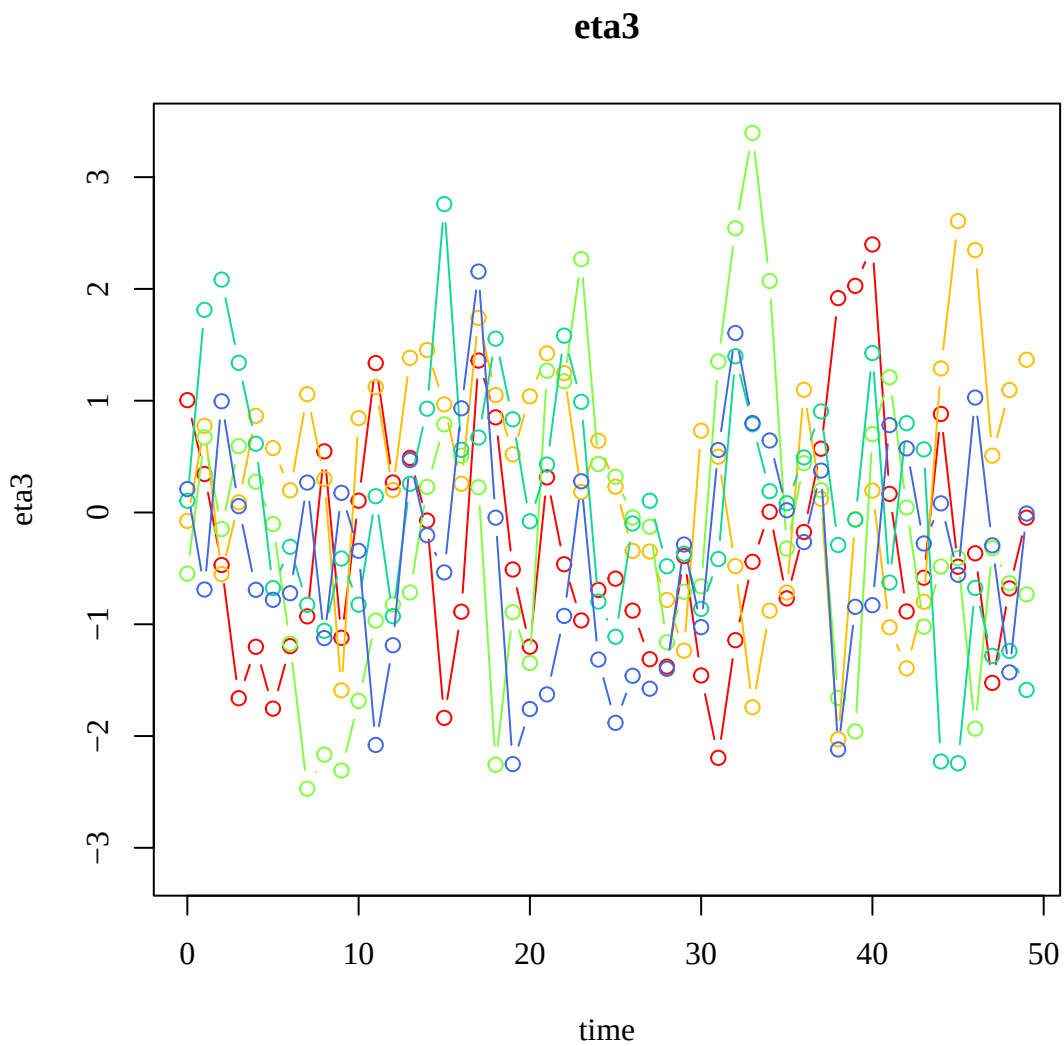
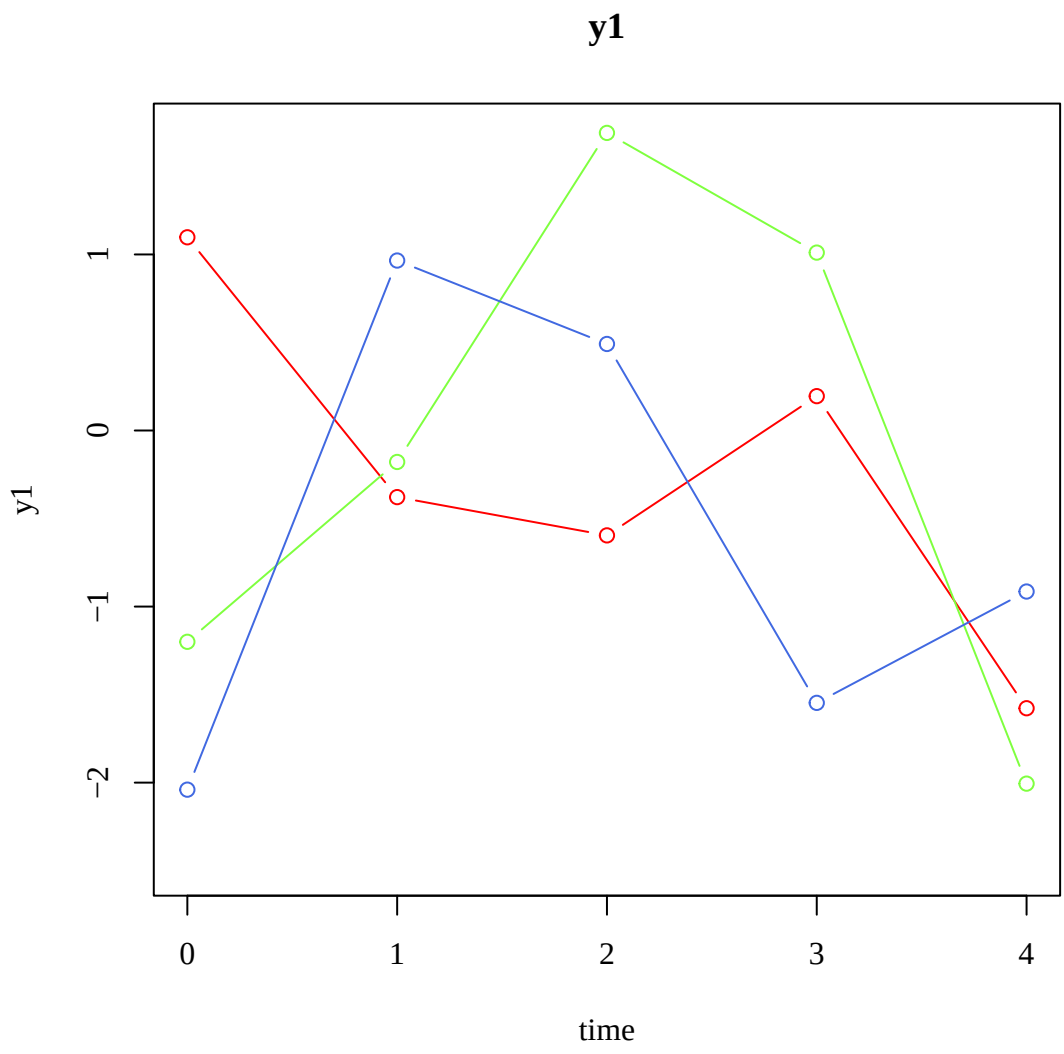
1
0
−1
−2
−3
0
10
20
30
40
50
time

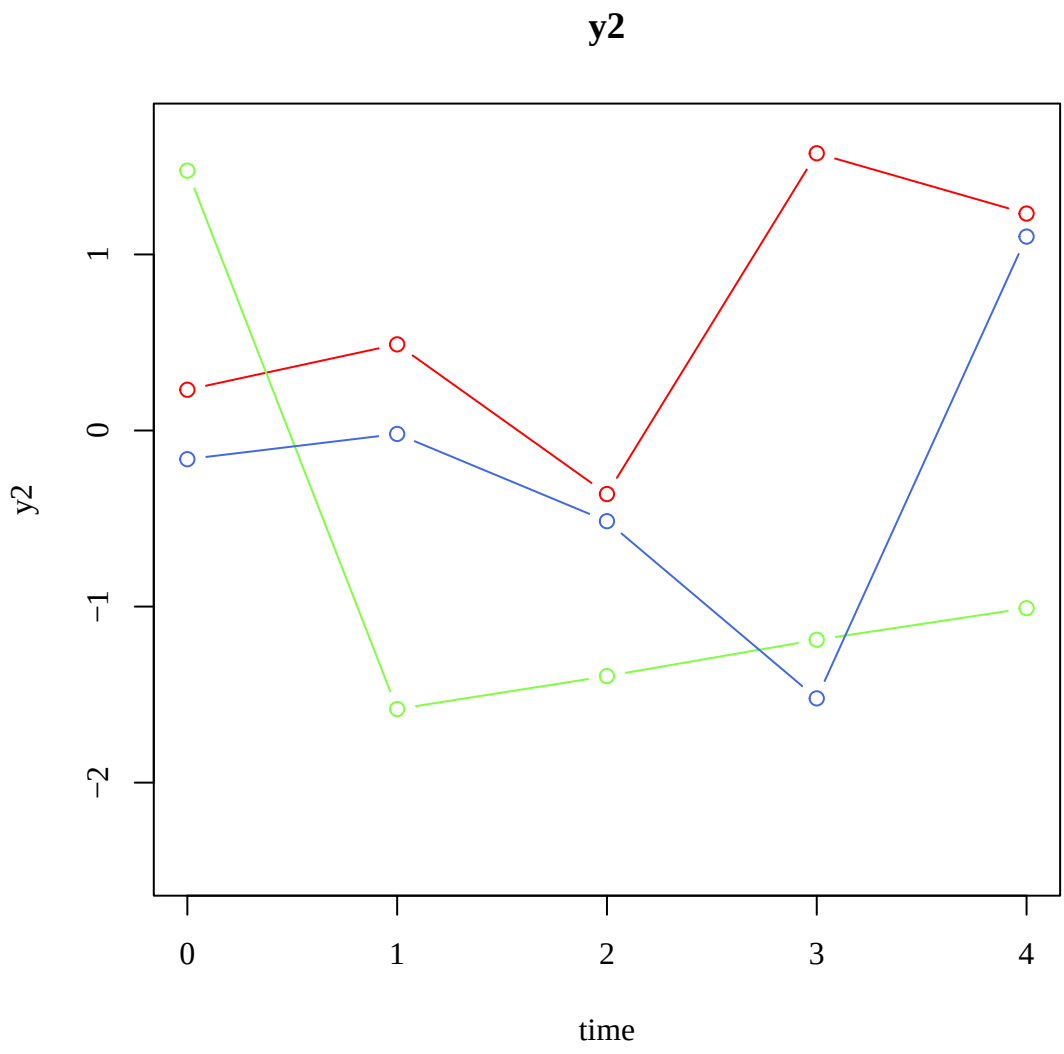


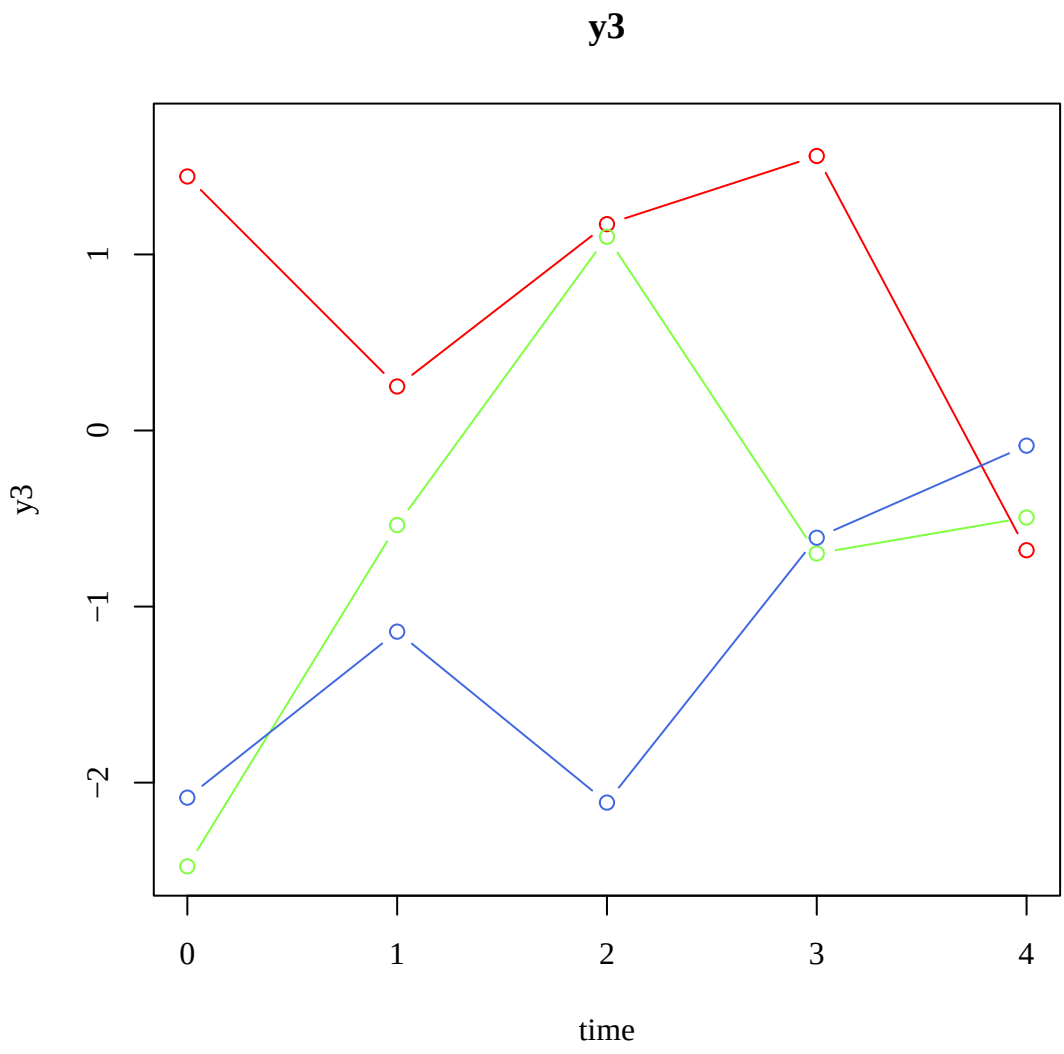

```
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-sim-ssm-i-vary
```

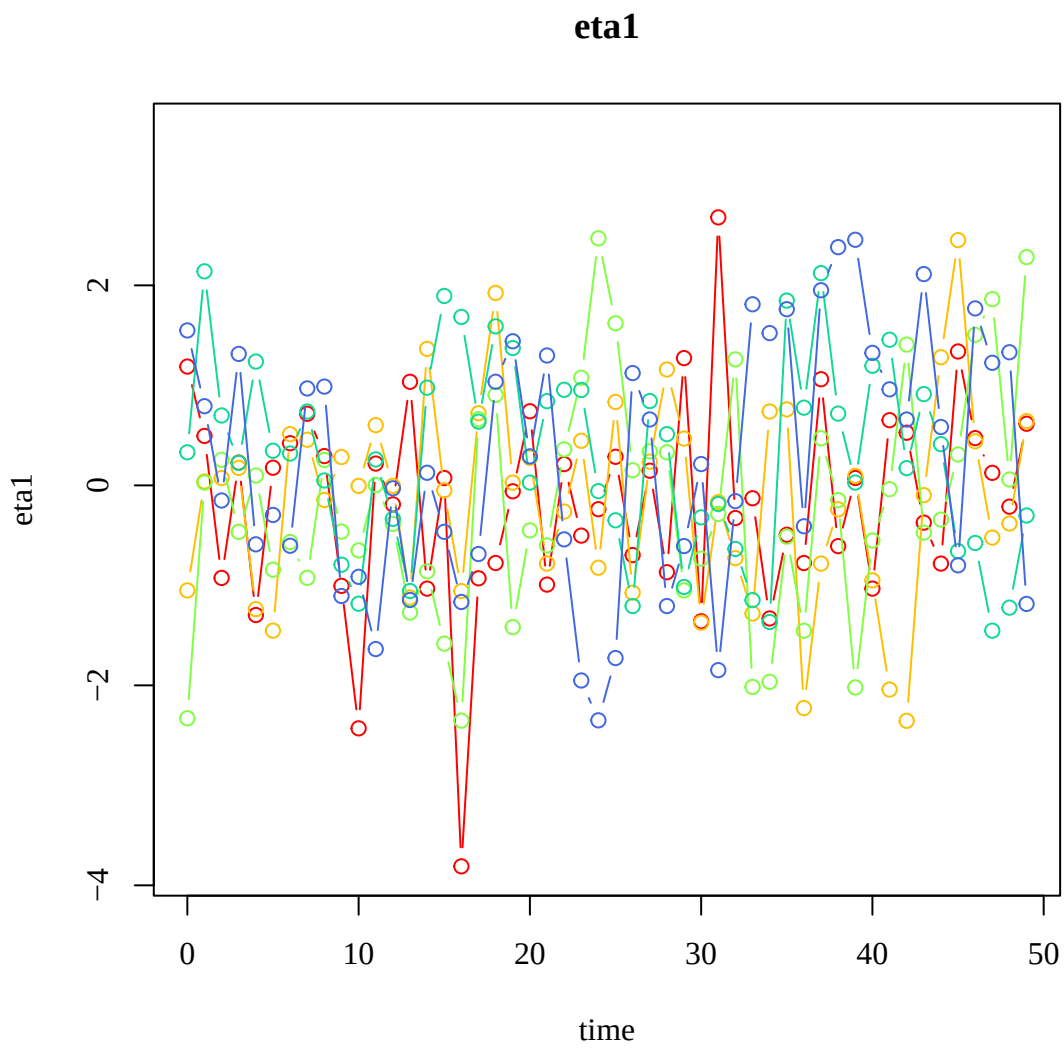


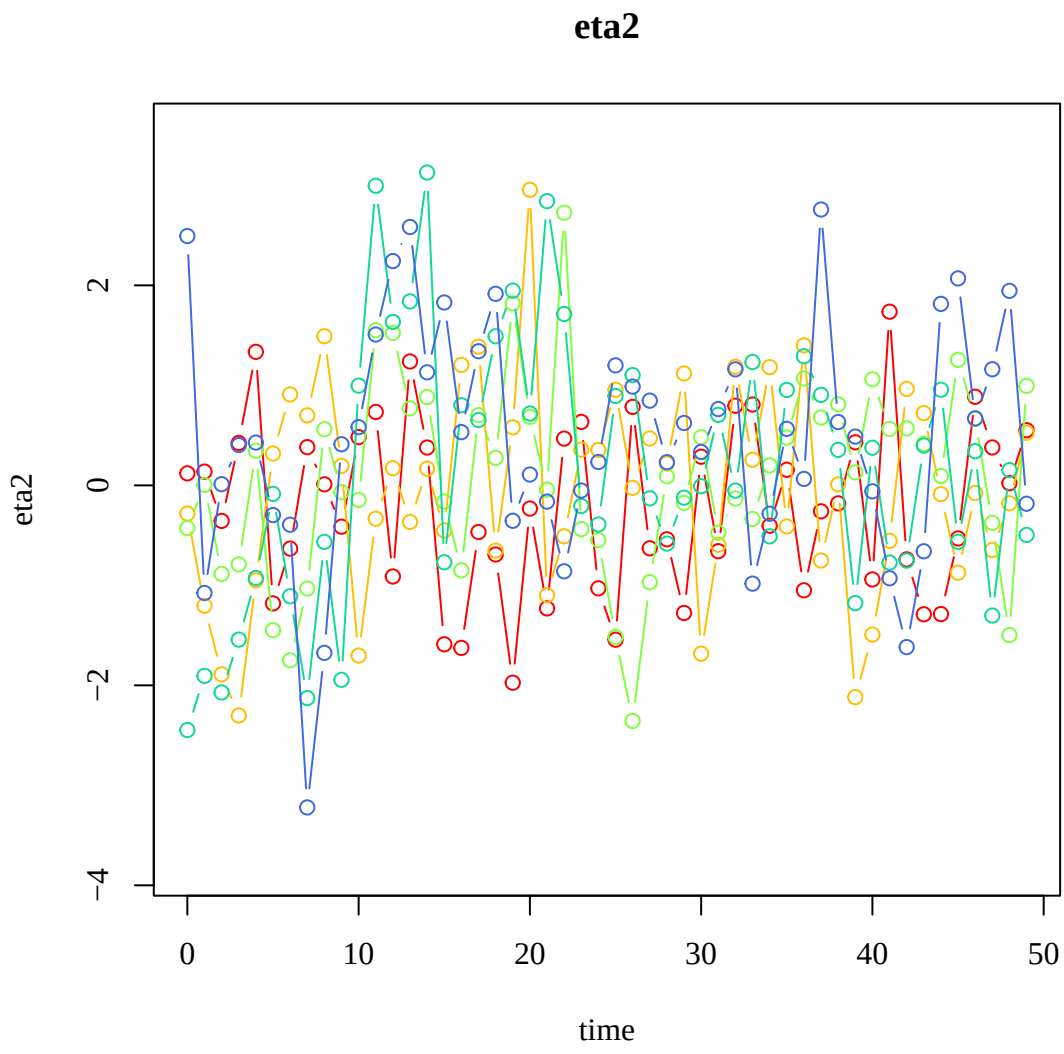

```
#> Call:
#> simStateSpace::SimSSMIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, nu = nu, lambda = lambda,
#>     theta = theta, type = 0, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMIVary`
#> to convert it to a data frame or a matrix.
```

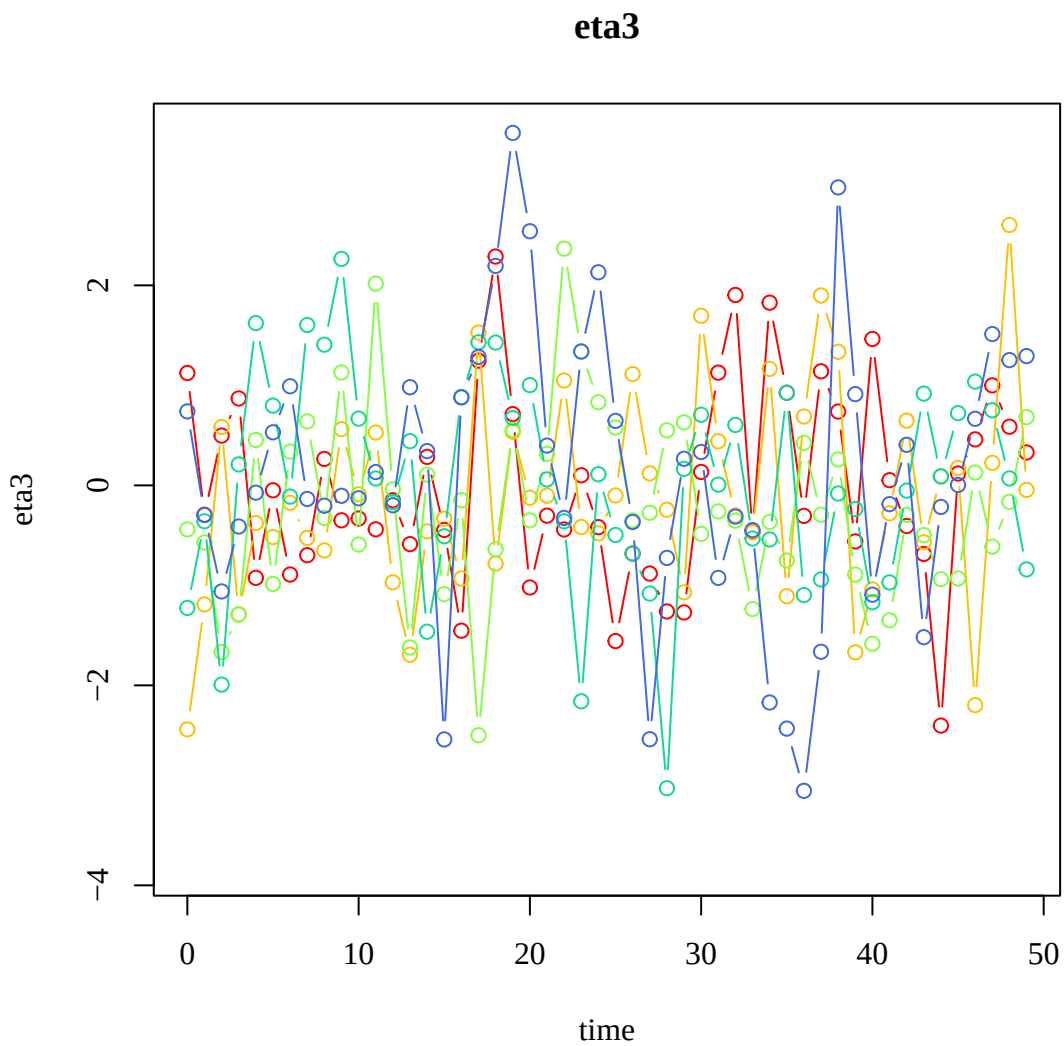

y1
y1
1
0
−1
−2
0
1
2
3
4
time


y2
y2
1
0
−1
−2
0
1
2
3
4
time

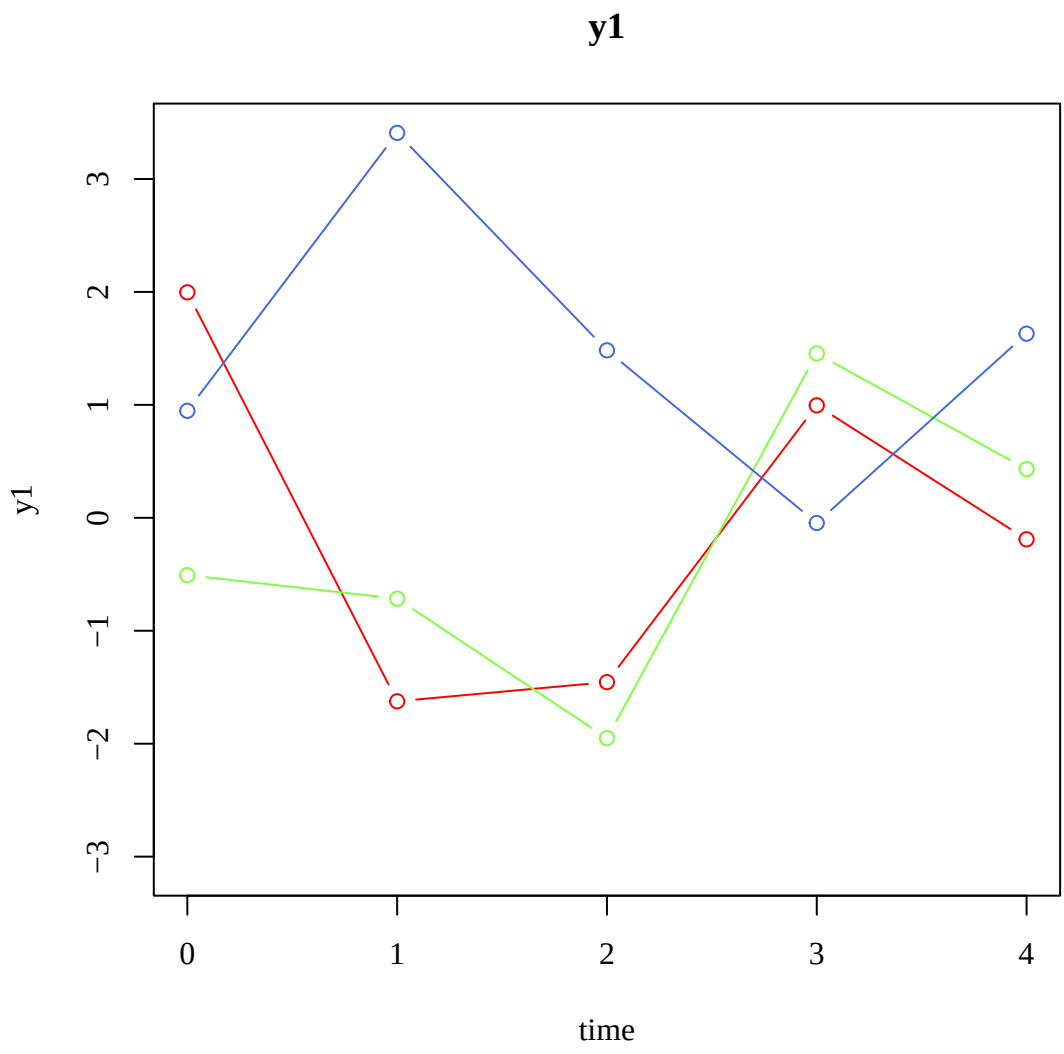

y3
y3
1
0
−1
−2
0
1
2
3
4
time

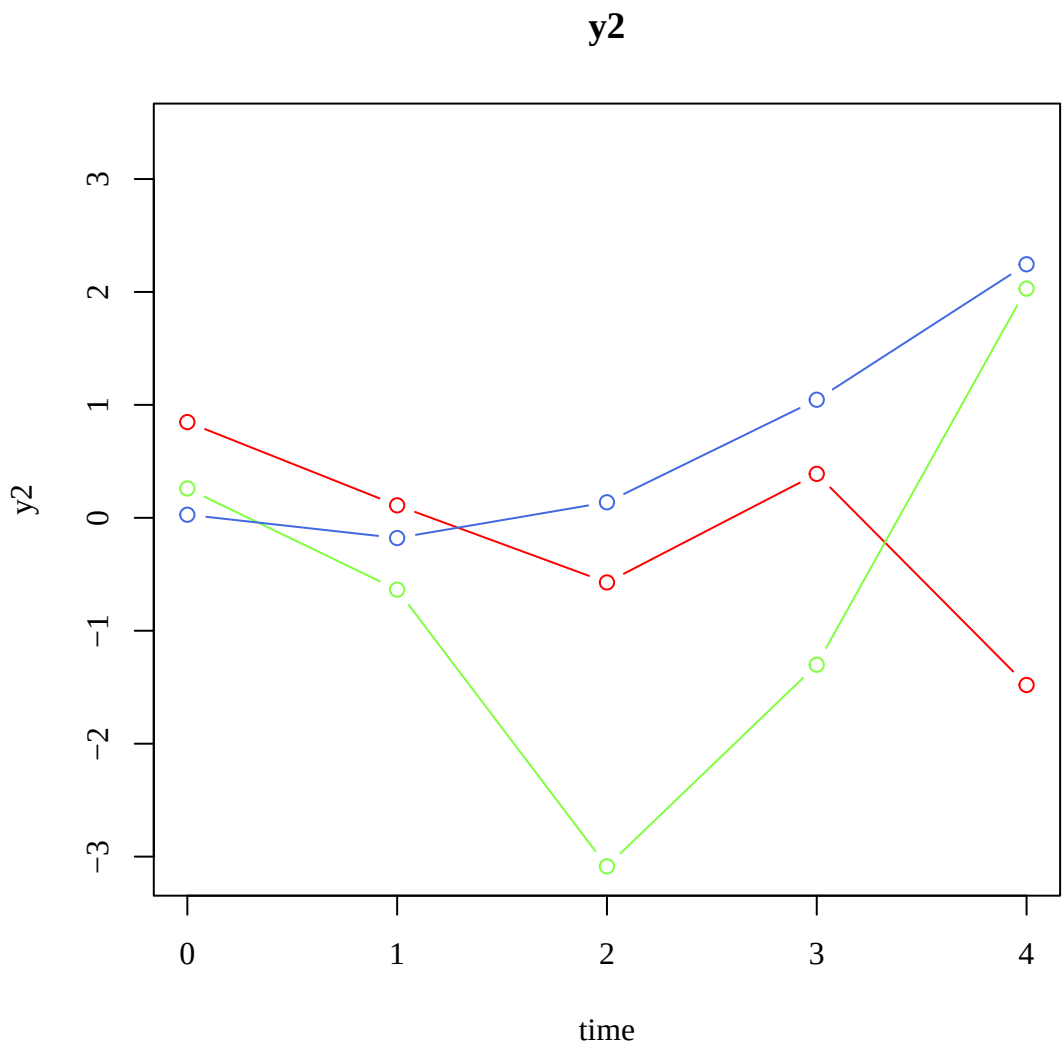

eta1
eta1
2
0
−2
−4
0
10
20
30
40
50
time

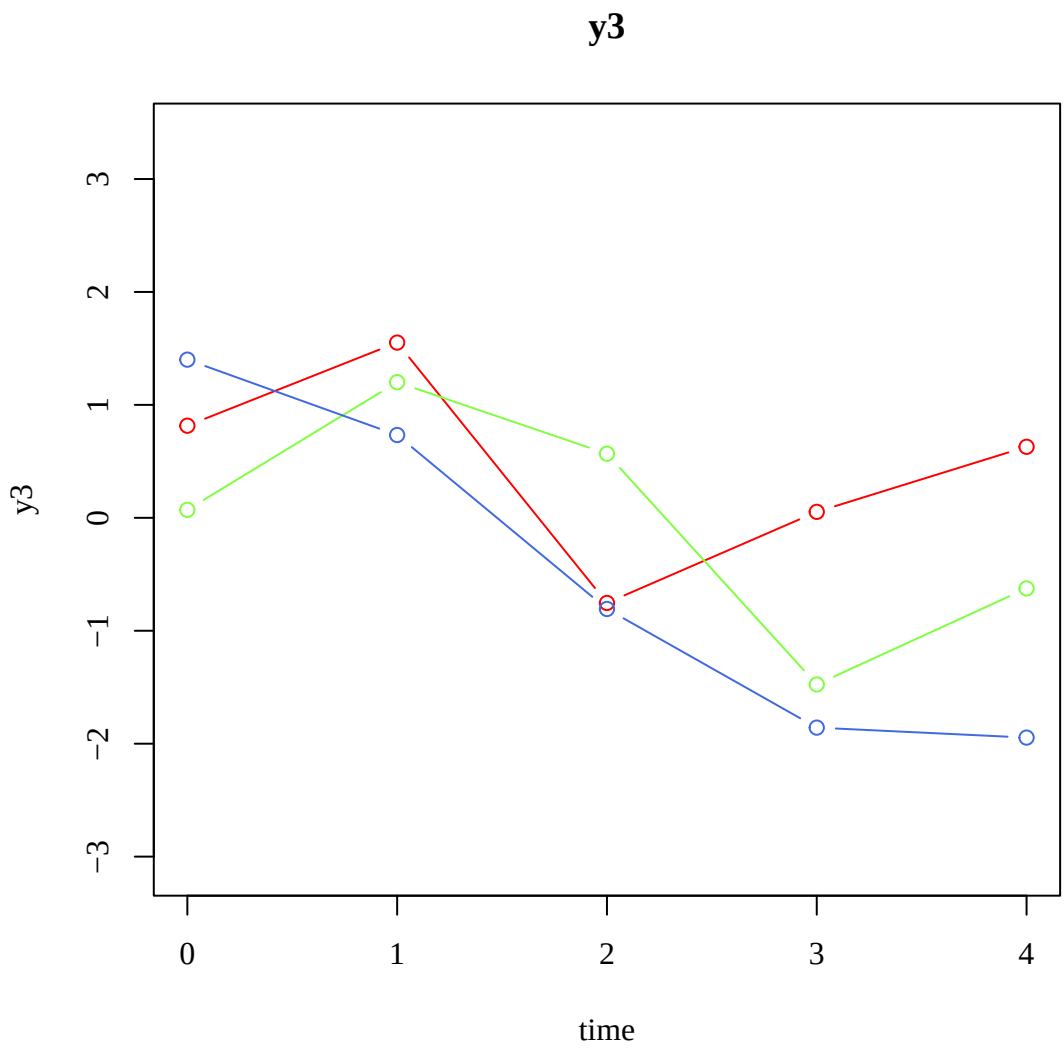

eta2
eta2
2
0
−2
−4
0
10
20
30
40
50
time

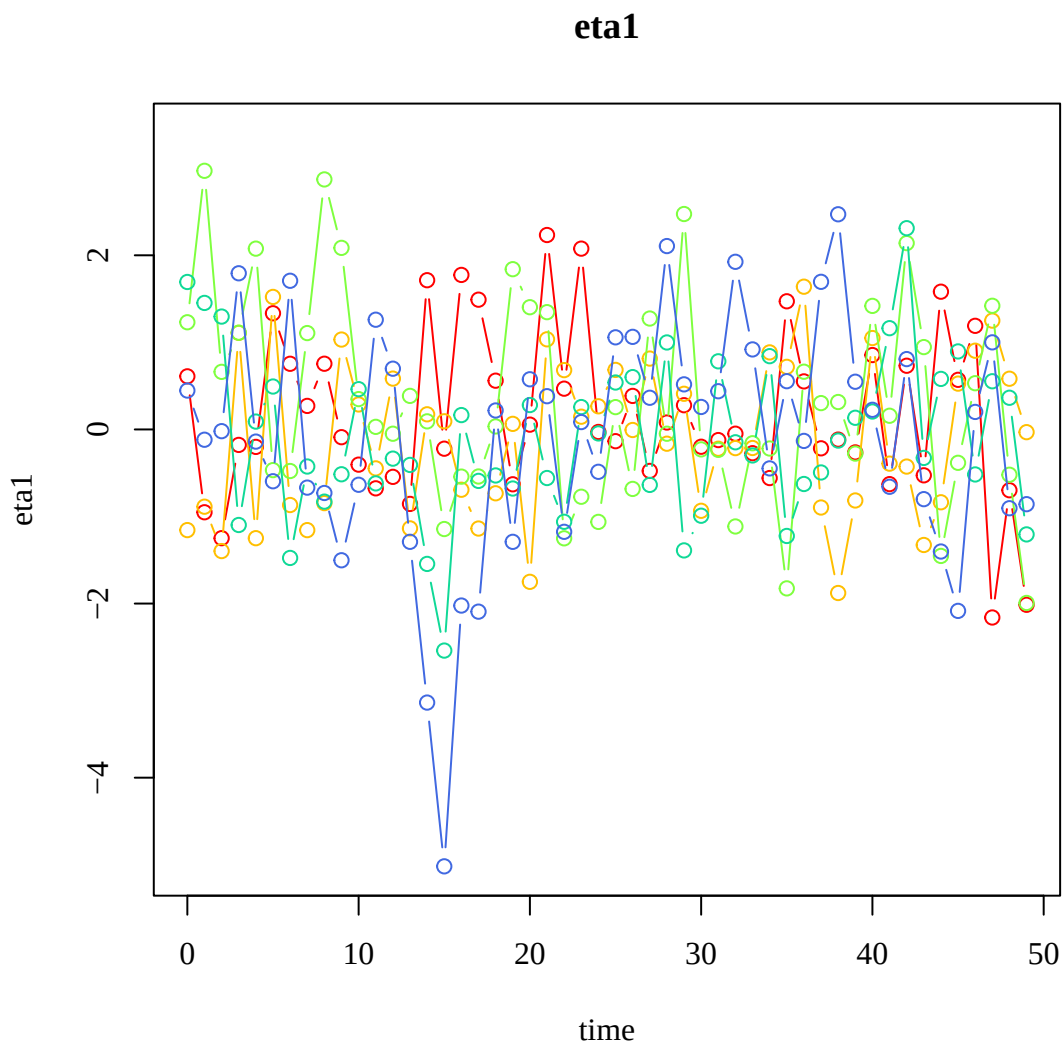


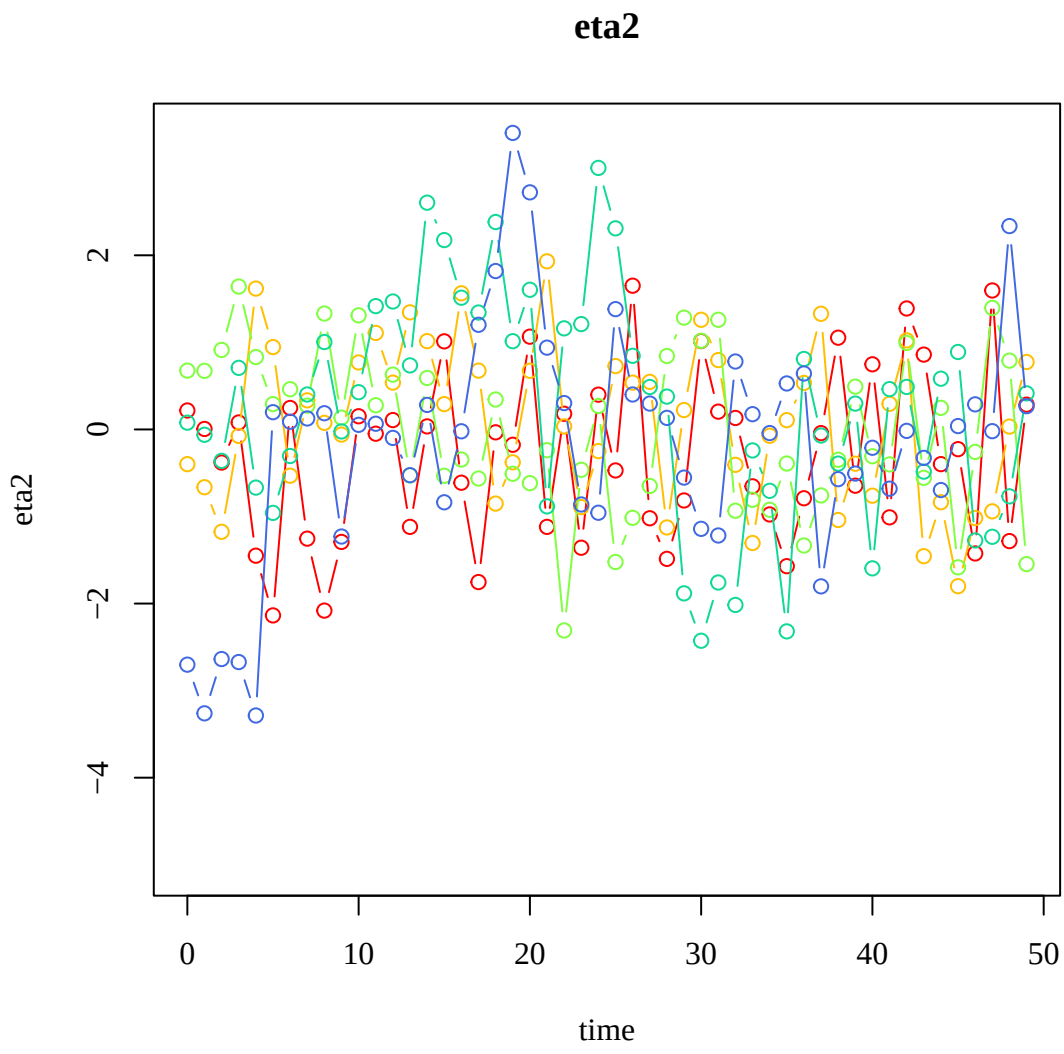

```
#> Call:
#> simStateSpace::SimSSMIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, nu = nu, lambda = lambda,
#>     theta = theta, gamma_eta = gamma_eta, x = x, type = 1, time = time,
#>     burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMIVary`
#> to convert it to a data frame or a matrix.
```

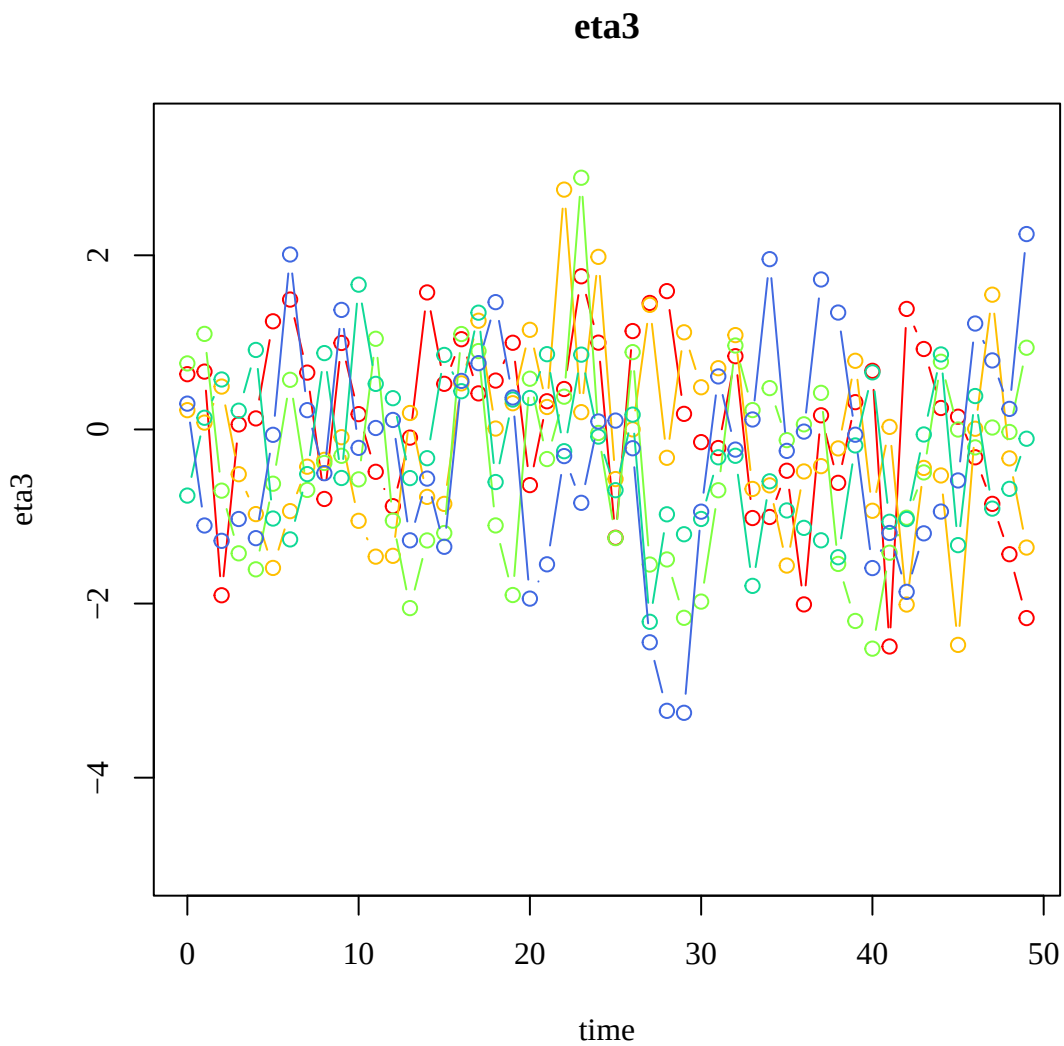

y1
y1
3
2
1
0
−1
−2
−3
0
1
2
3
4
time


y2
y2
3
2
1
0
−1
−2
−3
0
1
2
3
4
time

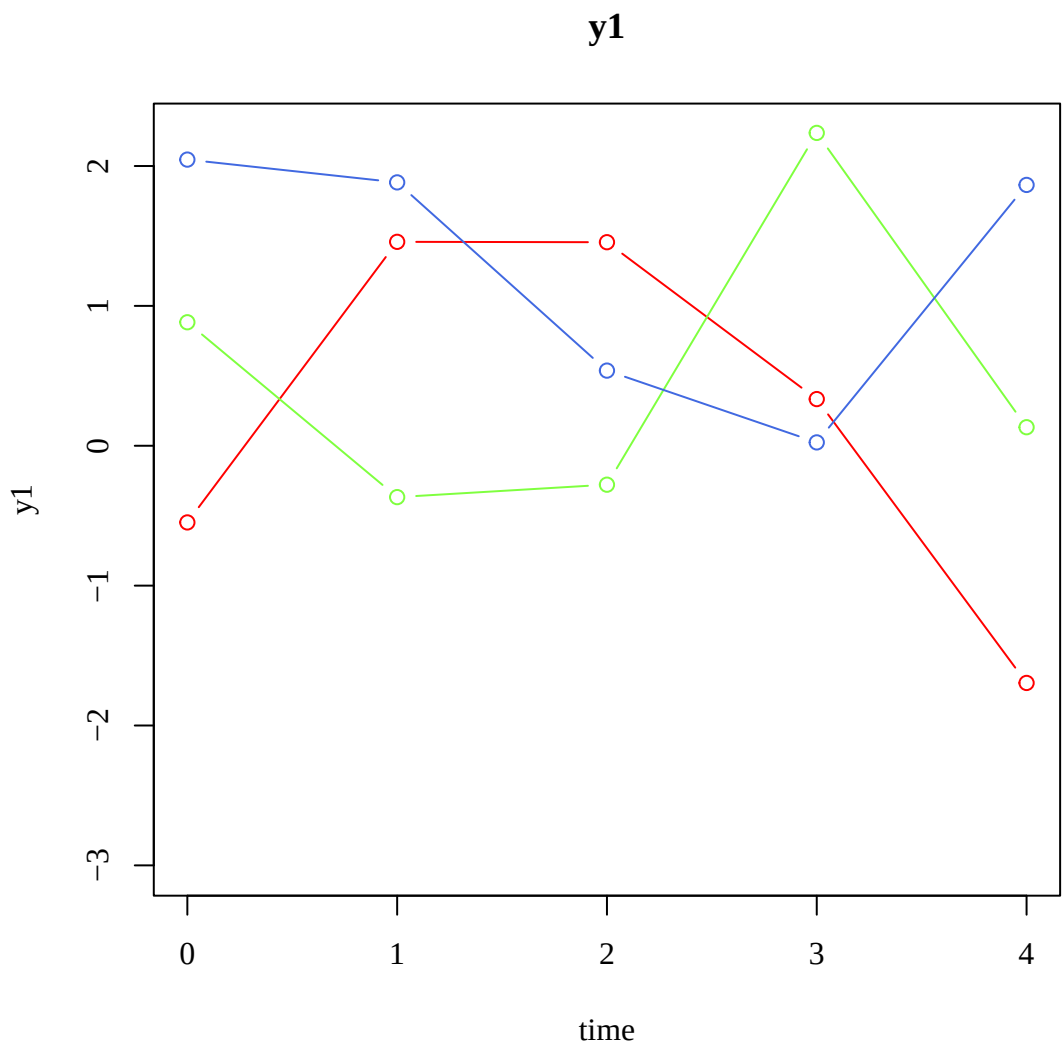

y3
y3
3
2
1
0
−1
−2
−3
0
1
2
3
4
time

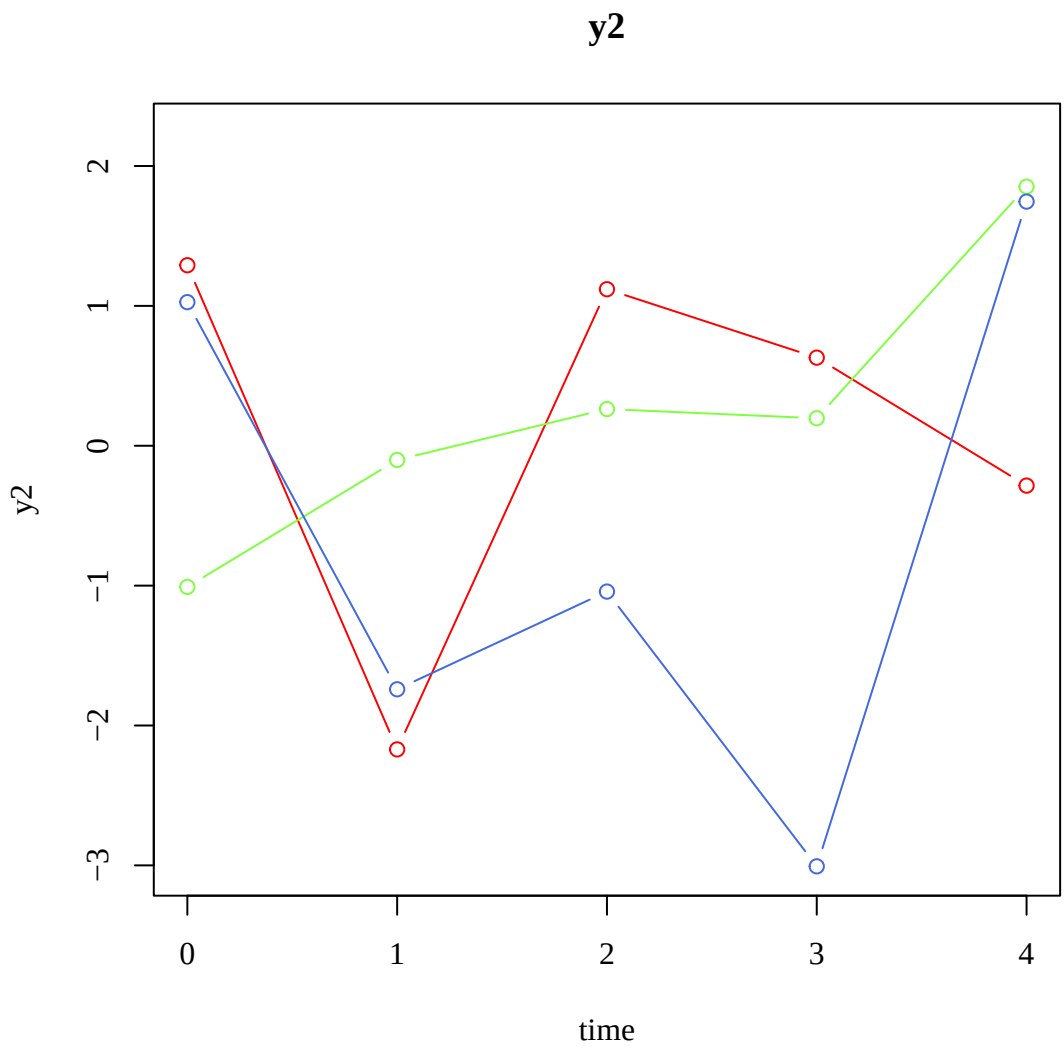

eta1
eta1
2
0
−2
−4
0
10
20
30
40
50
time

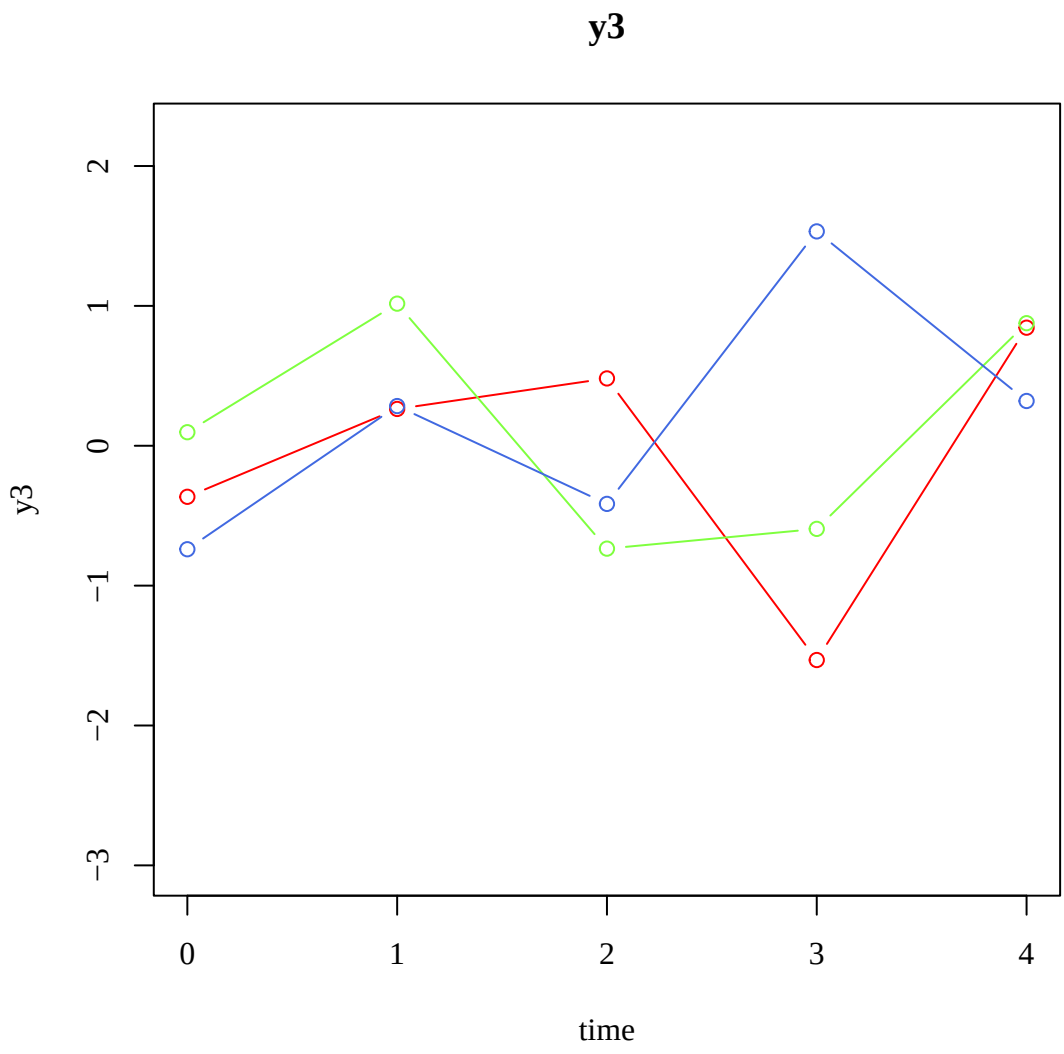

eta2
eta2
2
0
−2
−4
0
10
20
30
40
50
time

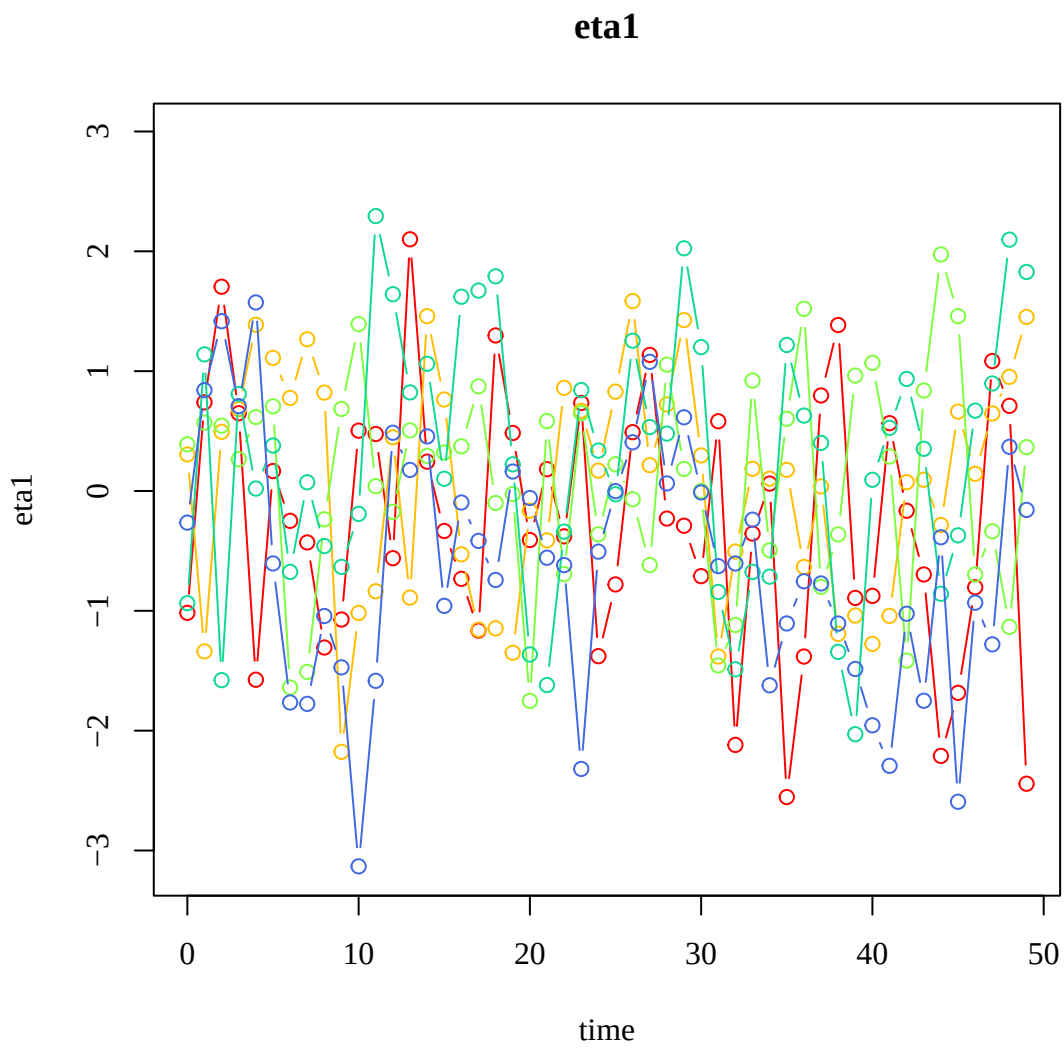


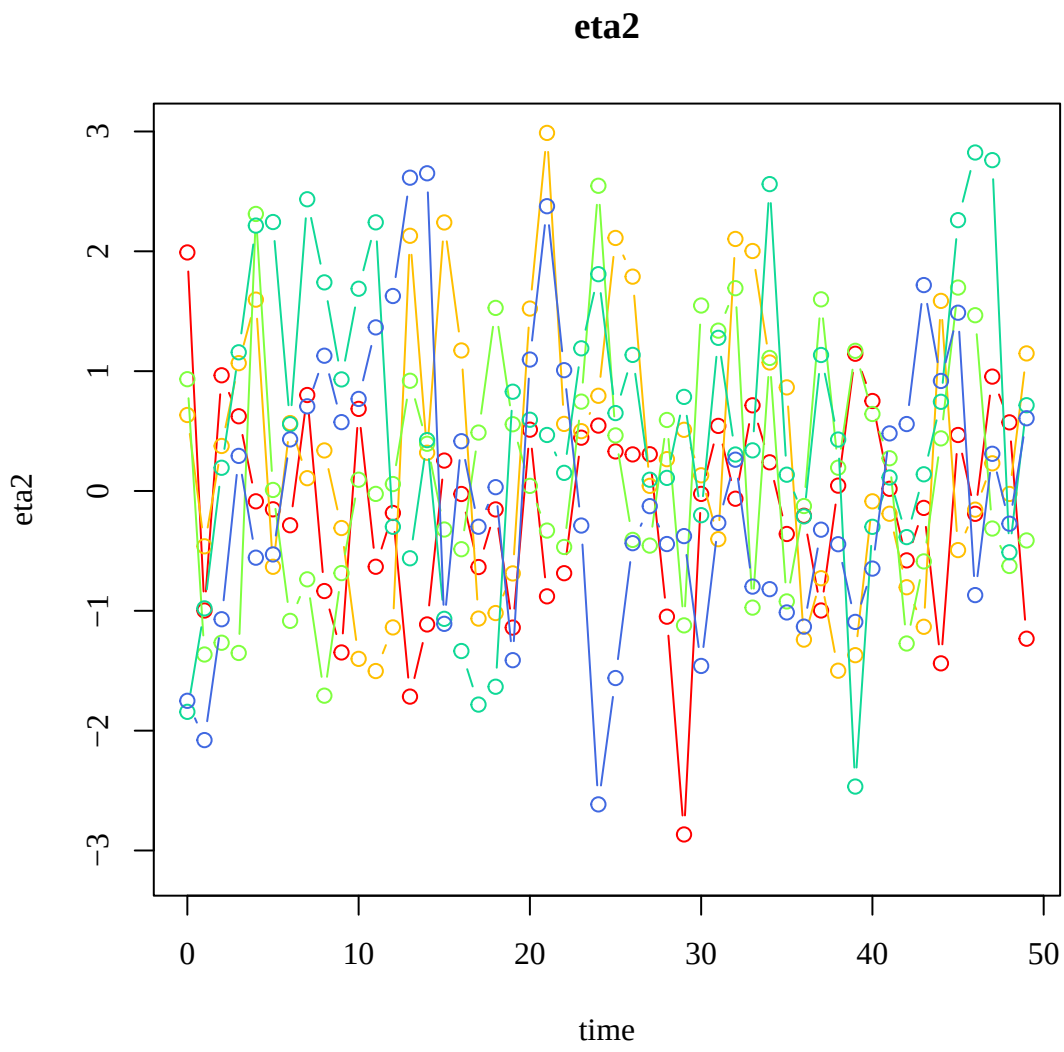

```
#> Call:
#> simStateSpace::SimSSMIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, nu = nu, lambda = lambda,
#>     theta = theta, gamma_y = gamma_y, gamma_eta = gamma_eta,
#>     x = x, type = 2, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMIVary`
#> to convert it to a data frame or a matrix.
```


y1
y1
2
1
0
−1
−2
−3
0
1
2
3
4
time

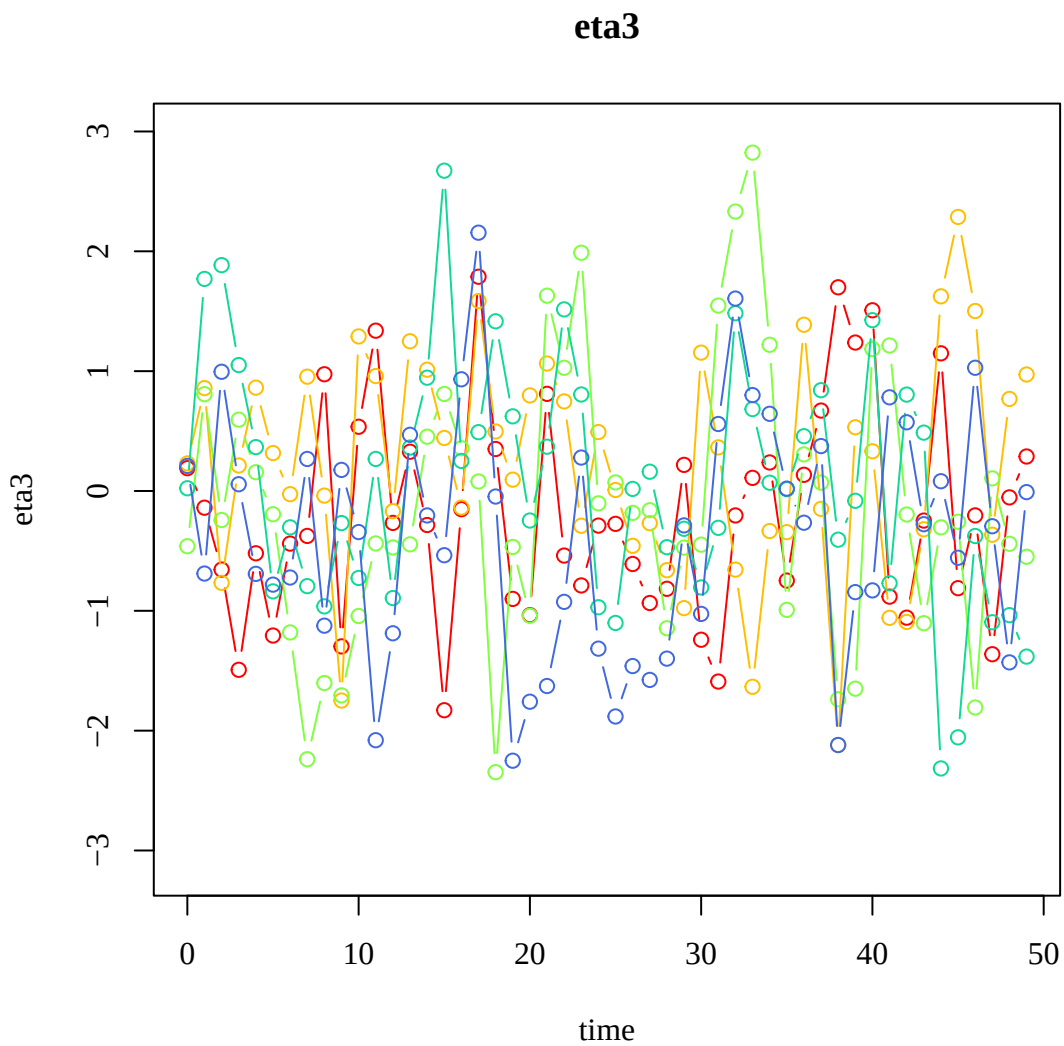

y2
y2
2
1
0
−1
−2
−3
0
1
2
3
4
time


y3
2
1
0
−1
−2
−3
y3
0
1
2
3
4
time

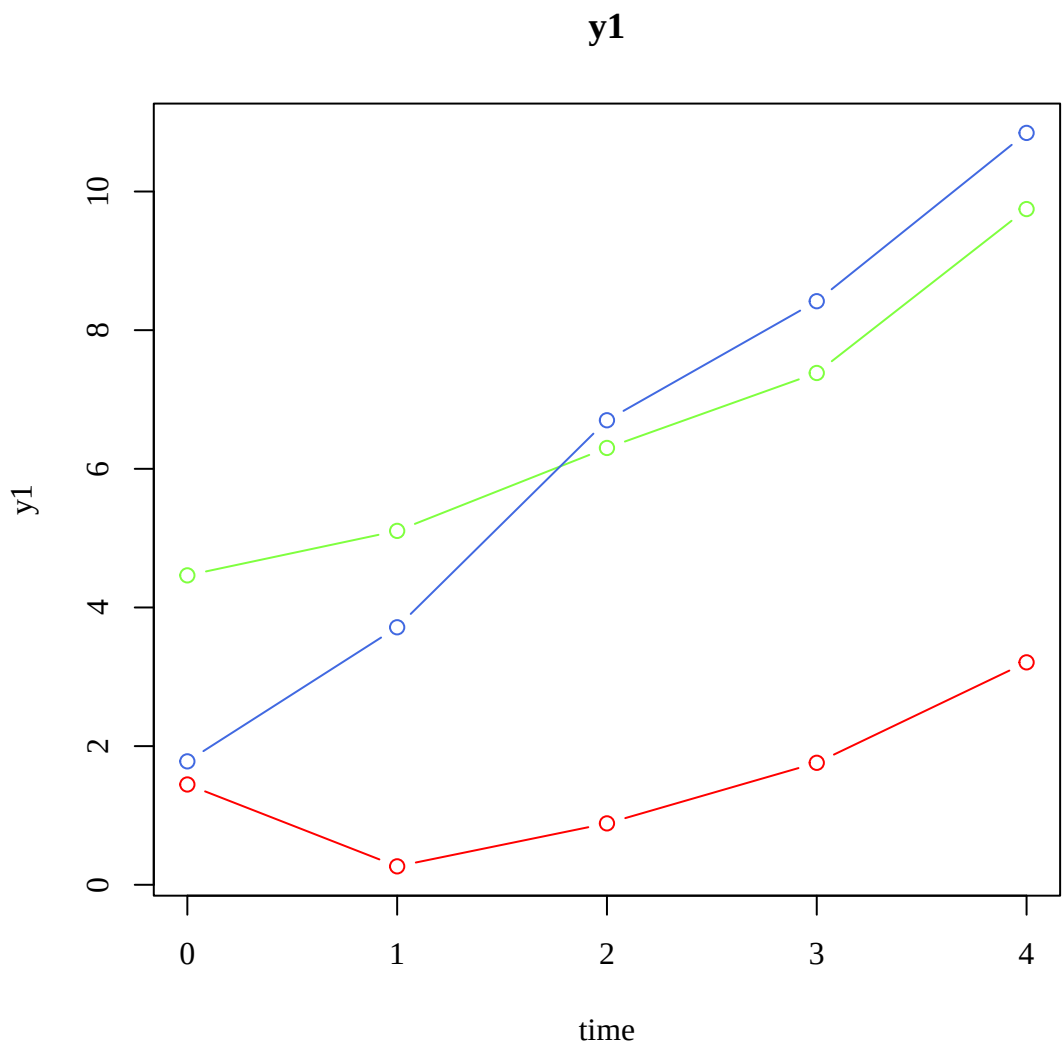

eta1
eta1
time
3
2
1
0
−1
−2
−3
0
10
20
30
40
50

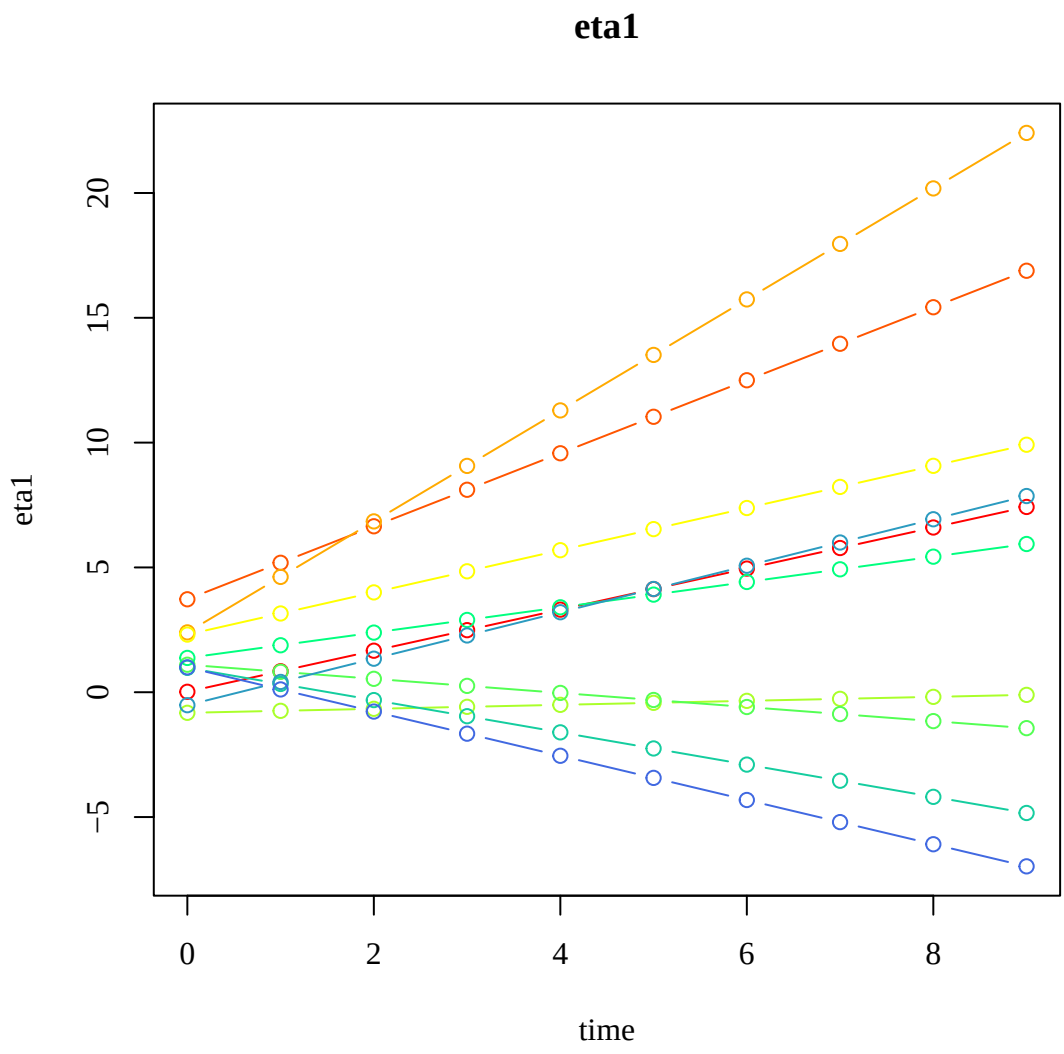


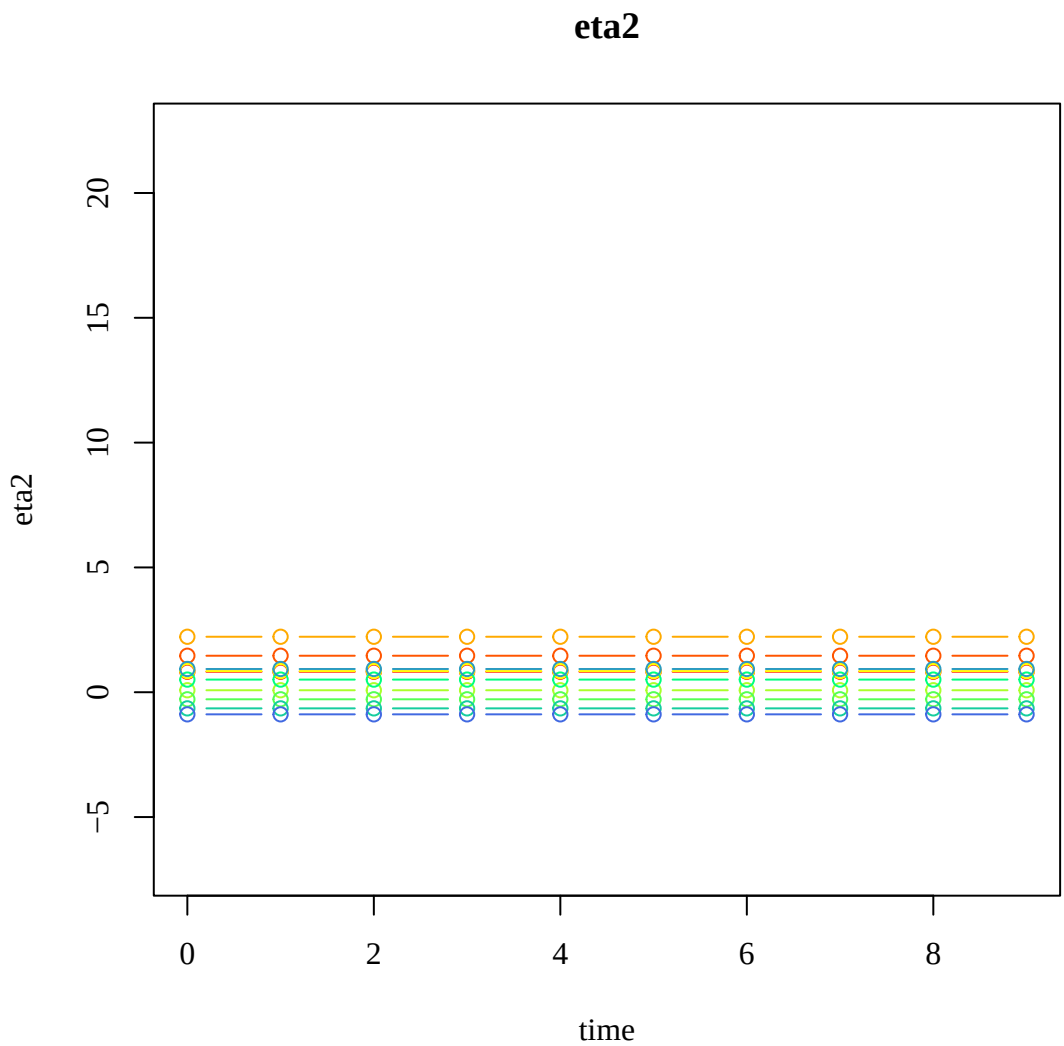

```
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-sim-ssm-lin-growth-i-vary
```



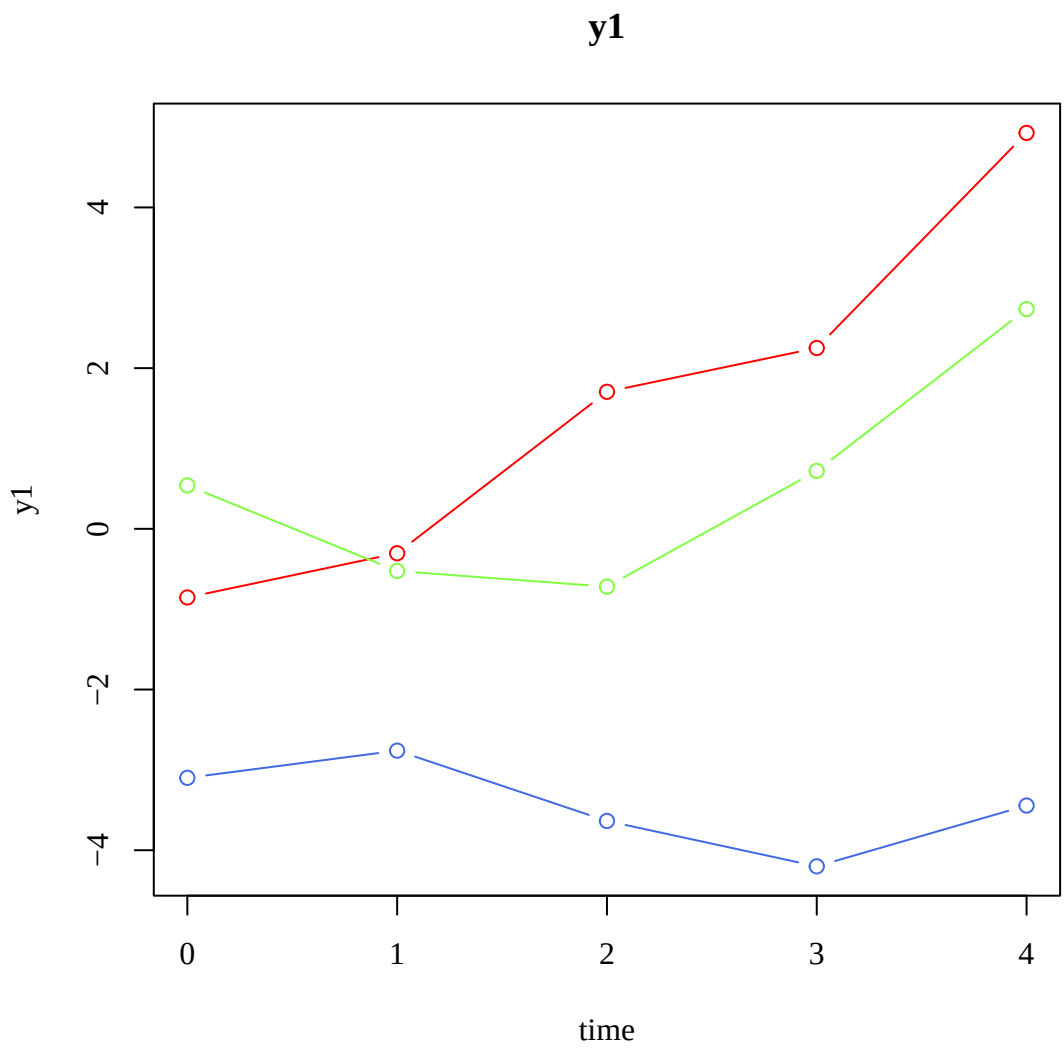

```
#> Call:
#> simStateSpace::SimSSMLinGrowthIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     theta = theta, type = 0, time = time)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMLinGrowthIVary`
#> to convert it to a data frame or a matrix.
```

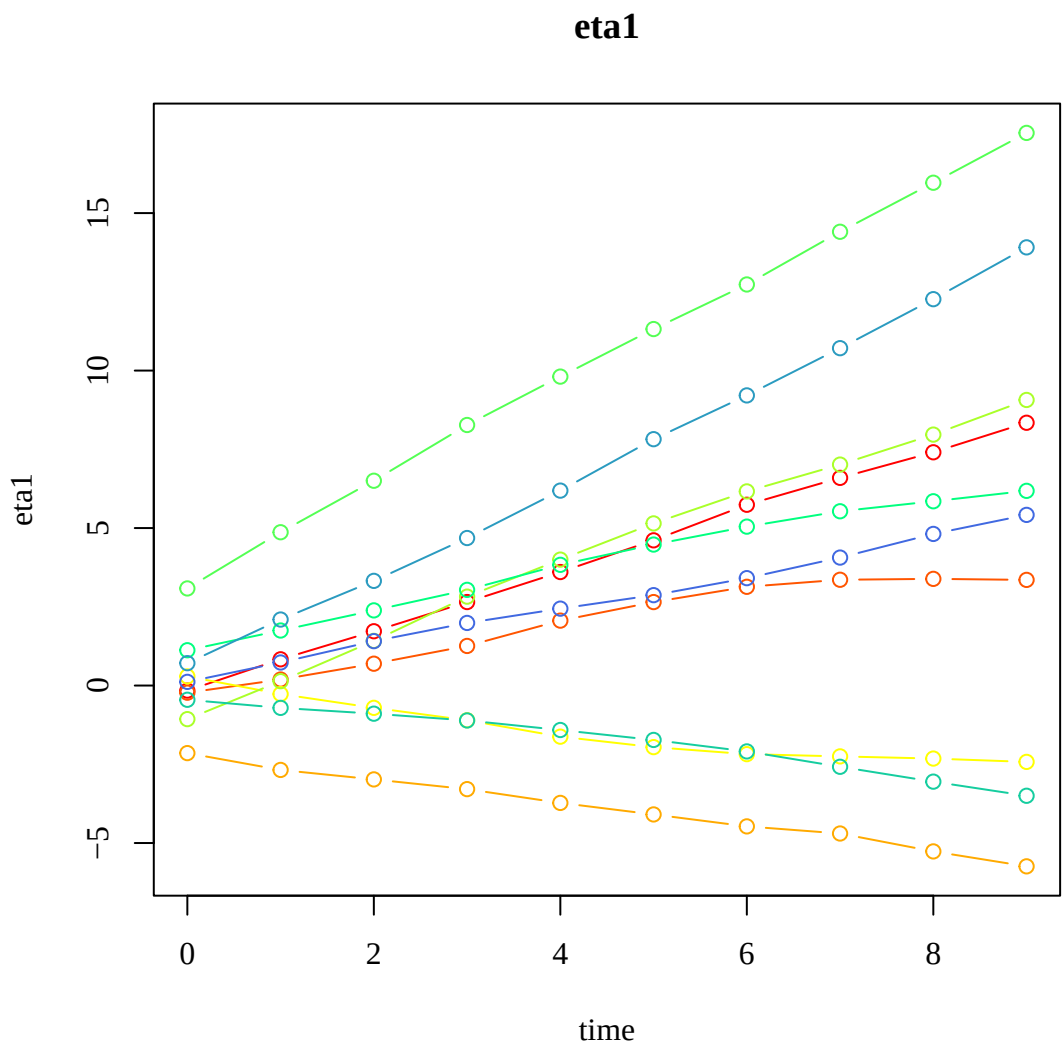

y1
y1
0
2
4
6
8
10
0
1
2
3
4
time

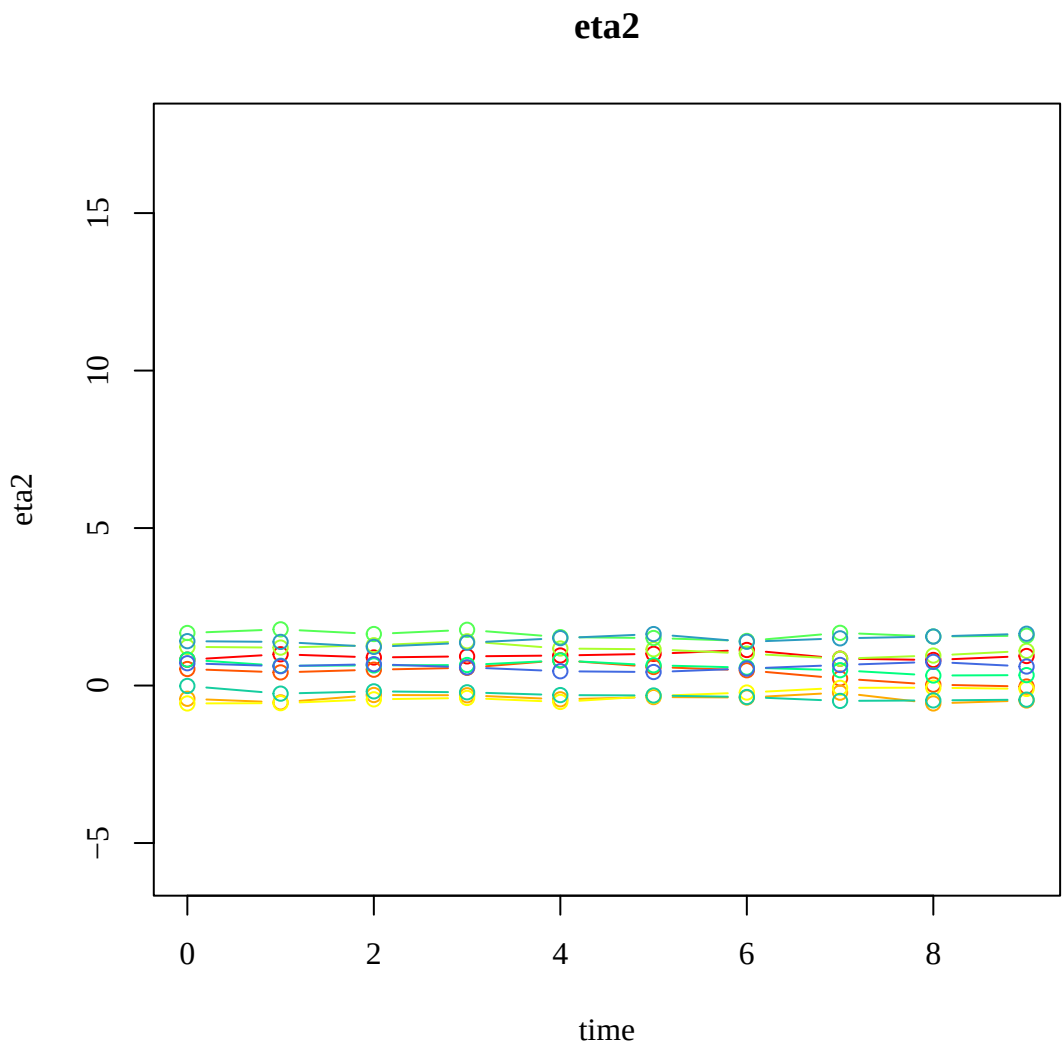

eta1
eta1
20
15
10
5
0
−5
0
2
4
6
8
time



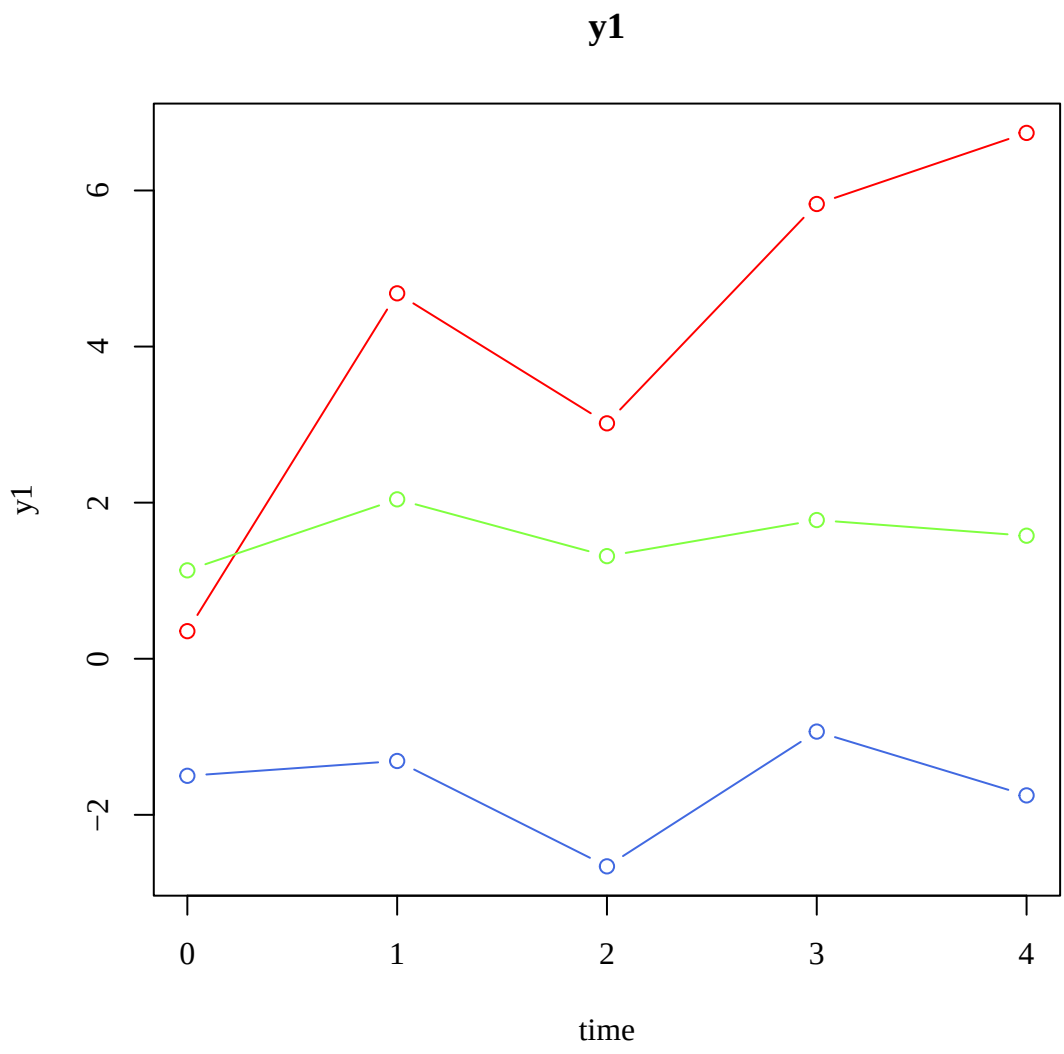

```
#> Call:
#> simStateSpace::SimSSMLinGrowthIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     theta = theta, gamma_eta = gamma_eta, x = x, type = 1, time = time)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMLinGrowthIVary`
#> to convert it to a data frame or a matrix.
```

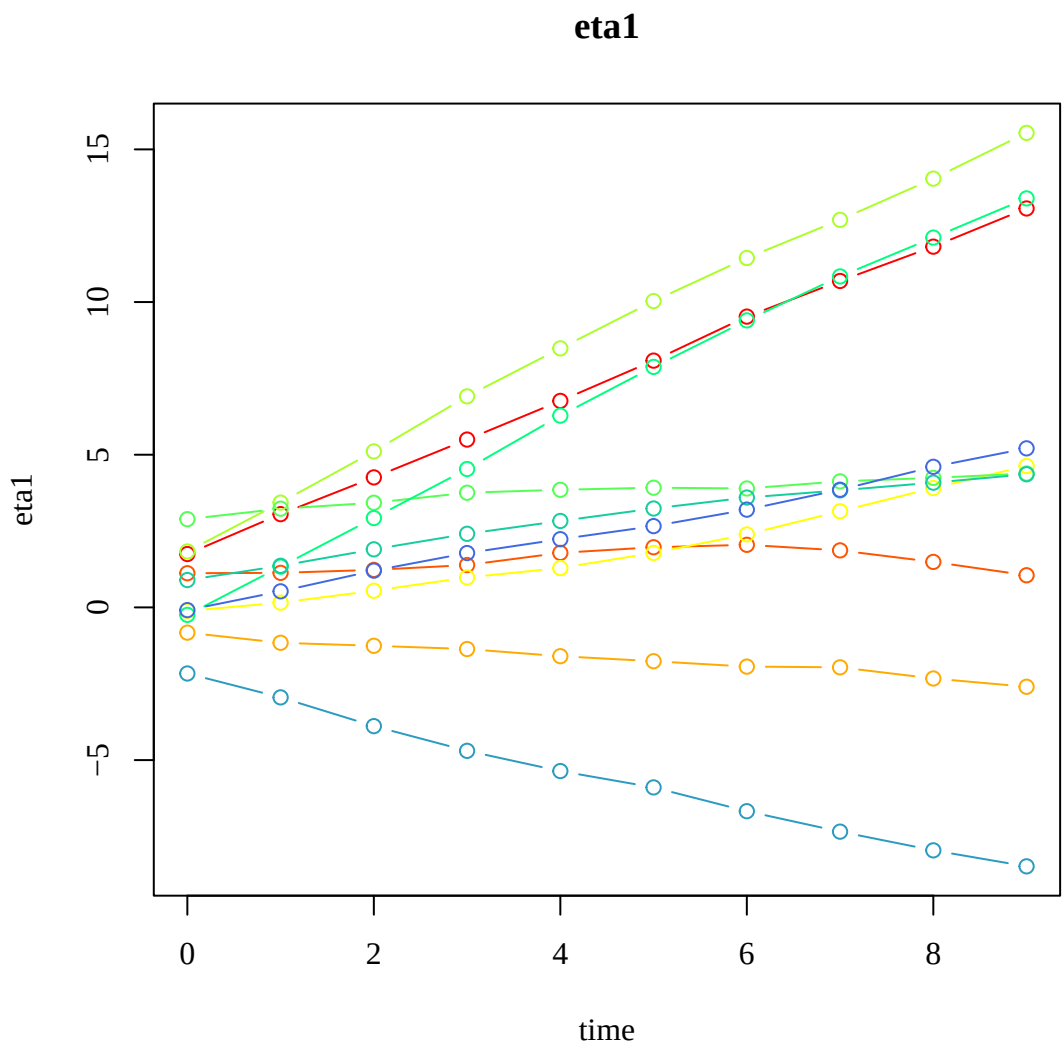

y1
y1
4
2
0
−2
−4
0
1
2
3
4
time


eta1
eta1
15
10
5
0
−5
0
2
4
6
8
time

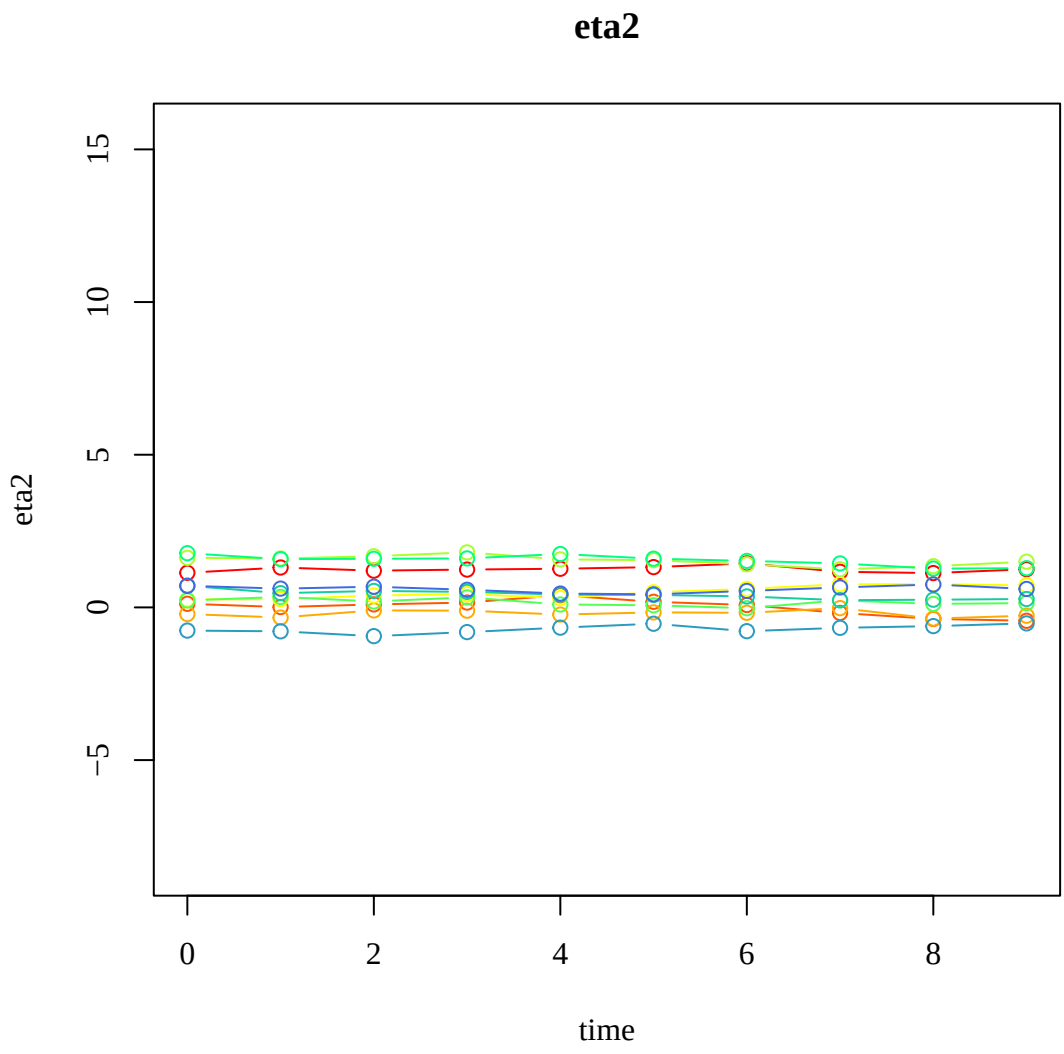



```
#> Call:
#> simStateSpace::SimSSMLinGrowthIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     theta = theta, gamma_y = gamma_y, gamma_eta = gamma_eta,
#>     x = x, type = 2, time = time)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMLinGrowthIVary`
#> to convert it to a data frame or a matrix.
```

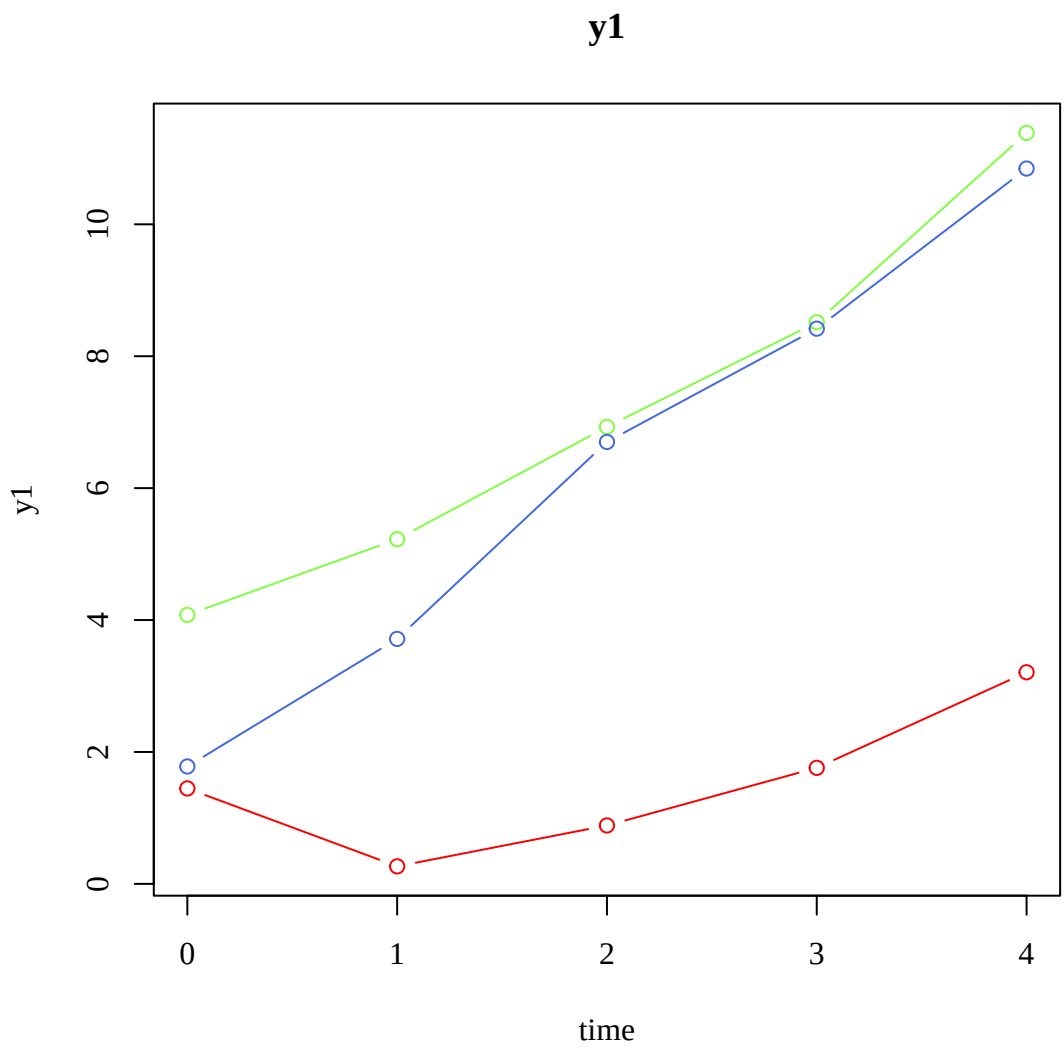

y1
y1
6
4
2
0
−2
0
1
2
3
4
time

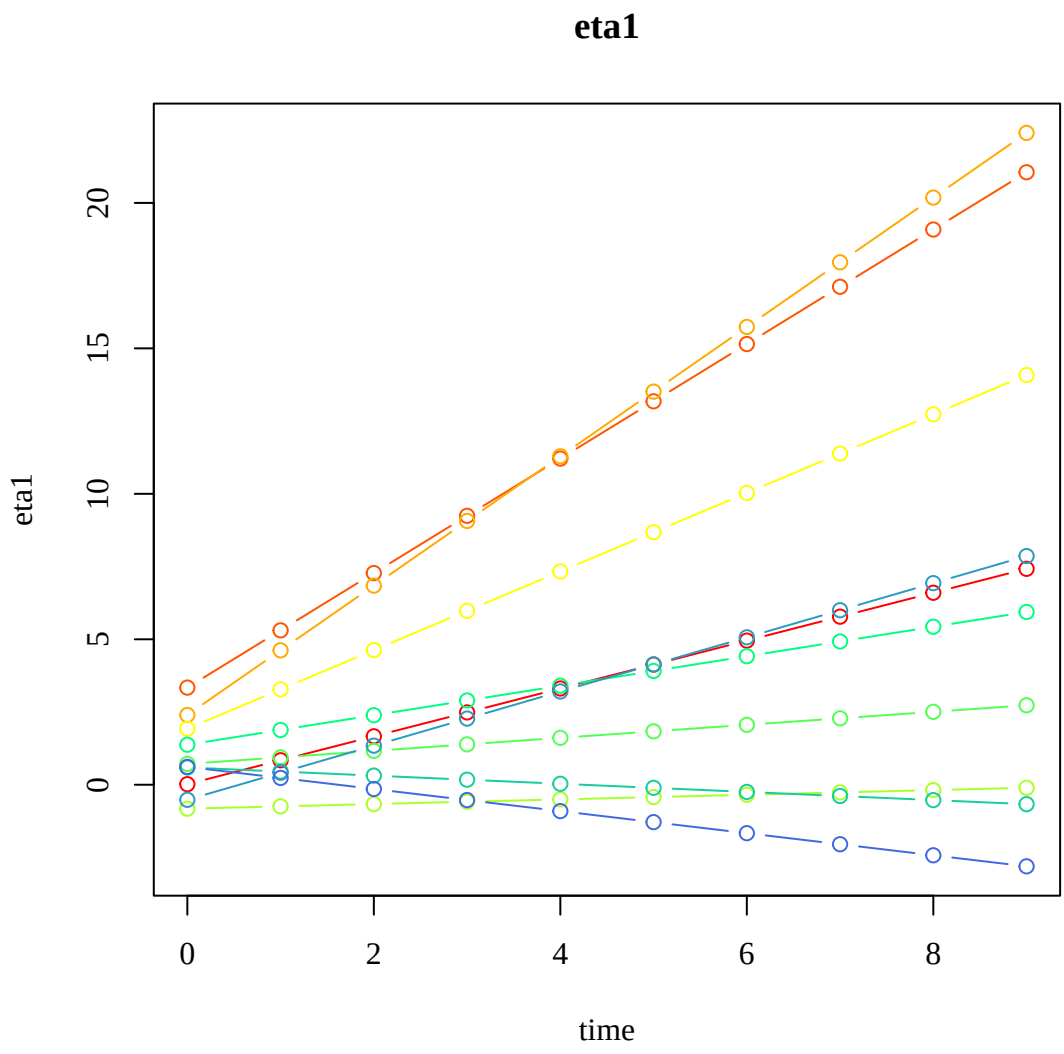


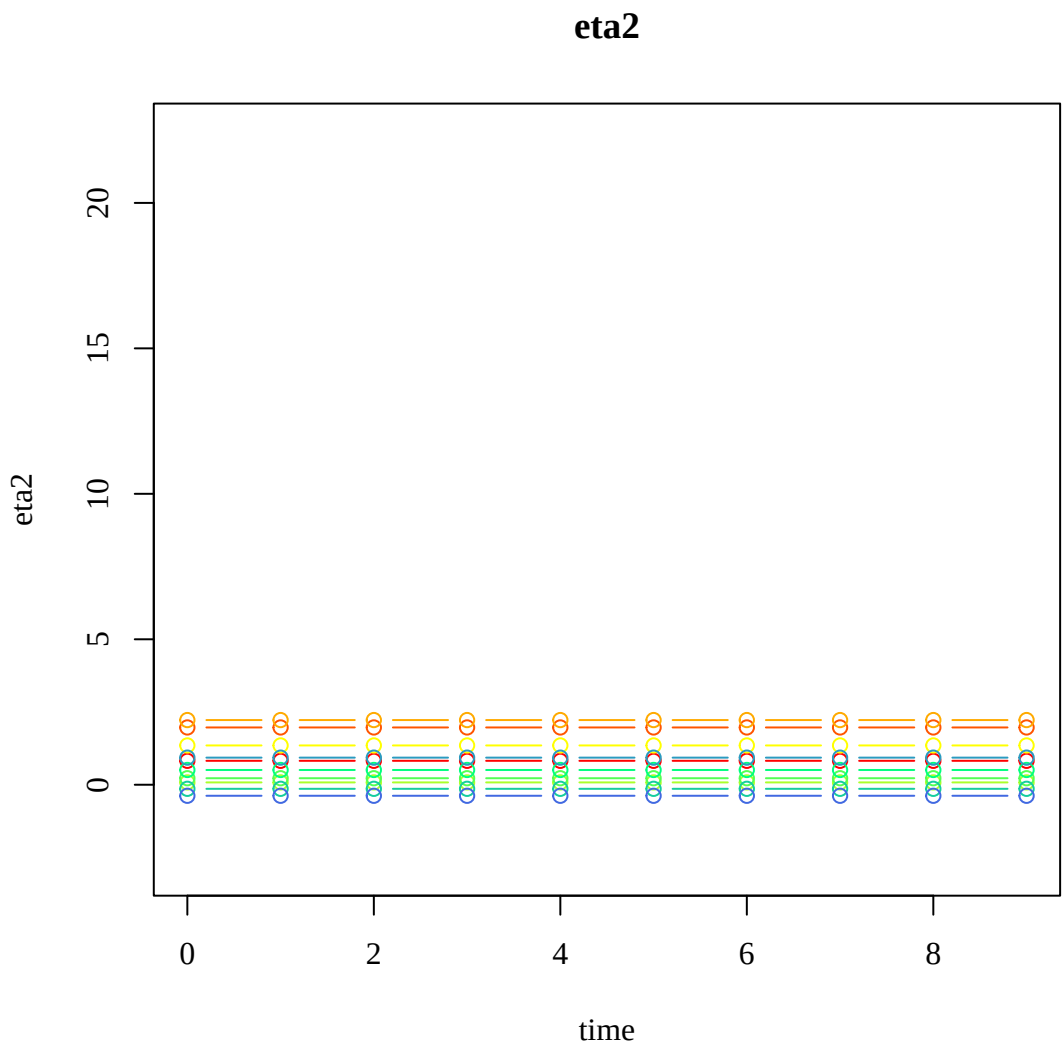

```
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-sim-ssm-lin-growth
```



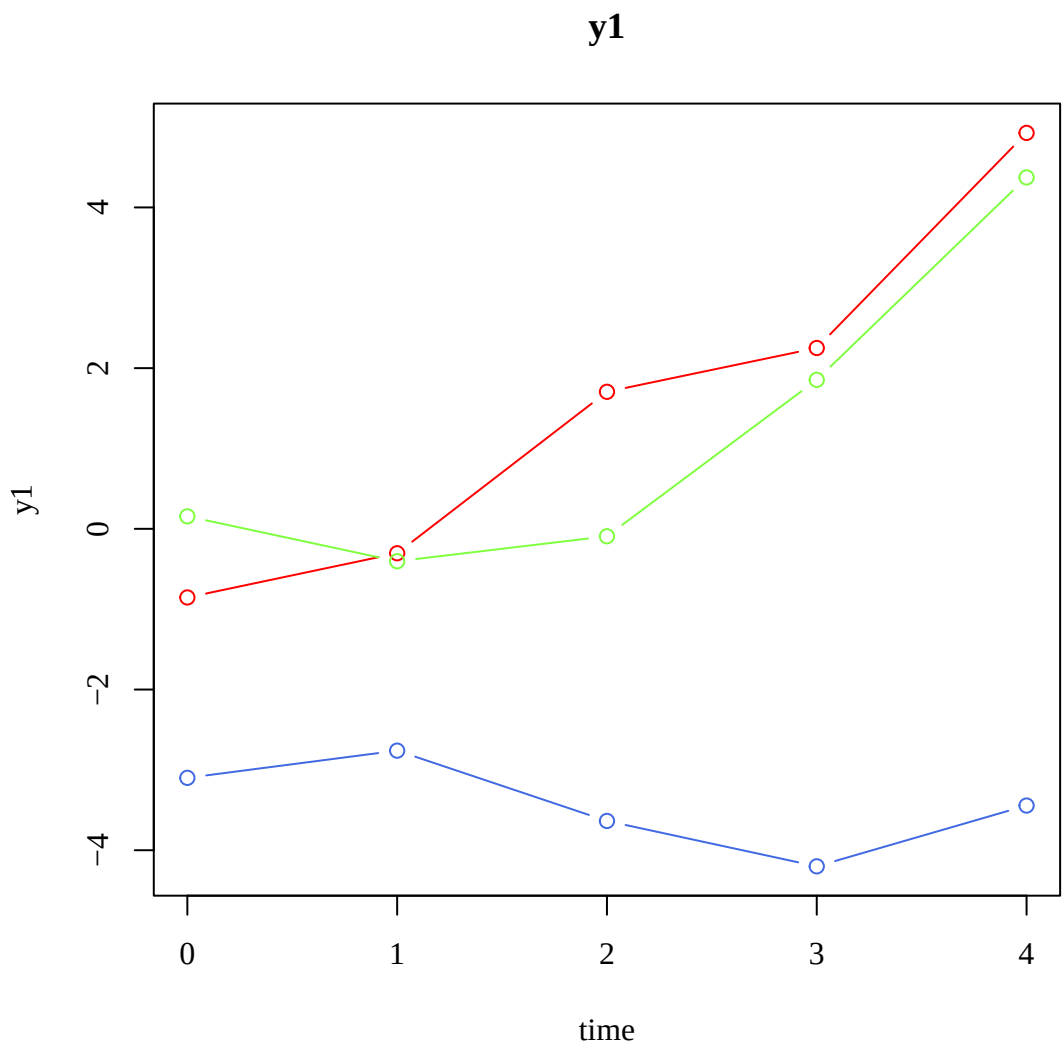

```
#> Call:
#> simStateSpace::SimSSMLinGrowth(n = n, mu0 = mu0, sigma0 = sigma0,
#>     theta = theta, type = 0, time = time)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMLinGrowth`
#> to convert it to a data frame or a matrix.
```

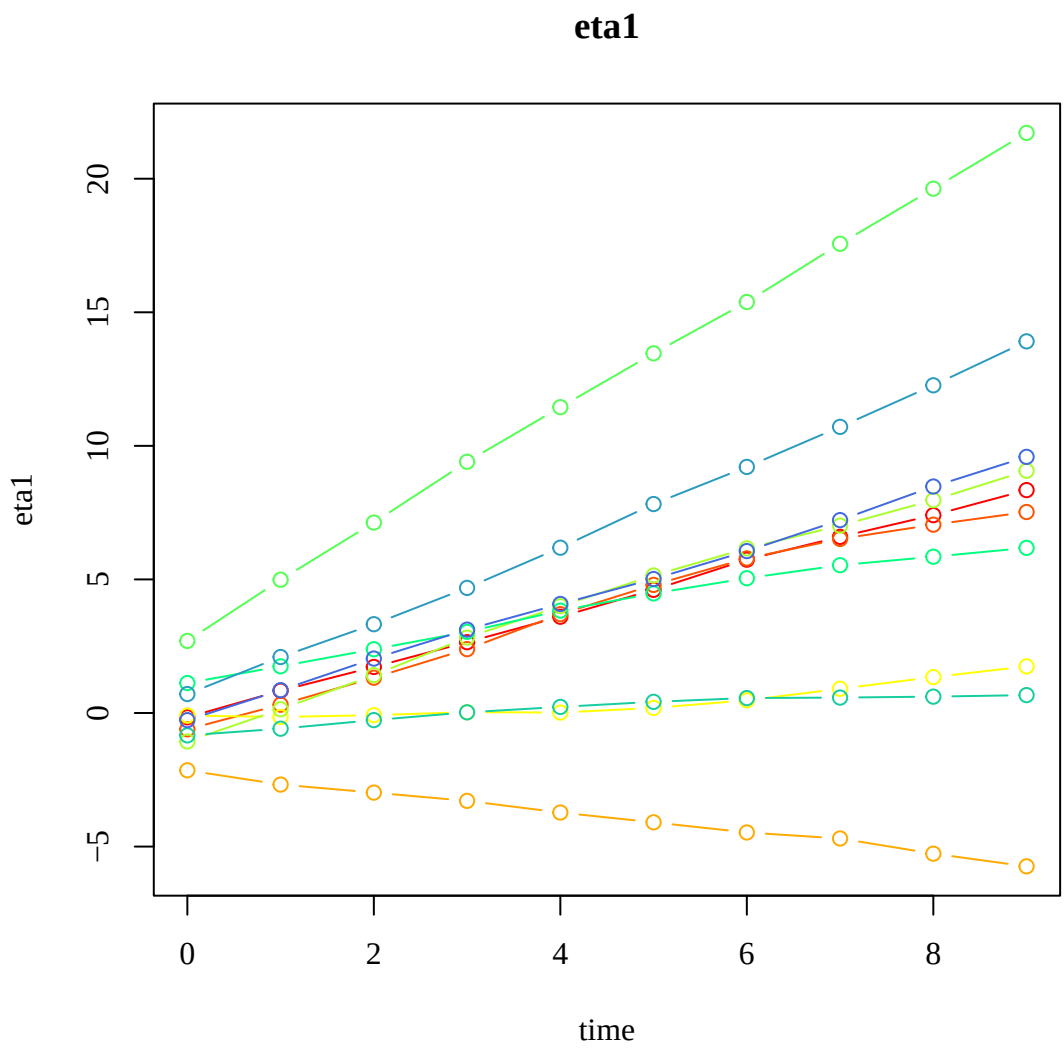

y1
y1
0
2
4
6
8
10
0
1
2
3
4
time

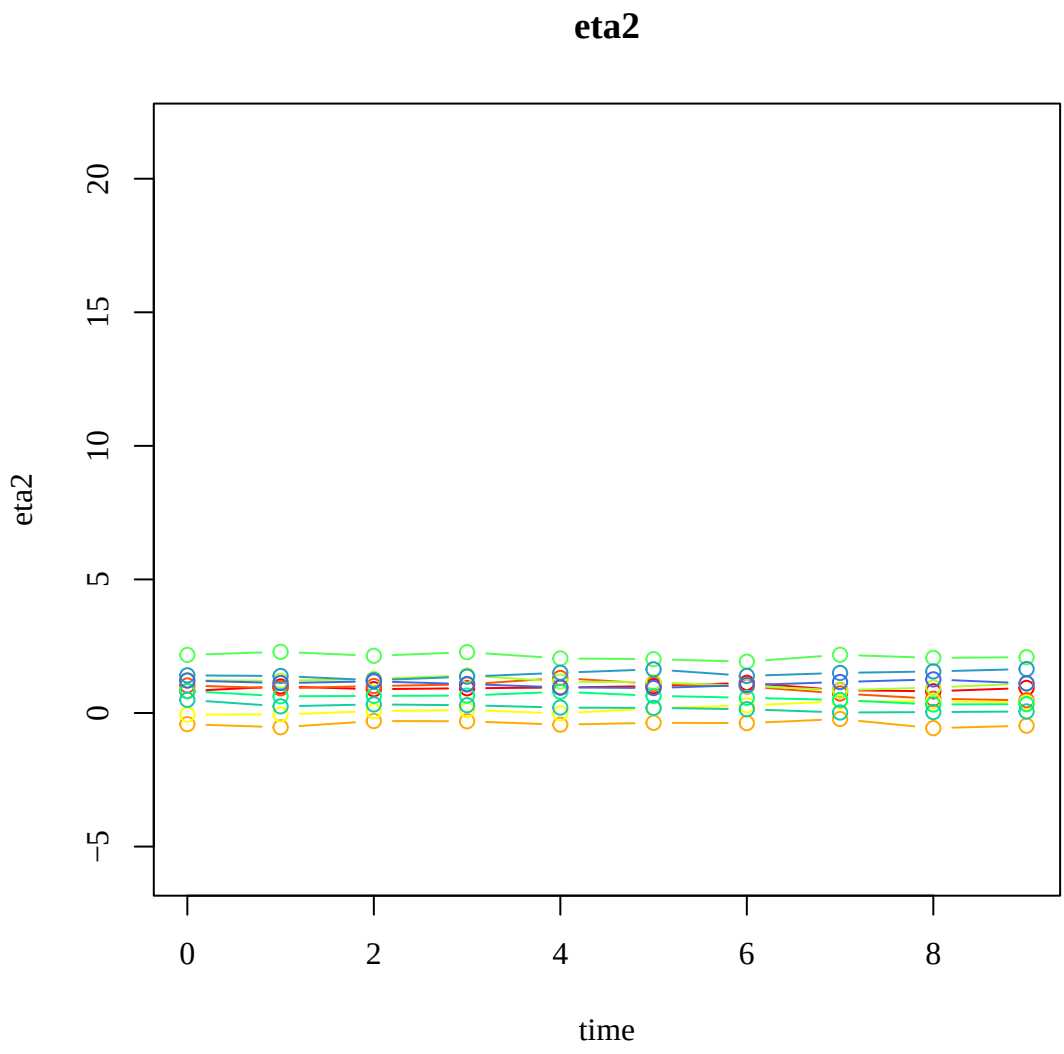

eta1
eta1
0
5
10
15
20
0
2
4
6
8
time



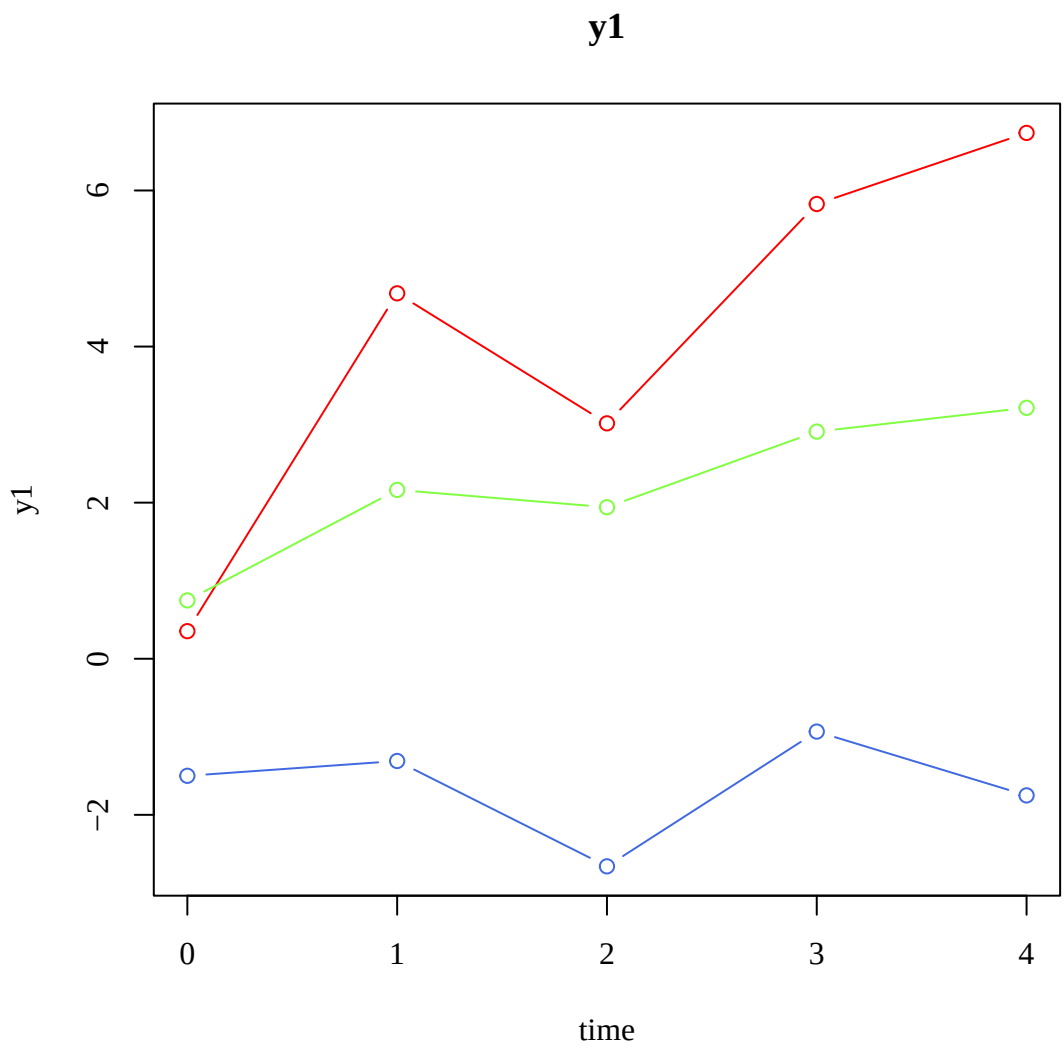

```
#> Call:
#> simStateSpace::SimSSMLinGrowth(n = n, mu0 = mu0, sigma0 = sigma0,
#>     theta = theta, gamma_eta = gamma_eta, x = x, type = 1, time = time)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMLinGrowth`
#> to convert it to a data frame or a matrix.
```

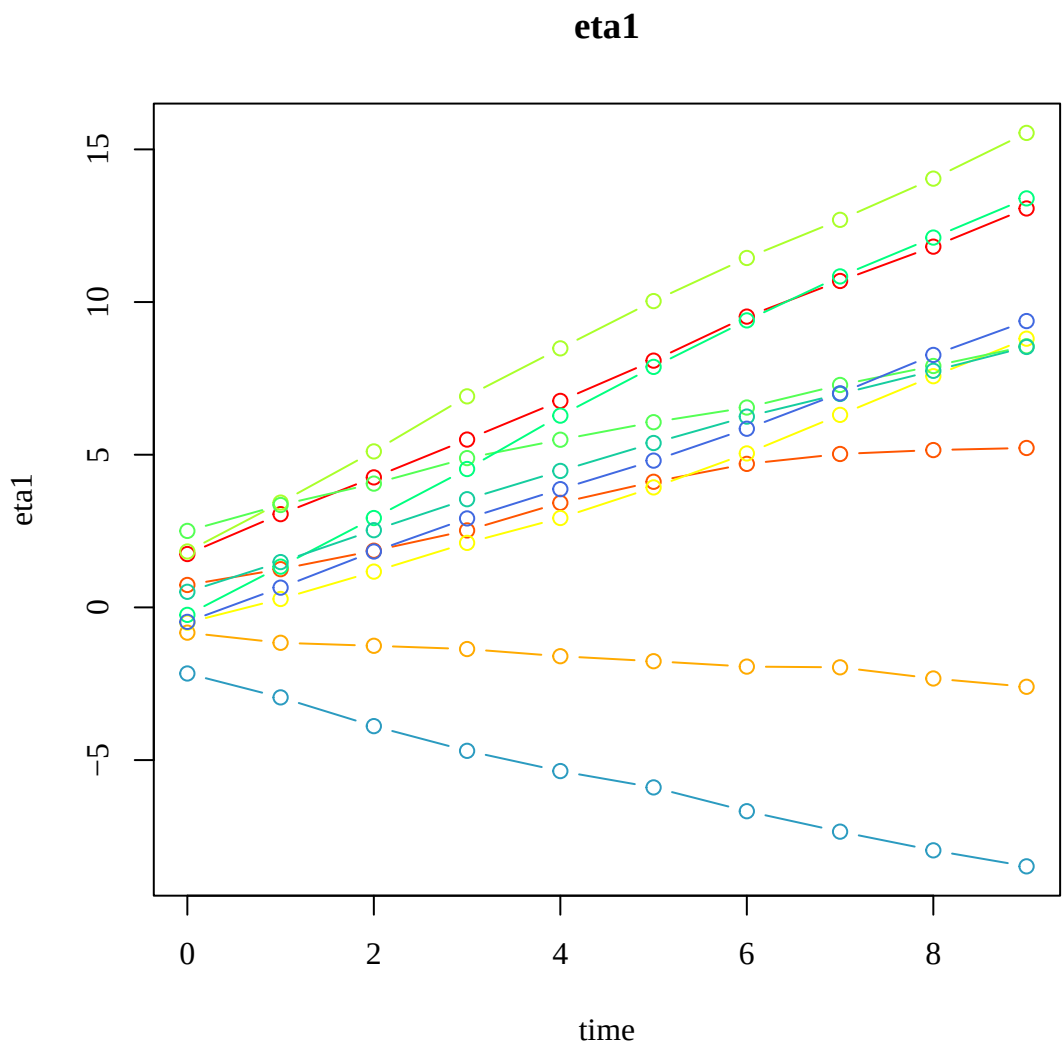

y1
y1
4
2
0
−2
−4
0
1
2
3
4
time


eta1
eta1
20
15
10
5
0
−5
0
2
4
6
8
time

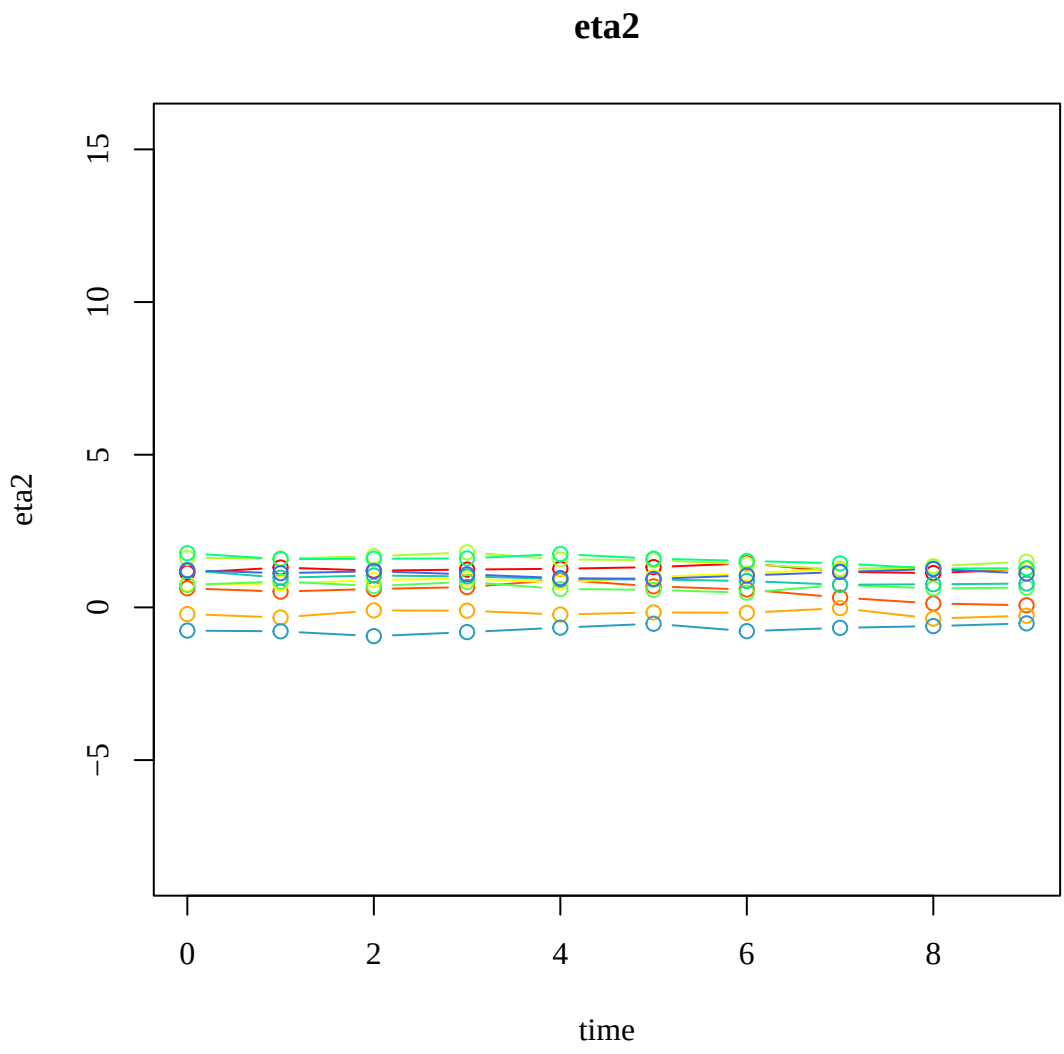



```
#> Call:
#> simStateSpace::SimSSMLinGrowth(n = n, mu0 = mu0, sigma0 = sigma0,
#>     theta = theta, gamma_y = gamma_y, gamma_eta = gamma_eta,
#>     x = x, type = 2, time = time)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMLinGrowth`
#> to convert it to a data frame or a matrix.
```

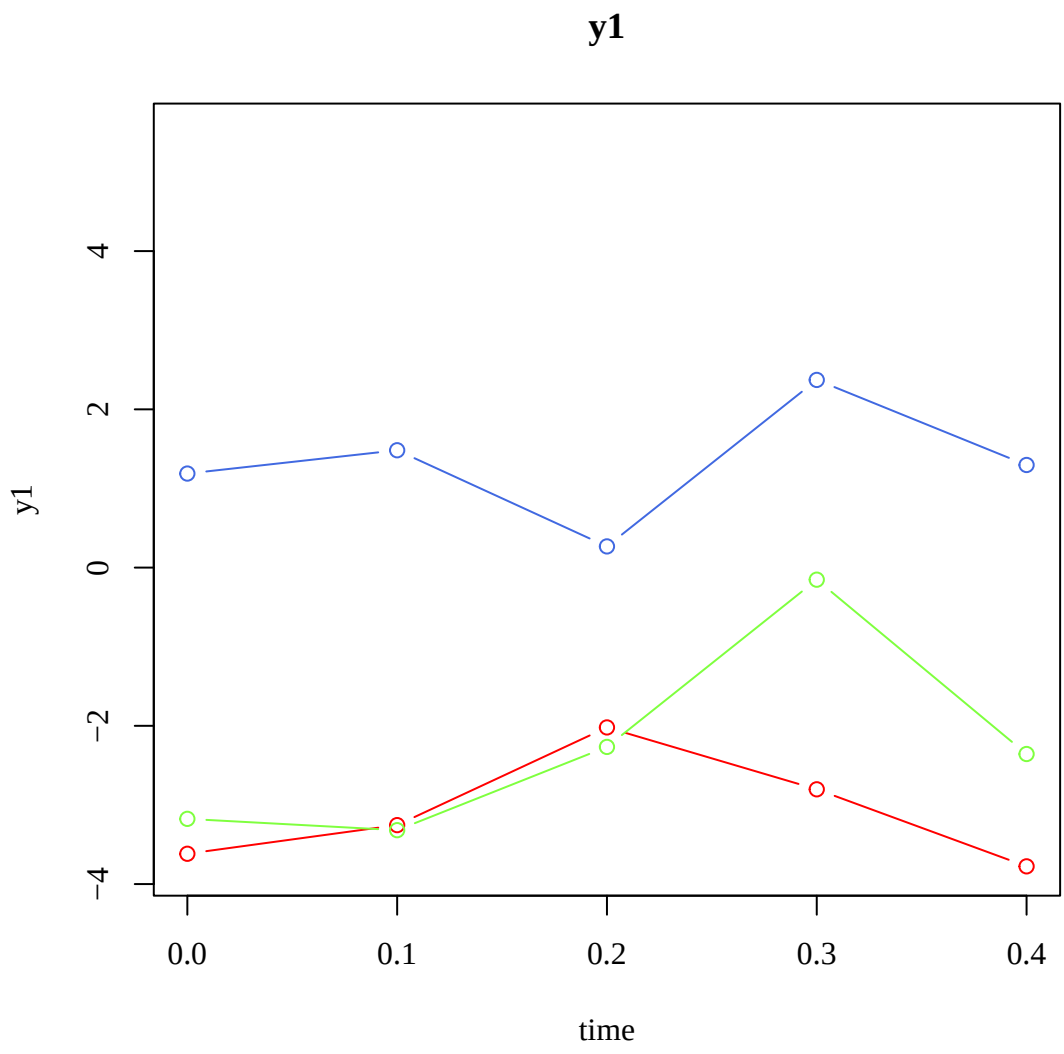

y1
y1
6
4
2
0
−2
0
1
2
3
4
time

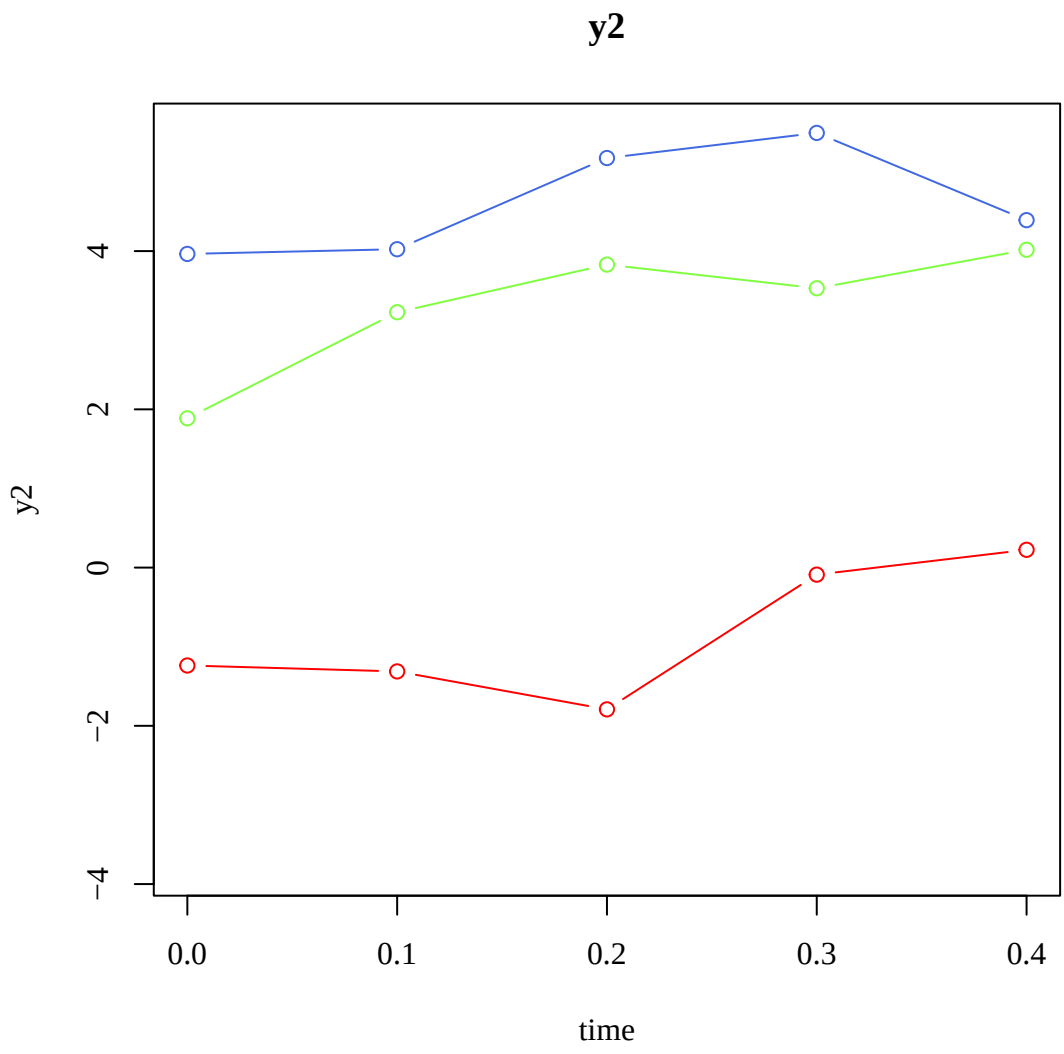


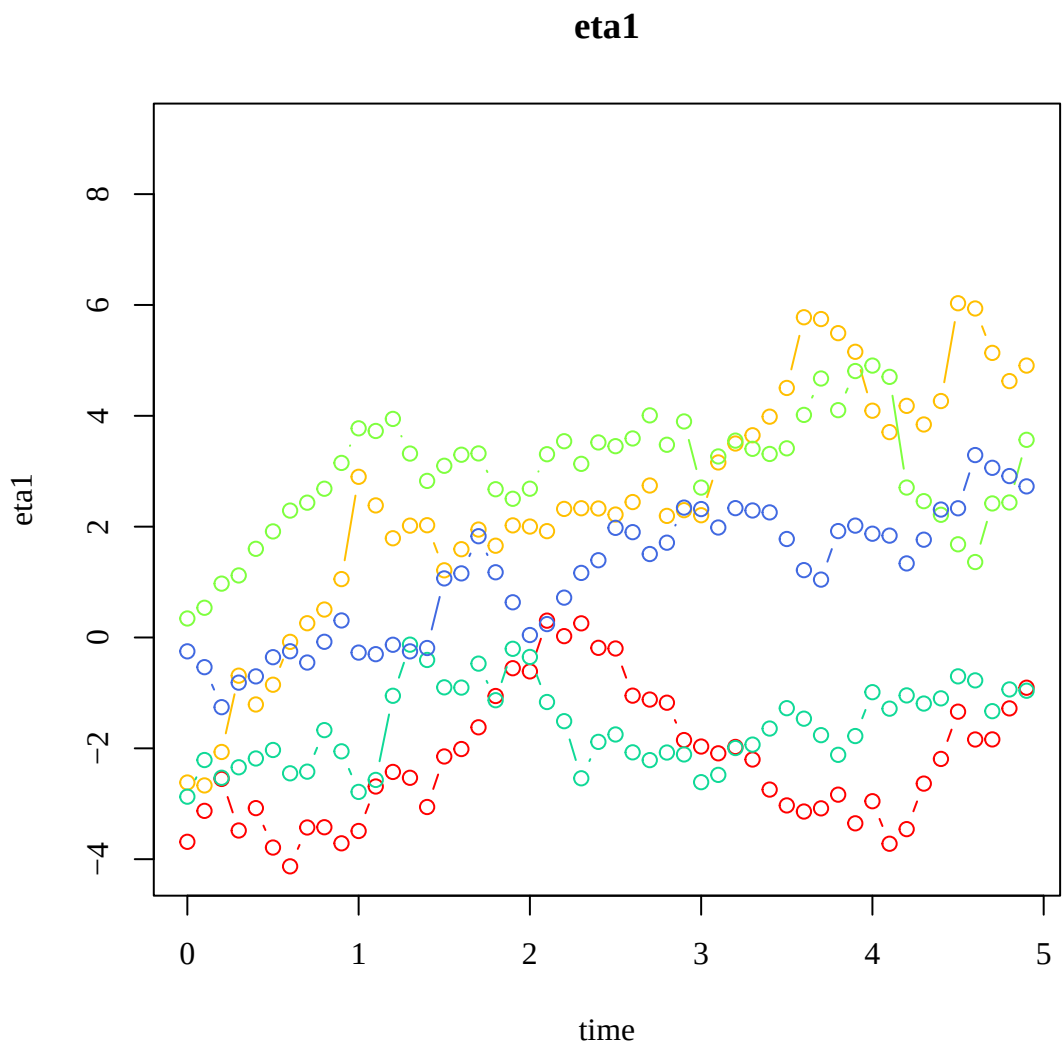

```
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-sim-ssm-ou-fixed
```

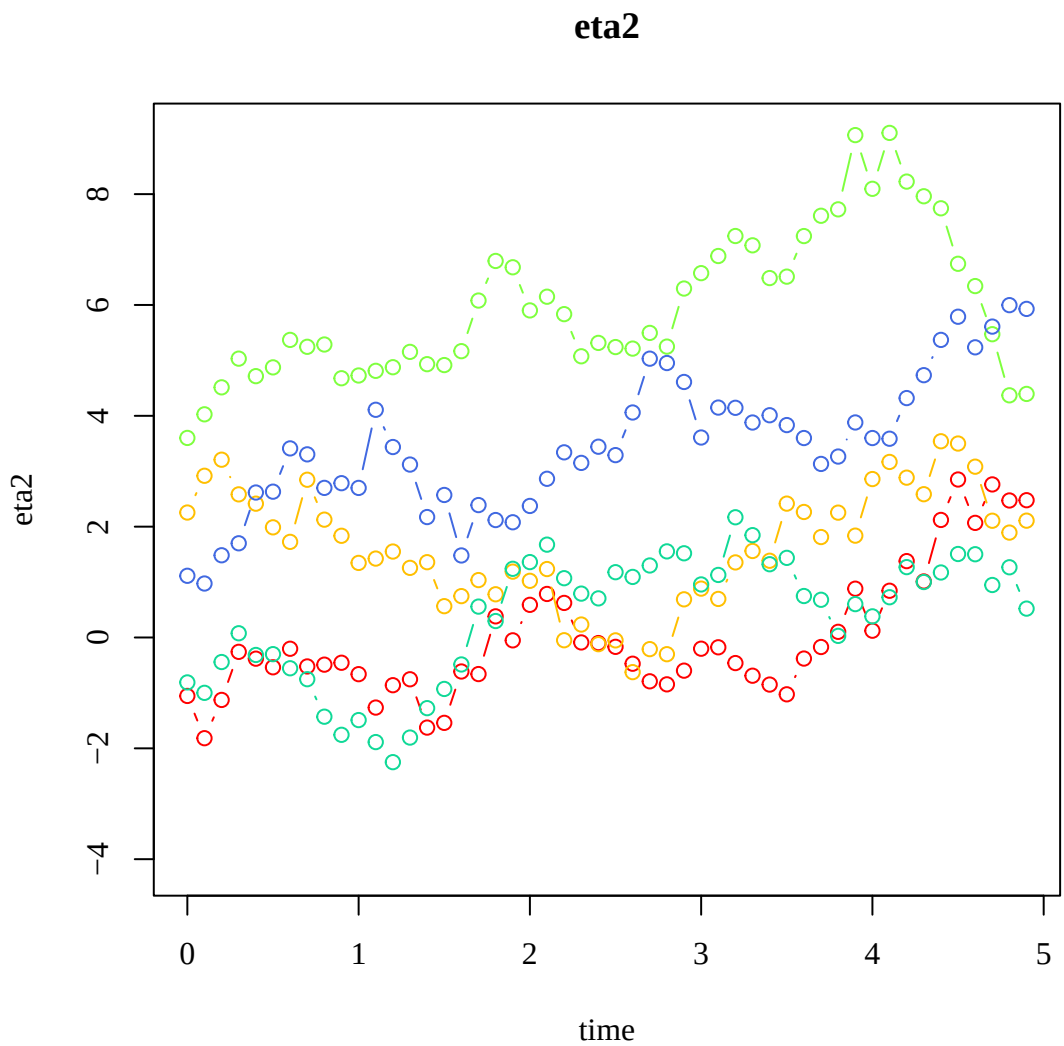



```
#> Call:
#> simStateSpace::SimSSMOUFixed(n = n, mu0 = mu0, sigma0 = sigma0,
#>     mu = mu, phi = phi, sigma = sigma, nu = nu, lambda = lambda,
#>     theta = theta, type = 0, delta_t = delta_t, time = time,
#>     burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUFixed`
#> to convert it to a data frame or a matrix.
```

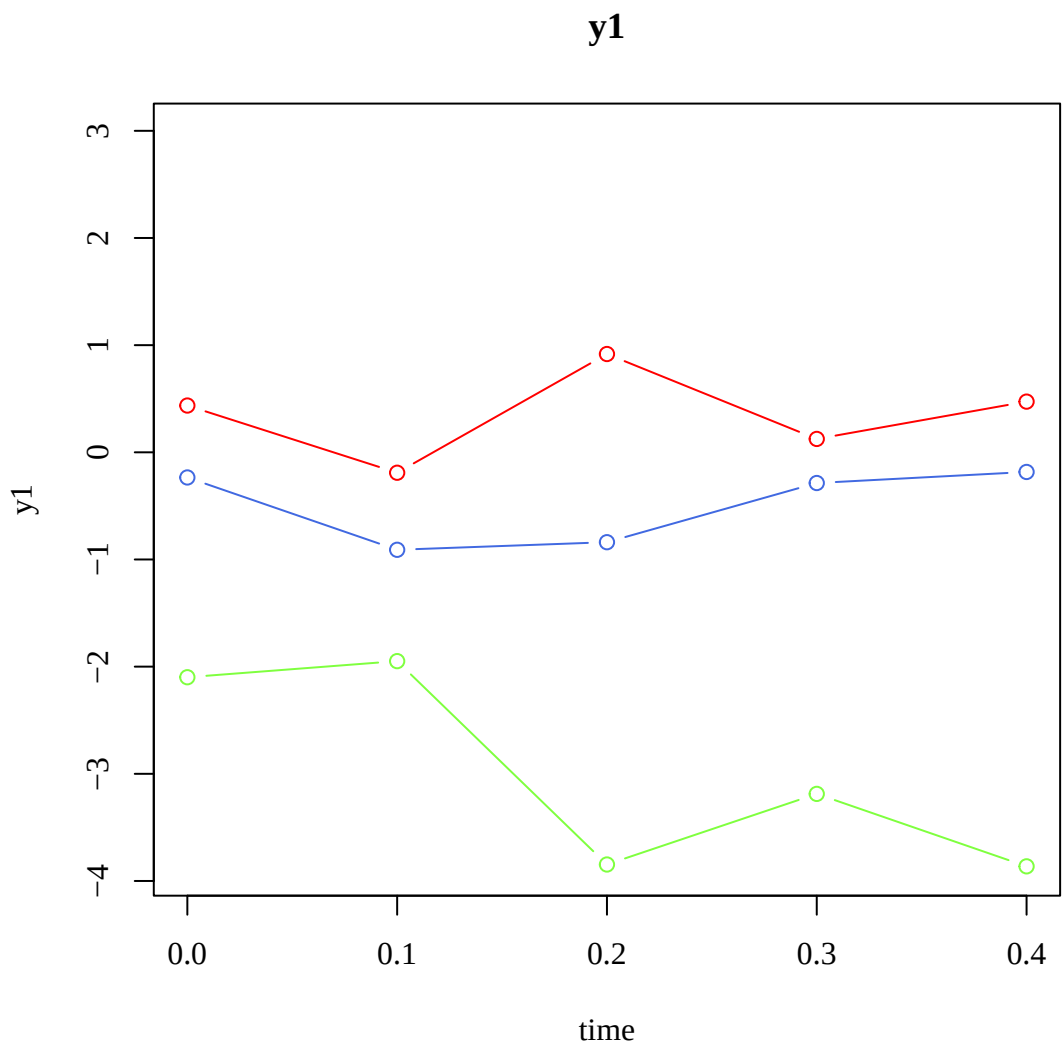

y1
y1
4
2
0
−2
−4
0.0
0.1
0.2
0.3
0.4
time

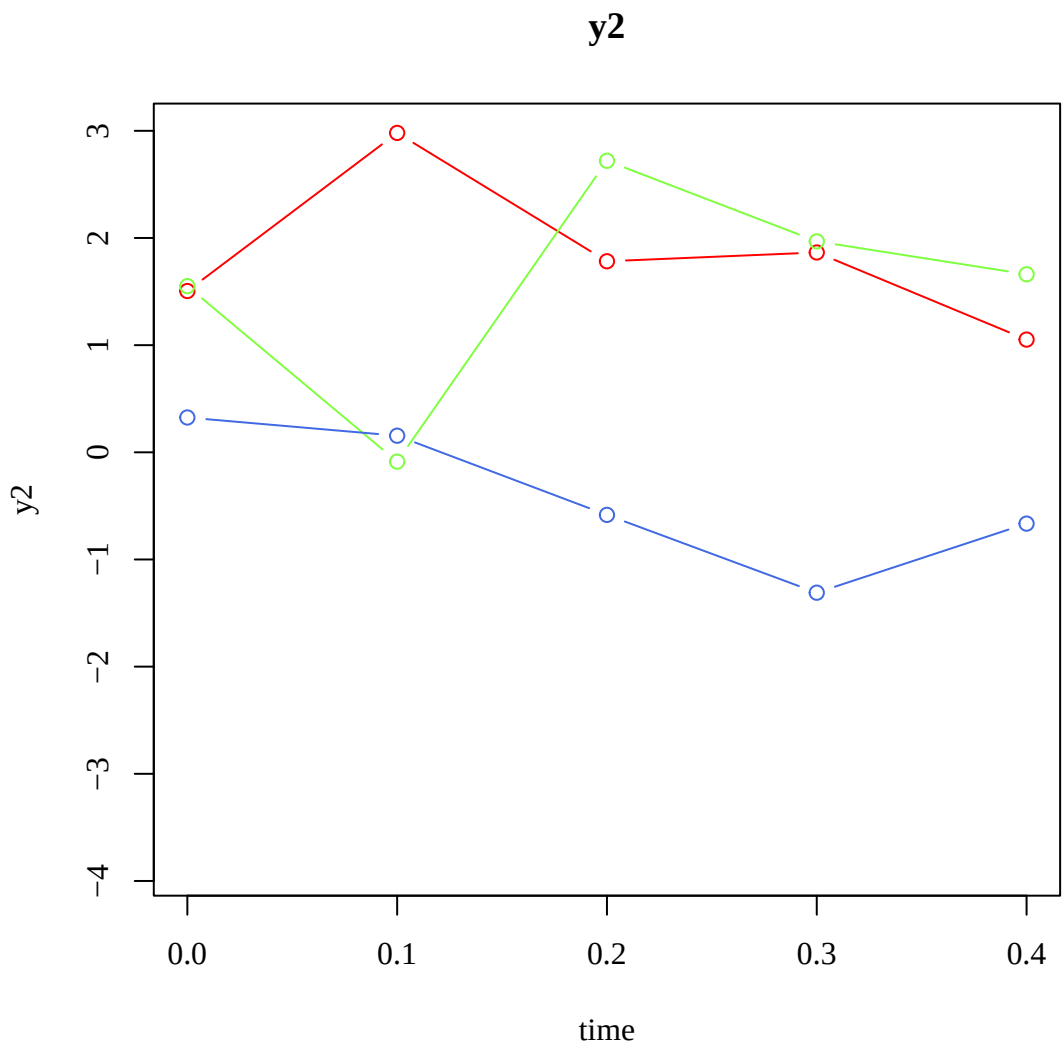

y2
y2
4
2
0
−2
−4
0.0
0.1
0.2
0.3
0.4
time

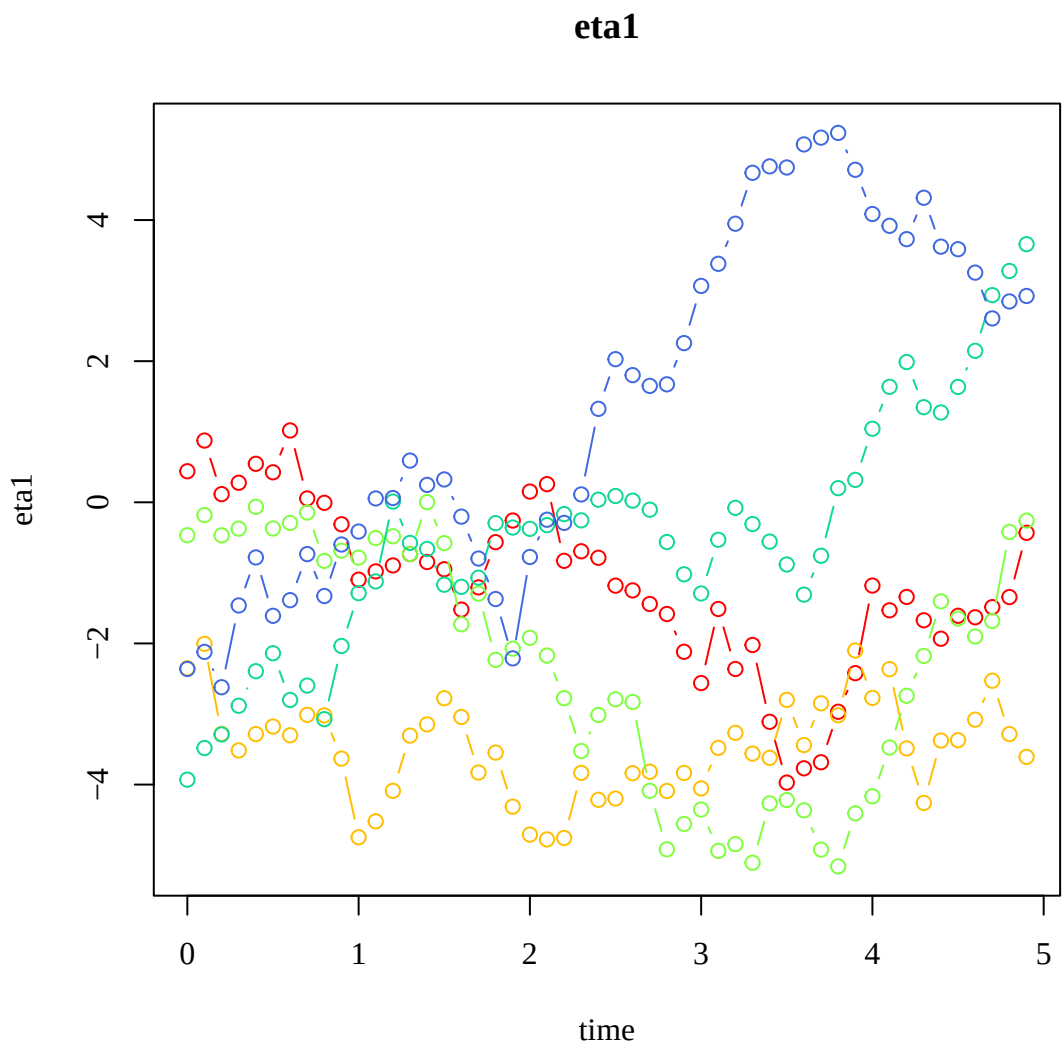

eta1
eta1
8
6
4
2
0
−2
−4
0
1
2
3
4
5
time

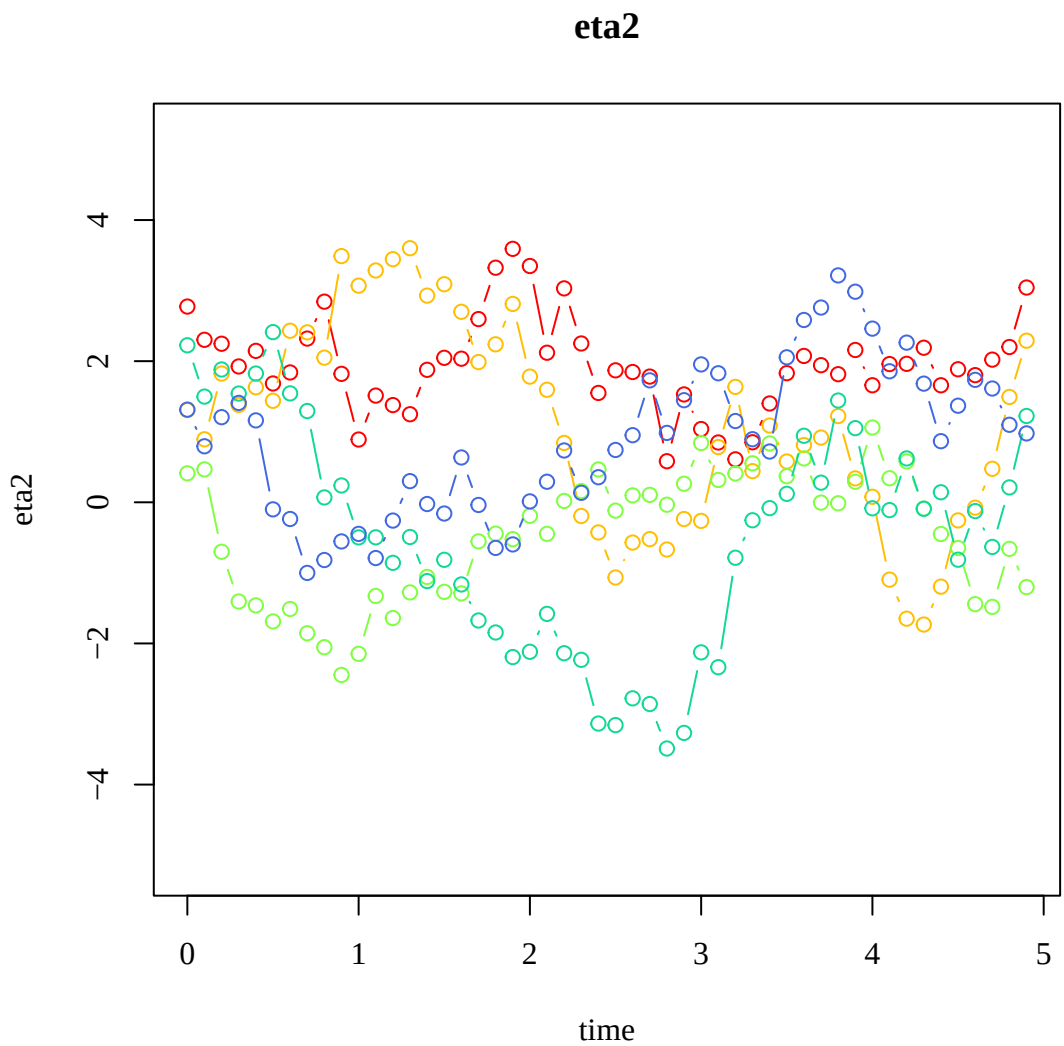



```
#> Call:
#> simStateSpace::SimSSMOUFixed(n = n, mu0 = mu0, sigma0 = sigma0,
#>     mu = mu, phi = phi, sigma = sigma, nu = nu, lambda = lambda,
#>     theta = theta, gamma_eta = gamma_eta, x = x, type = 1, delta_t = delta_t,
#>     time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUFixed`
#> to convert it to a data frame or a matrix.
```

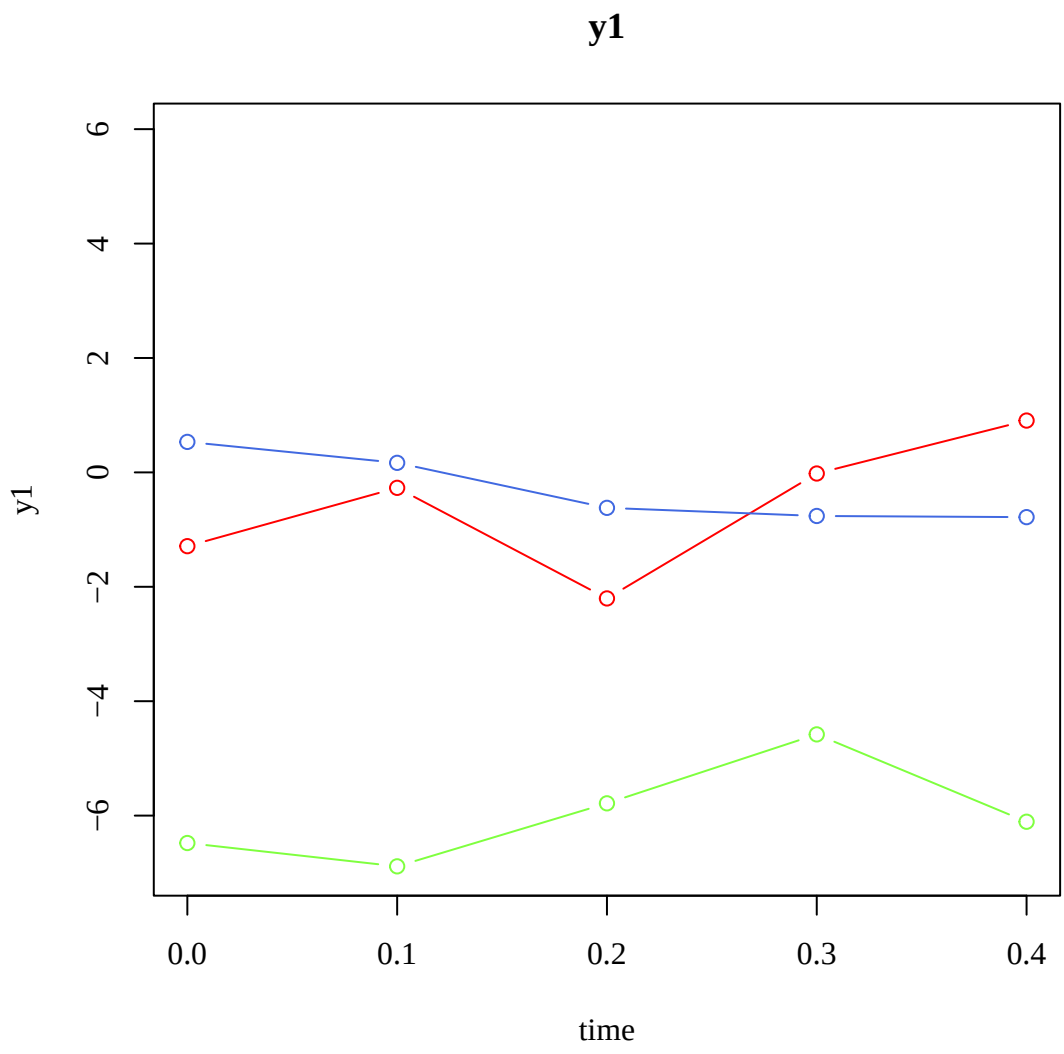

y1
3
2
1
0
−1
−2
−3
−4
y1
0.0
0.1
0.2
0.3
0.4
time

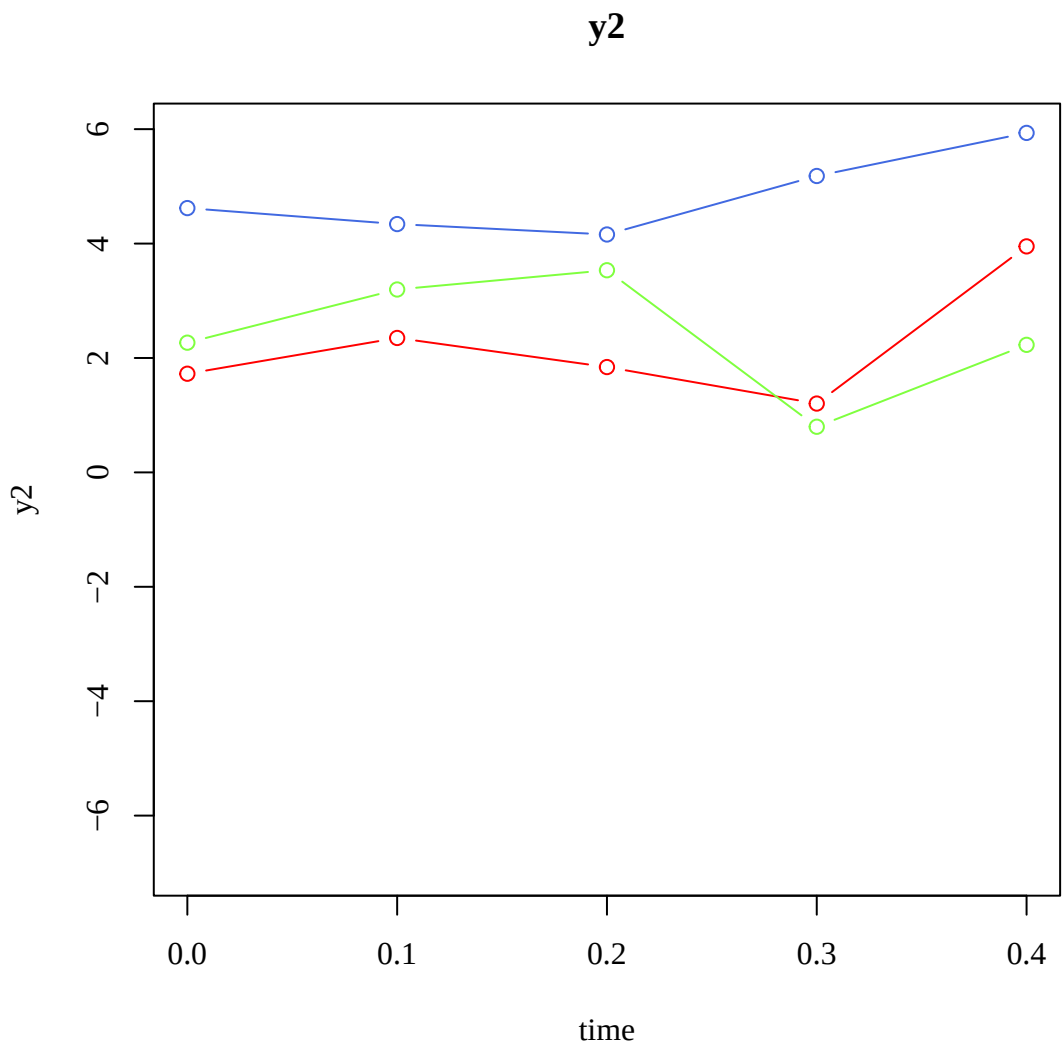

y2
y2
3
2
1
0
−1
−2
−3
−4
0.0
0.1
0.2
0.3
0.4
time

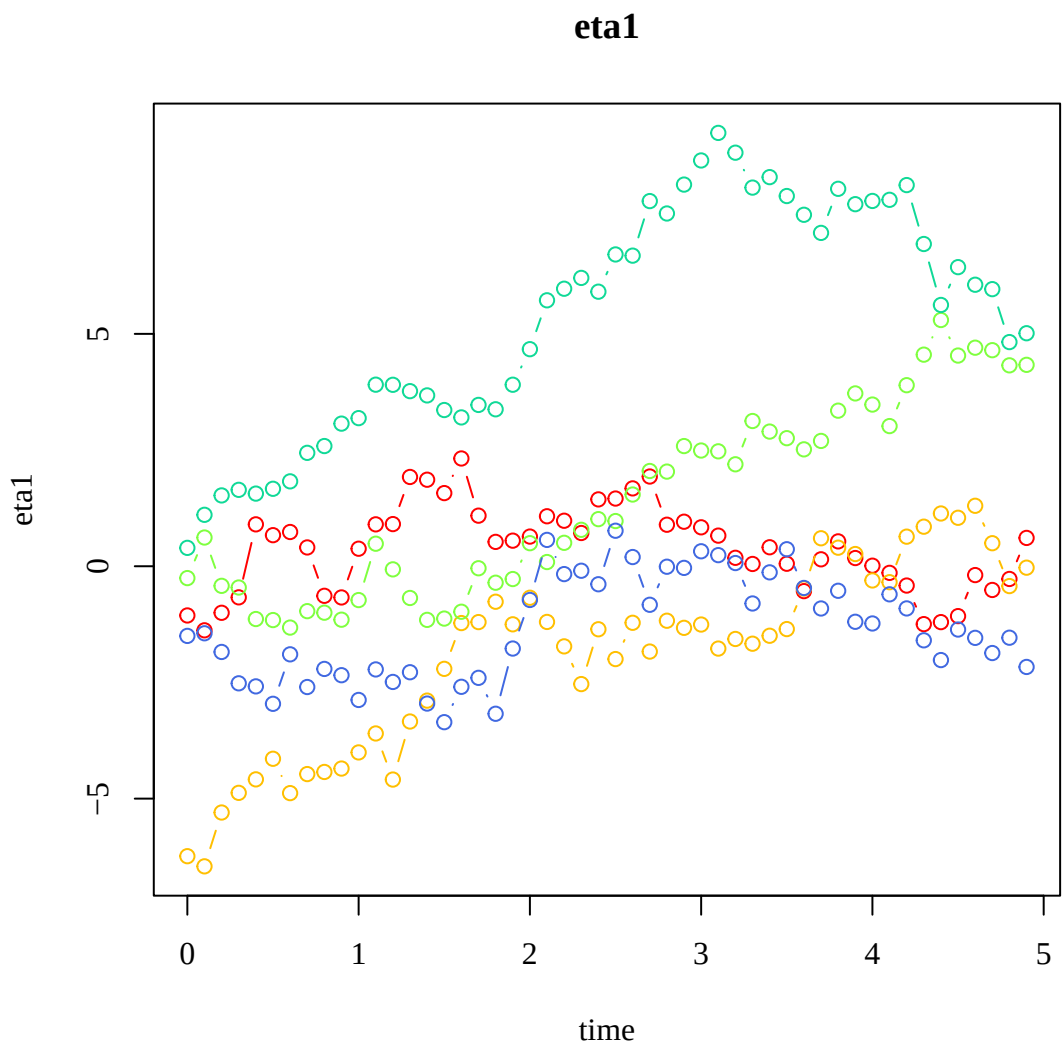

eta1
eta1
4
2
0
−2
−4
0
1
2
3
4
5
time



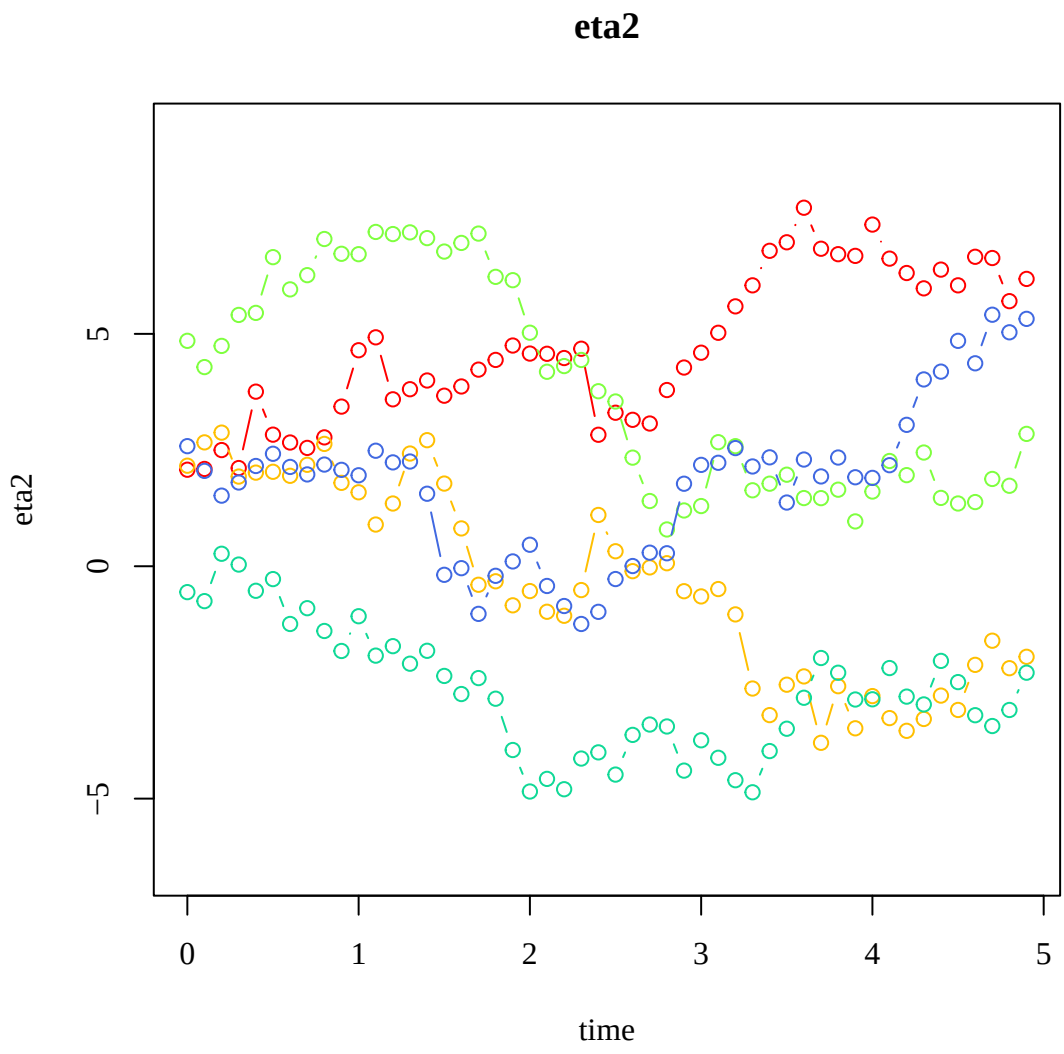

```
#> Call:
#> simStateSpace::SimSSMOUFixed(n = n, mu0 = mu0, sigma0 = sigma0,
#>     mu = mu, phi = phi, sigma = sigma, nu = nu, lambda = lambda,
#>     theta = theta, gamma_y = gamma_y, gamma_eta = gamma_eta,
#>     x = x, type = 2, delta_t = delta_t, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUFixed`
#> to convert it to a data frame or a matrix.
```


y1
6
4
2
0
−2
−4
−6
y1
0.0
0.1
0.2
0.3
0.4
time

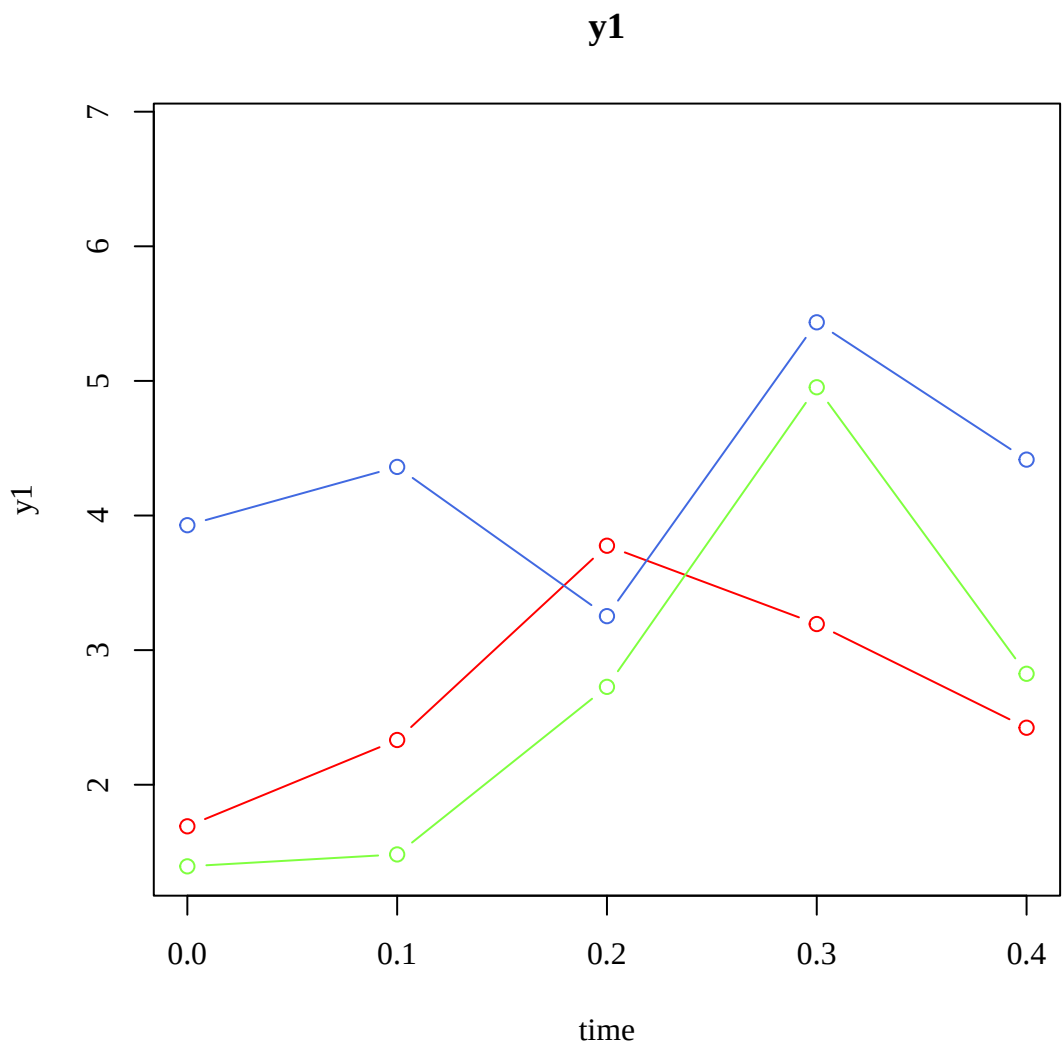

y2
6
4
2
0
−2
−4
−6
y2
0.0
0.1
0.2
0.3
0.4
time

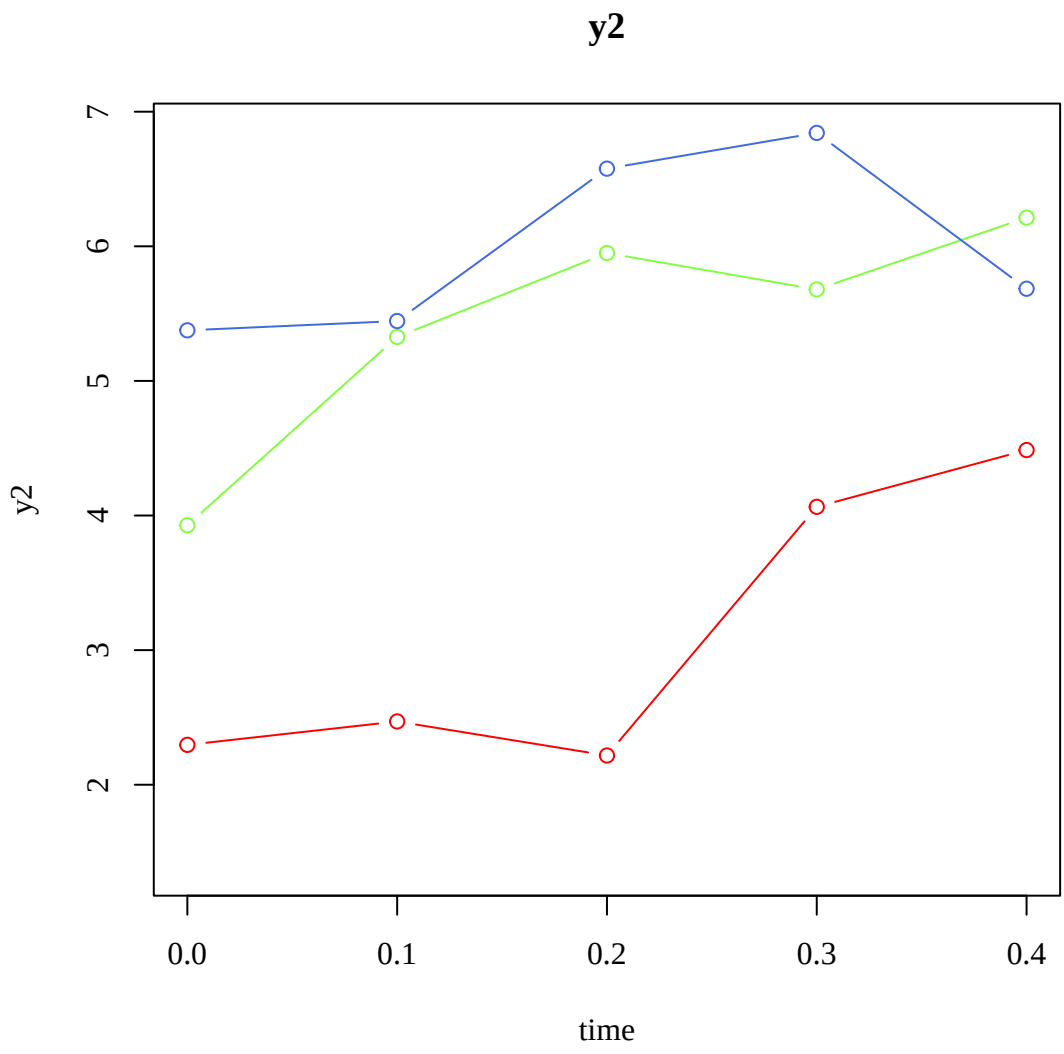


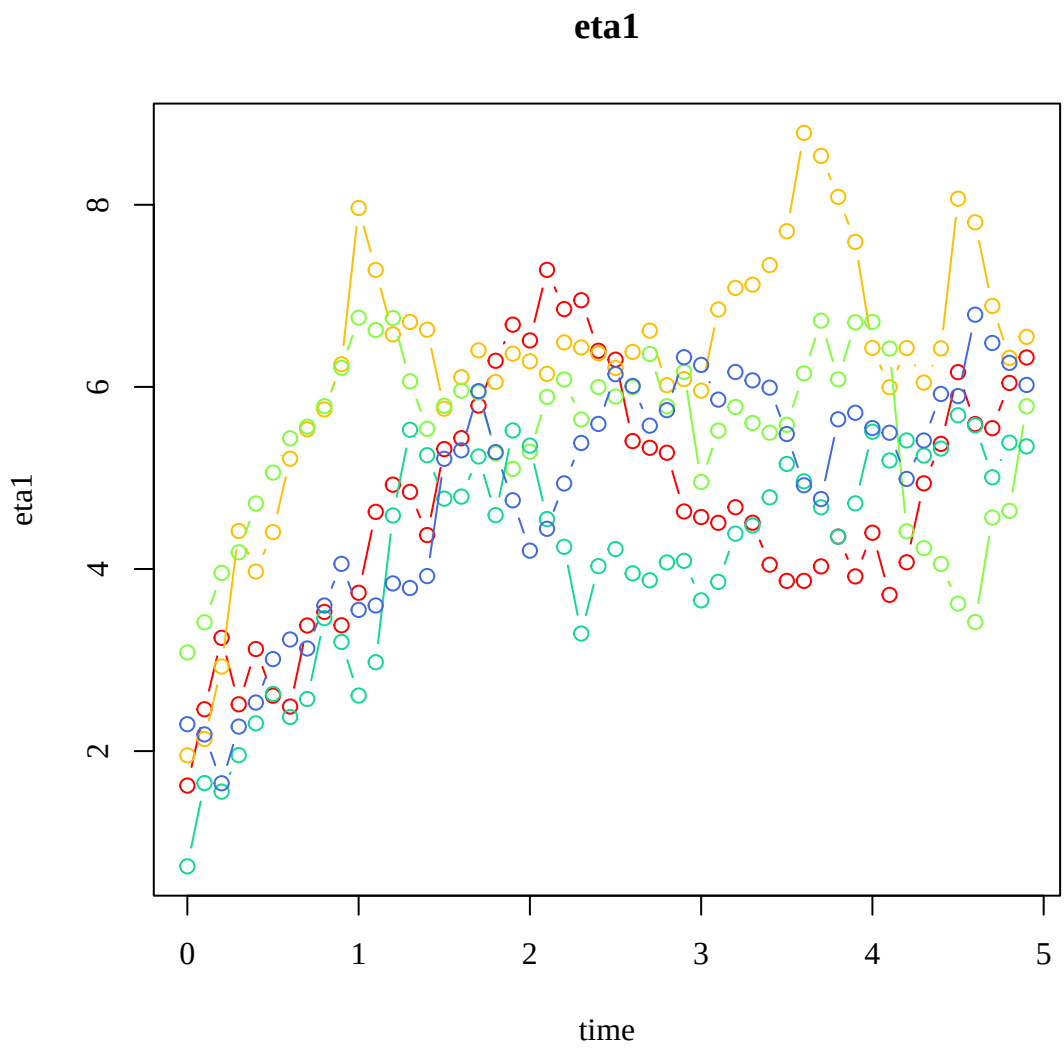

```
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-sim-ssm-ou-i-vary
```

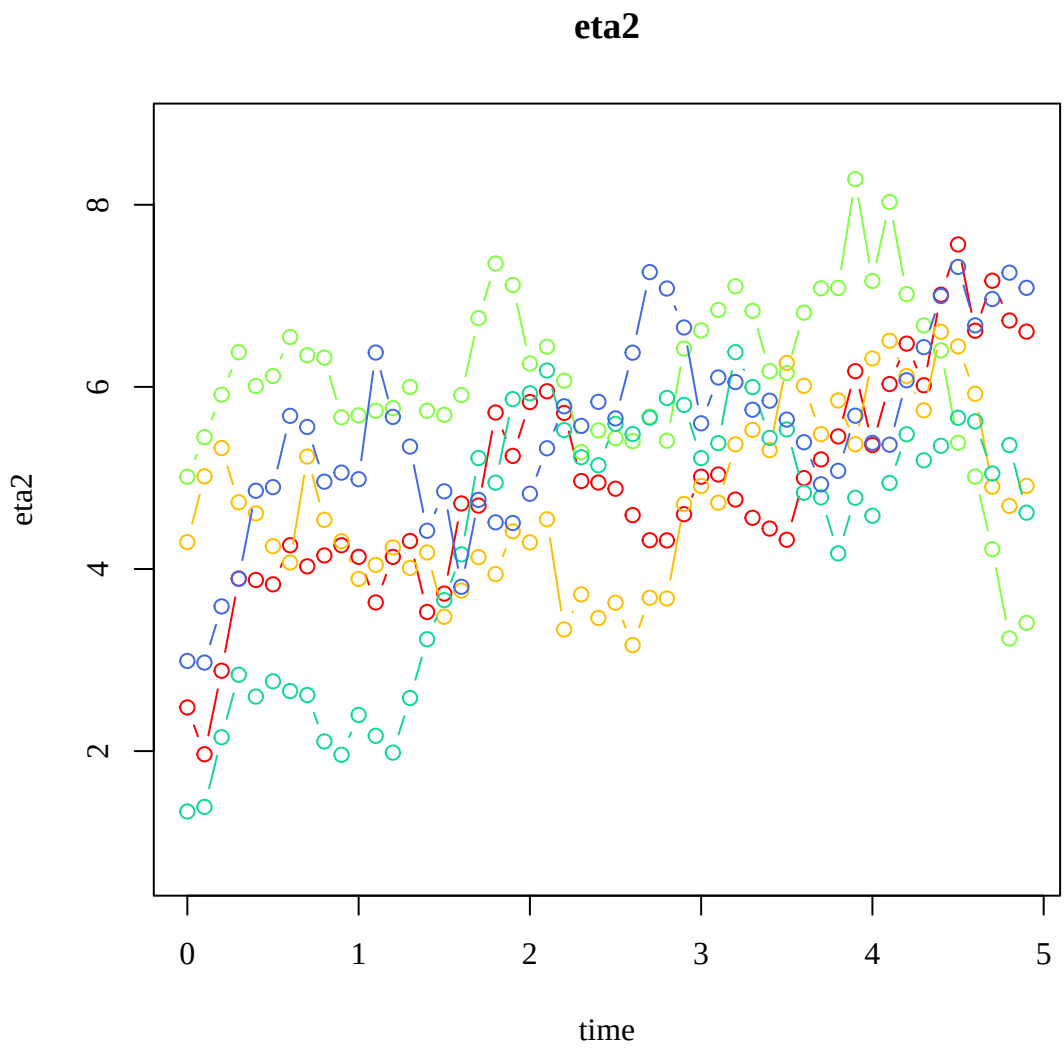



```
#> Call:
#> simStateSpace::SimSSMOUIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     mu = mu, phi = phi, sigma = sigma, nu = nu, lambda = lambda,
#>     theta = theta, type = 0, delta_t = delta_t, time = time,
#>     burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUIVary`
#> to convert it to a data frame or a matrix.
```

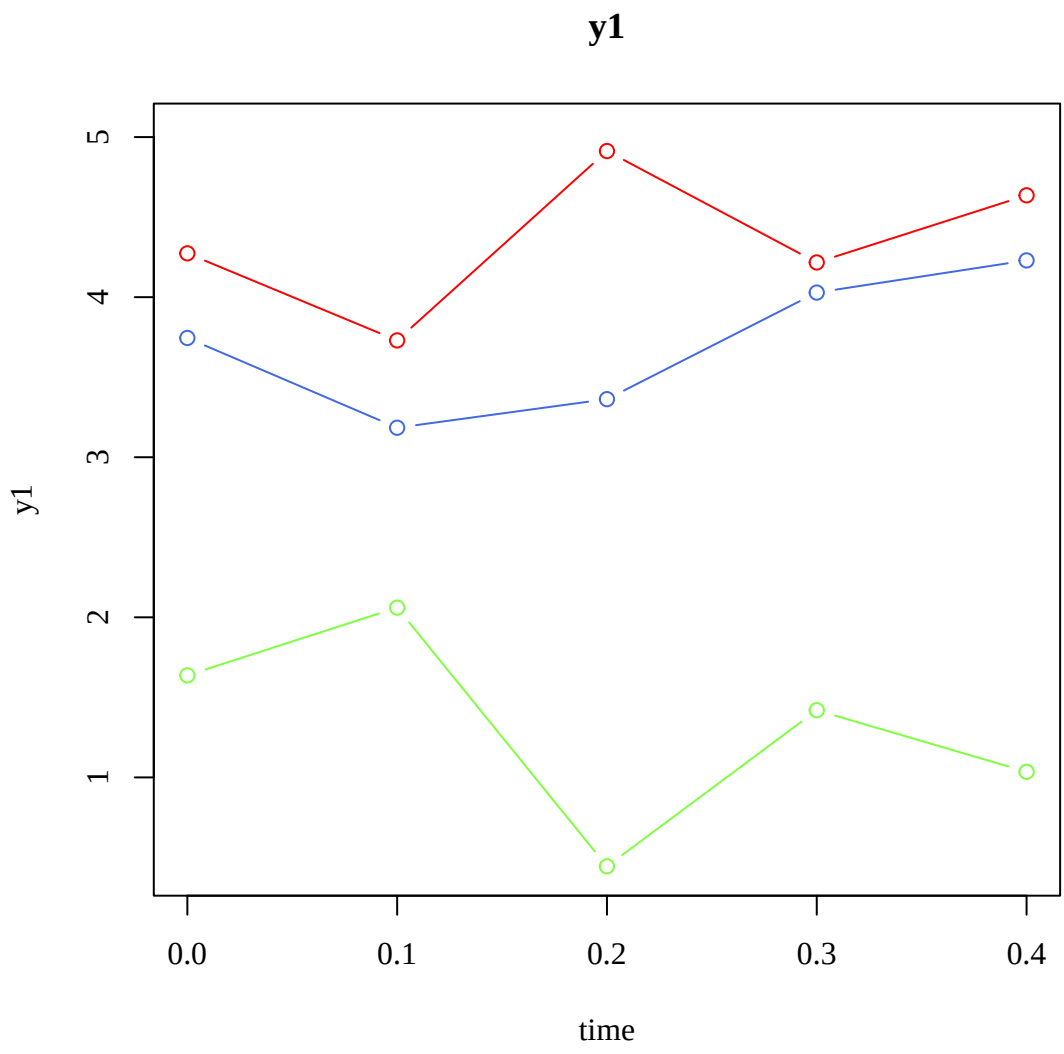

y1
y1
7
6
5
4
3
2
0.0
0.1
0.2
0.3
0.4
time

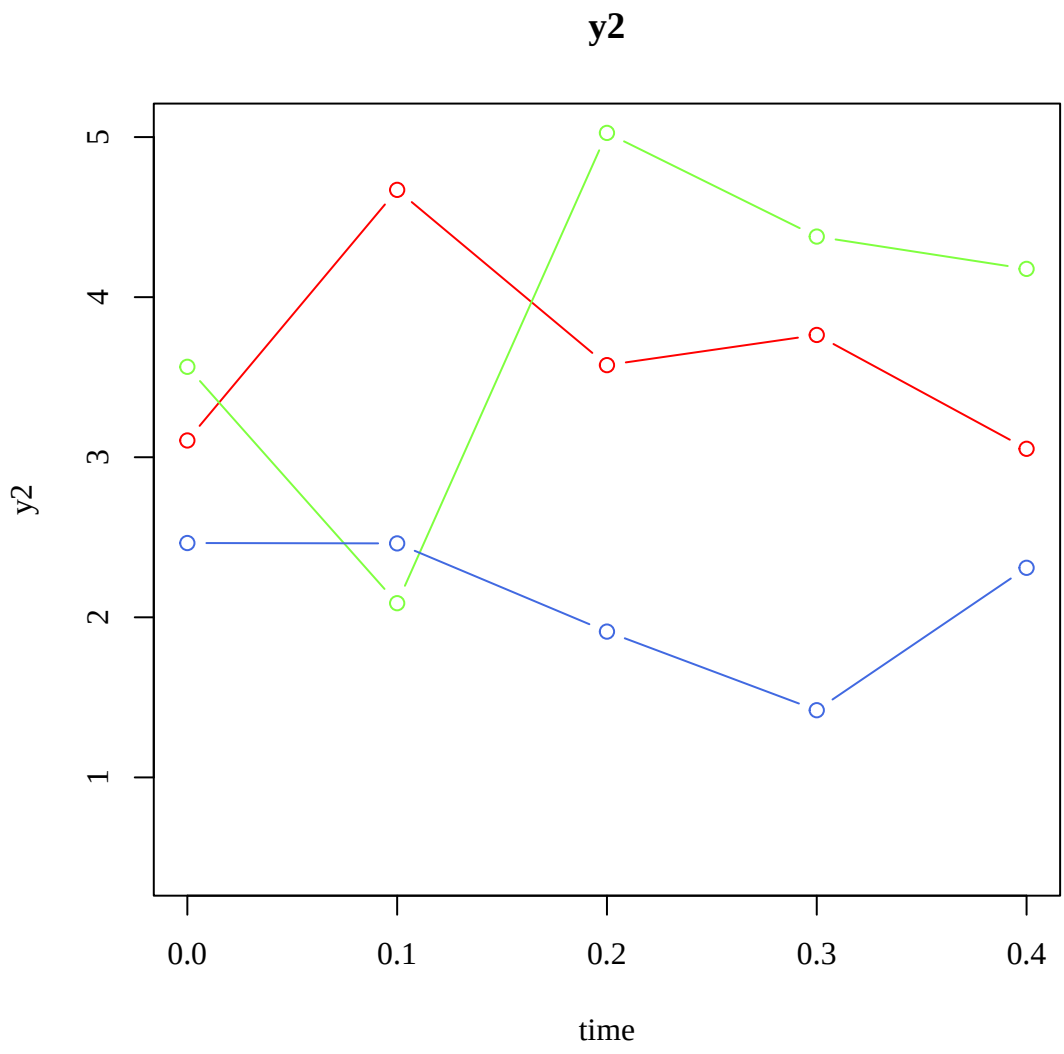

y2
y2
7
6
5
4
3
2
0.0
0.1
0.2
0.3
0.4
time

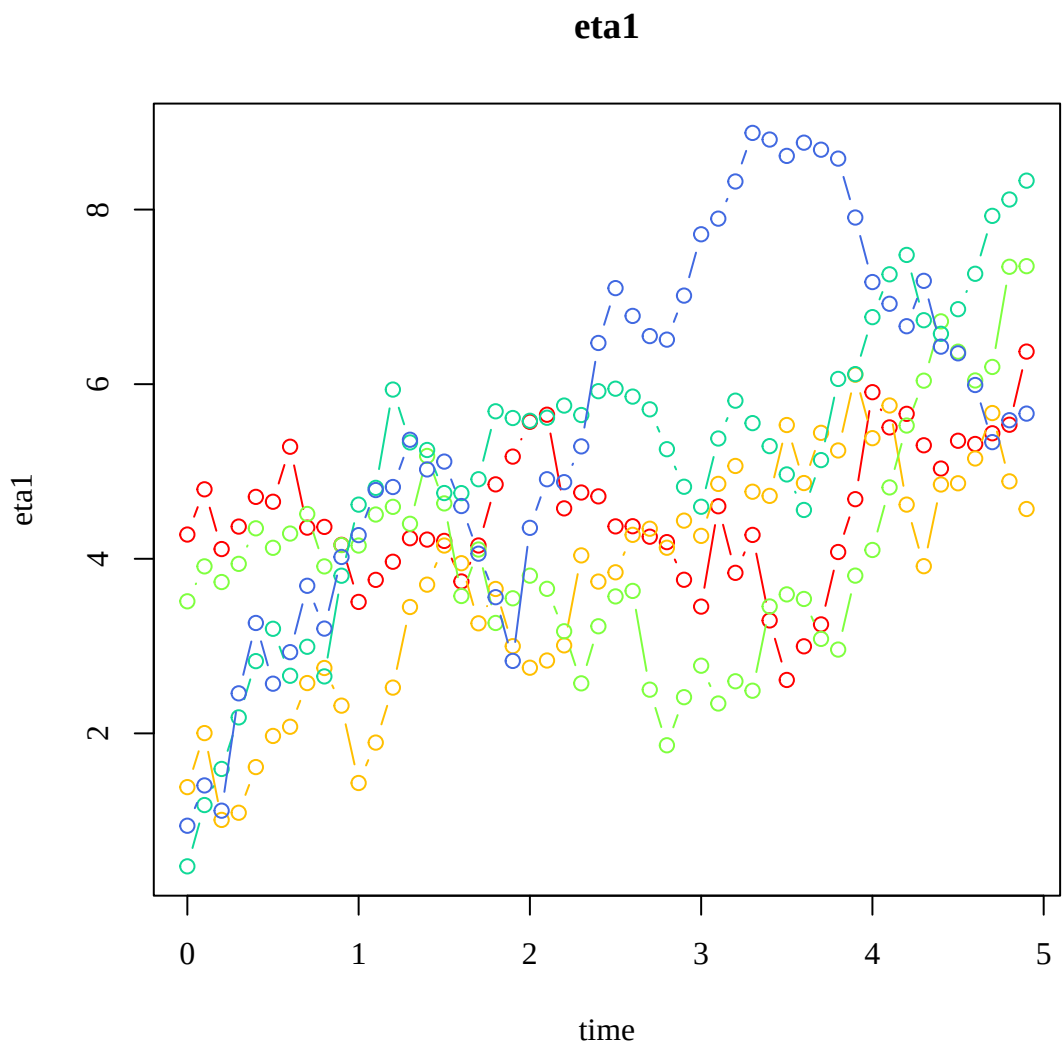

eta1
eta1
2
4
6
8
0
1
2
3
4
5
time

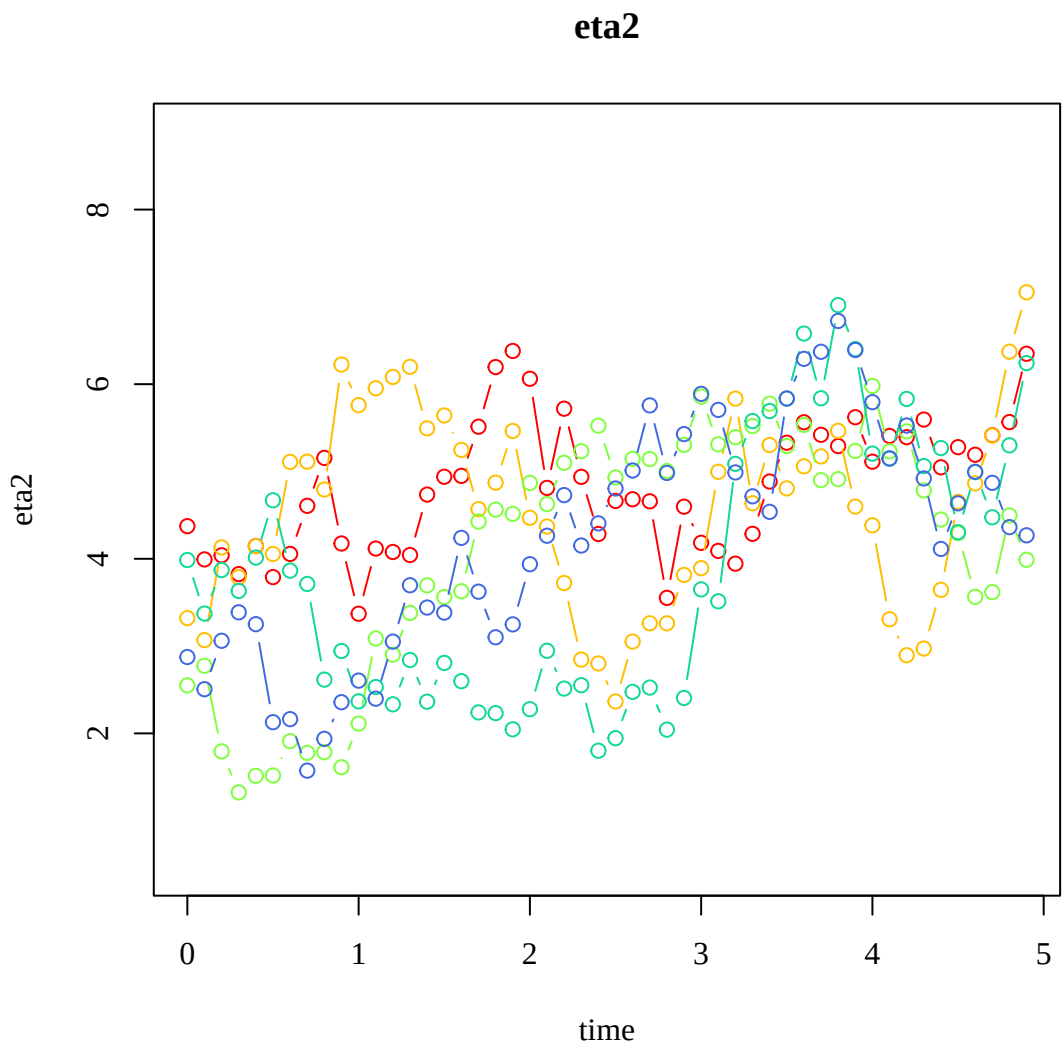



```
#> Call:
#> simStateSpace::SimSSMOUIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     mu = mu, phi = phi, sigma = sigma, nu = nu, lambda = lambda,
#>     theta = theta, gamma_eta = gamma_eta, x = x, type = 1, delta_t = delta_t,
#>     time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUIVary`
#> to convert it to a data frame or a matrix.
```

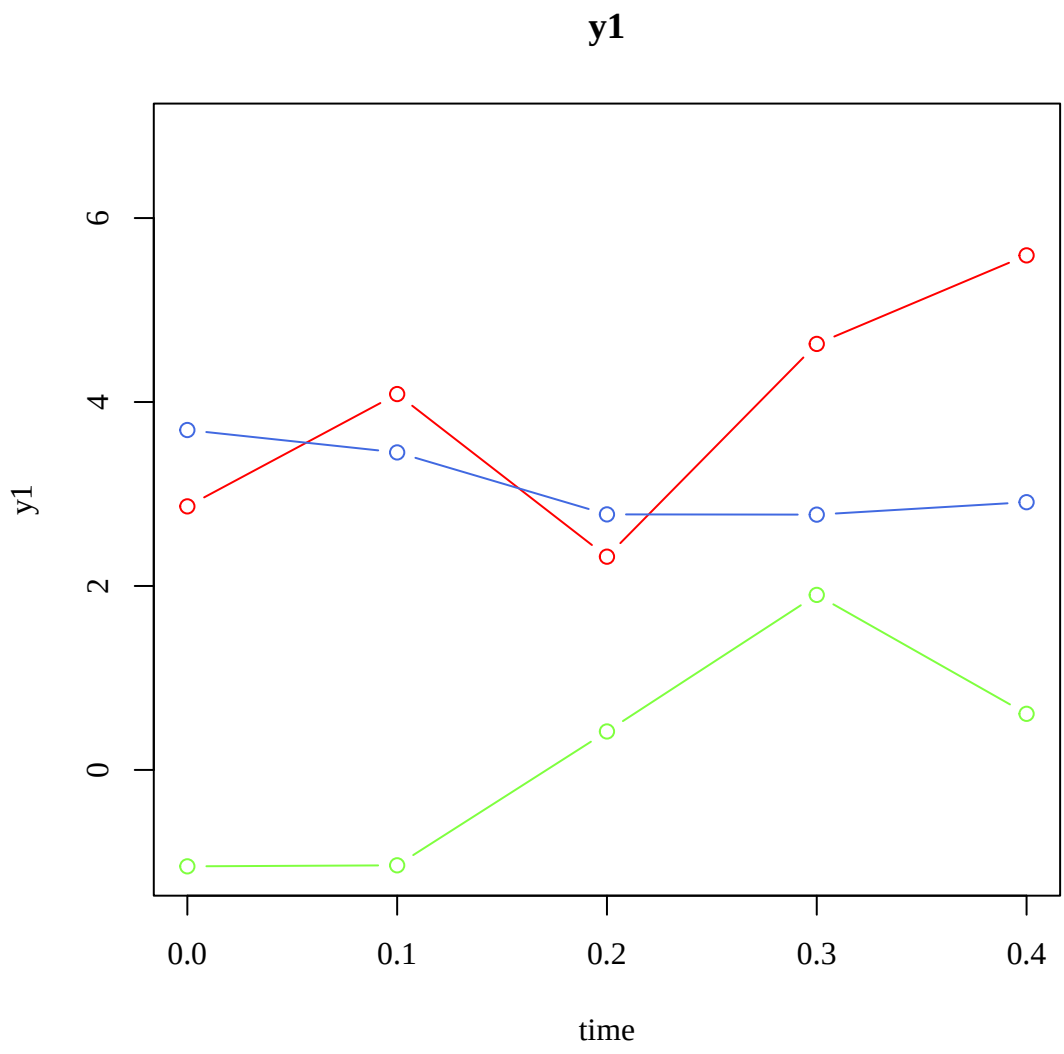

y1
y1
5
4
3
2
1
0.0
0.1
0.2
0.3
0.4
time

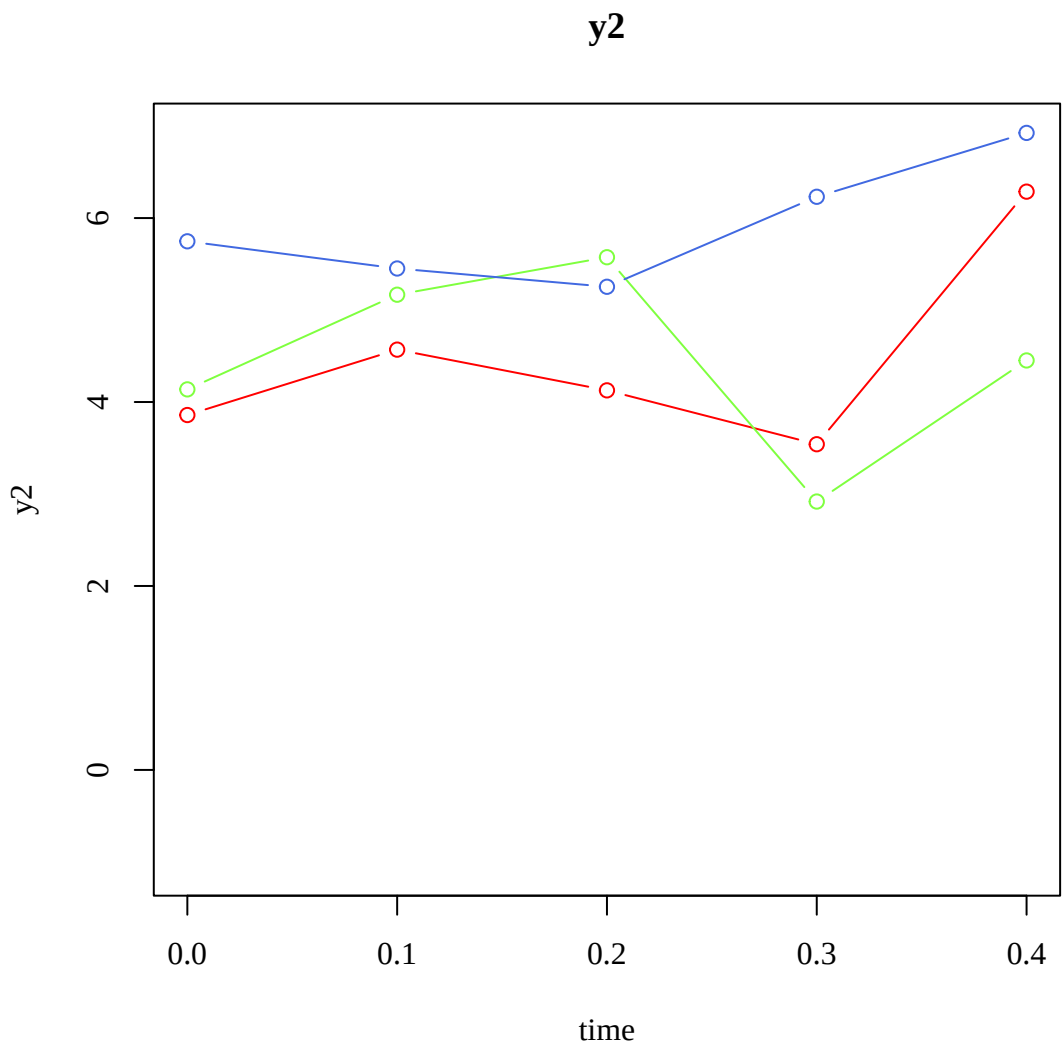

y2
y2
1
2
3
4
5
0.0
0.1
0.2
0.3
0.4
time

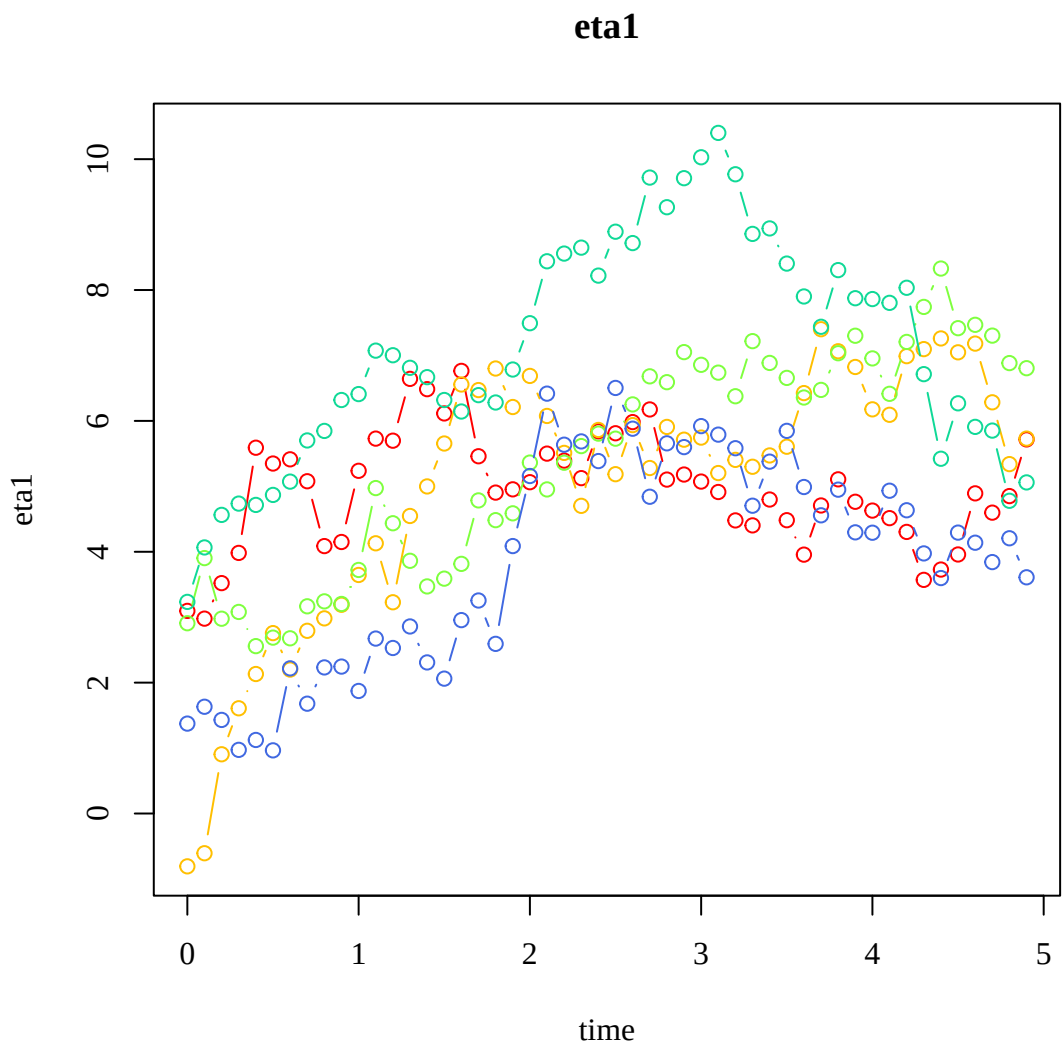

eta1
eta1
2
4
6
8
0
1
2
3
4
5
time



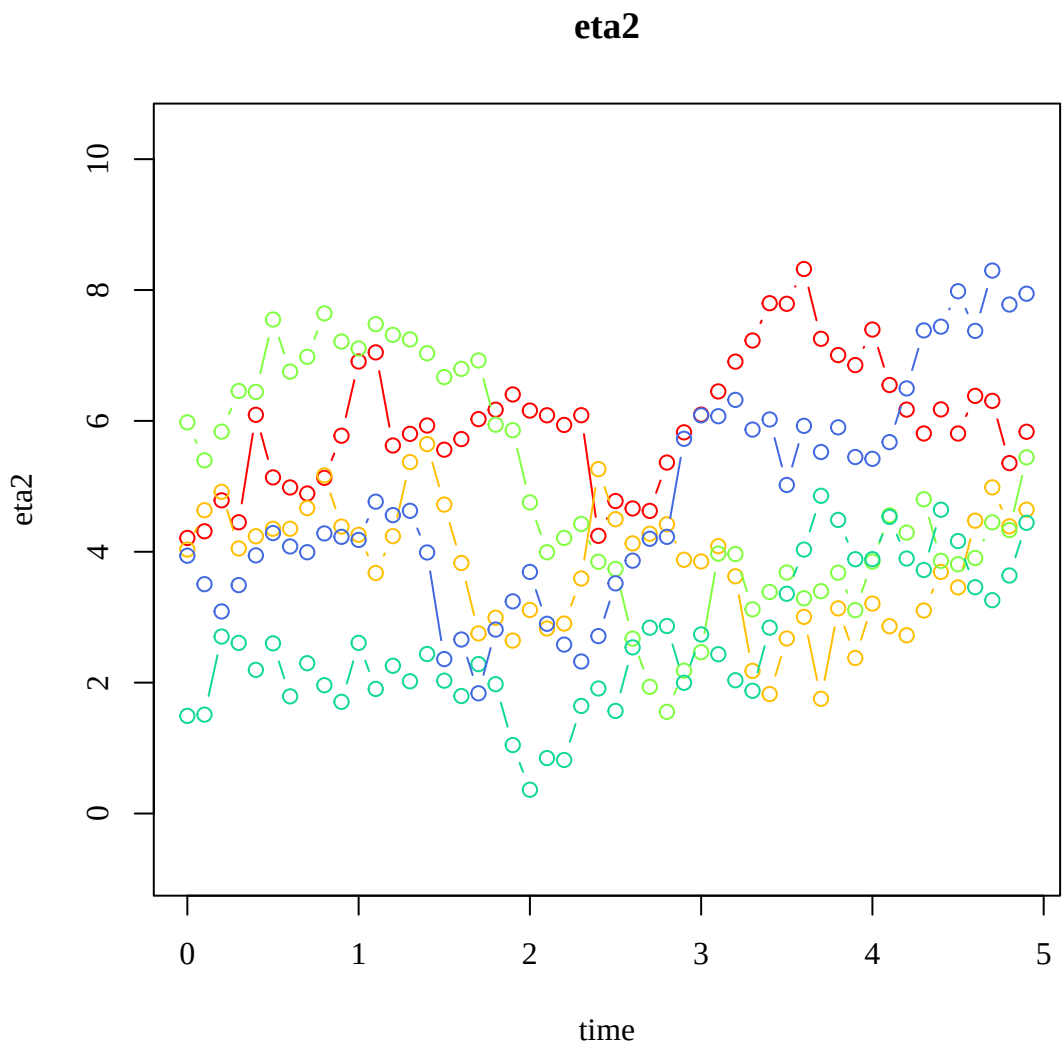

```
#> Call:
#> simStateSpace::SimSSMOUIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     mu = mu, phi = phi, sigma = sigma, nu = nu, lambda = lambda,
#>     theta = theta, gamma_y = gamma_y, gamma_eta = gamma_eta,
#>     x = x, type = 2, delta_t = delta_t, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUIVary`
#> to convert it to a data frame or a matrix.
```


y1
y1
0
2
4
6
0.0
0.1
0.2
0.3
0.4
time

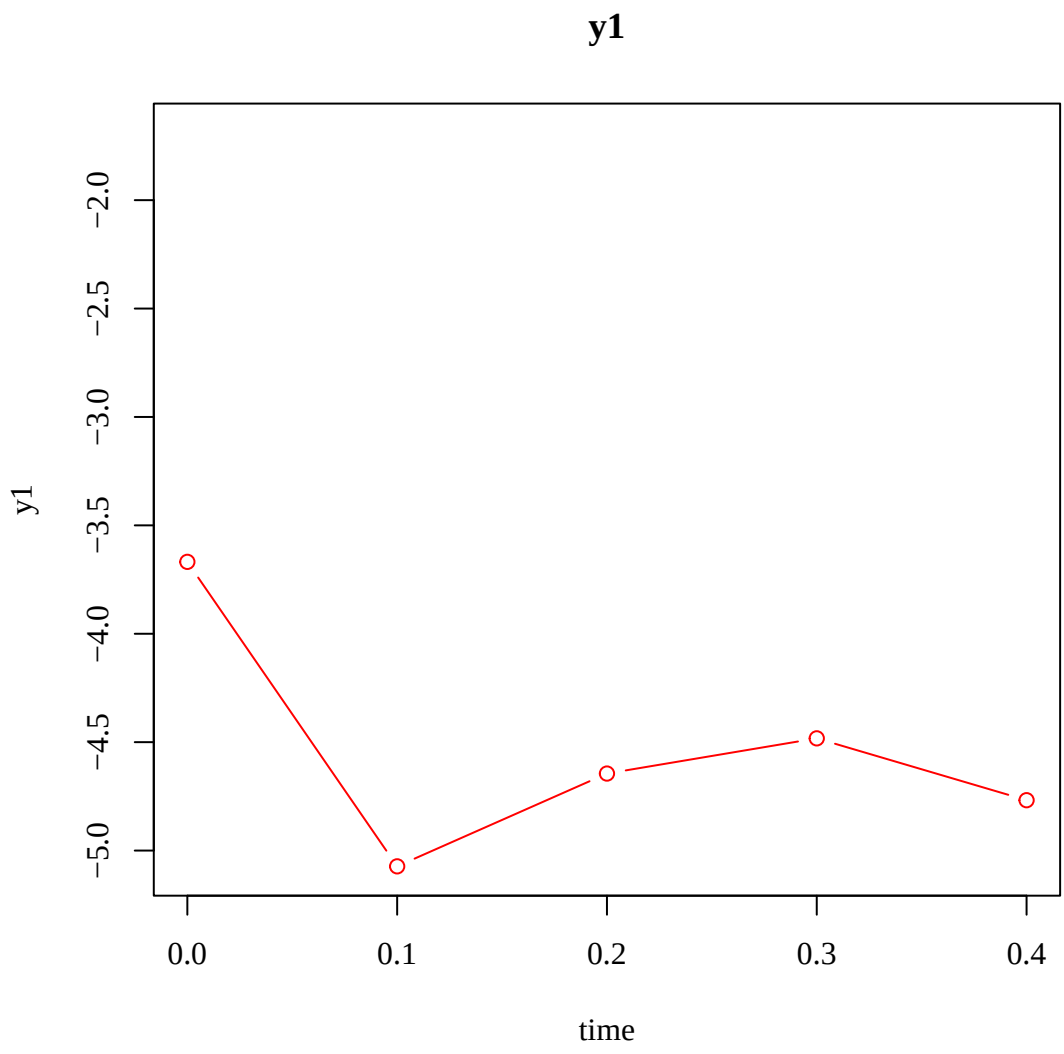

y2
y2
0
2
4
6
0.0
0.1
0.2
0.3
0.4
time

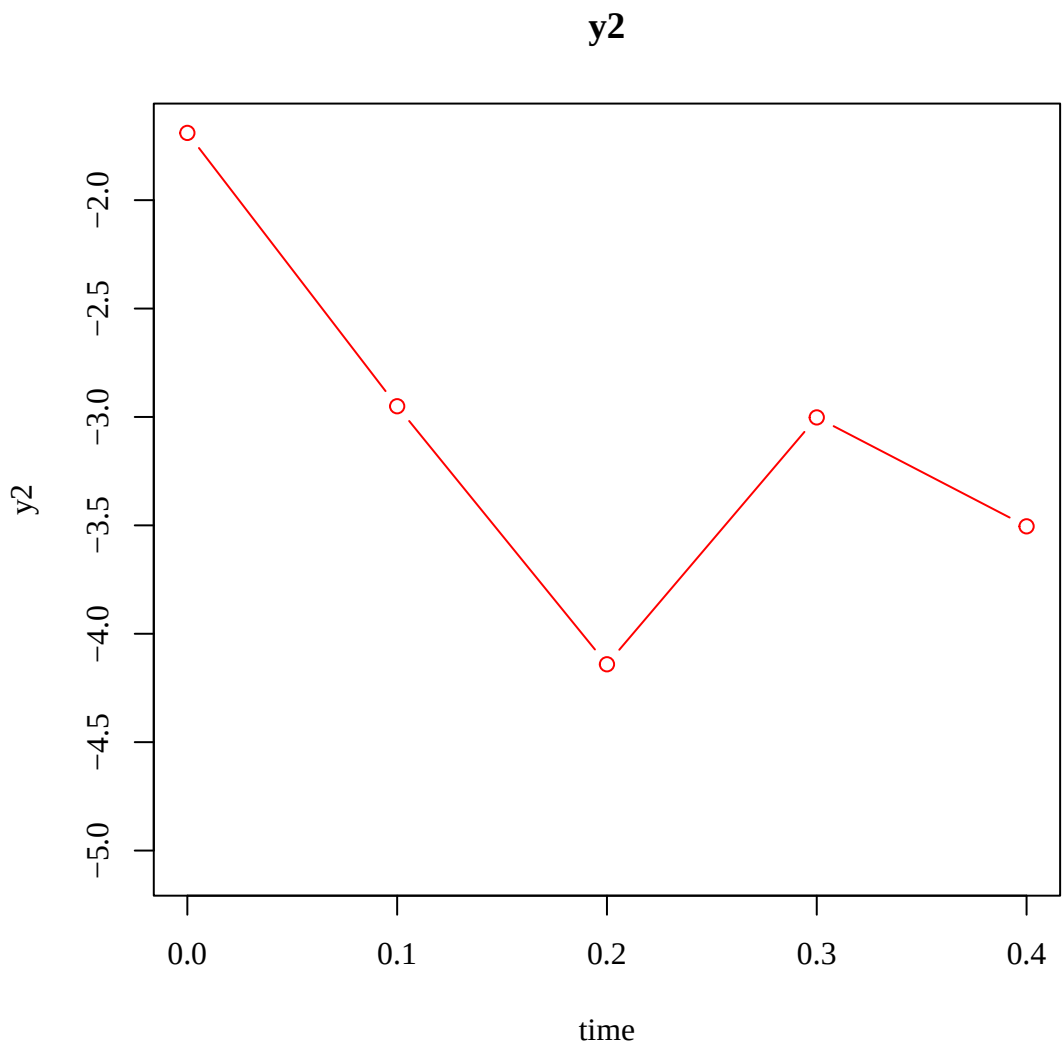


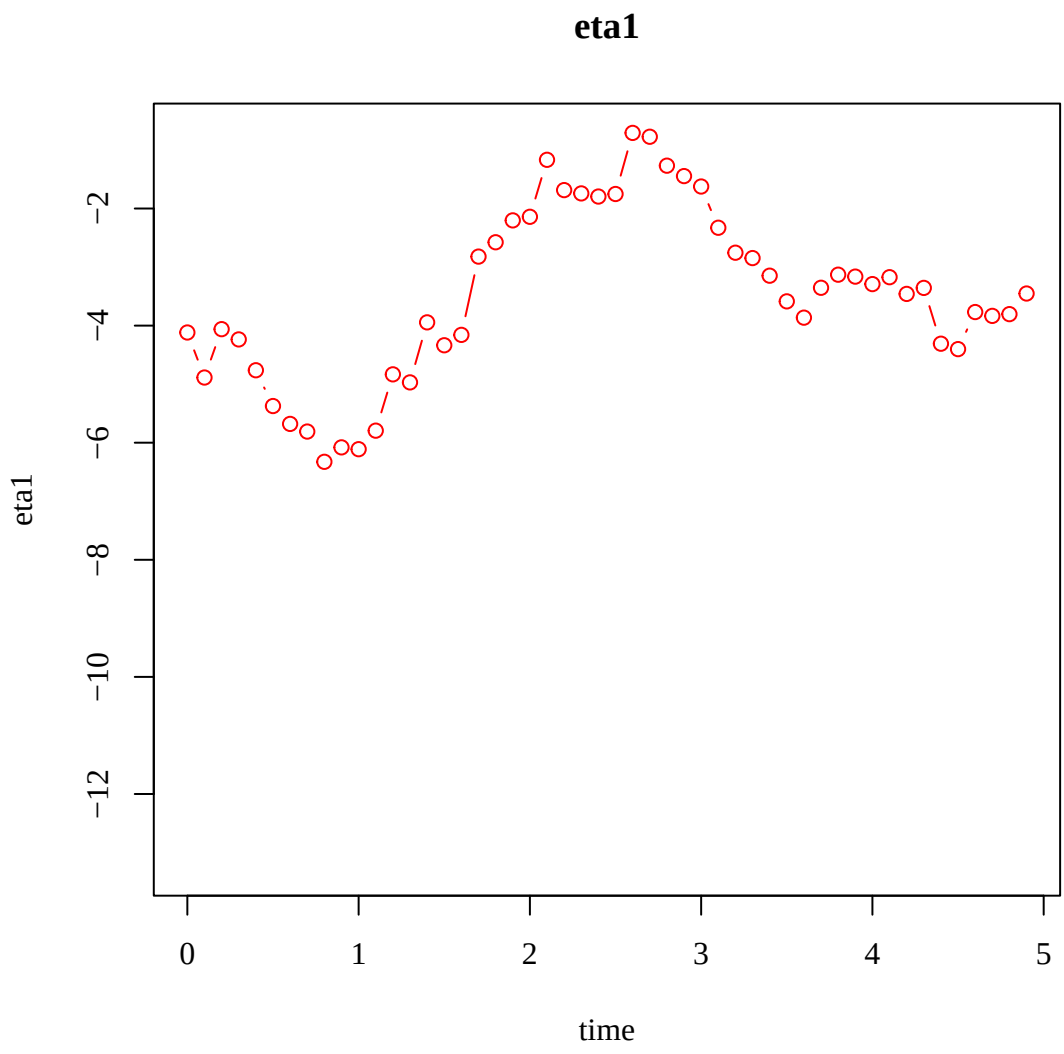

```
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-sim-ssm-ou
```

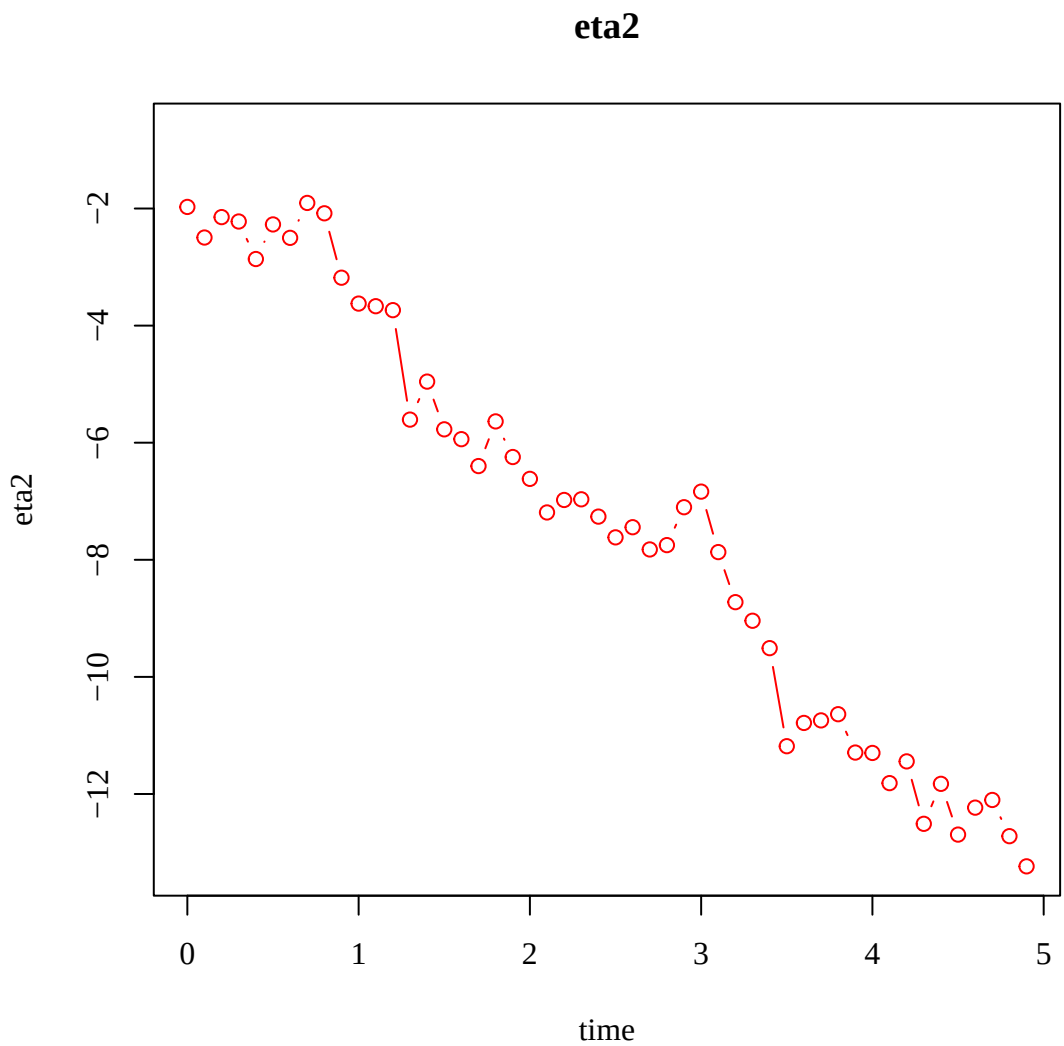



```
#> Call:
#> simStateSpace::SimSSMOU(mu0 = mu0, sigma0 = sigma0, mu = mu,
#>     phi = phi, sigma = sigma, nu = nu, lambda = lambda, theta = theta,
#>     type = 0, delta_t = delta_t, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUIVary`
#> to convert it to a data frame or a matrix.
```

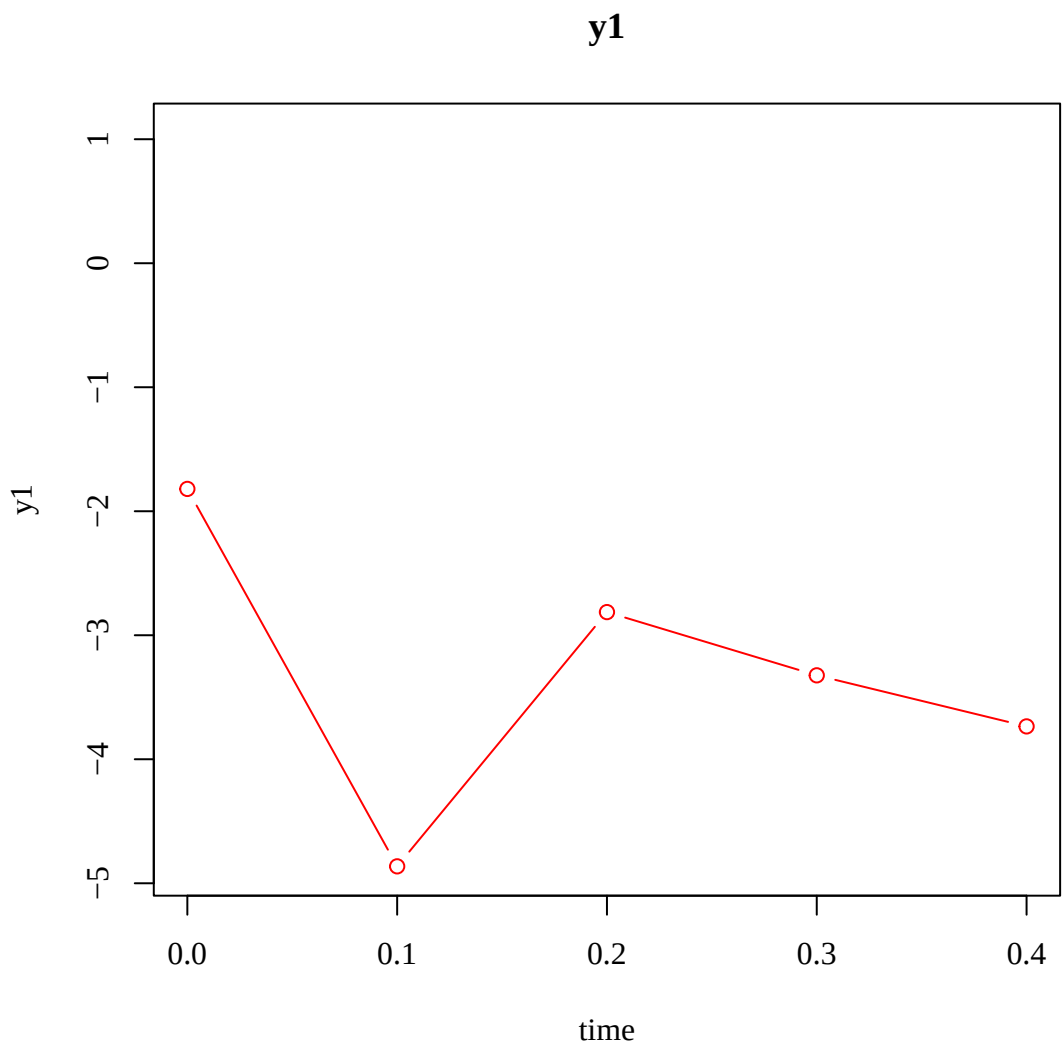

y1
−2.0
−2.5
−3.0
−3.5
−4.0
−4.5
−5.0
y1
0.0
0.1
0.2
0.3
0.4
time

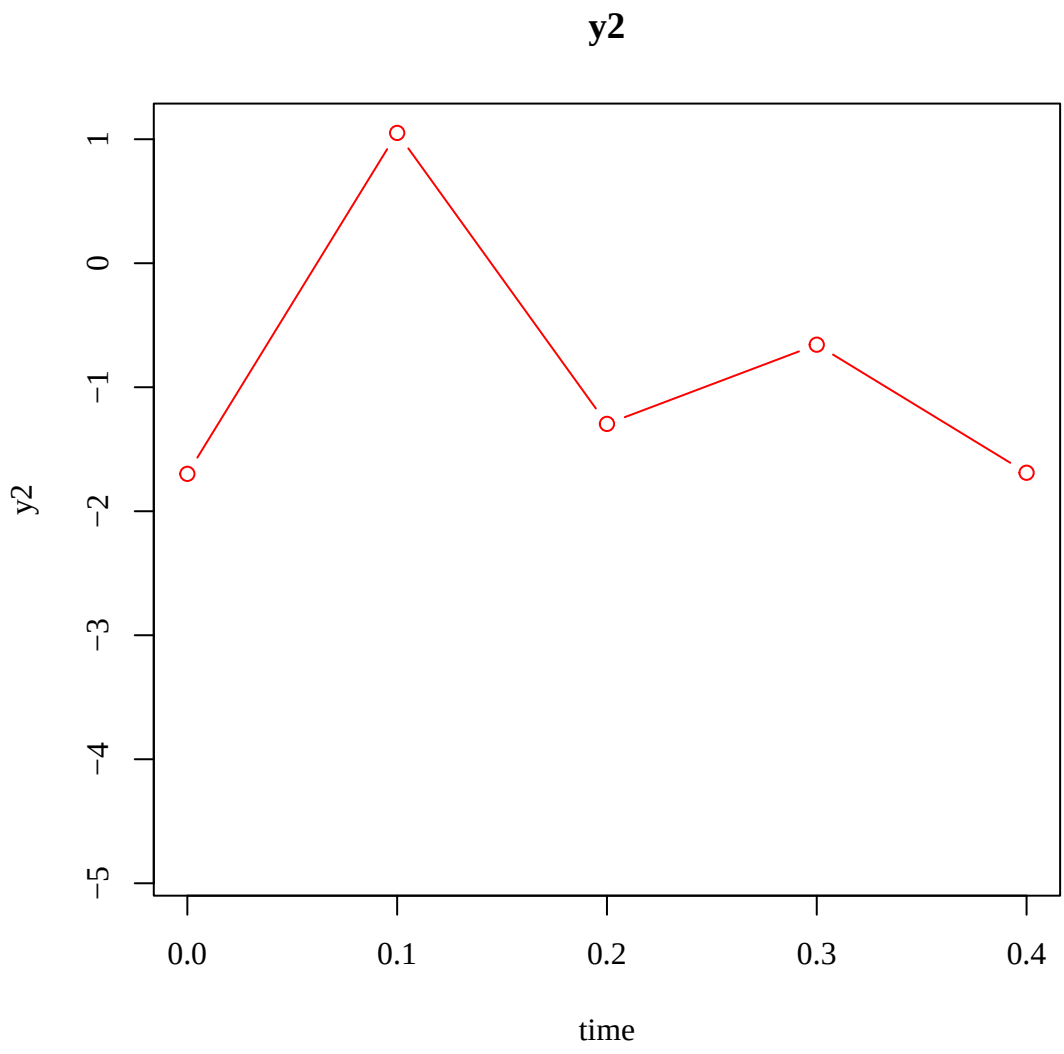

y2
y2
−2.0
−2.5
−3.0
−3.5
−4.0
−4.5
−5.0
0.0
0.1
0.2
0.3
0.4
time

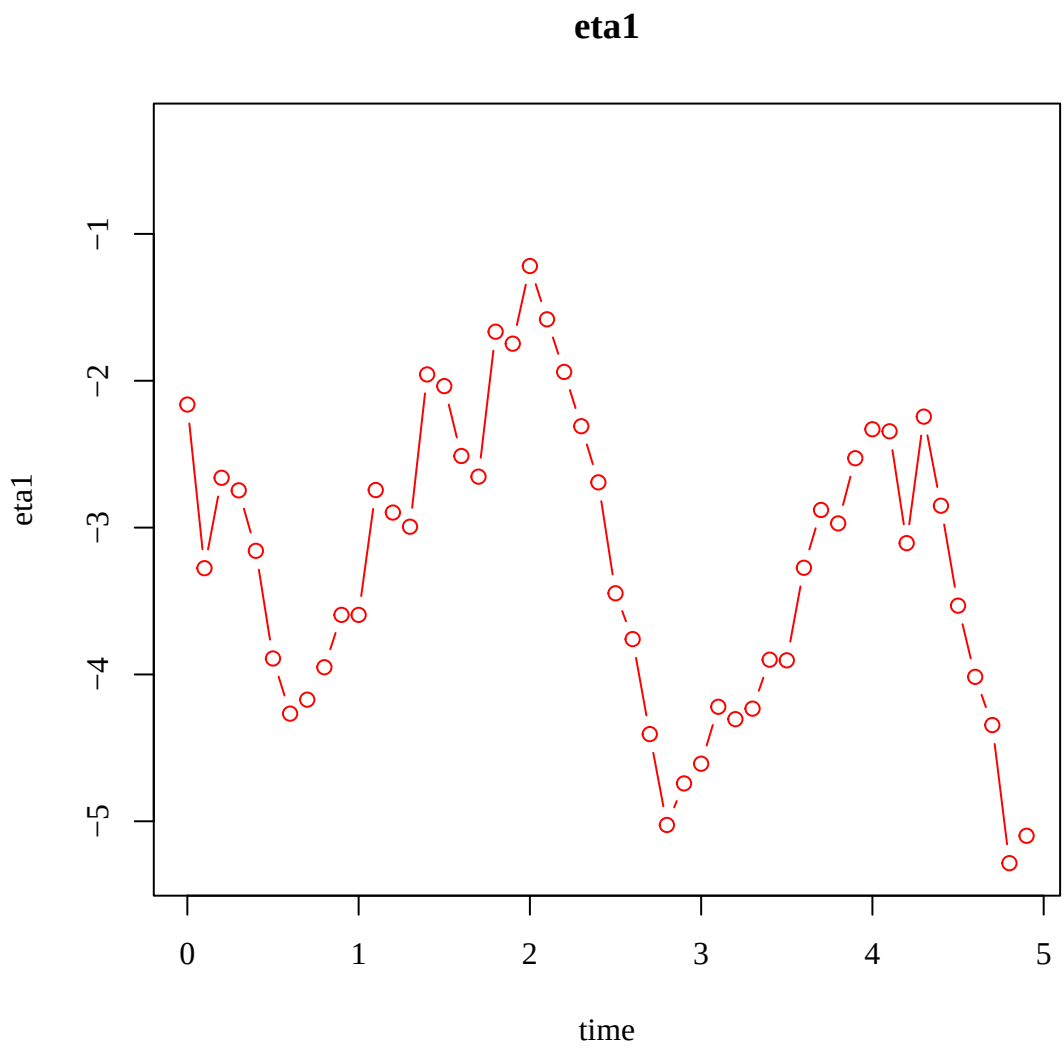

eta1
eta1
−2
−4
−6
−8
−10
−12
0
1
2
3
4
5
time

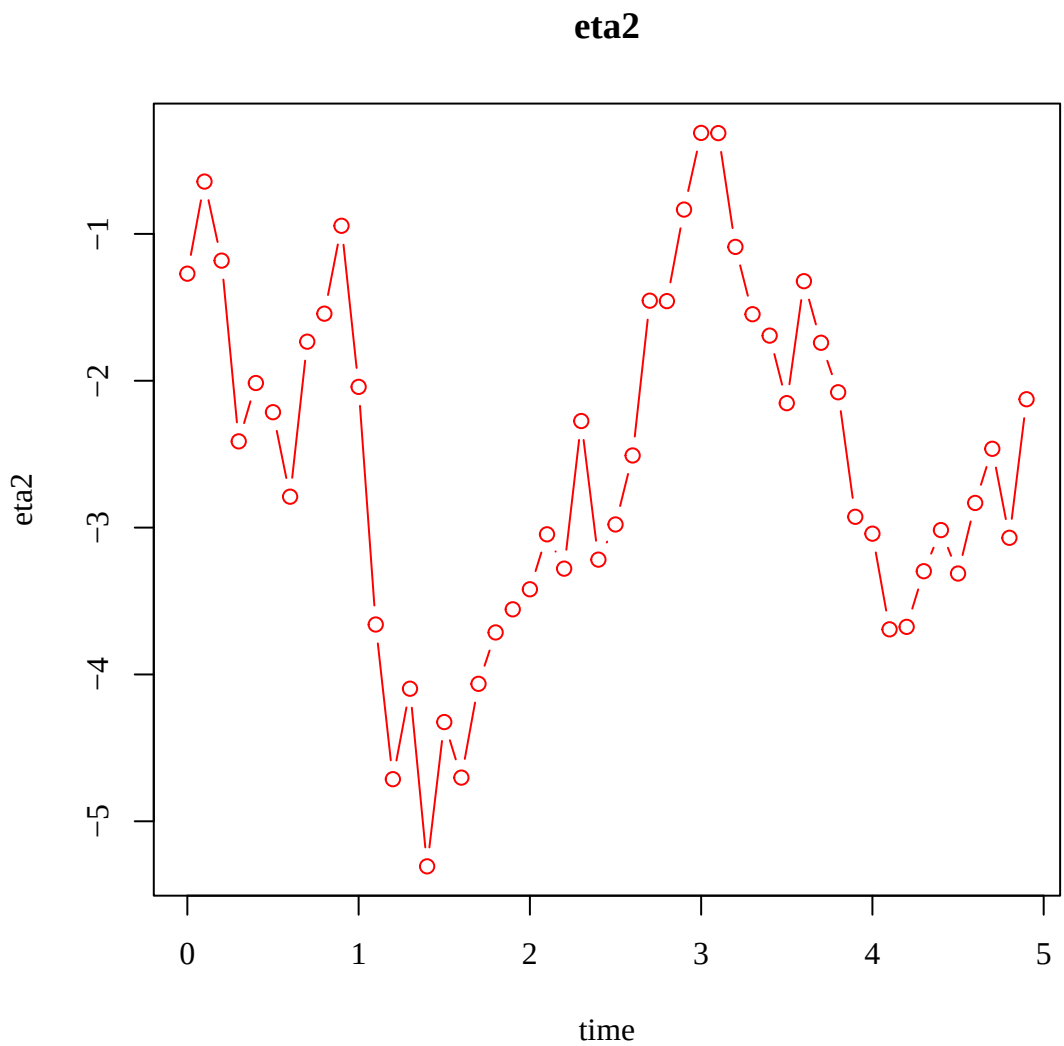



```
#> Call:
#> simStateSpace::SimSSMOU(mu0 = mu0, sigma0 = sigma0, mu = mu,
#>     phi = phi, sigma = sigma, nu = nu, lambda = lambda, theta = theta,
#>     gamma_eta = gamma_eta, x = x, type = 1, delta_t = delta_t,
#>     time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUIVary`
#> to convert it to a data frame or a matrix.
```

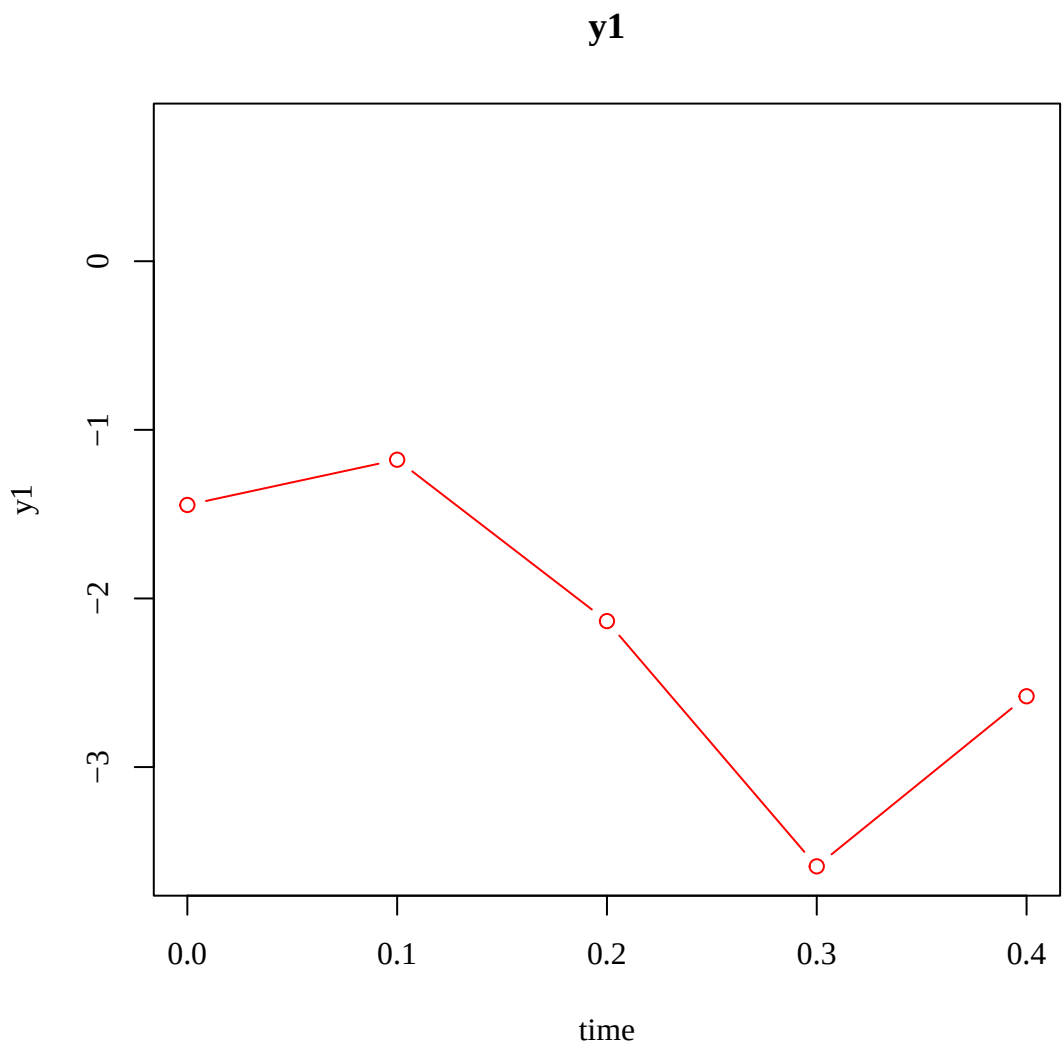

y1
1
0
−1
−2
−3
−4
−5
y1
0.0
0.1
0.2
0.3
0.4
time

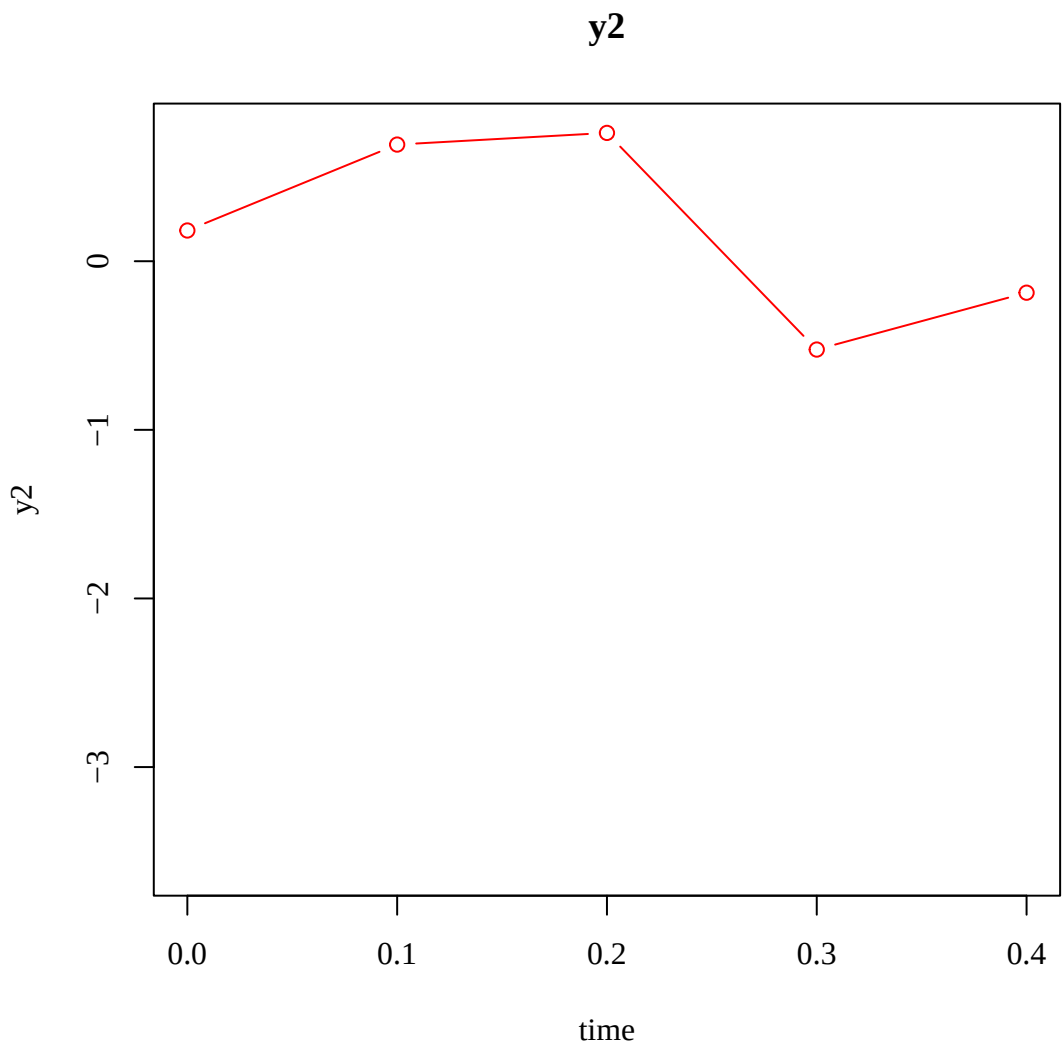

y2
1
0
−1
−2
−3
−4
−5
y2
0.0
0.1
0.2
0.3
0.4
time

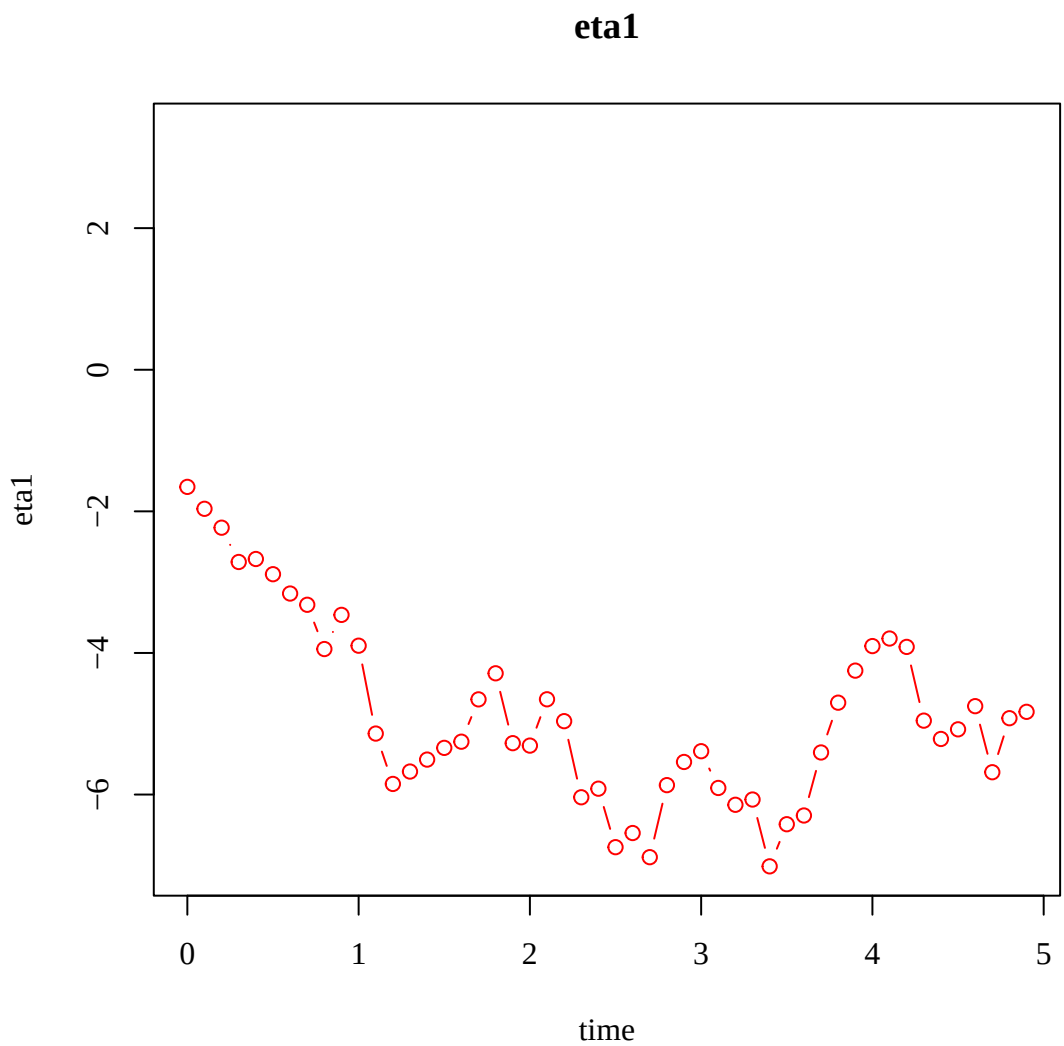

eta1
eta1
−1
−2
−3
−4
−5
0
1
2
3
4
5
time



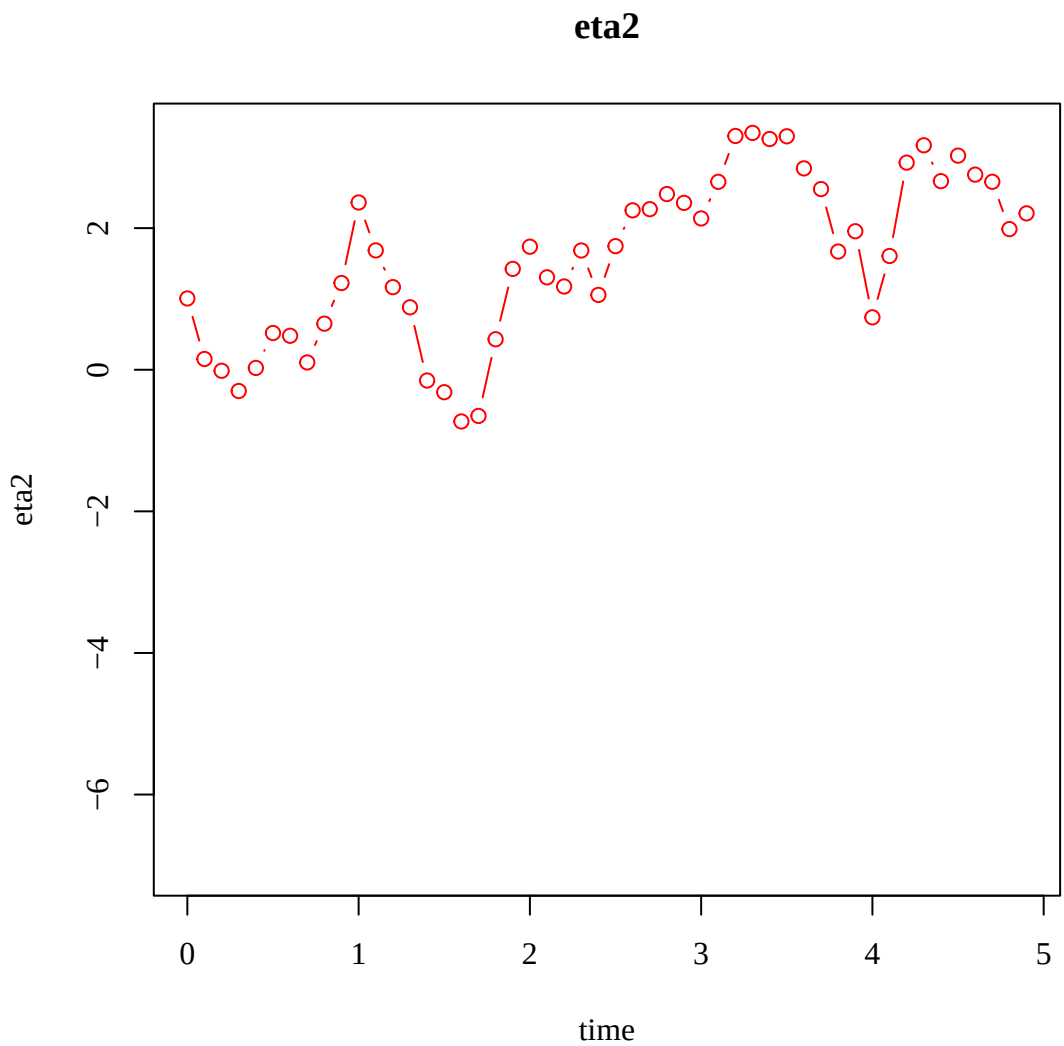

```
#> Call:
#> simStateSpace::SimSSMOU(mu0 = mu0, sigma0 = sigma0, mu = mu,
#>     phi = phi, sigma = sigma, nu = nu, lambda = lambda, theta = theta,
#>     gamma_y = gamma_y, gamma_eta = gamma_eta, x = x, type = 2,
#>     delta_t = delta_t, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMOUIVary`
#> to convert it to a data frame or a matrix.
```


y1
0
−1
−2
−3
y1
0.0
0.1
0.2
0.3
0.4
time

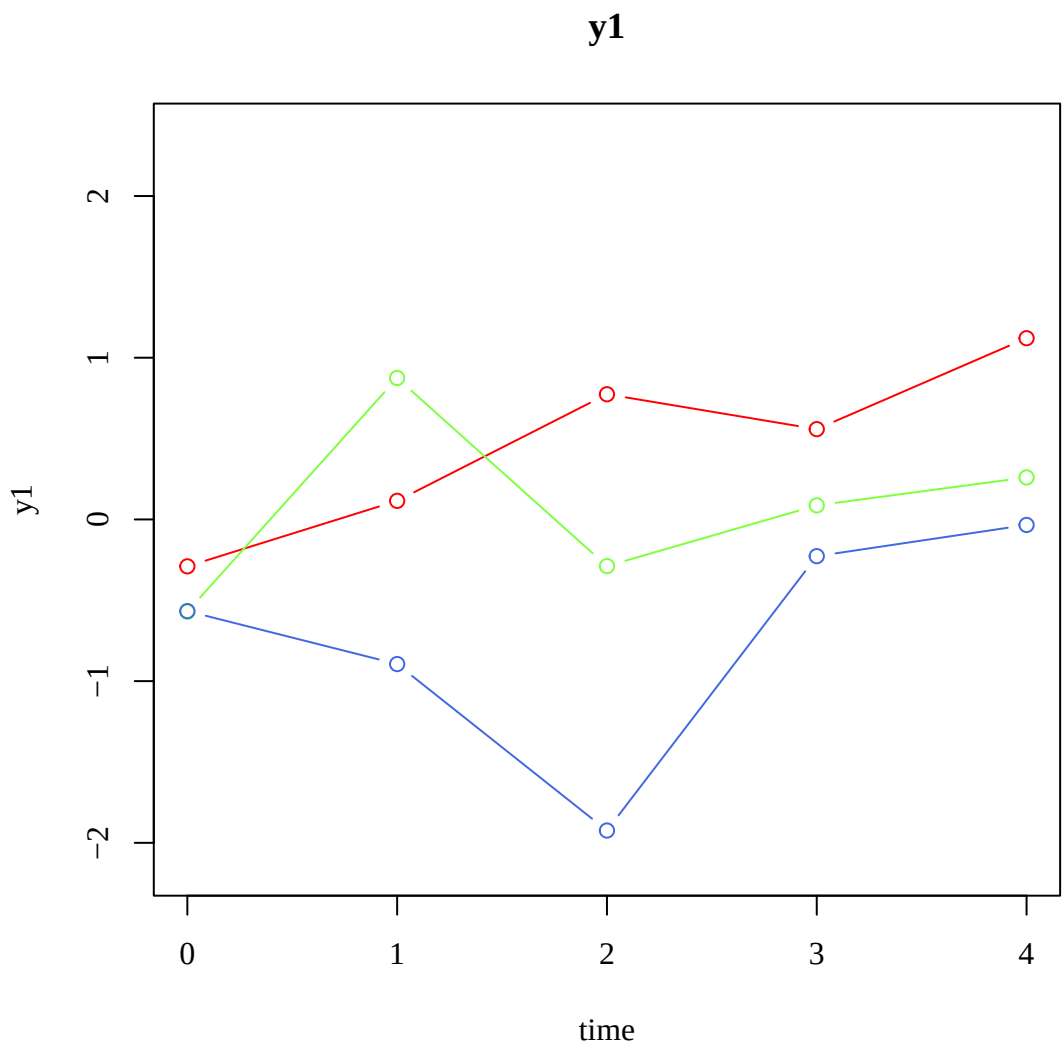

y2
0
−1
−2
−3
y2
0.0
0.1
0.2
0.3
0.4
time

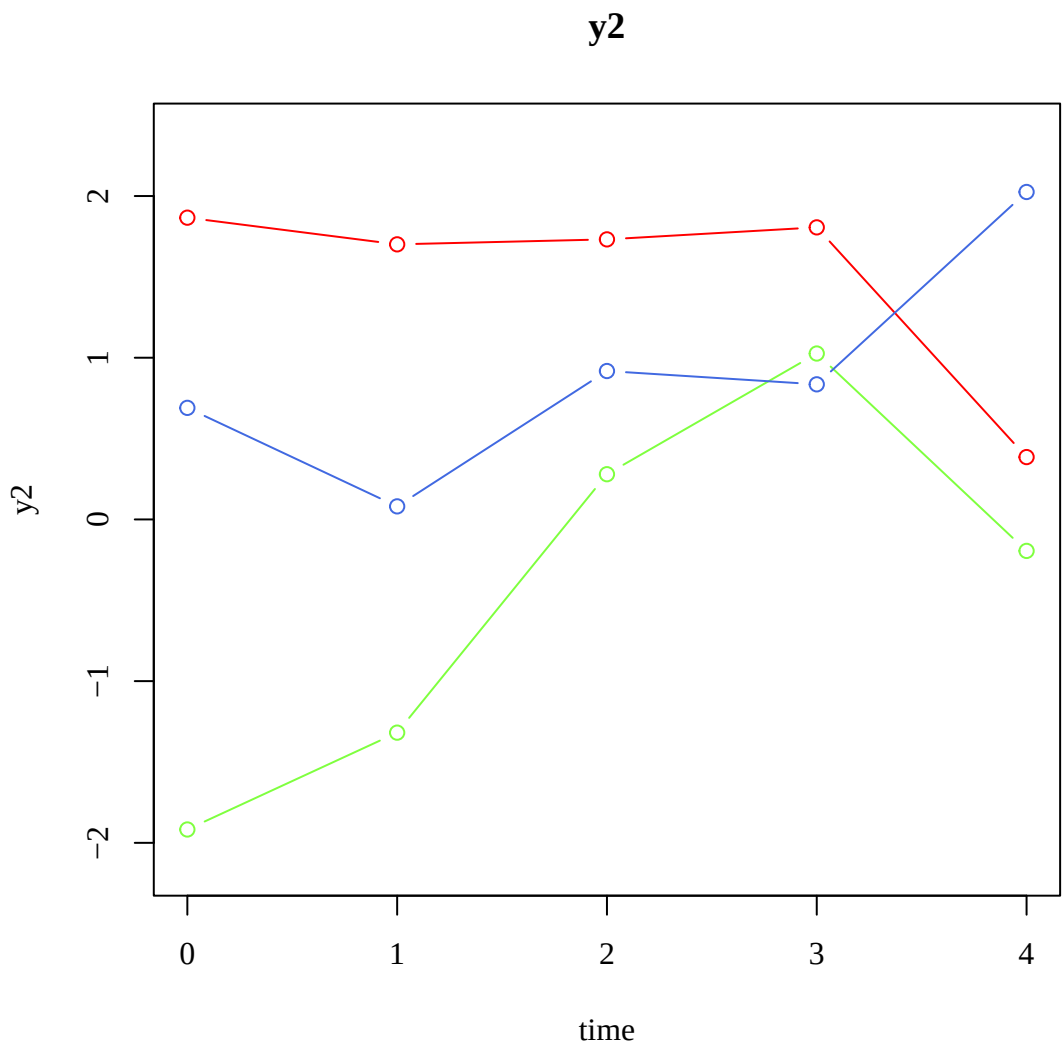


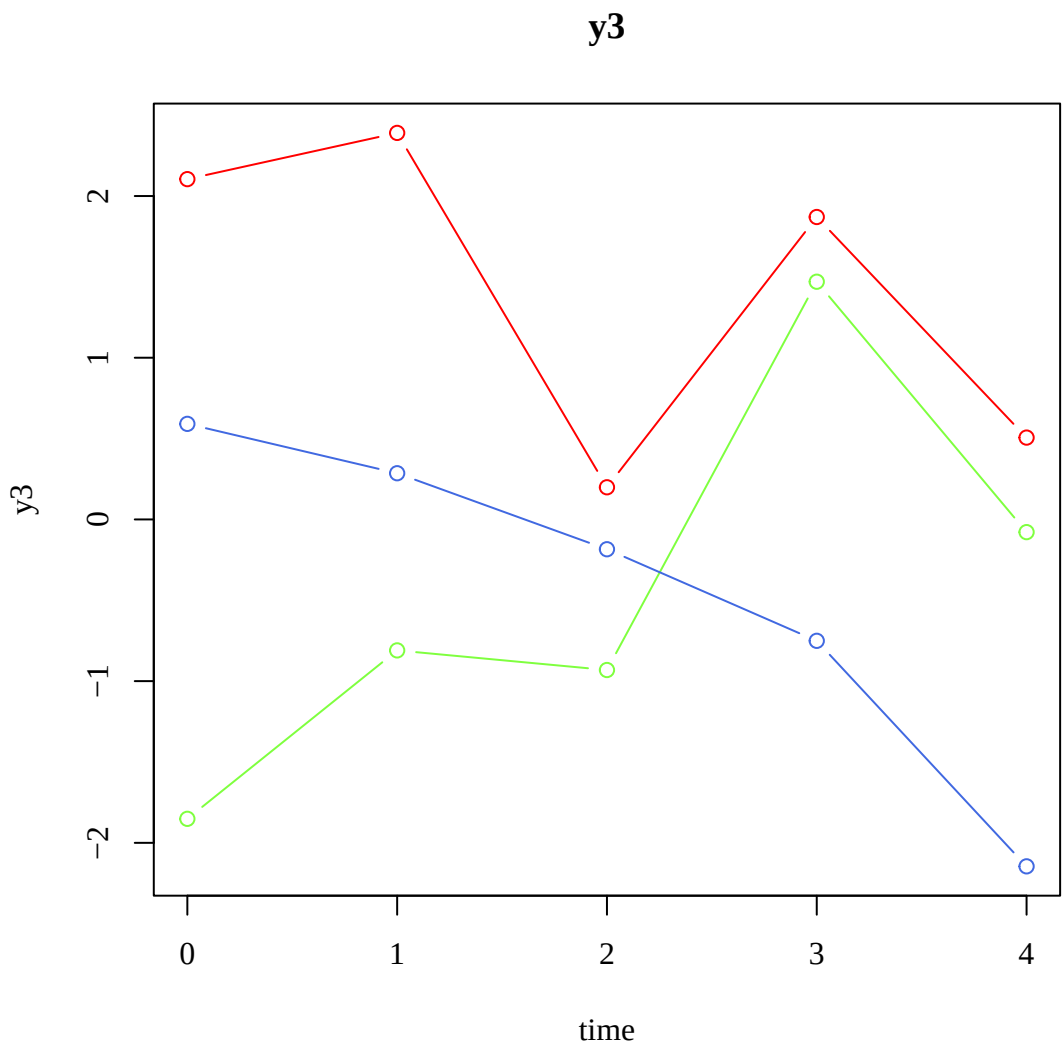

```
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-sim-ssm-var-fixed
```

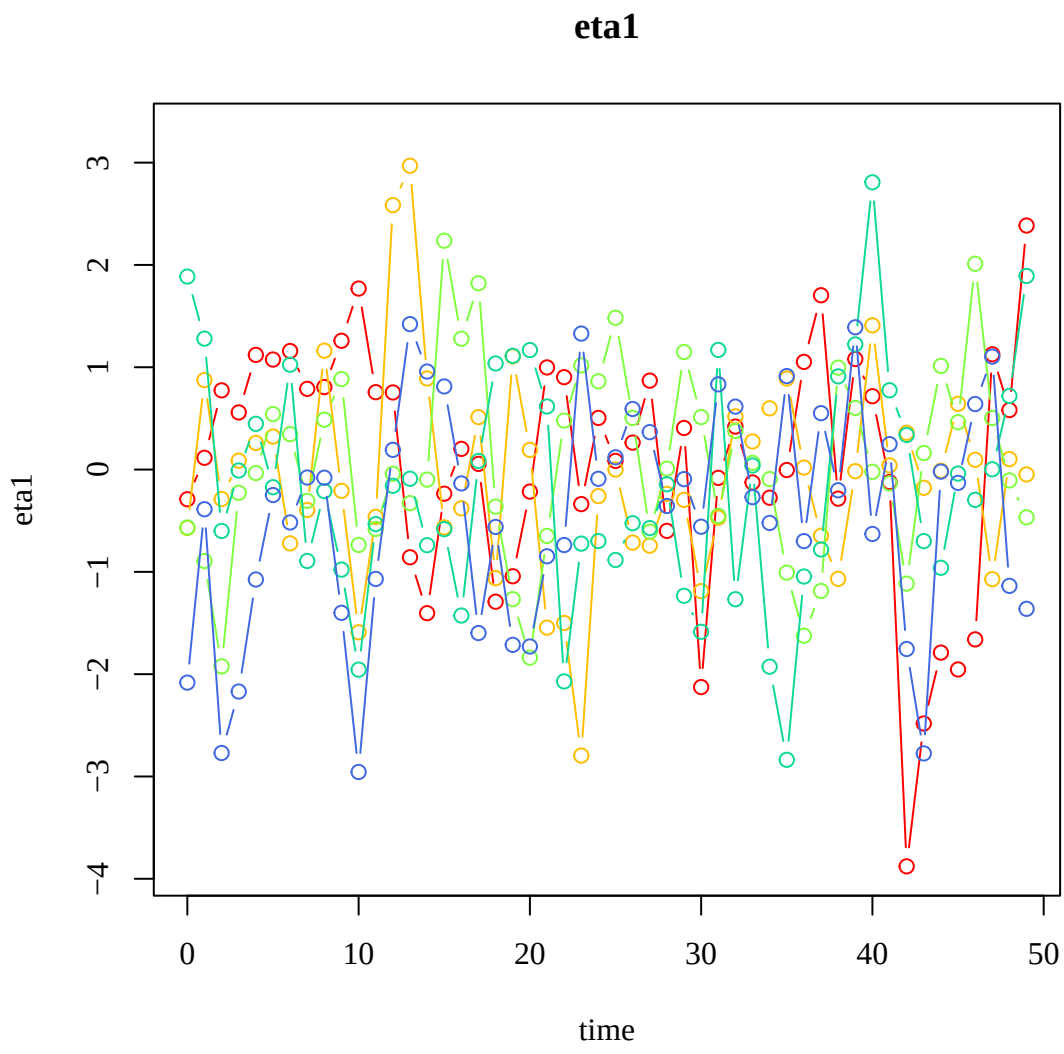


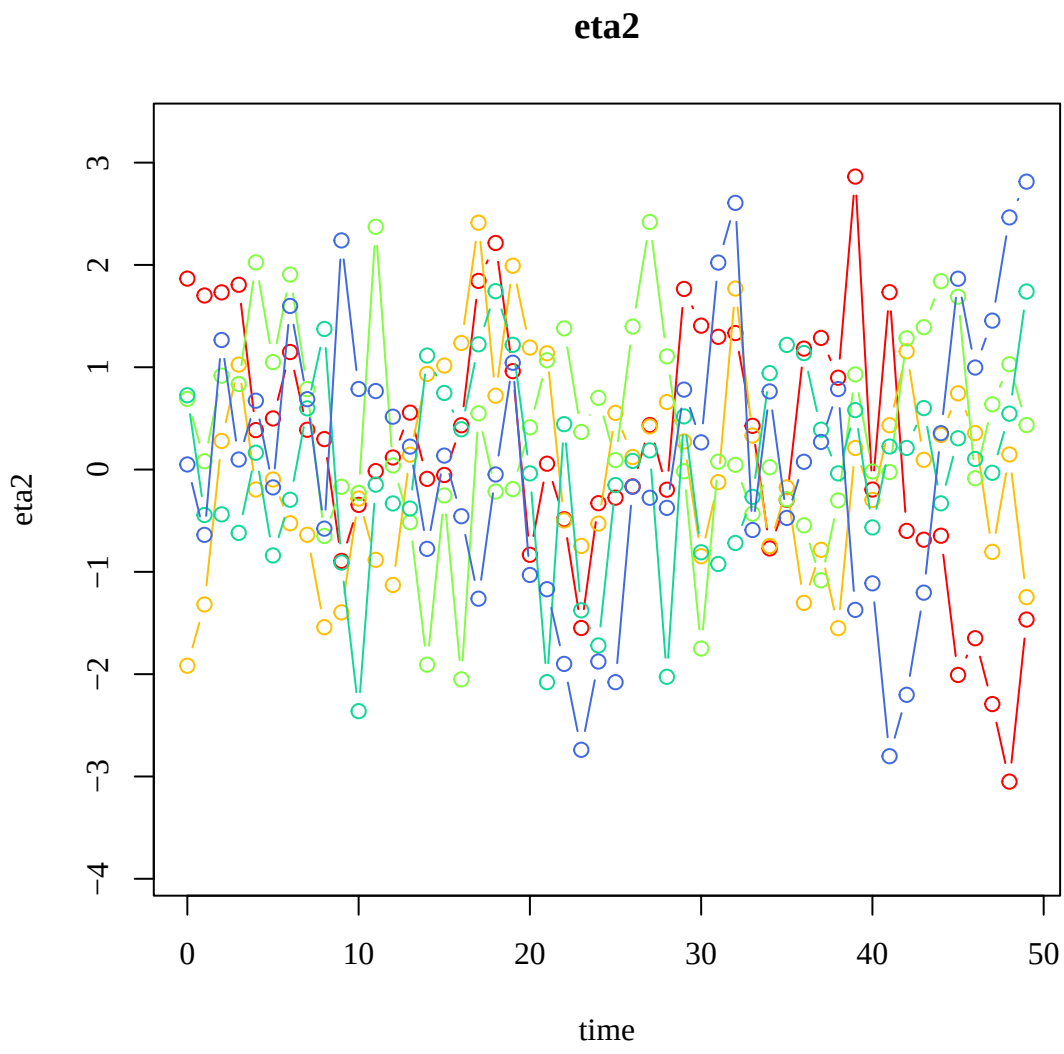

```
#> Call:
#> simStateSpace::SimSSMVARFixed(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMVARFixed`
#> to convert it to a data frame or a matrix.
```

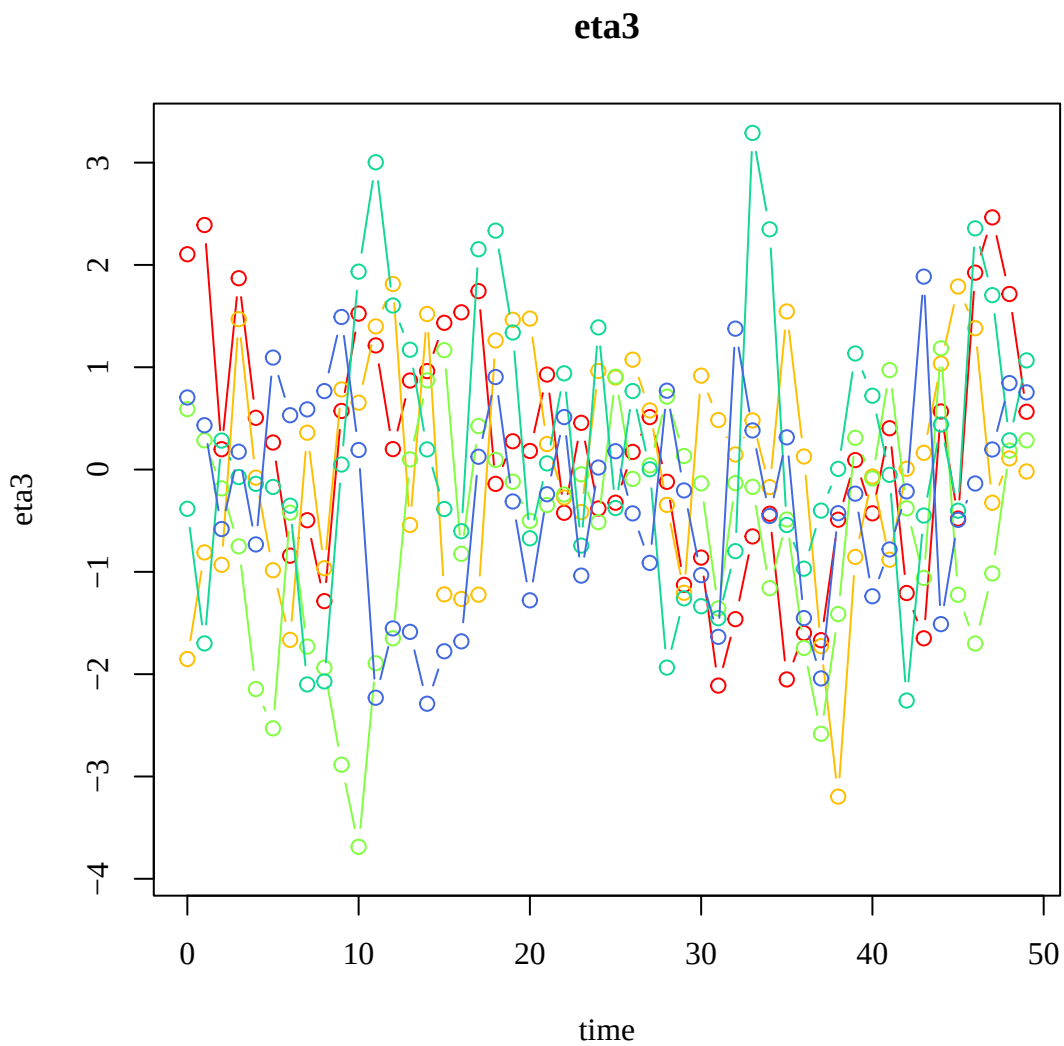

y1
y1
2
1
0
−1
−2
0
1
2
3
4
time


y2
y2
2
1
0
−1
−2
0
1
2
3
4
time

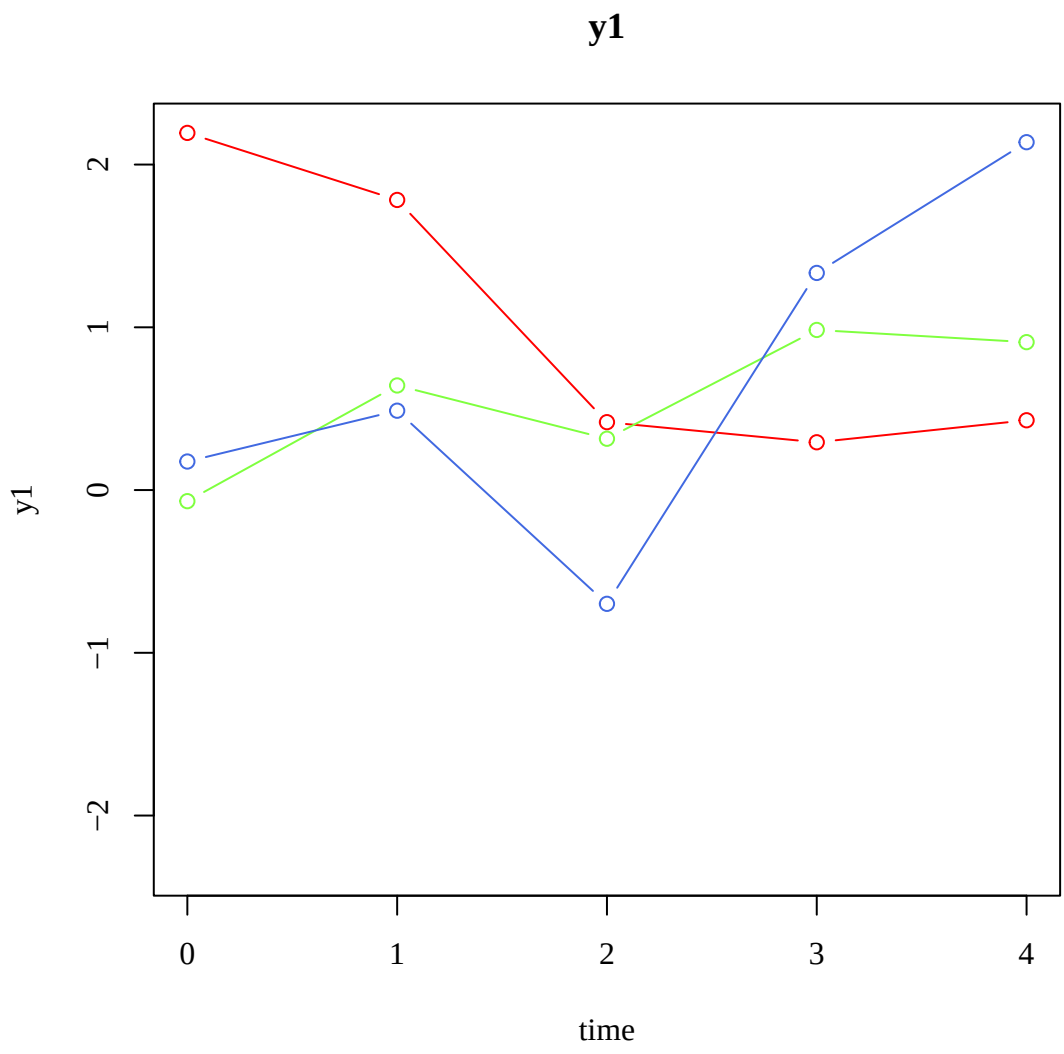

y3
y3
2
1
0
−1
−2
0
1
2
3
4
time

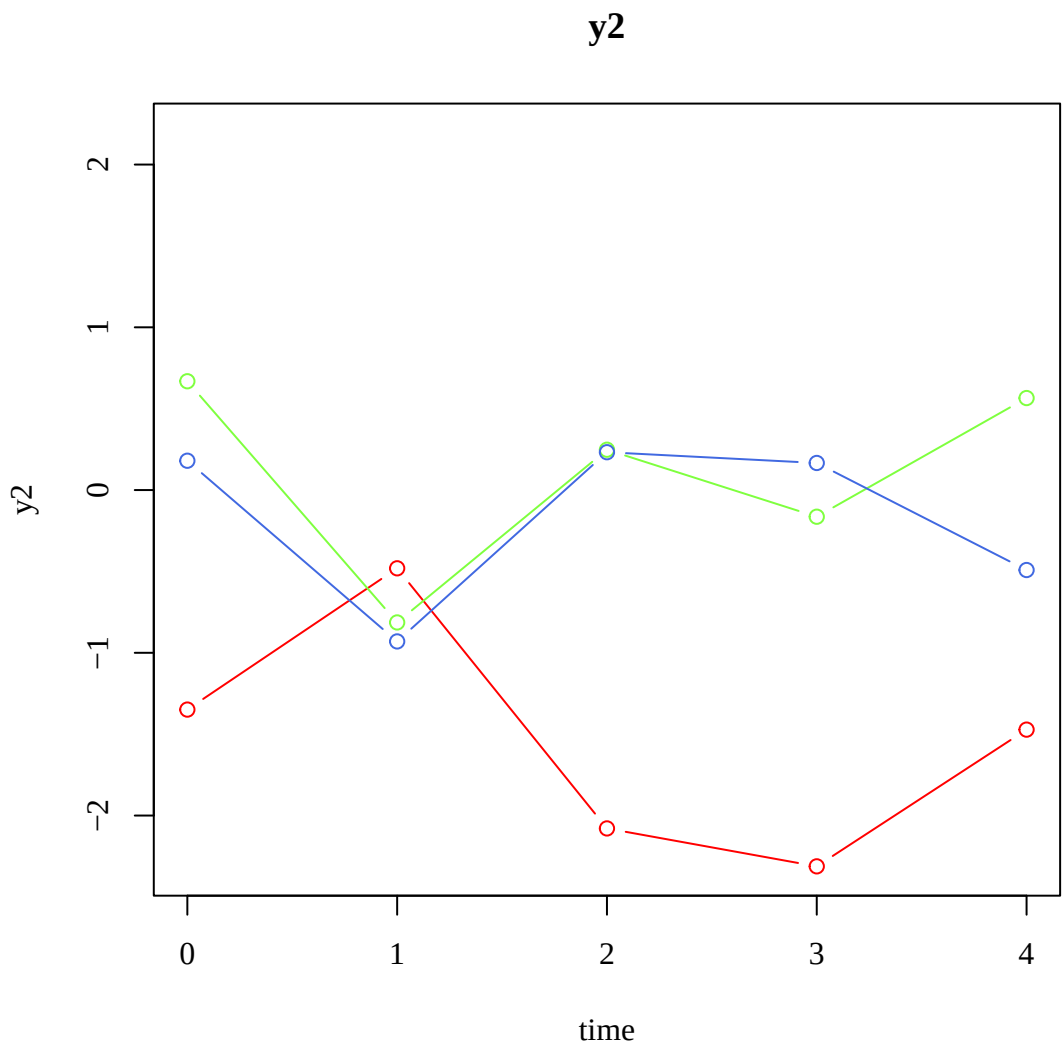

eta1
eta1
time
3
2
1
0
−1
−2
−3
−4
0
10
20
30
40
50

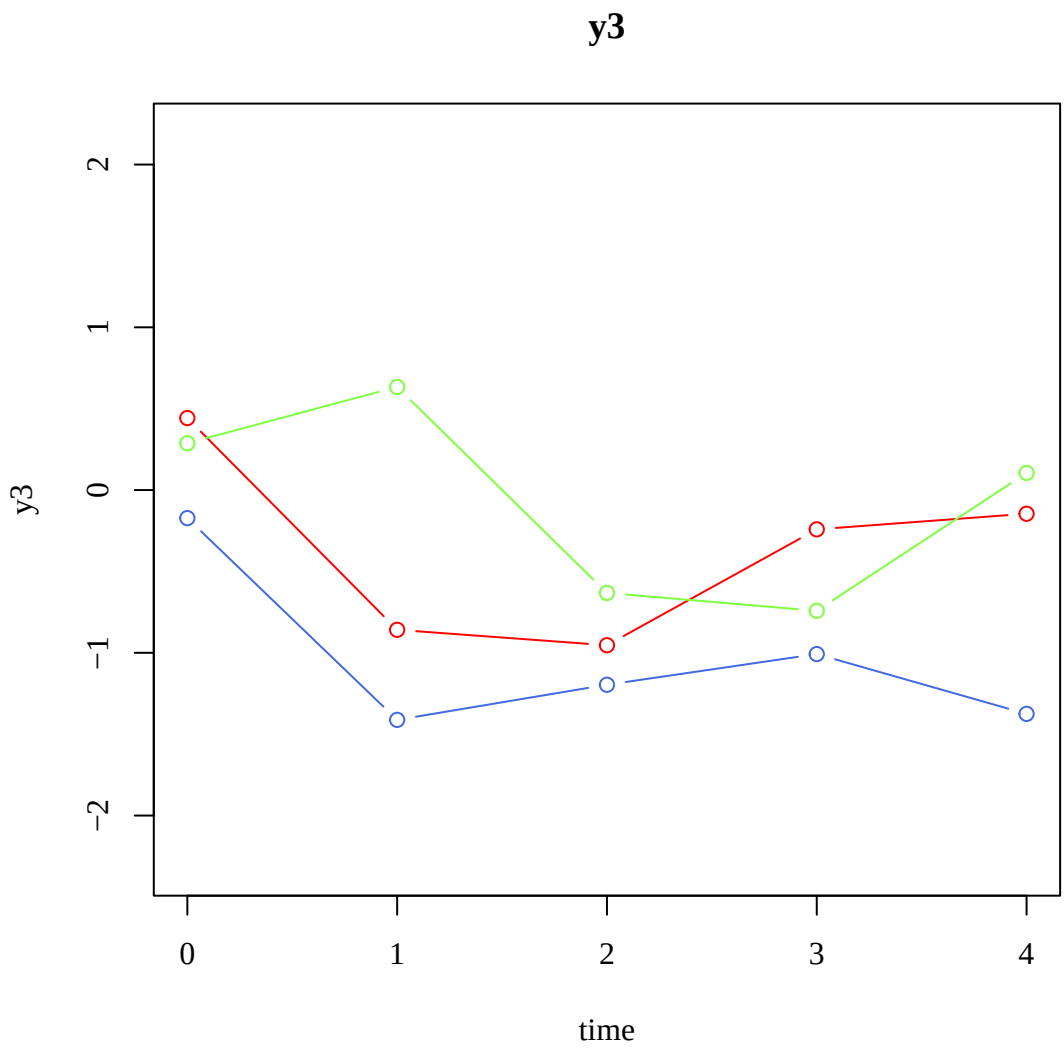

eta2
eta2
3
2
1
0
−1
−2
−3
−4
0
10
20
30
40
50
time

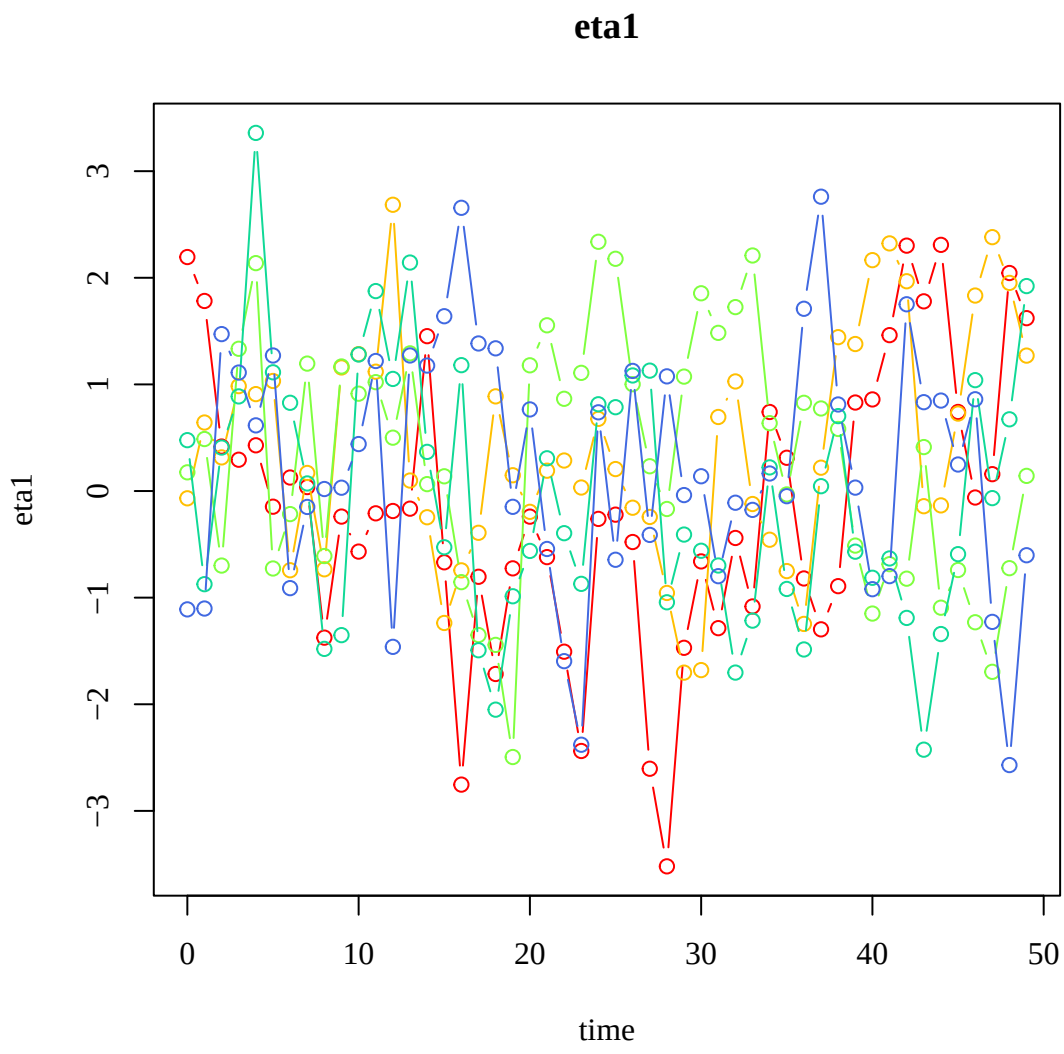


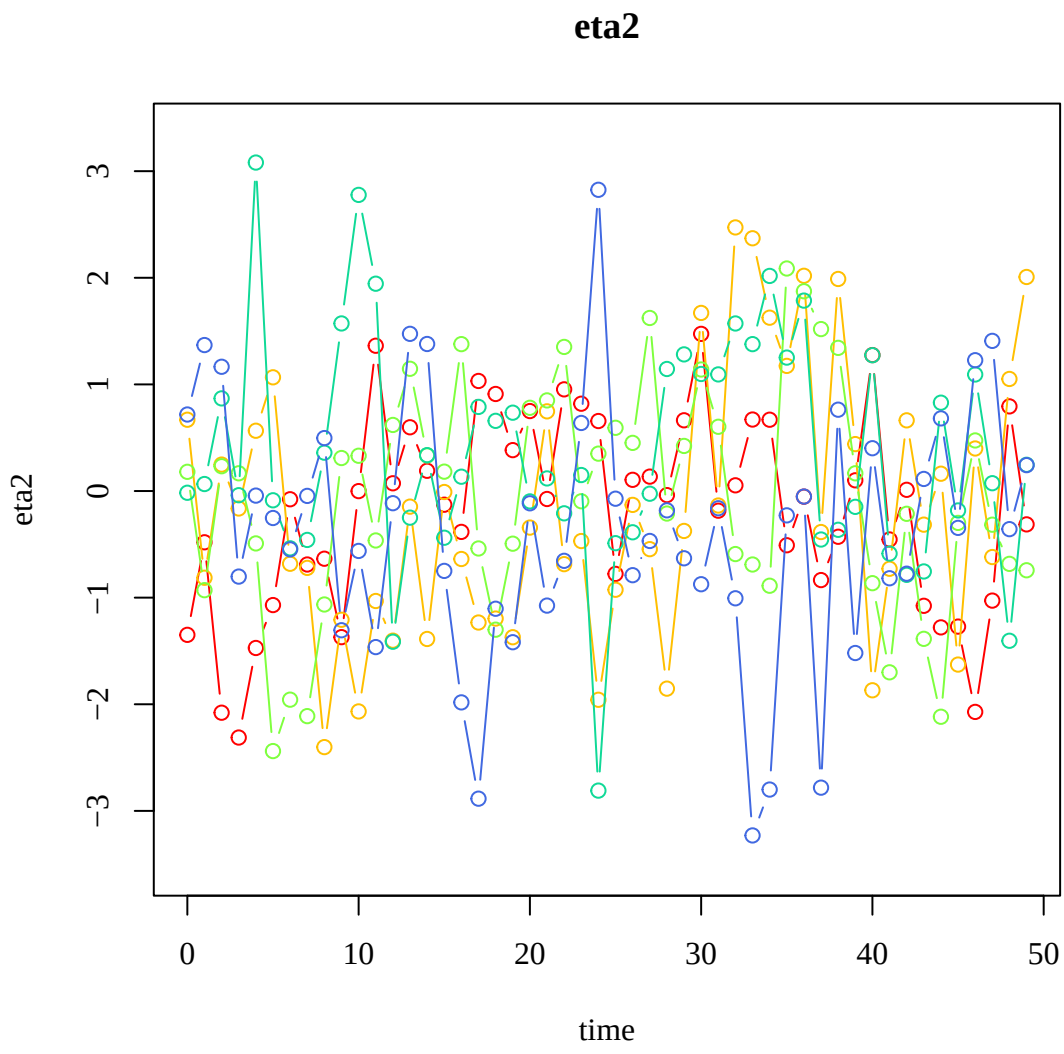

```
#> Call:
#> simStateSpace::SimSSMVARFixed(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, gamma_eta = gamma_eta,
#>     x = x, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMVARFixed`
#> to convert it to a data frame or a matrix.
```


y1
y1
2
1
0
−1
−2
0
1
2
3
4
time

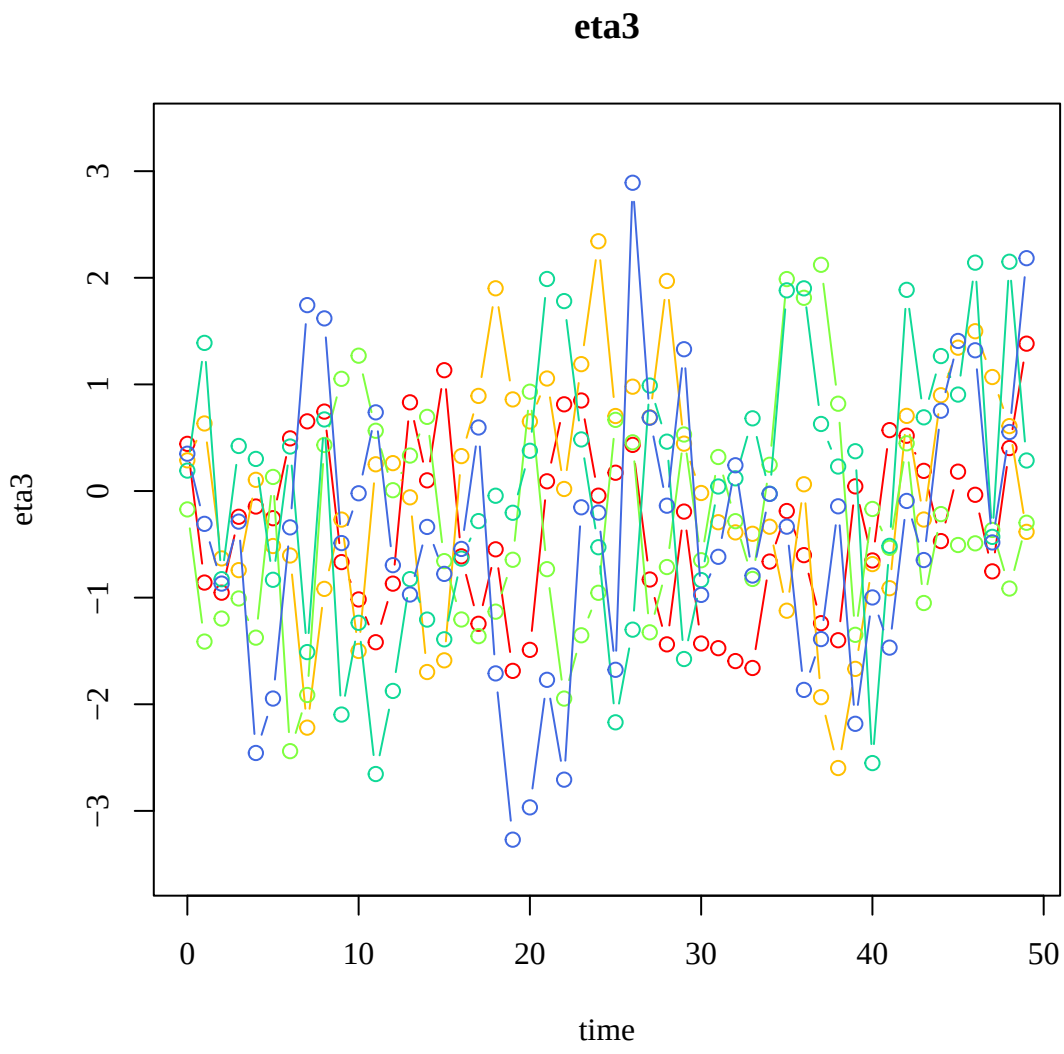

y2
y2
2
1
0
−1
−2
0
1
2
3
4
time


y3
y3
2
1
0
−1
−2
0
1
2
3
4
time

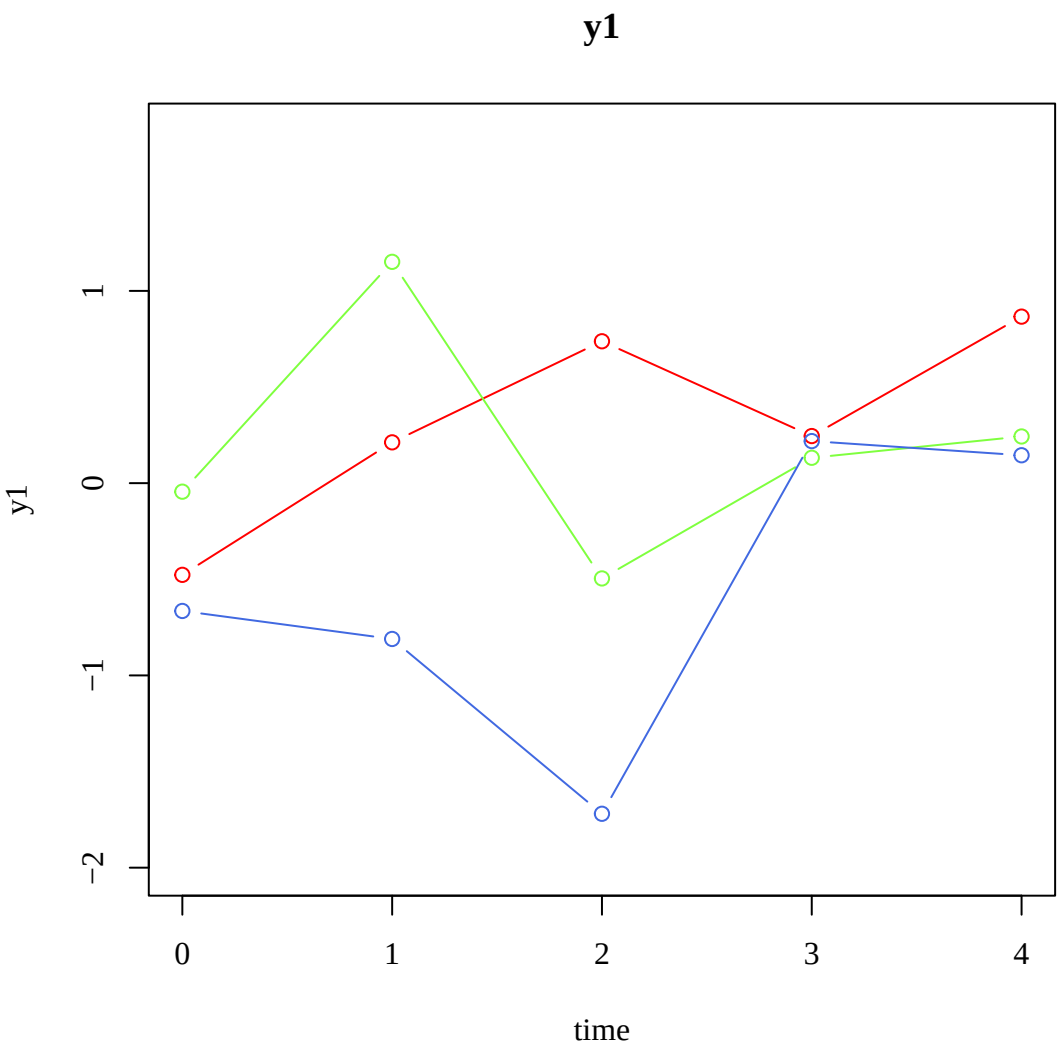

eta1
eta1
3
2
1
0
−1
−2
−3
0
10
20
30
40
50
time

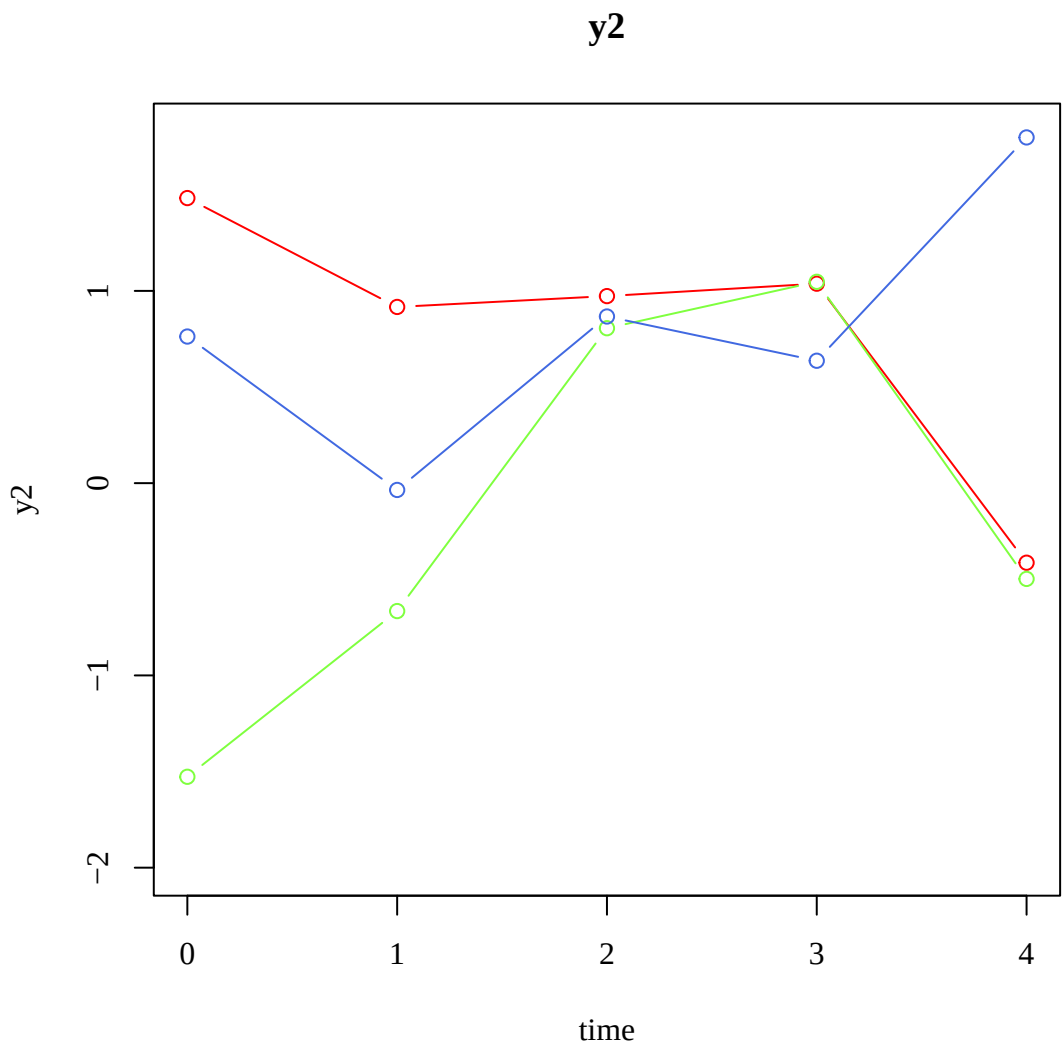


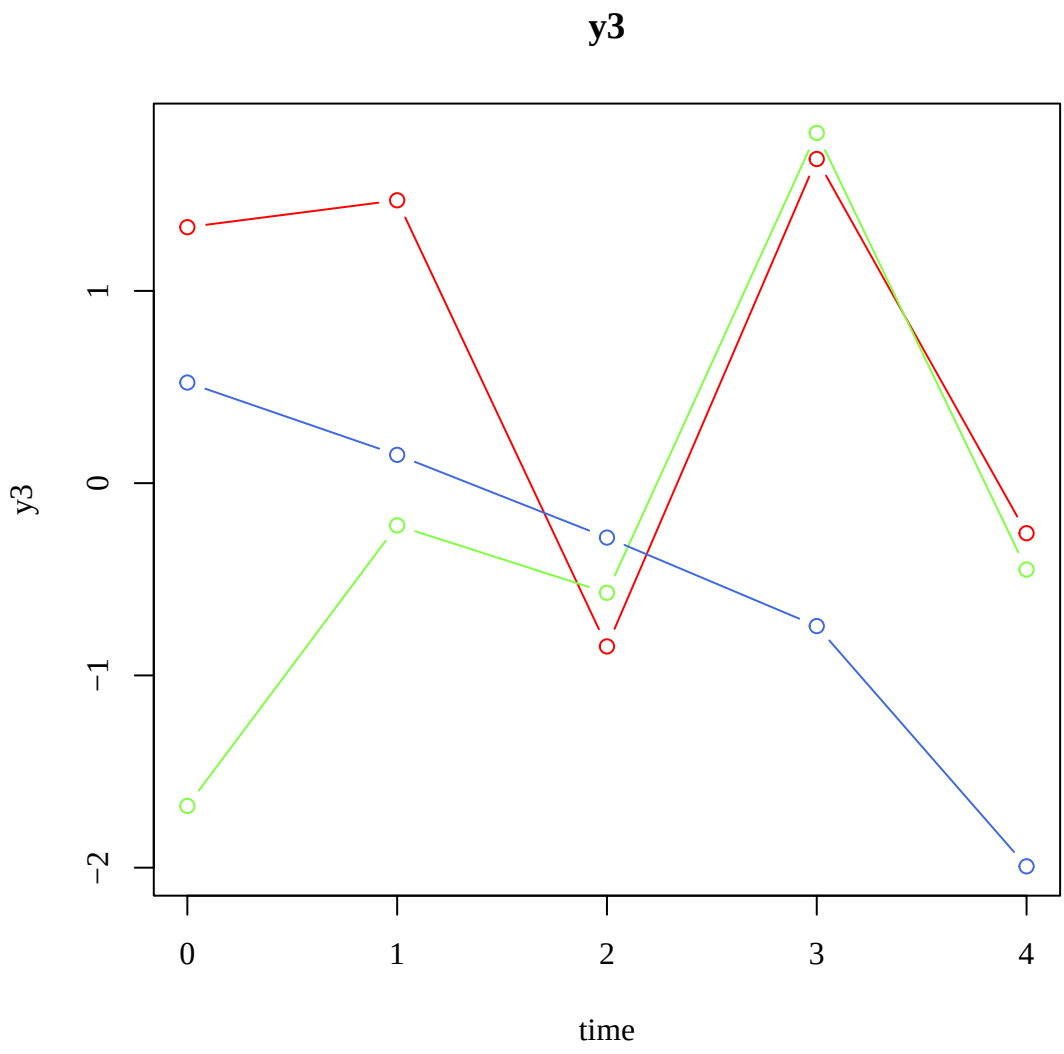

```
#> test-simStateSpace-sim-ssm-var-i-vary
```

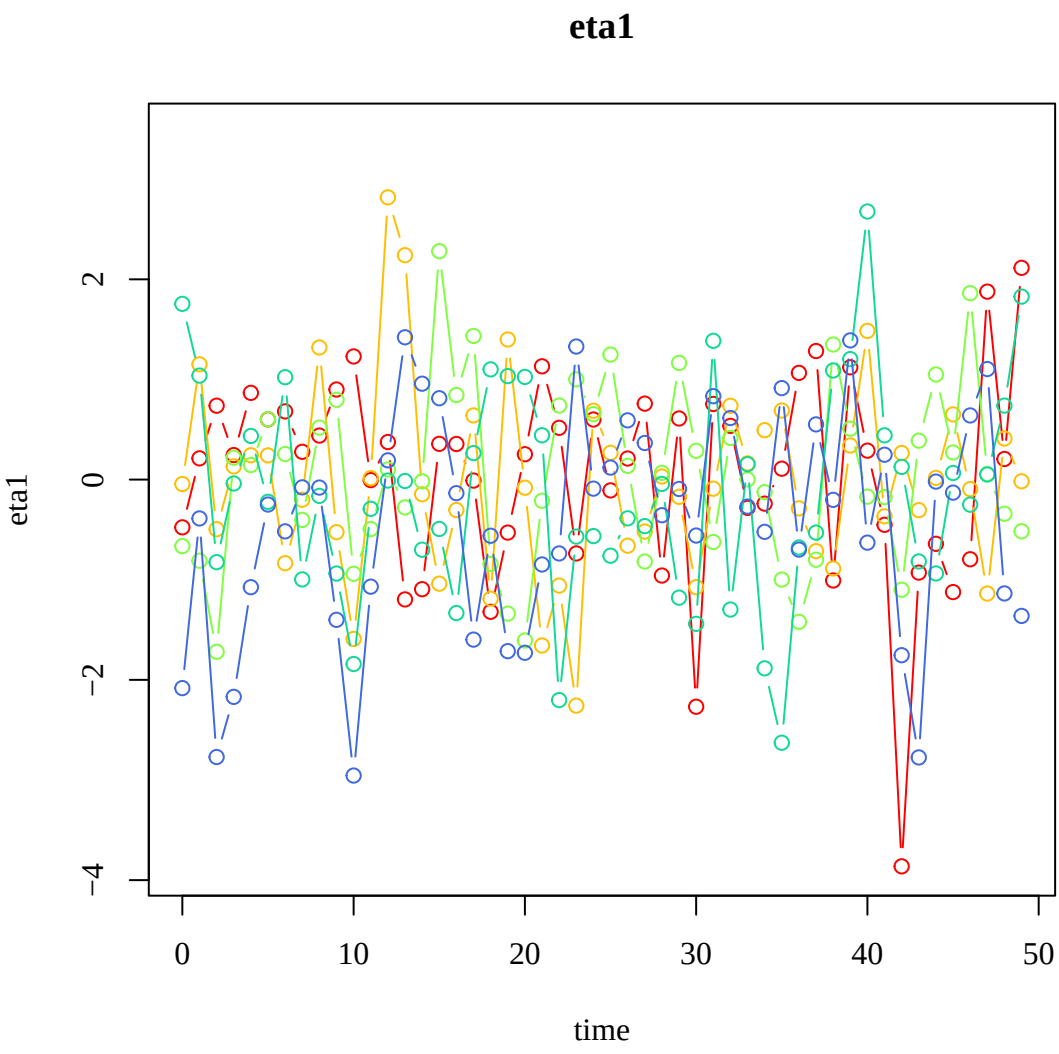


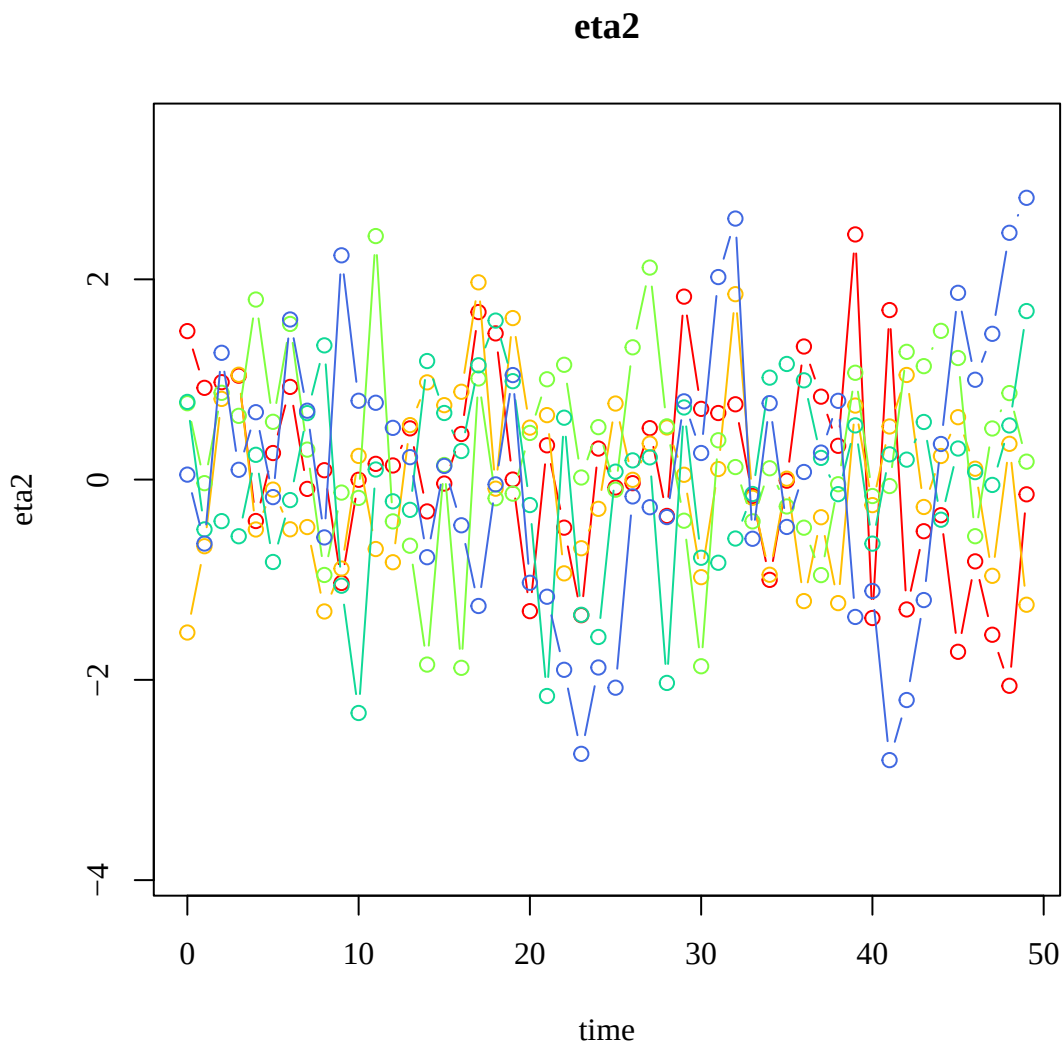

```
#> Call:
#> simStateSpace::SimSSMVARIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMVARIVary`
#> to convert it to a data frame or a matrix.
```

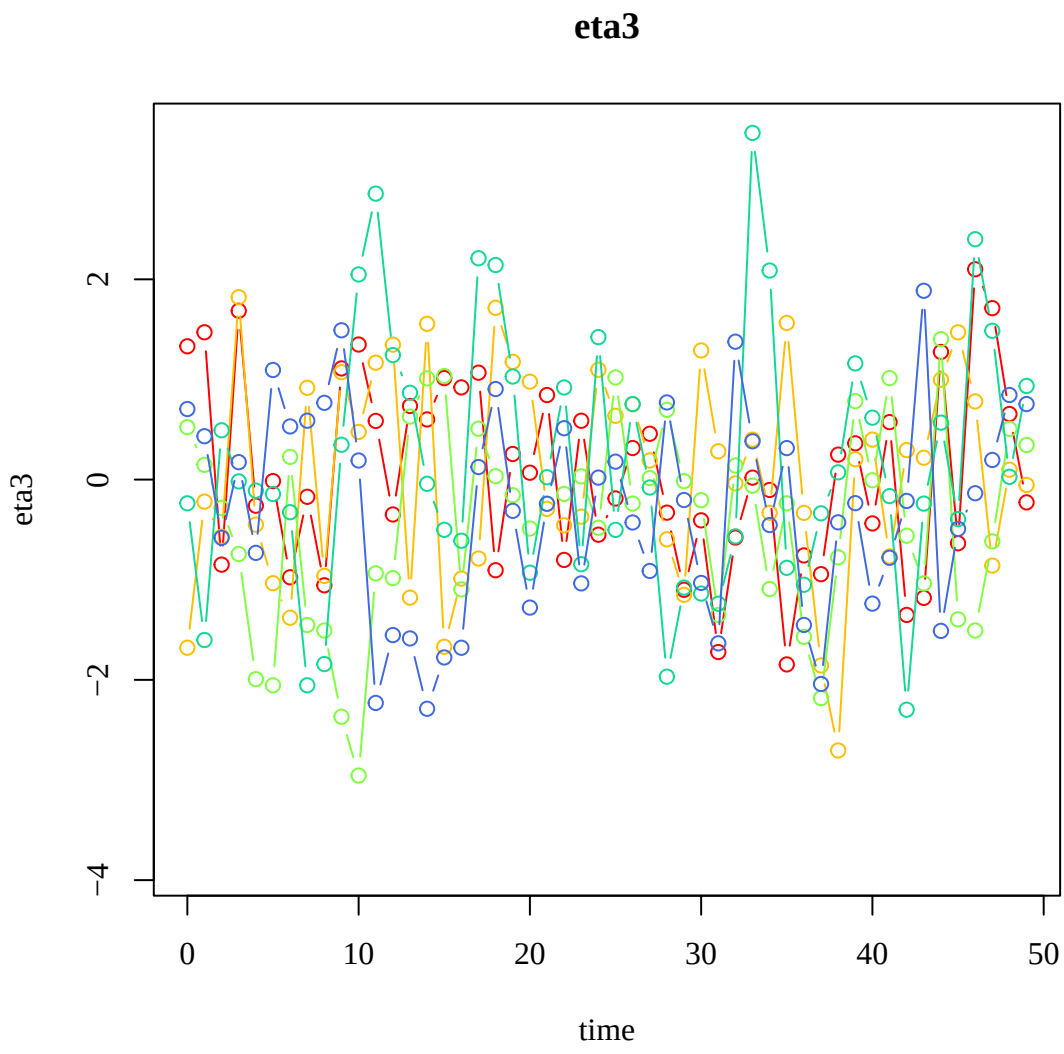

y1
y1
1
0
−1
−2
0
1
2
3
4
time


y2
y2
1
0
−1
−2
0
1
2
3
4
time

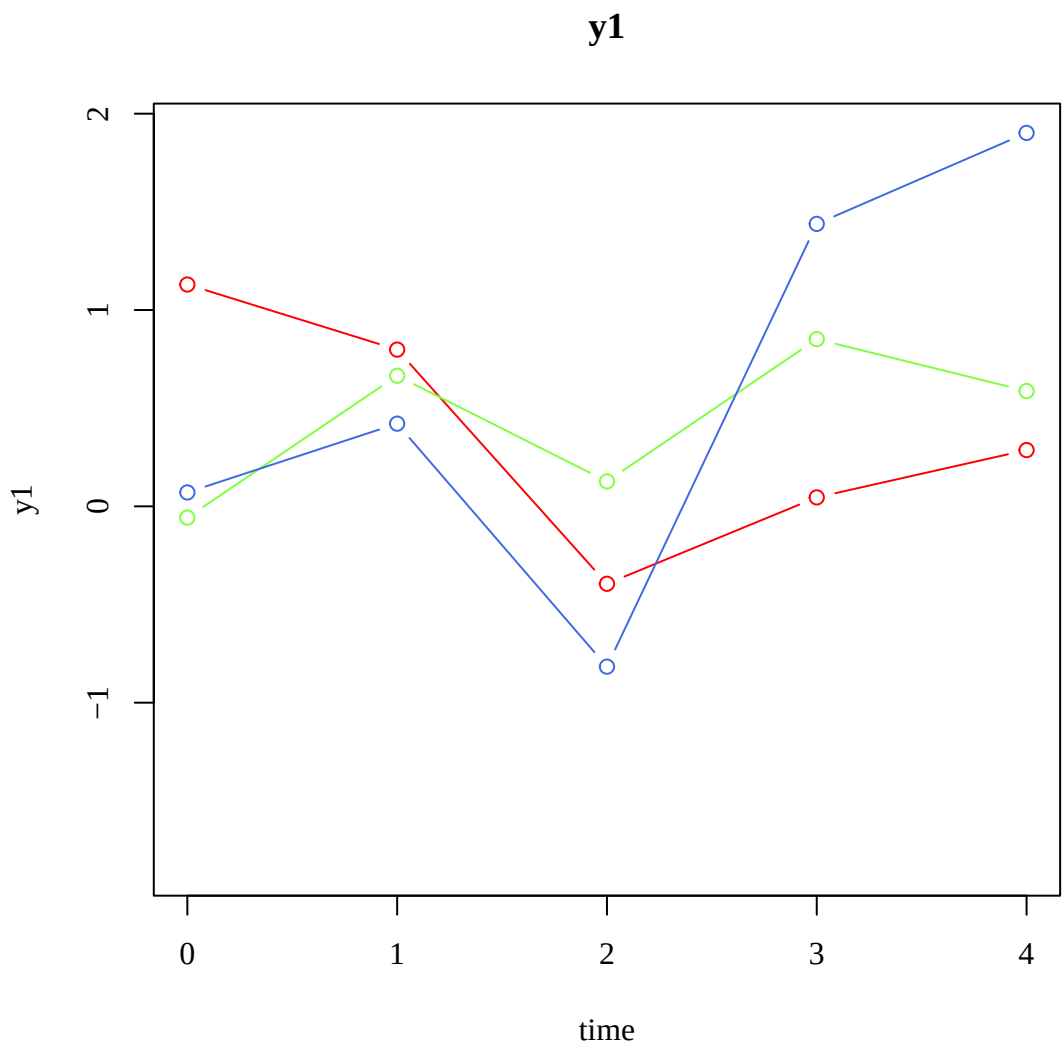

y3
y3
1
0
−1
−2
0
1
2
3
4
time

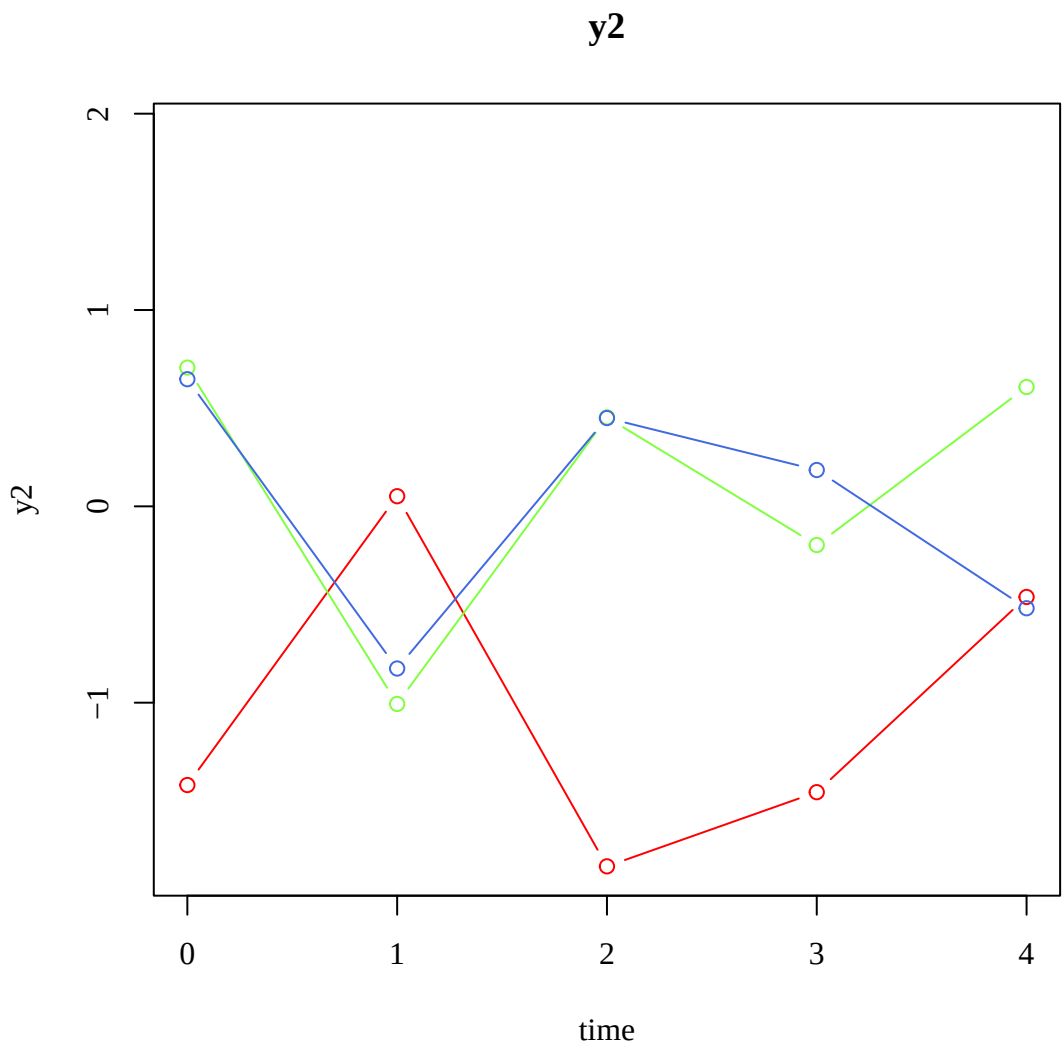

eta1
eta1
2
0
−2
−4
0
10
20
30
40
50
time

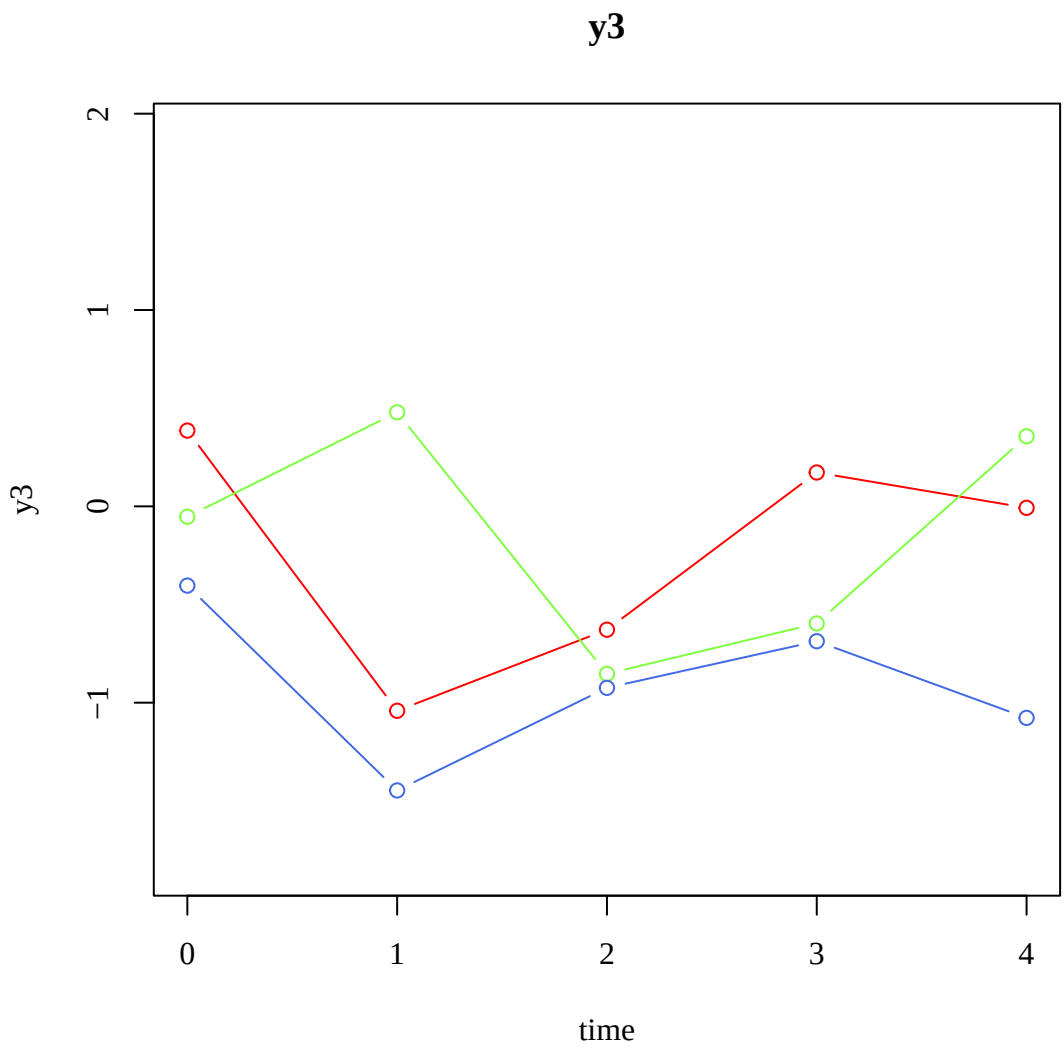

eta2
eta2
2
0
−2
−4
0
10
20
30
40
50
time

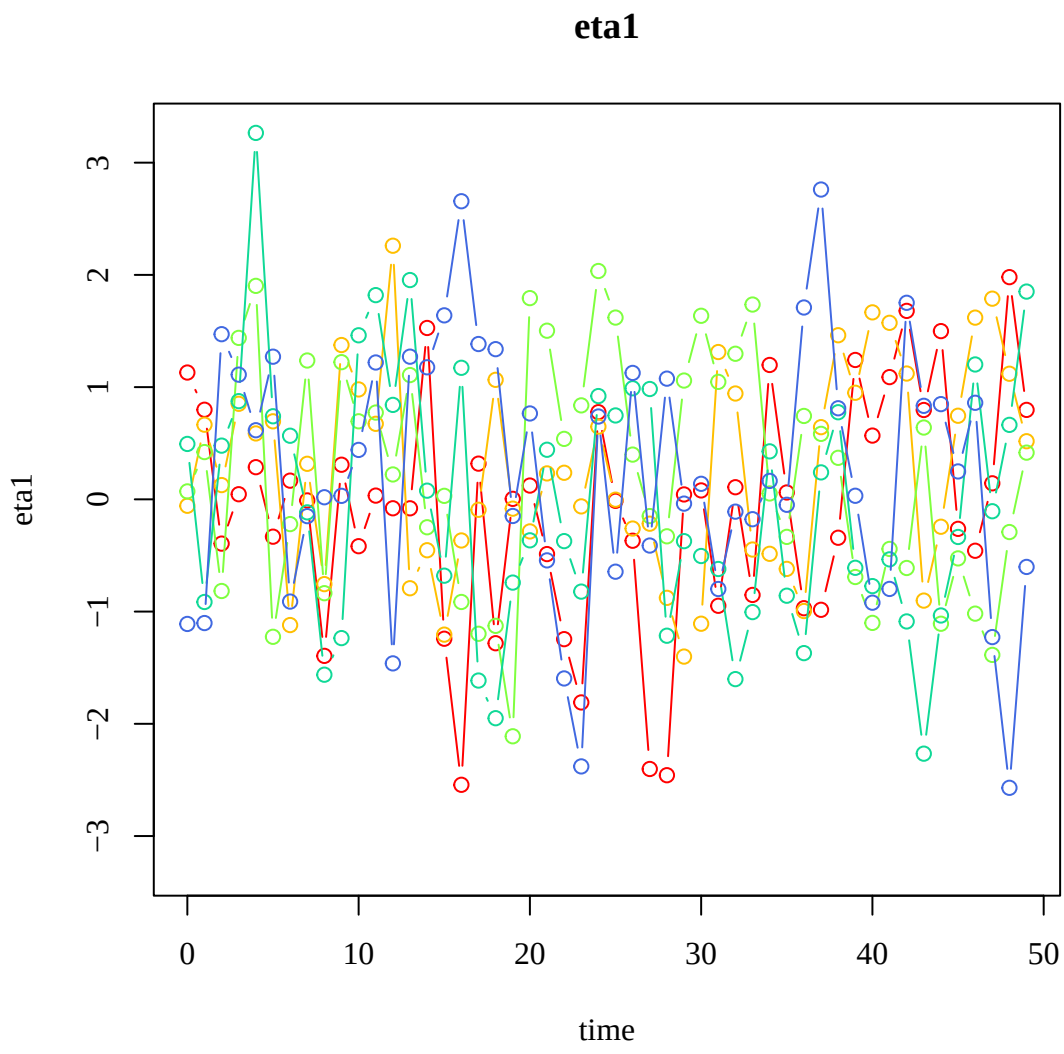


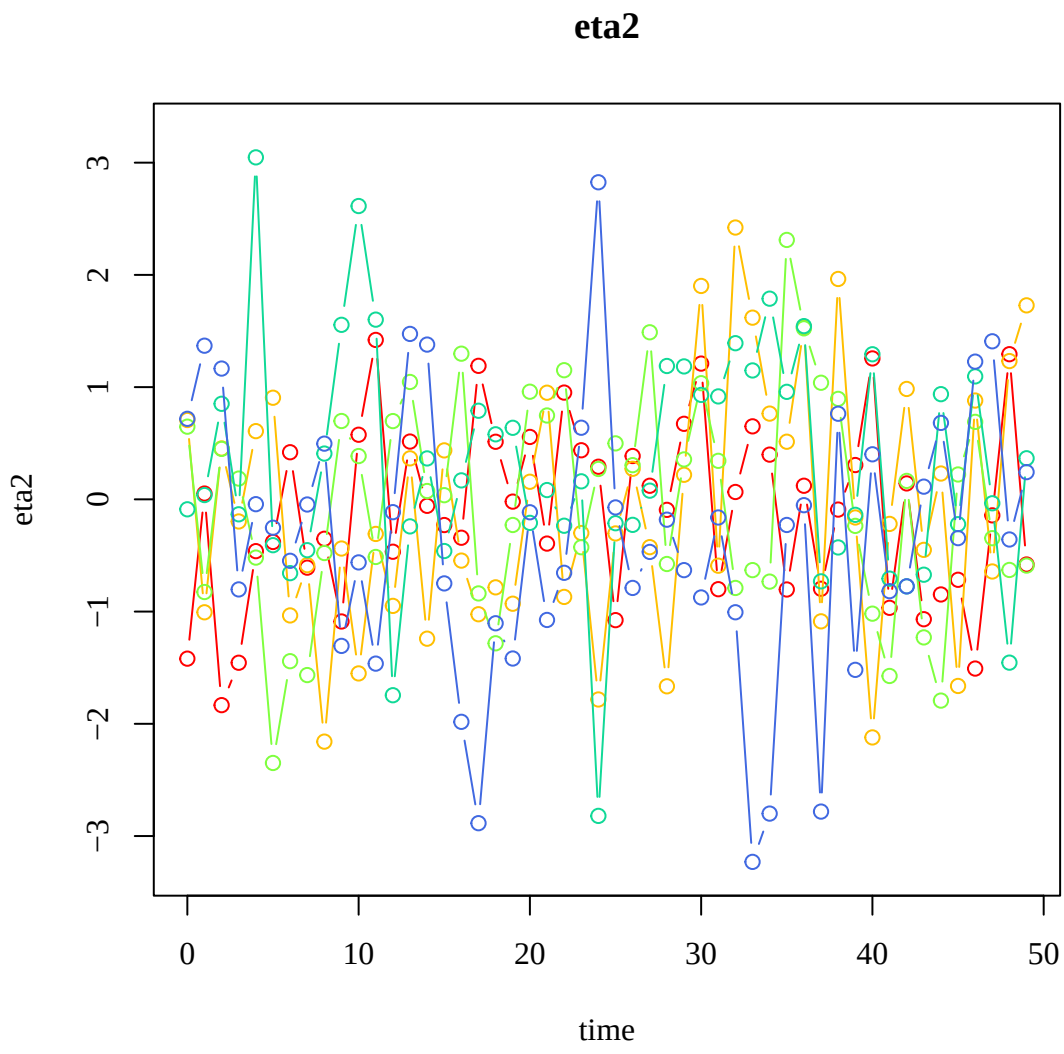

```
#> Call:
#> simStateSpace::SimSSMVARIVary(n = n, mu0 = mu0, sigma0 = sigma0,
#>     alpha = alpha, beta = beta, psi = psi, gamma_eta = gamma_eta,
#>     x = x, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMVARIVary`
#> to convert it to a data frame or a matrix.
```


y1
2
1
0
−1
y1
0
1
2
3
4
time

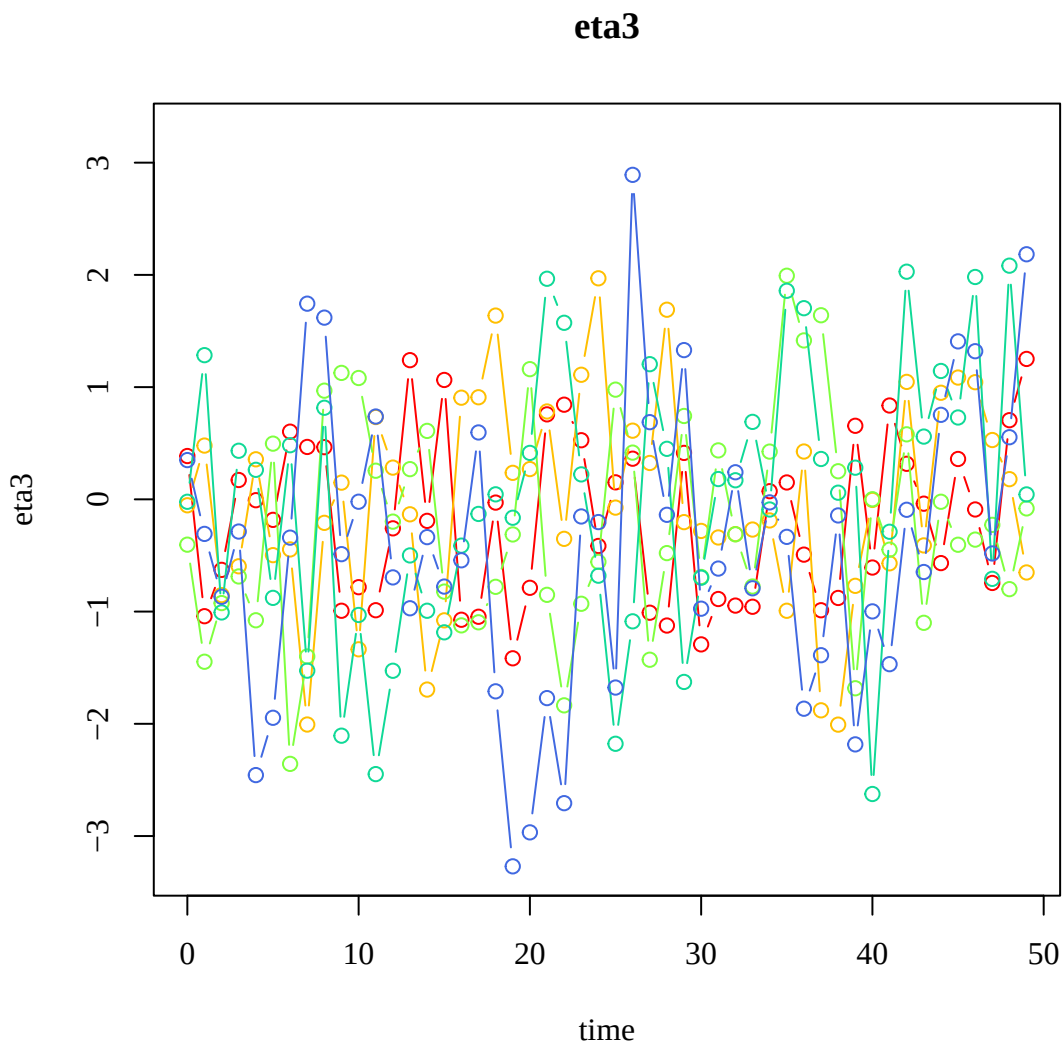

y2
2
1
0
−1
y2
0
1
2
3
4
time


y3
2
1
0
−1
y3
0
1
2
3
4
time

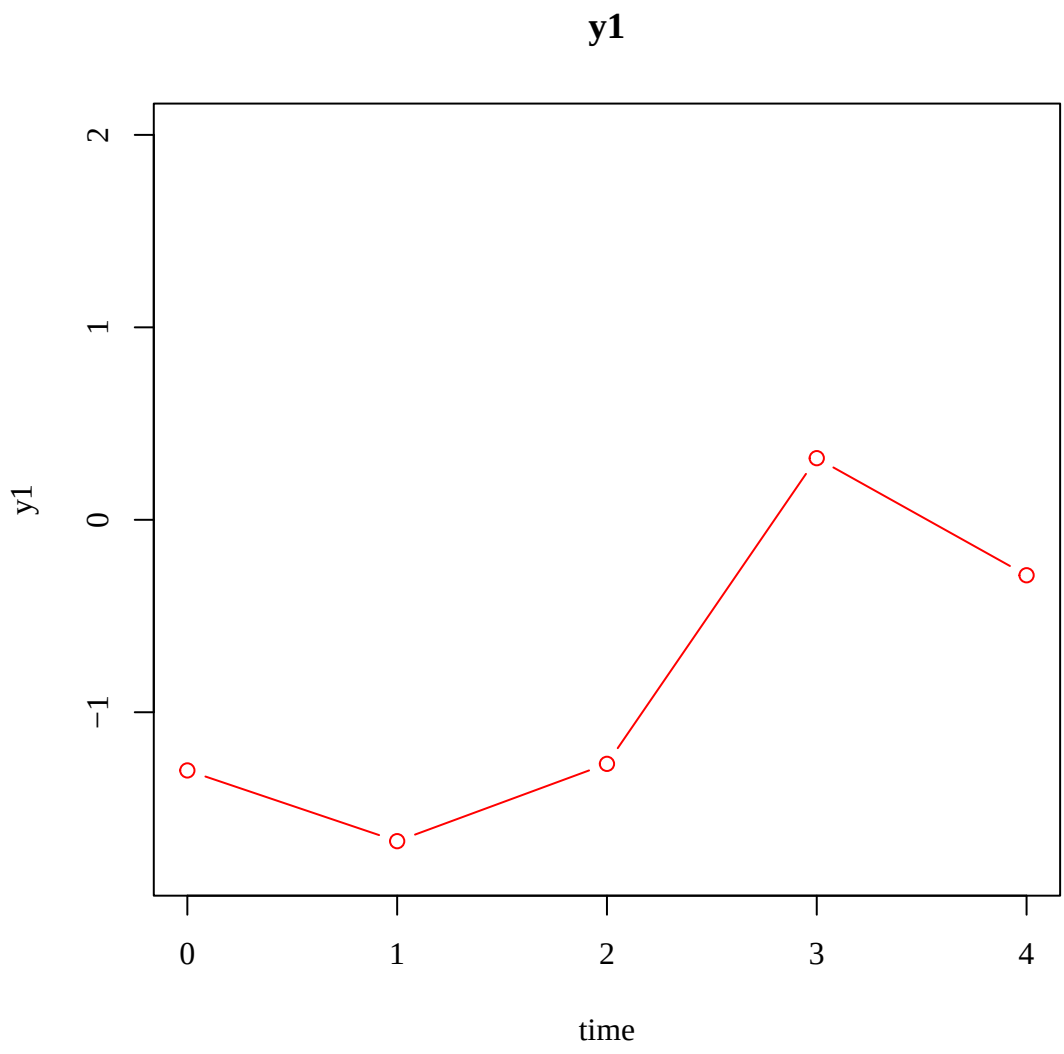

eta1
eta1
3
2
1
0
−1
−2
−3
0
10
20
30
40
50
time

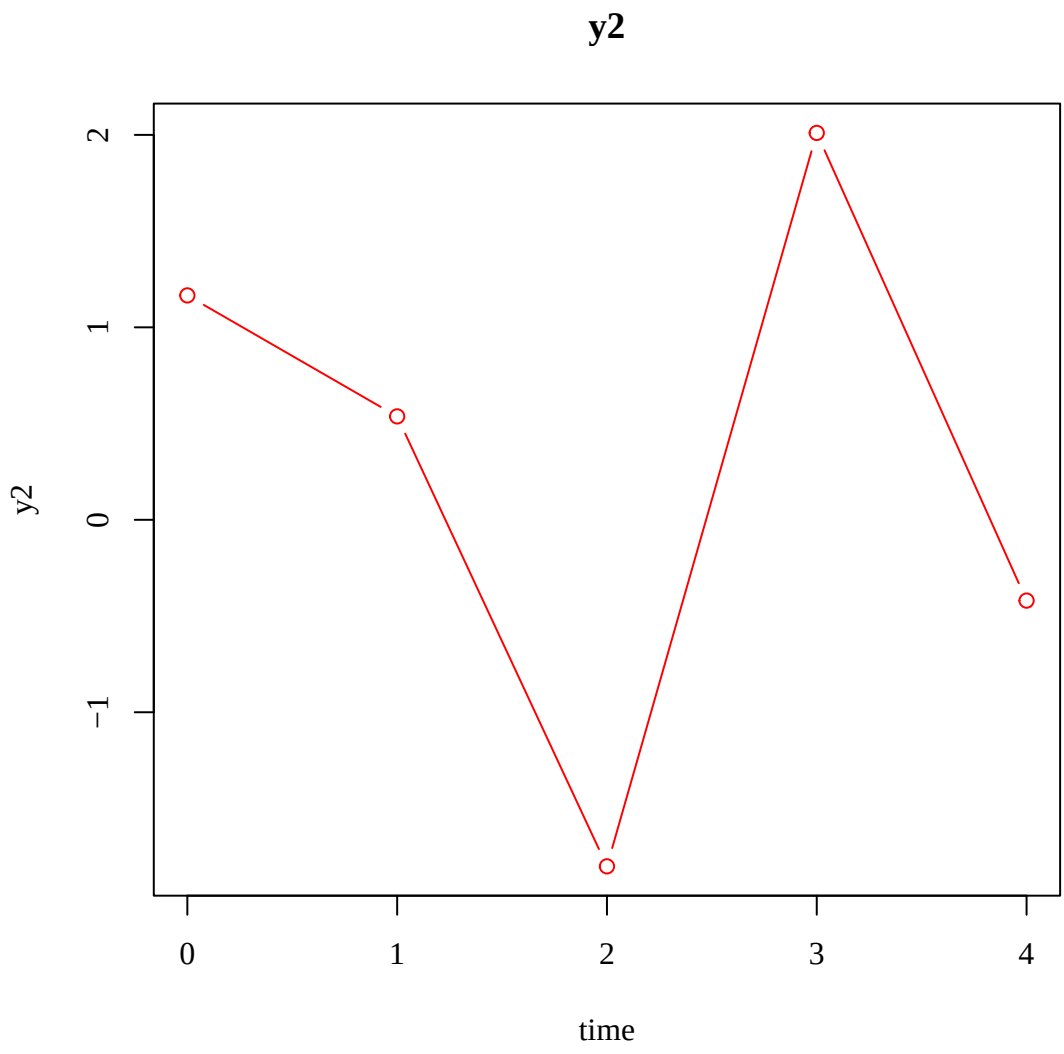


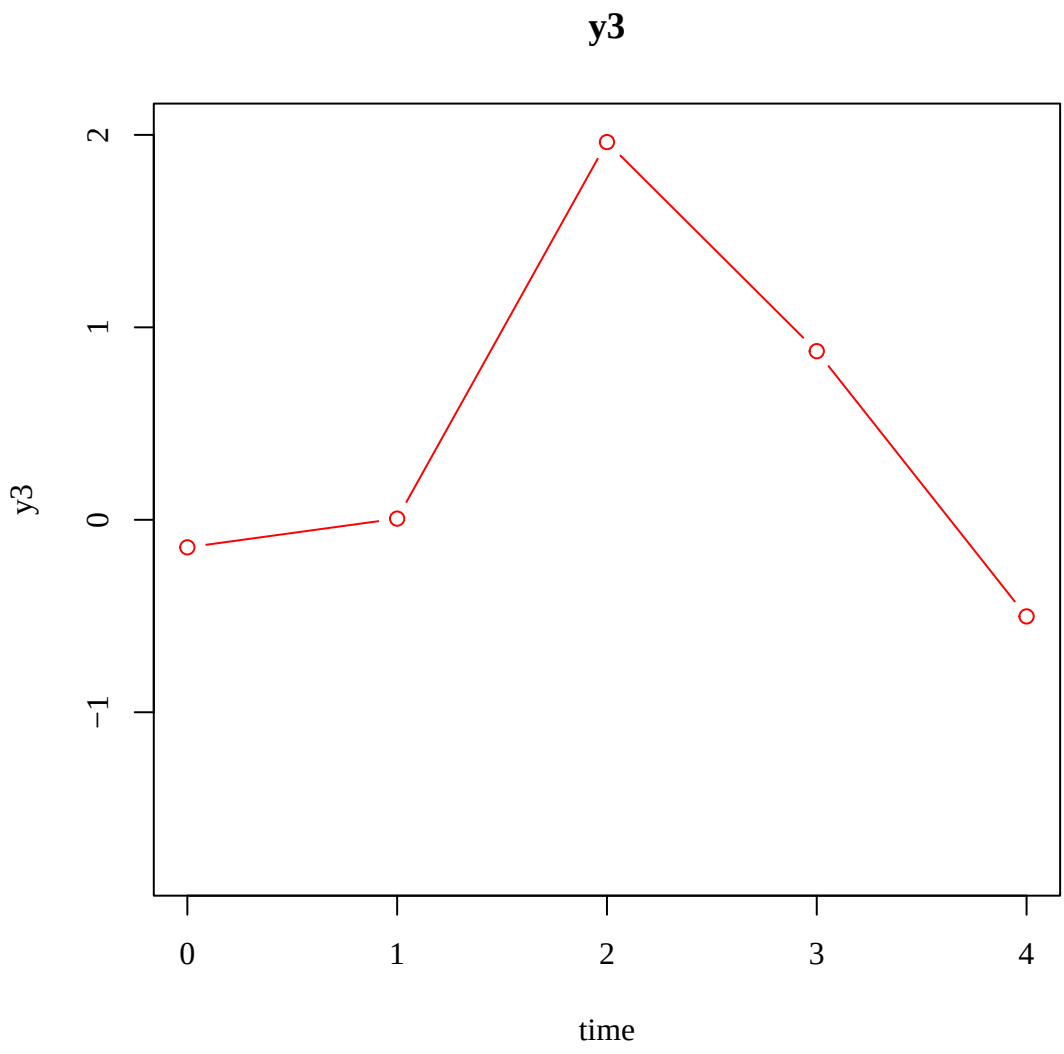

```
#> test-simStateSpace-sim-ssm-var
```

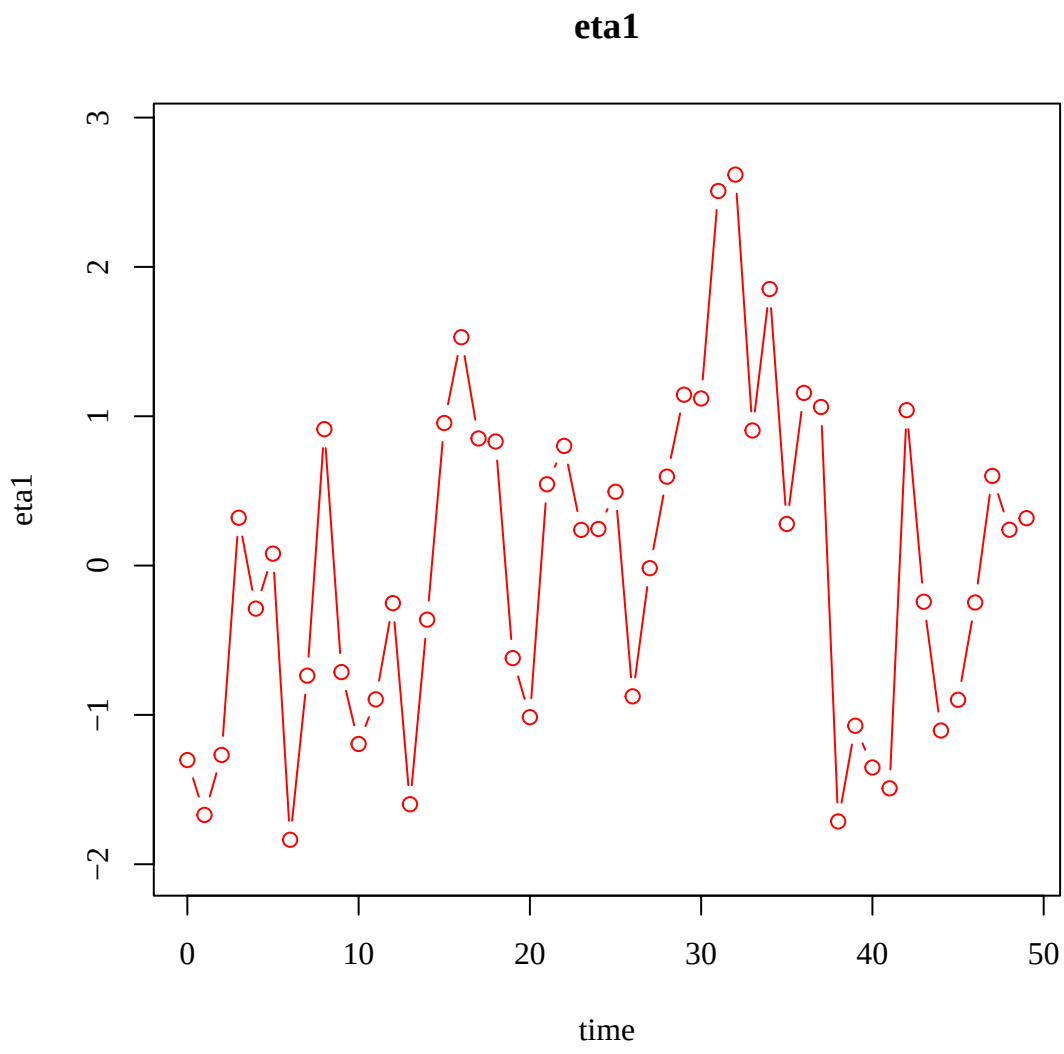


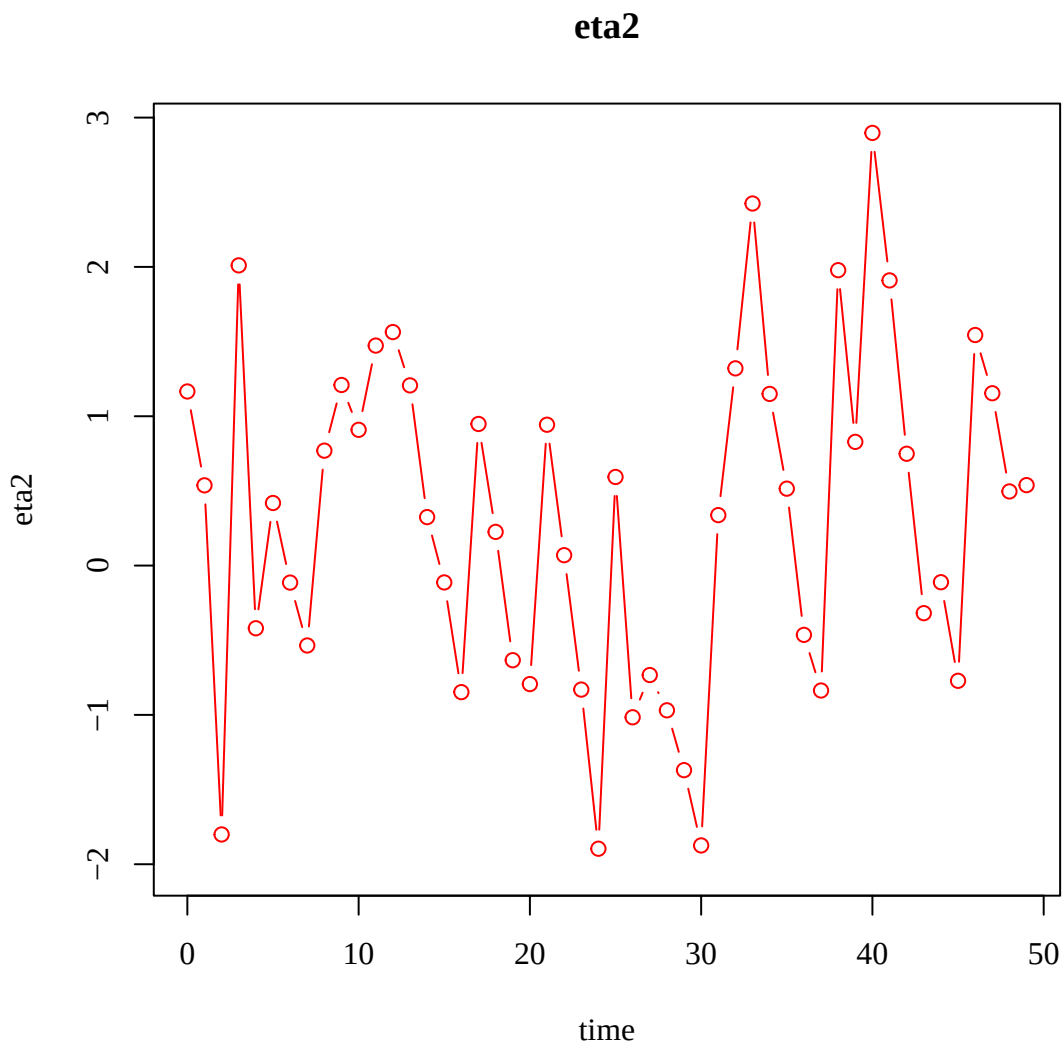

```
#> Call:
#> simStateSpace::SimSSMVAR(mu0 = mu0, sigma0 = sigma0, alpha = alpha,
#>     beta = beta, psi = psi, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMVAR`
#> to convert it to a data frame or a matrix.
```

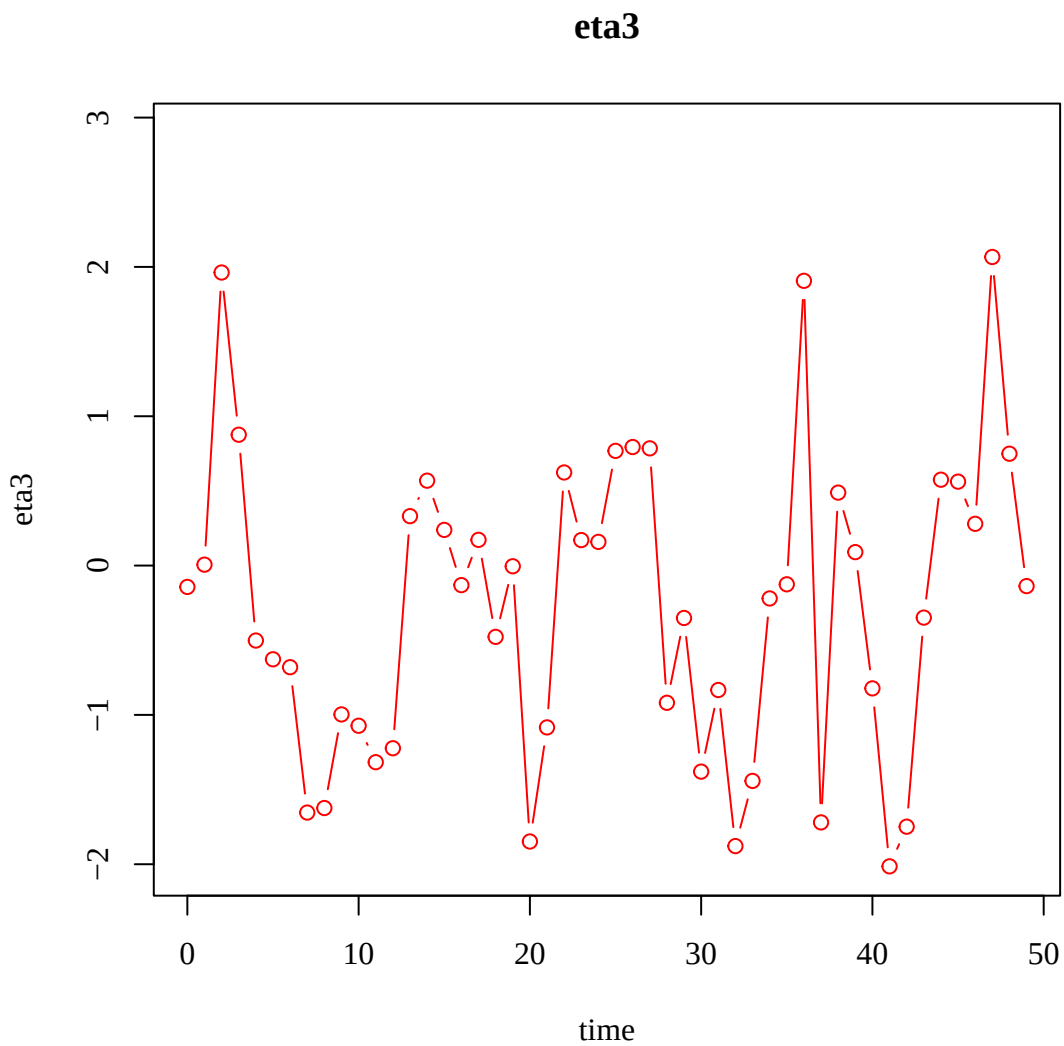

y1
2
1
0
−1
y1
0
1
2
3
4
time


y2
y2
2
1
0
−1
0
1
2
3
4
time

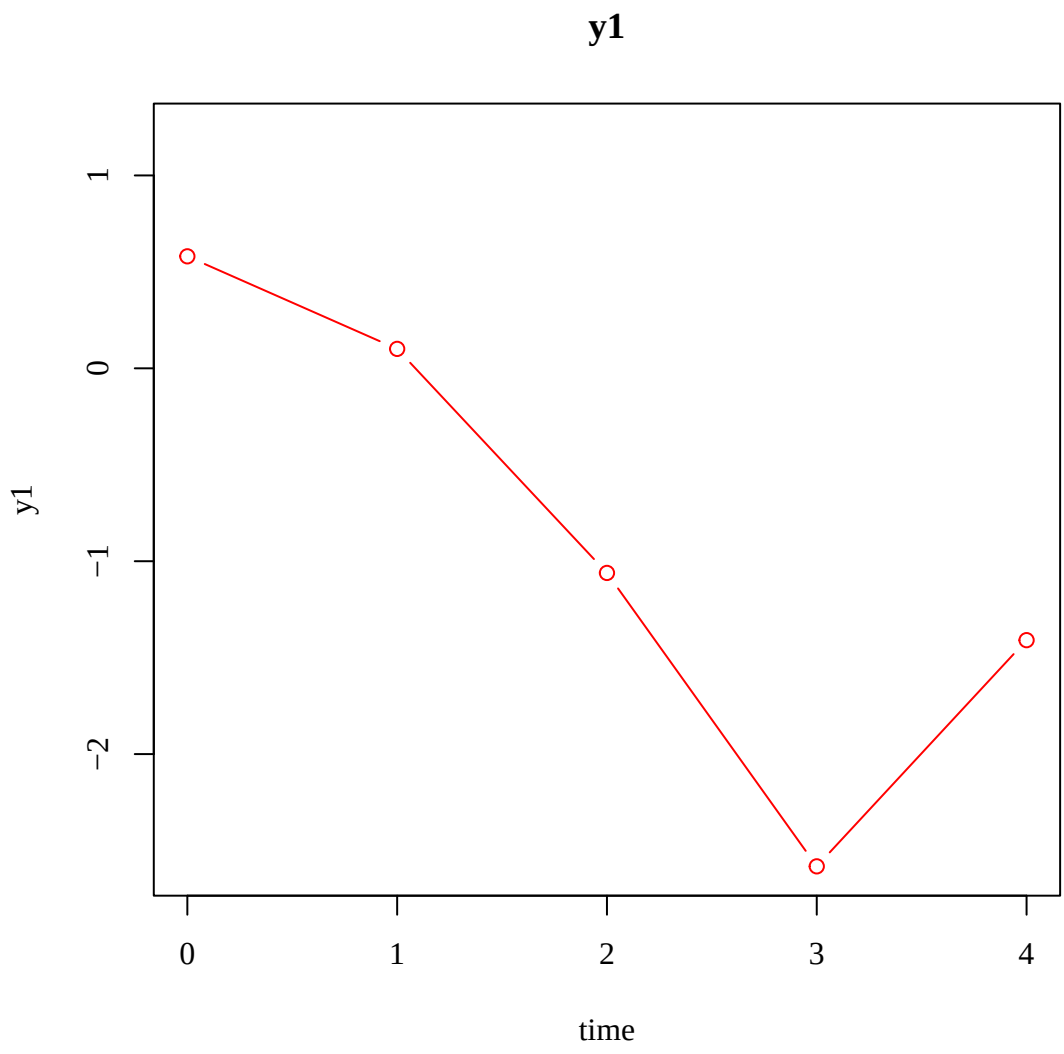

y3
2
1
0
−1
y3
0
1
2
3
4
time

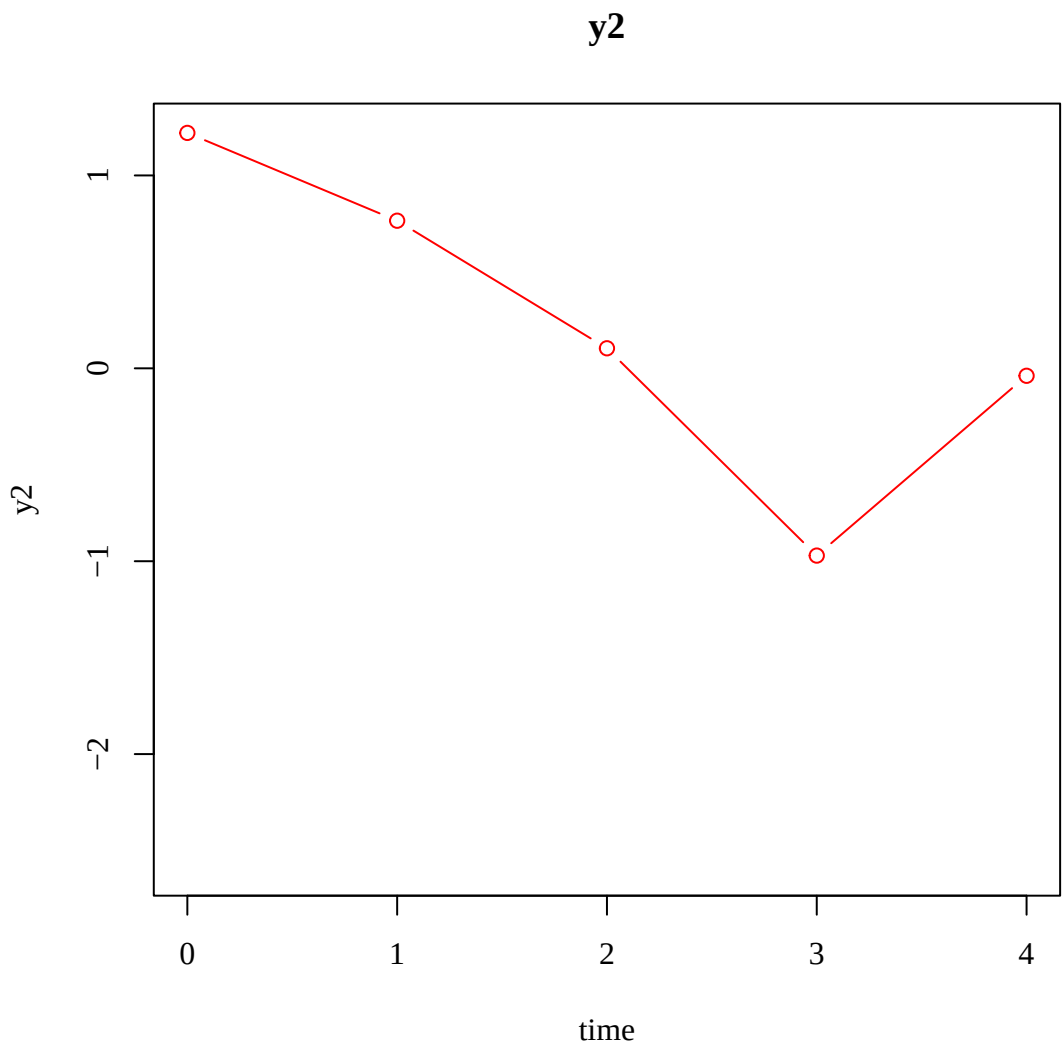

eta1
eta1
time
3
2
1
0
−1
−2
0
10
20
30
40
50

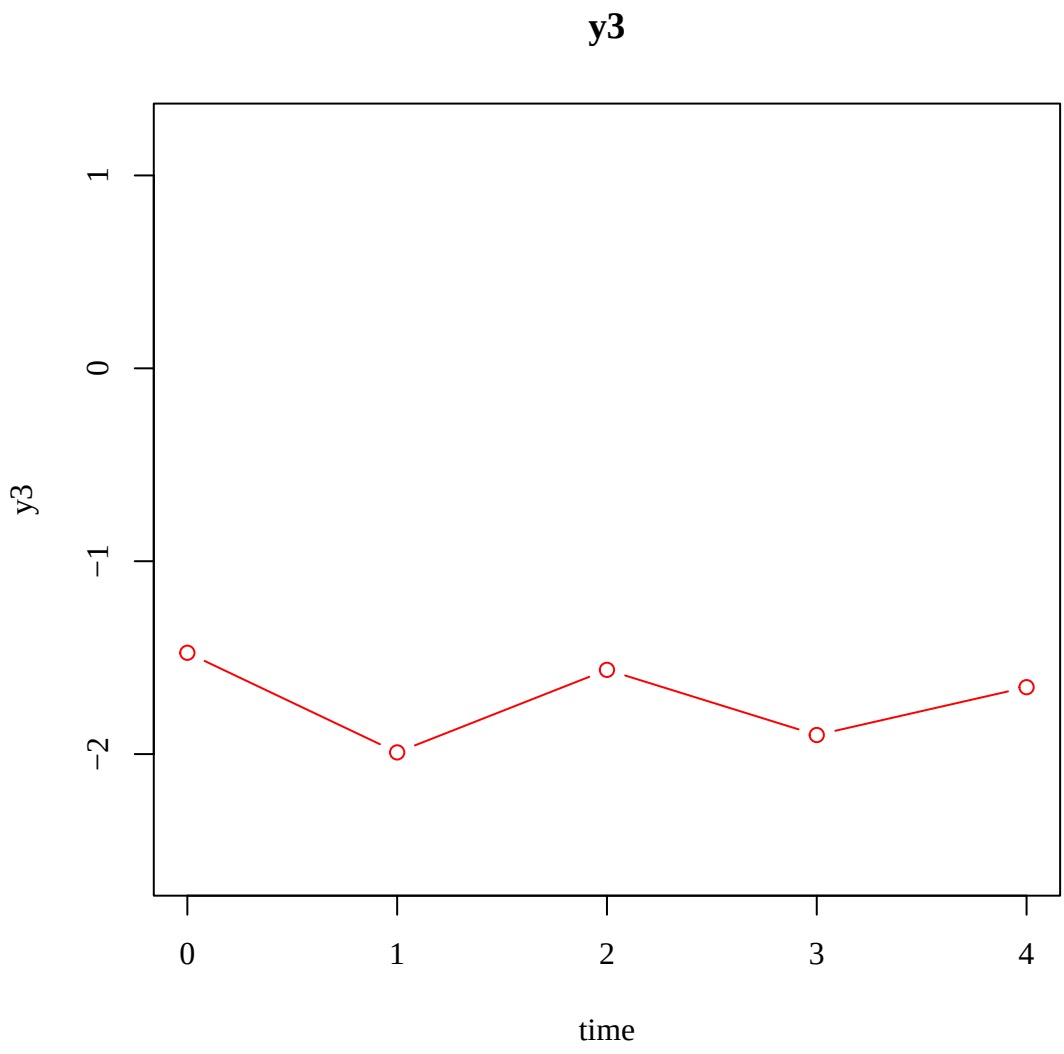

eta2
eta2
time
3
2
1
0
−1
−2
0
10
20
30
40
50

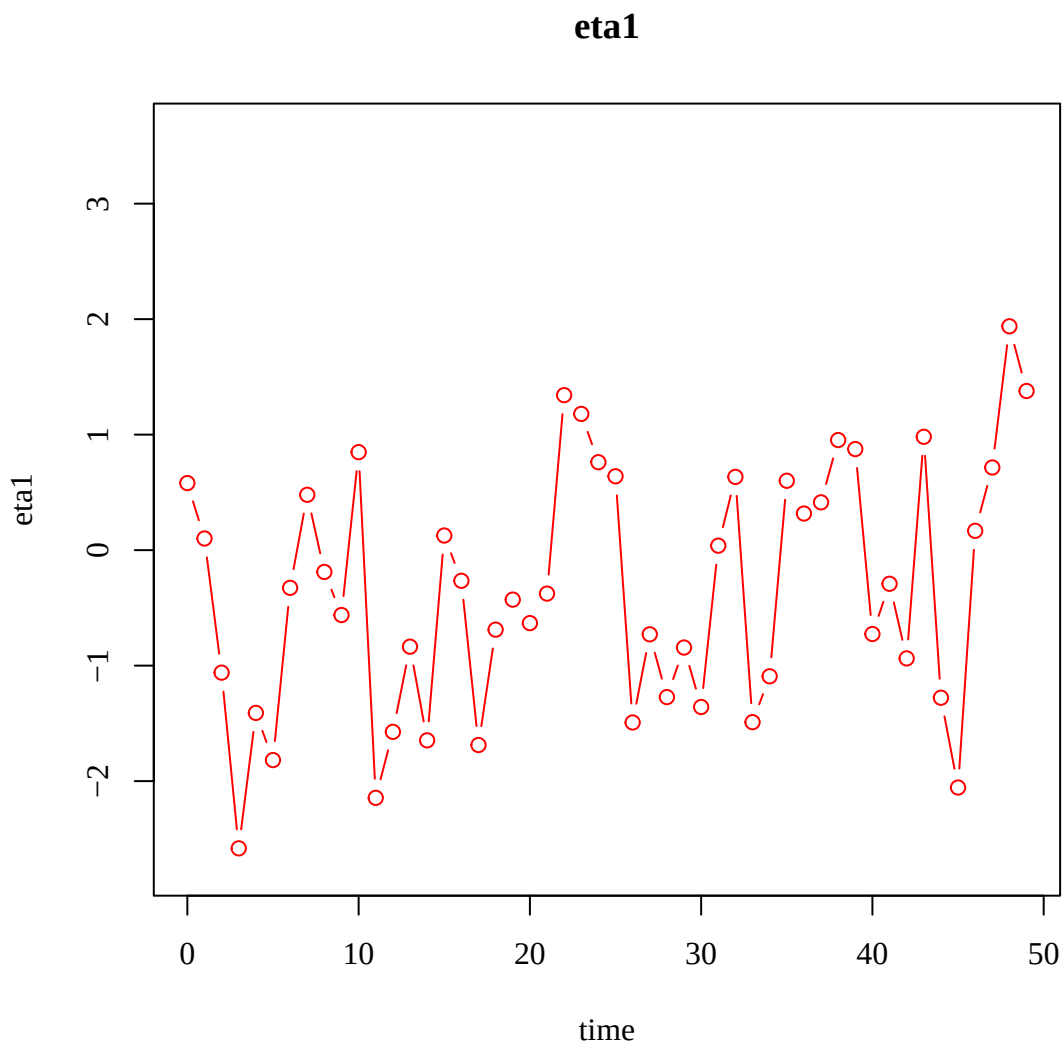


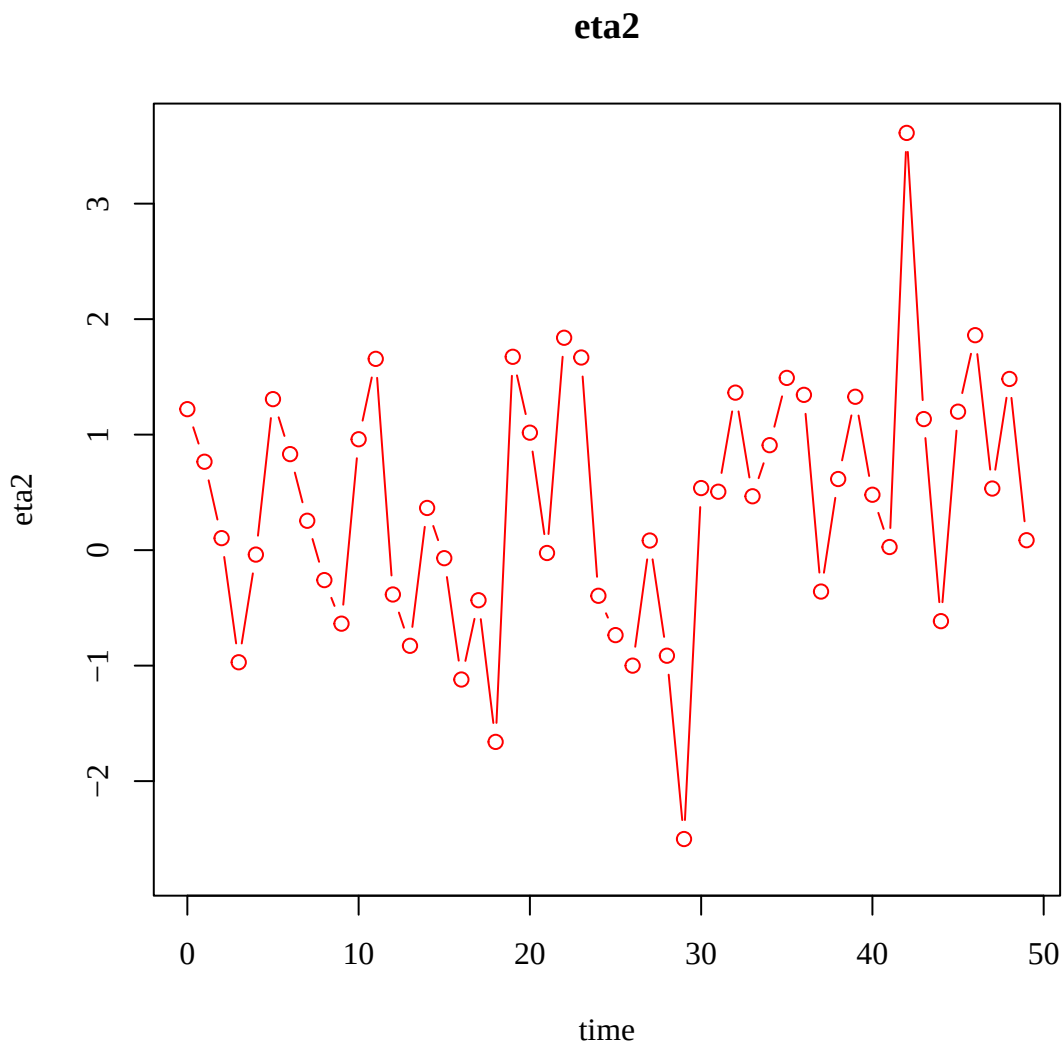

```
#> Call:
#> simStateSpace::SimSSMVAR(mu0 = mu0, sigma0 = sigma0, alpha = alpha,
#>     beta = beta, psi = psi, gamma_eta = gamma_eta, x = x, time = time,
#>     burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMVAR`
#> to convert it to a data frame or a matrix.
```

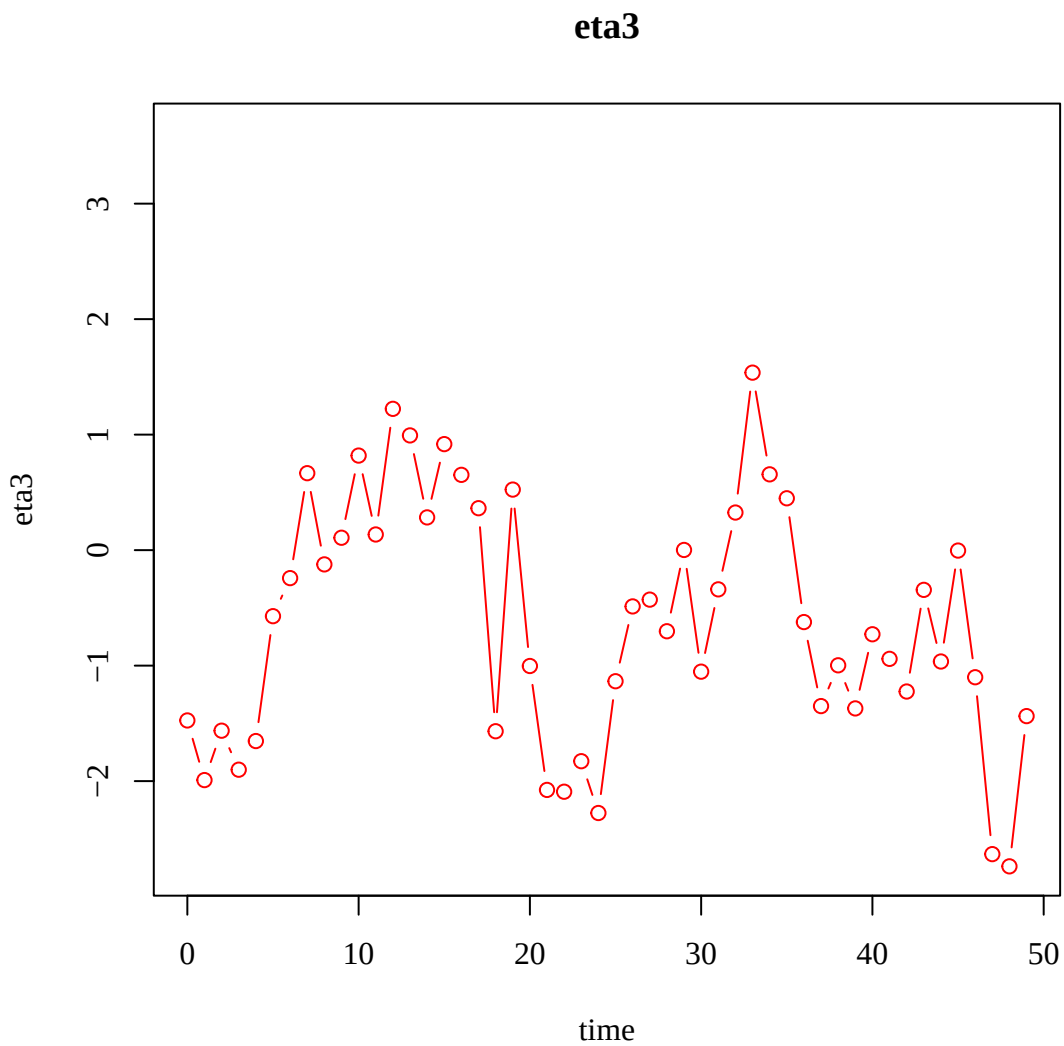

y1
1
0
−1
−2
y1
0
1
2
3
4
time


y2
1
0
−1
−2
y2
0
1
2
3
4
time

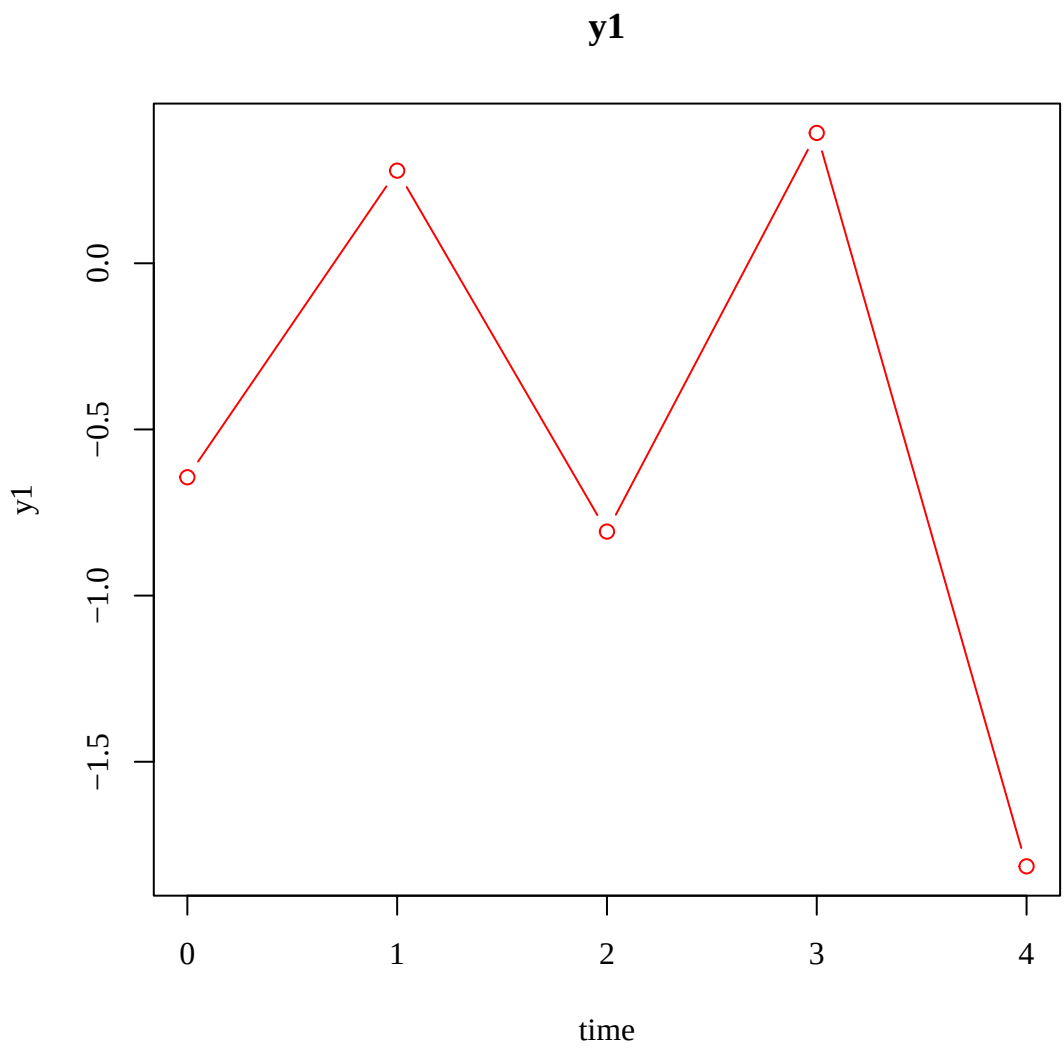

y3
1
0
−1
−2
y3
0
1
2
3
4
time


eta1
eta1
3
2
1
0
−1
−2
0
10
20
30
40
50
time

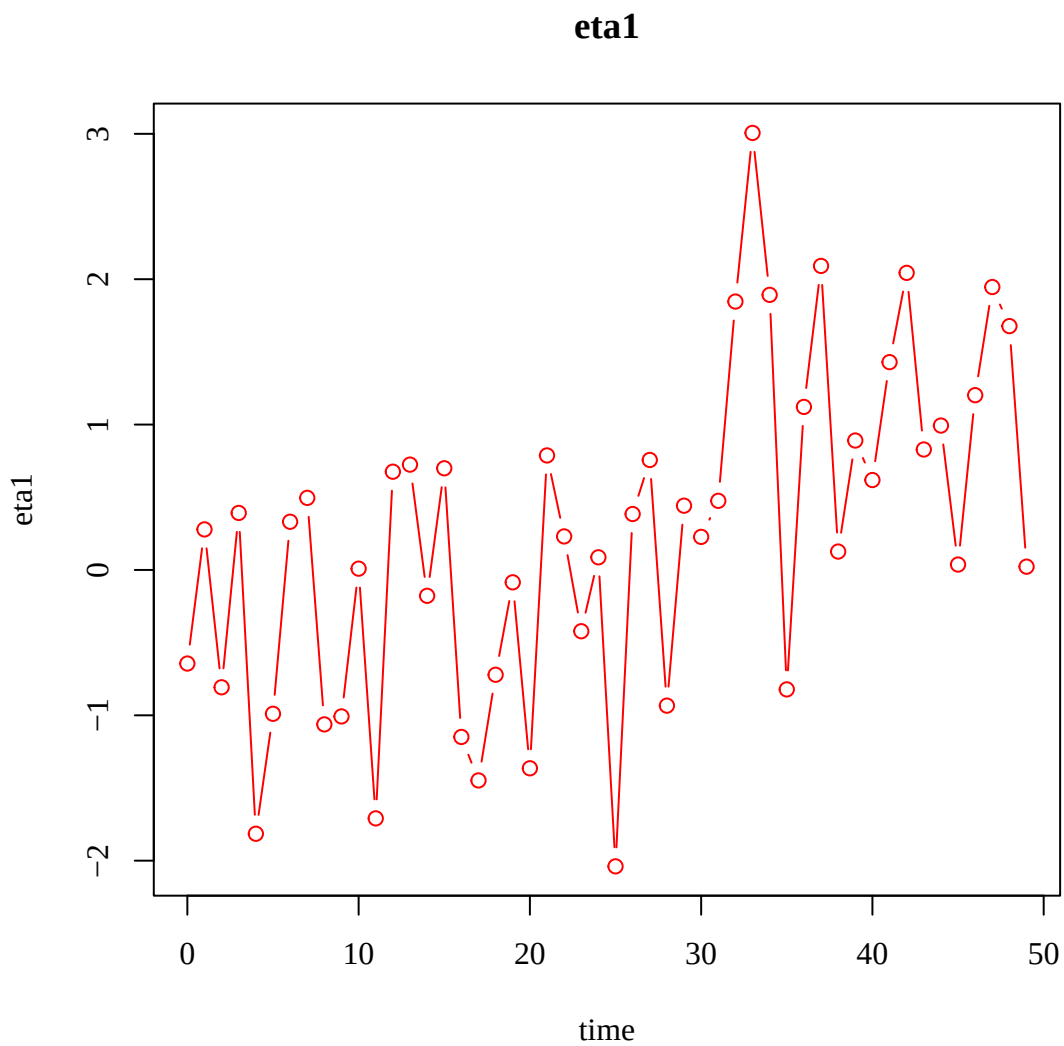

eta2
eta2
3
2
1
0
−1
−2
0
10
20
30
40
50
time



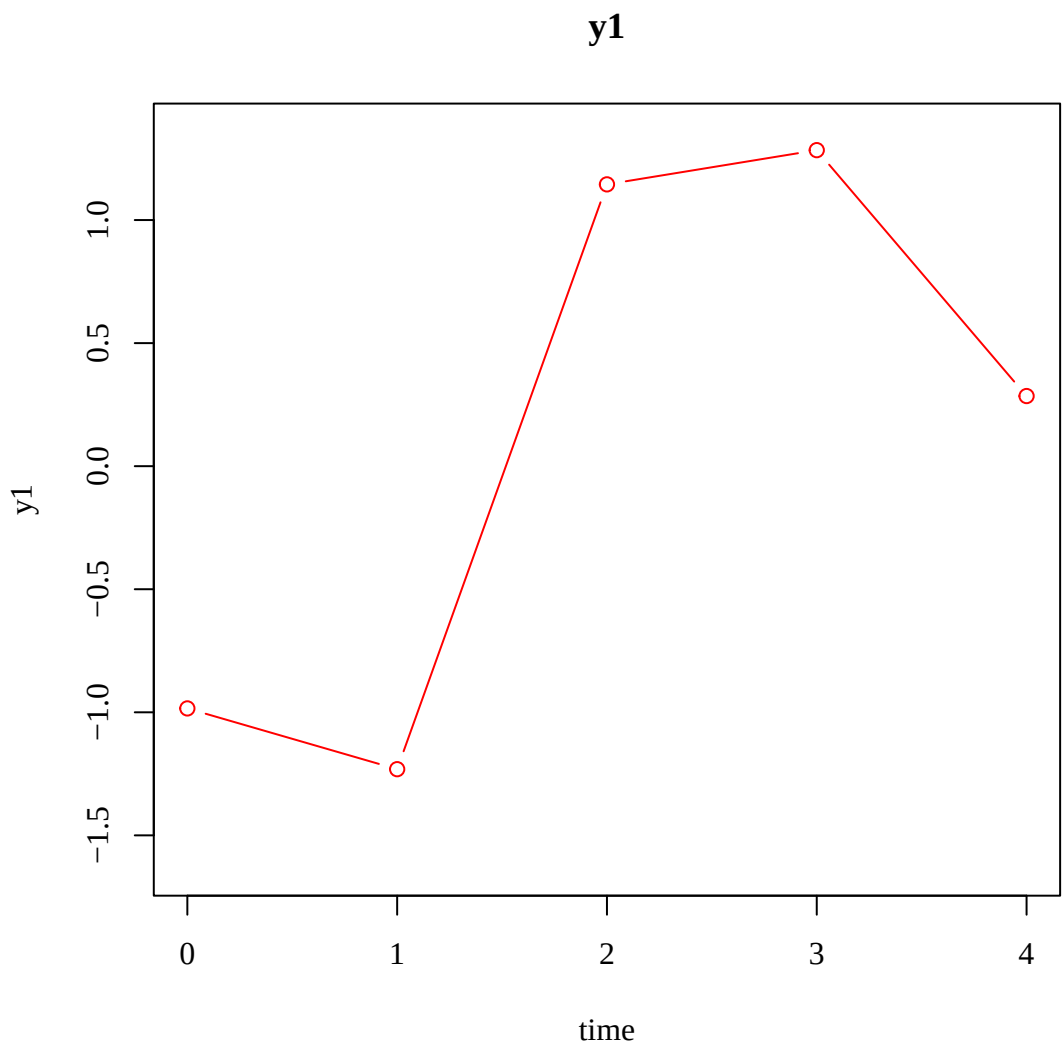

```
#> Call:
#> simStateSpace::SimSSMVAR(mu0 = mu0, sigma0 = sigma0, alpha = alpha,
#>     beta = beta, psi = psi, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSMVAR`
#> to convert it to a data frame or a matrix.
```

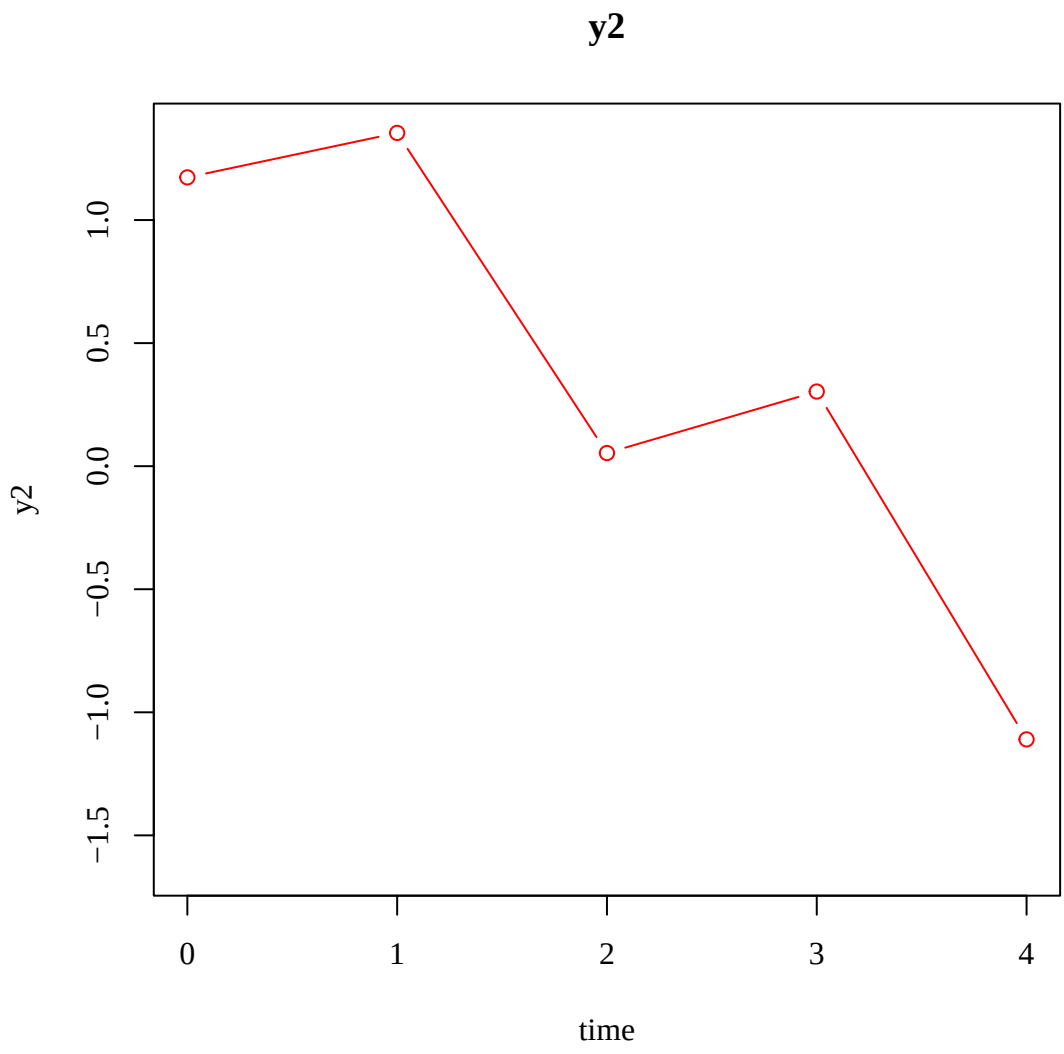


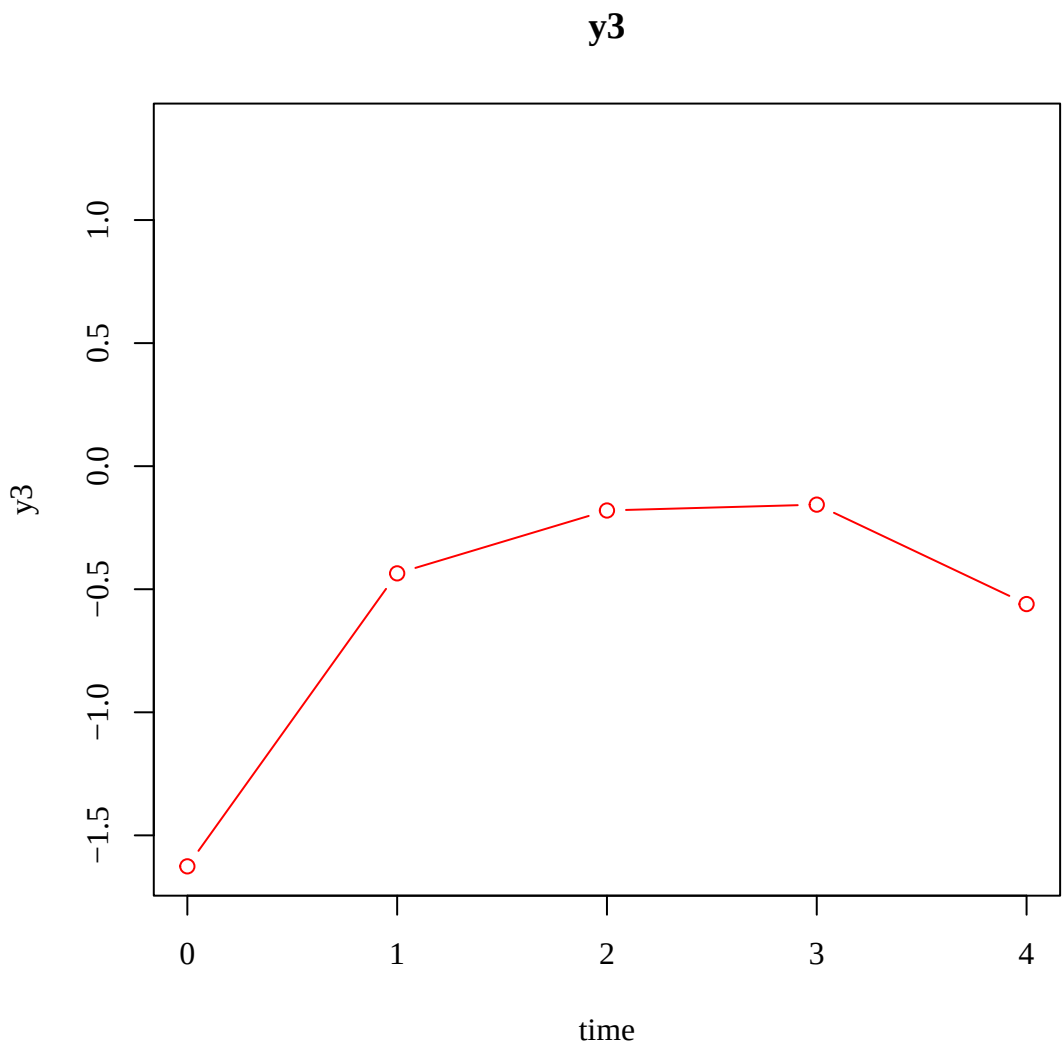

```
#> test-simStateSpace-sim-ssm
```

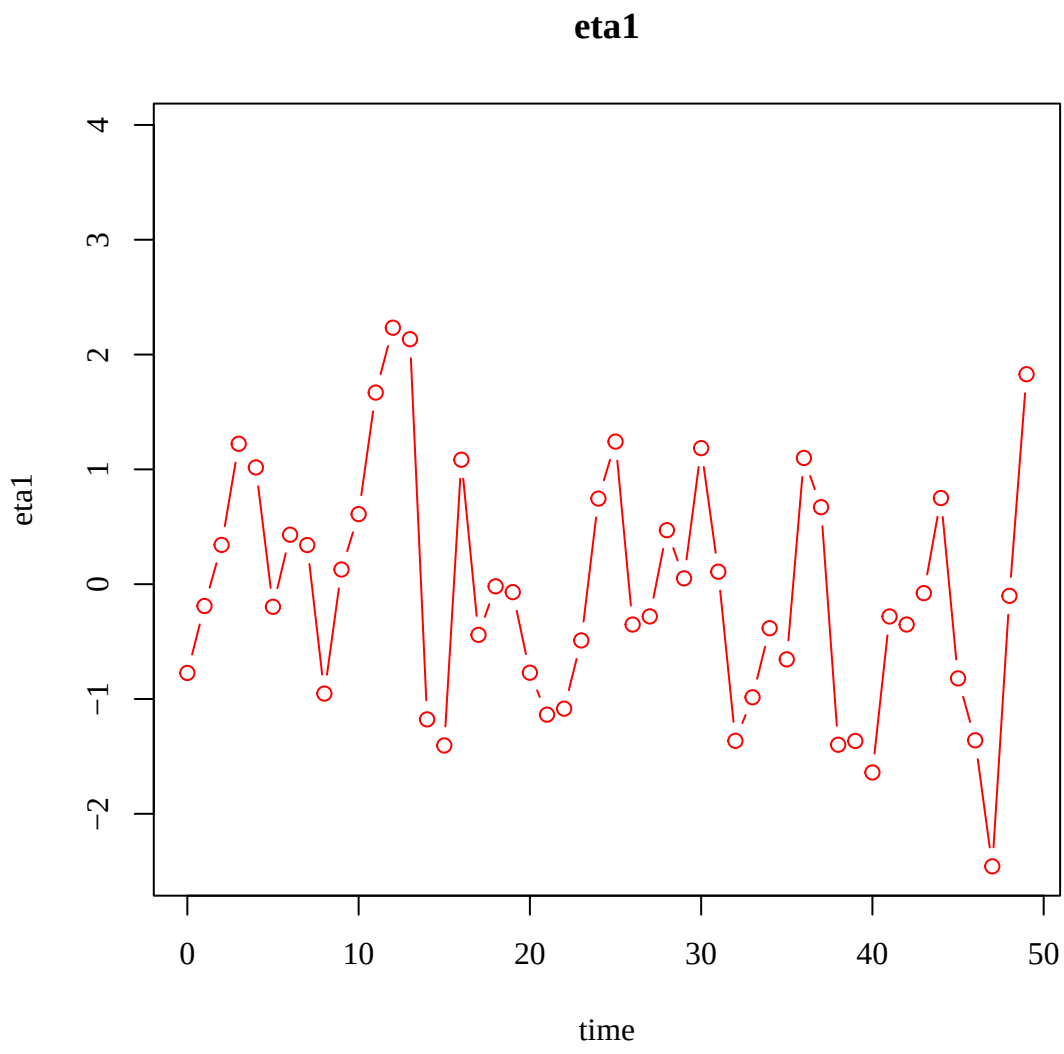


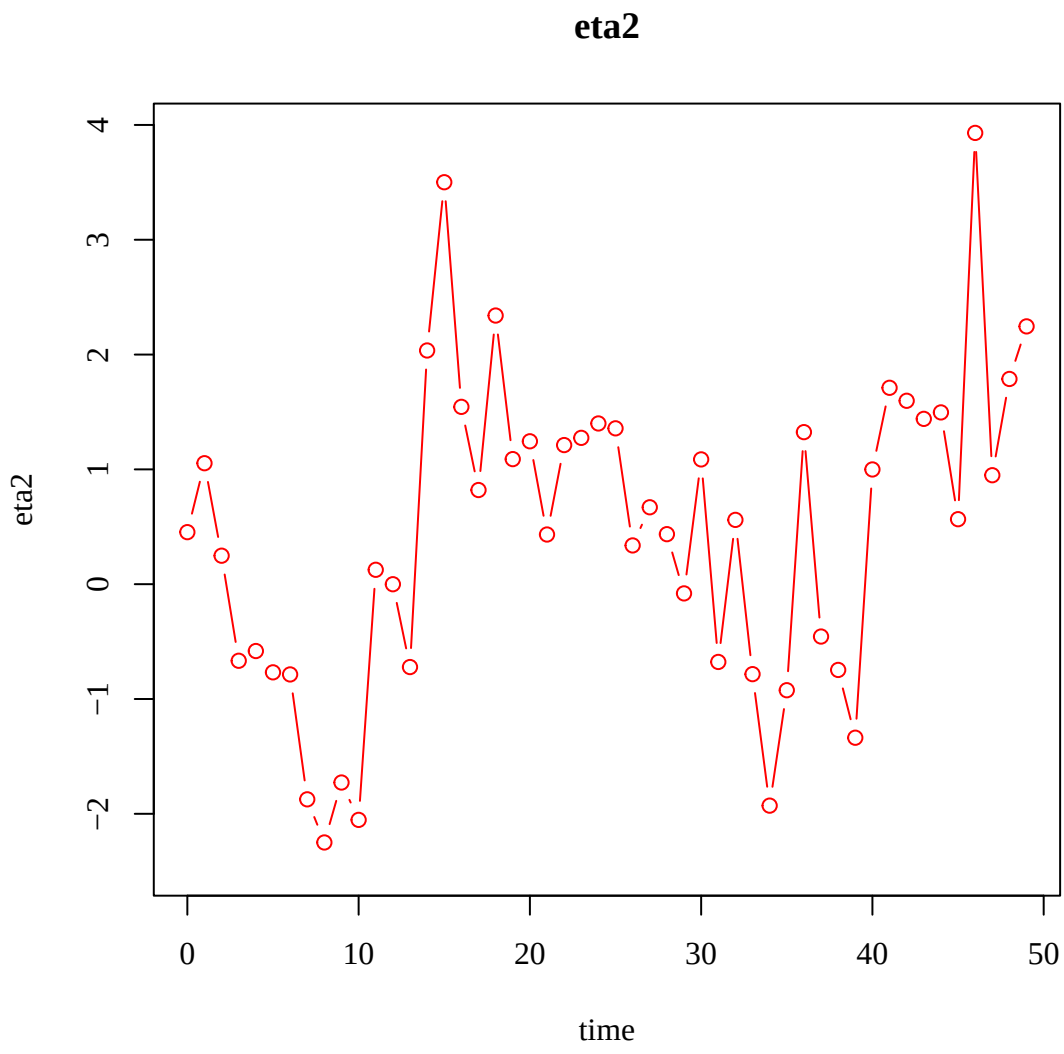

```
#> Call:
#> simStateSpace::SimSSM(mu0 = mu0, sigma0 = sigma0, alpha = alpha,
#>     beta = beta, psi = psi, nu = nu, lambda = lambda, theta = theta,
#>     type = 0, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSM`
#> to convert it to a data frame or a matrix.
```

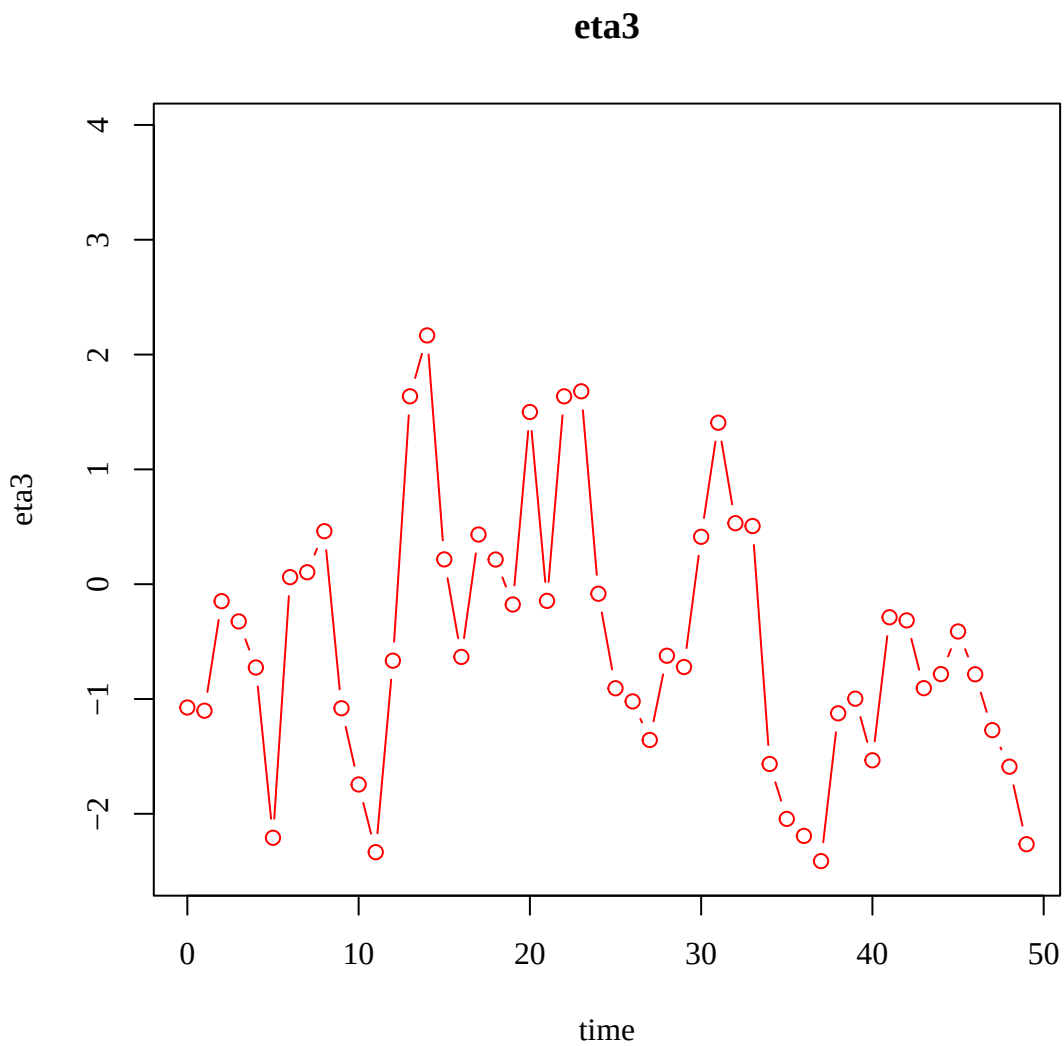

y1
y1
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
2
3
4
time


y2
y2
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
2
3
4
time

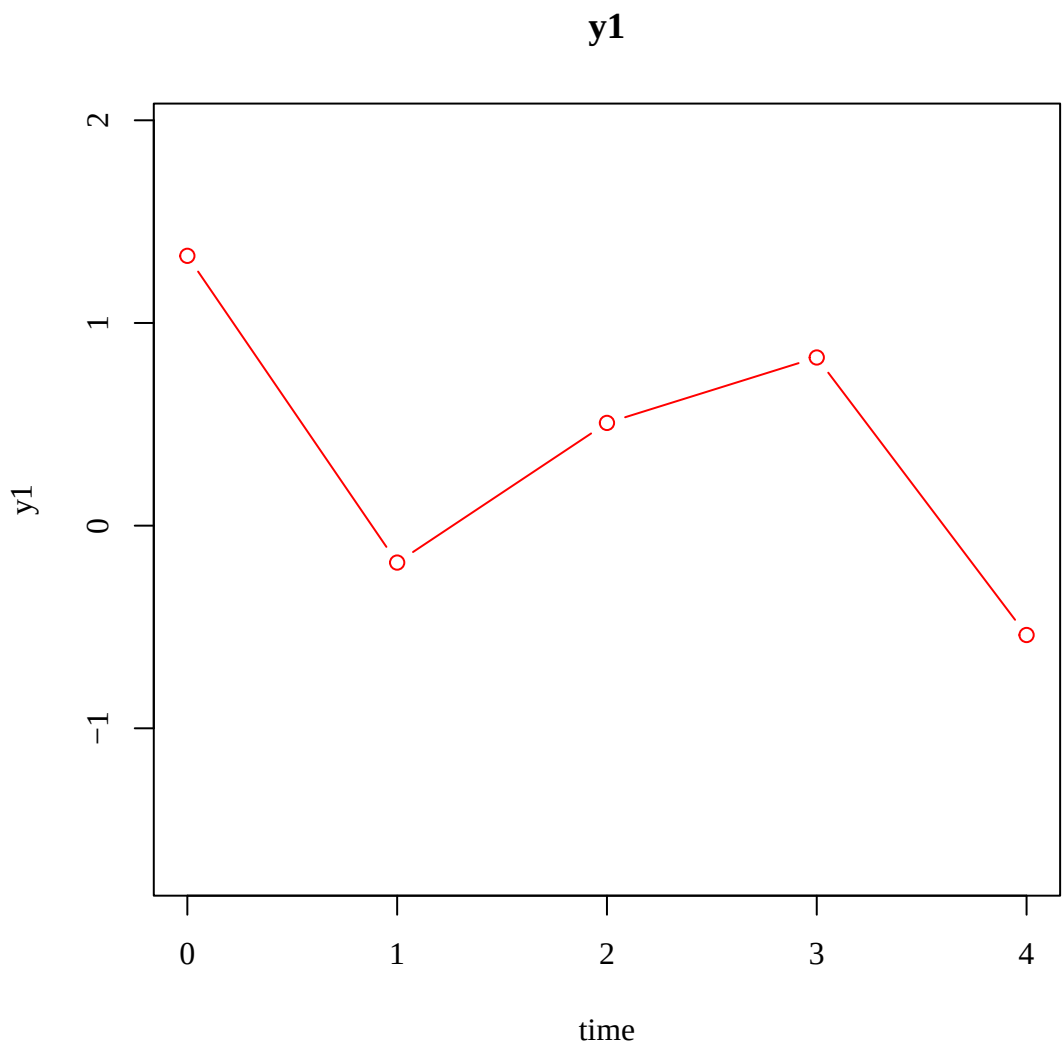

y3
y3
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
2
3
4
time

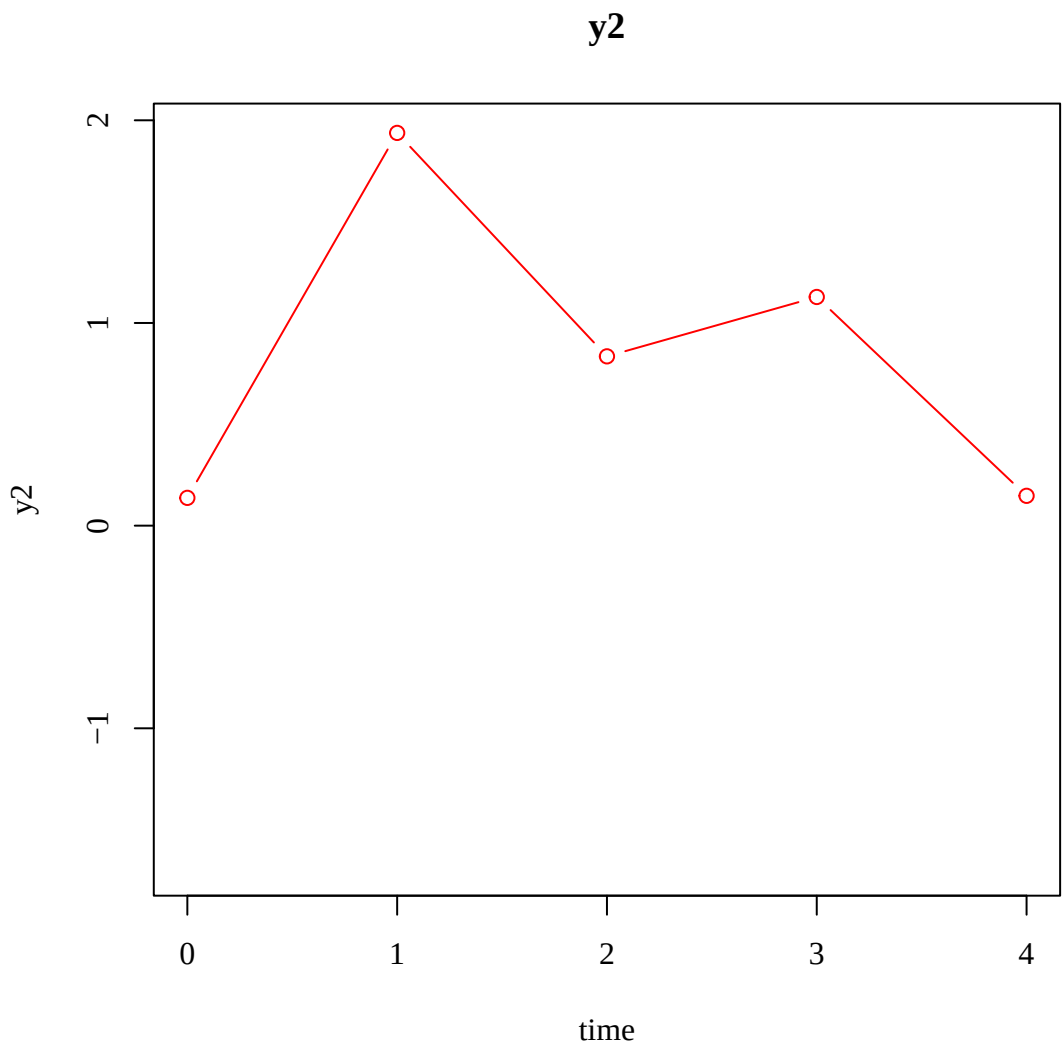

eta1
eta1
time
4
3
2
1
0
−1
−2
0
10
20
30
40
50

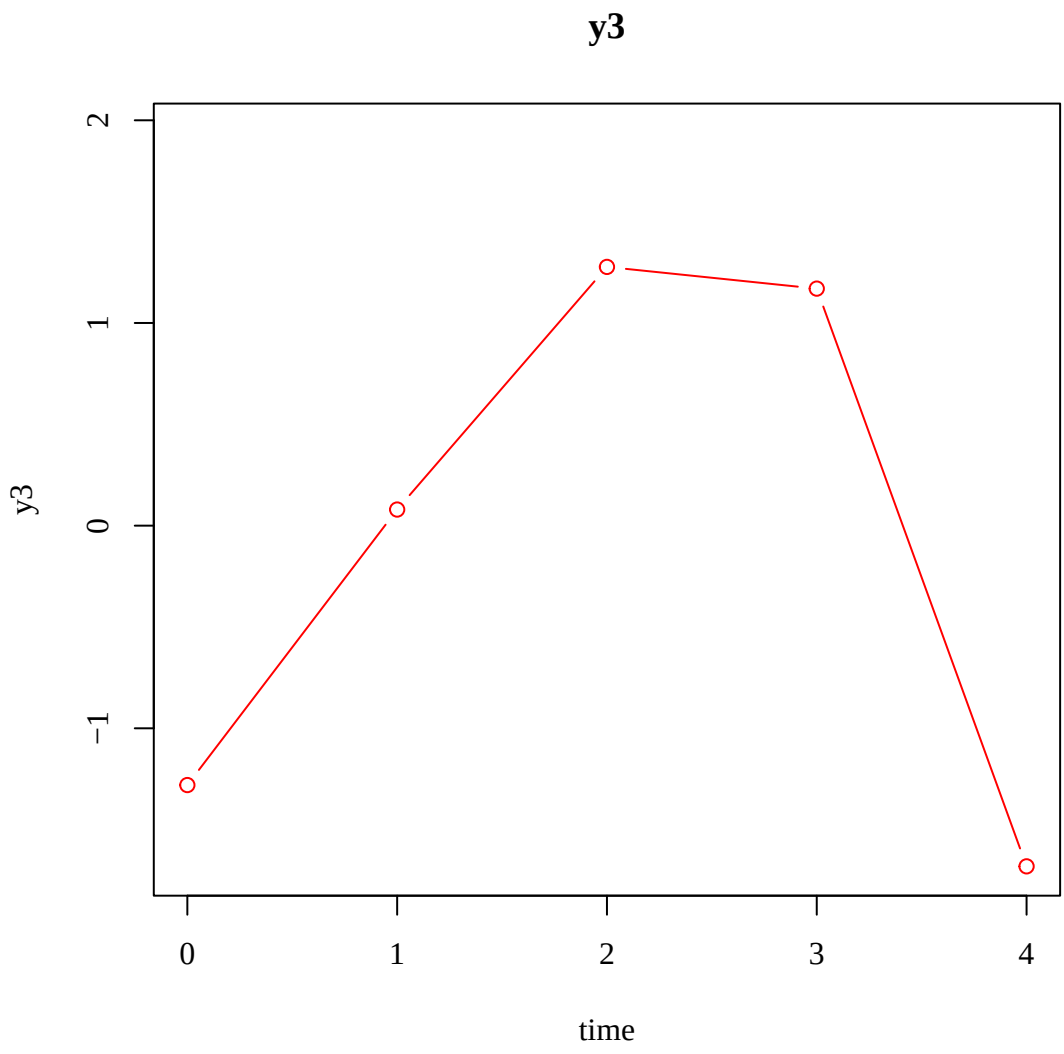

eta2
eta2
time
4
3
2
1
0
−1
−2
0
10
20
30
40
50

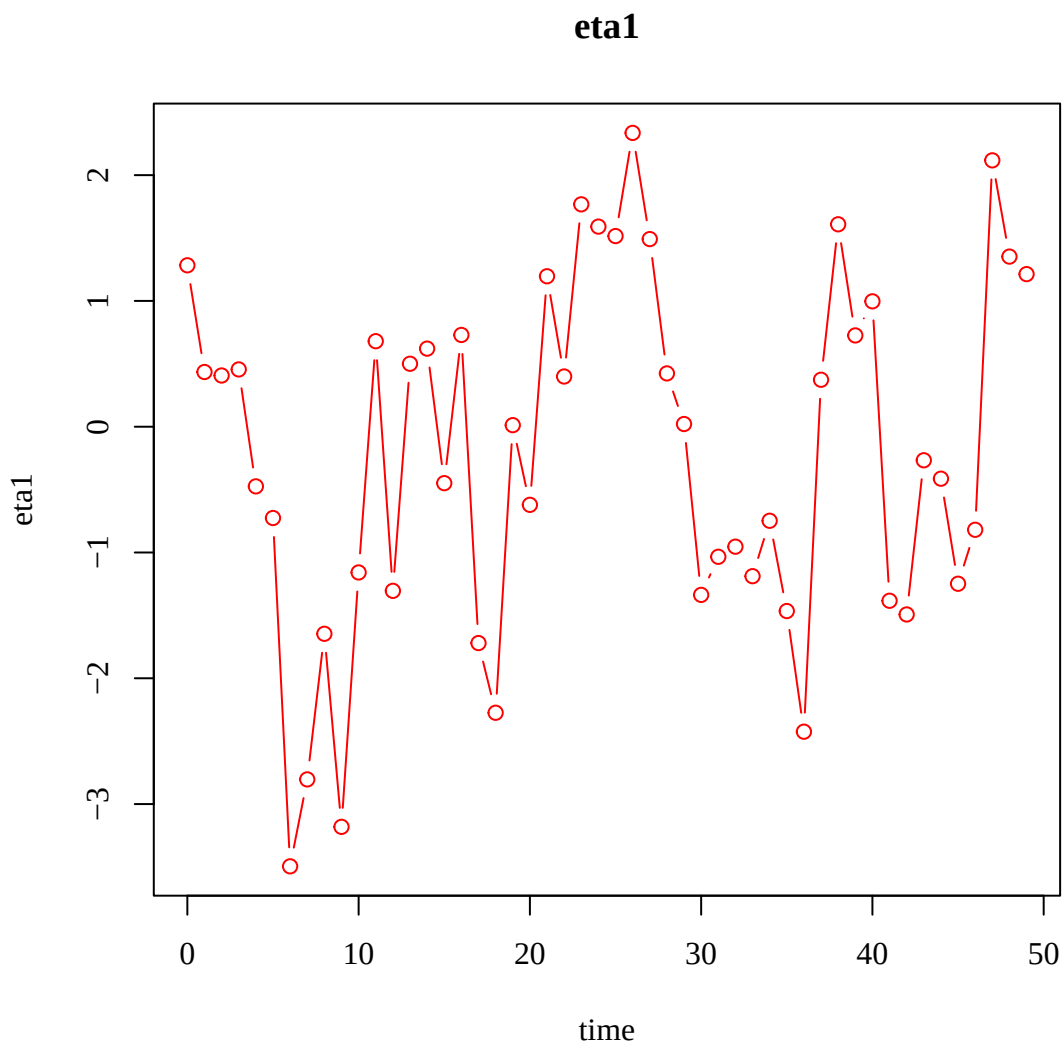


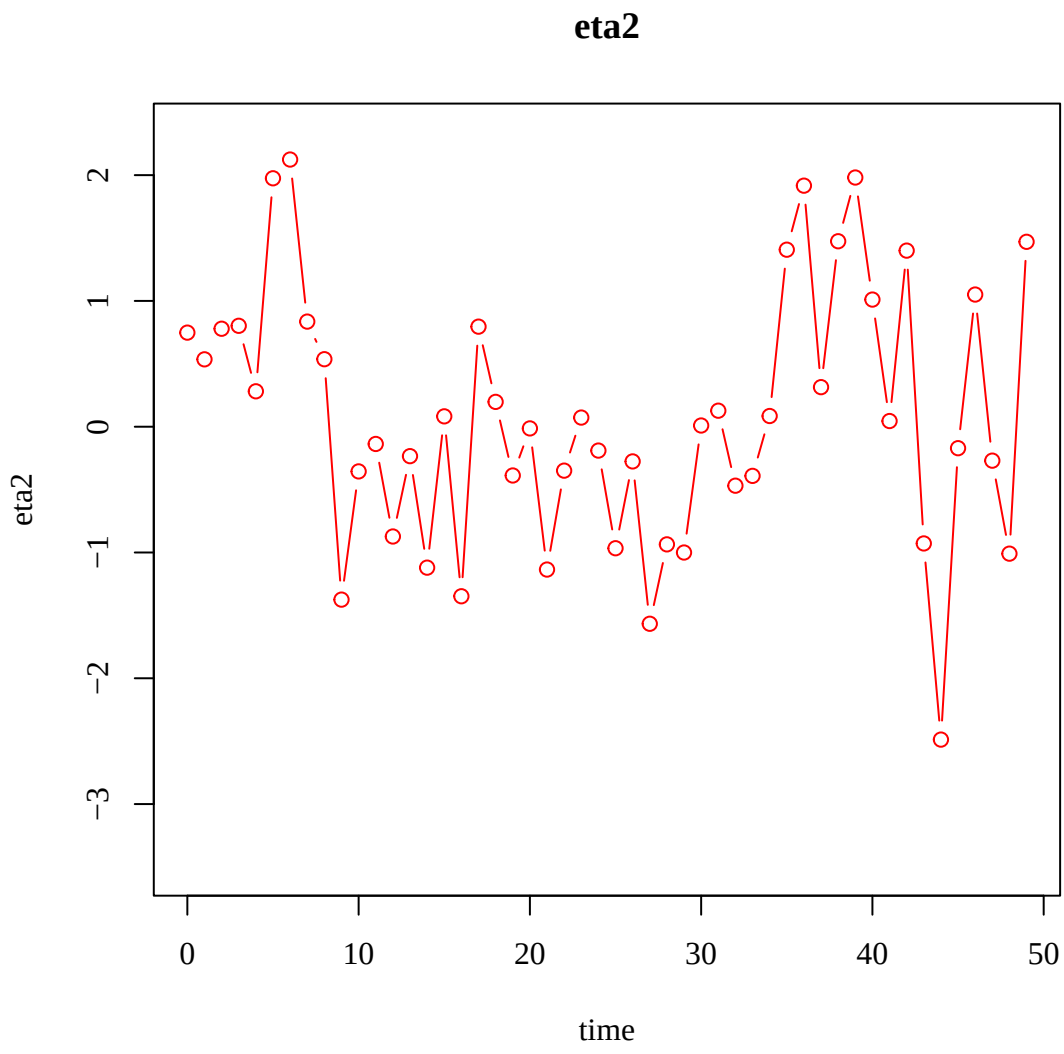

```
#> Call:
#> simStateSpace::SimSSM(mu0 = mu0, sigma0 = sigma0, alpha = alpha,
#>     beta = beta, psi = psi, nu = nu, lambda = lambda, theta = theta,
#>     gamma_eta = gamma_eta, x = x, type = 1, time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSM`
#> to convert it to a data frame or a matrix.
```

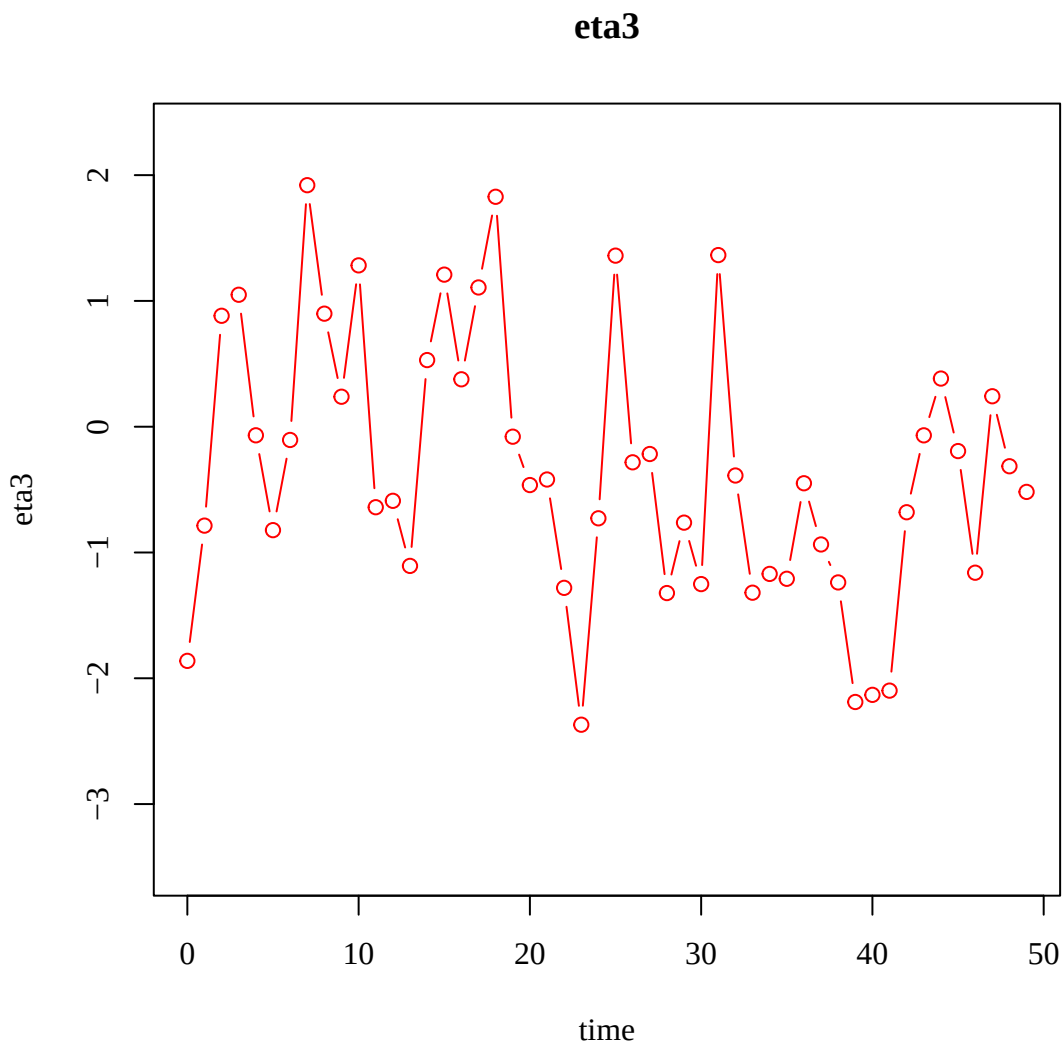

y1
2
1
0
−1
y1
0
1
2
3
4
time


y2
2
1
0
−1
y2
0
1
2
3
4
time

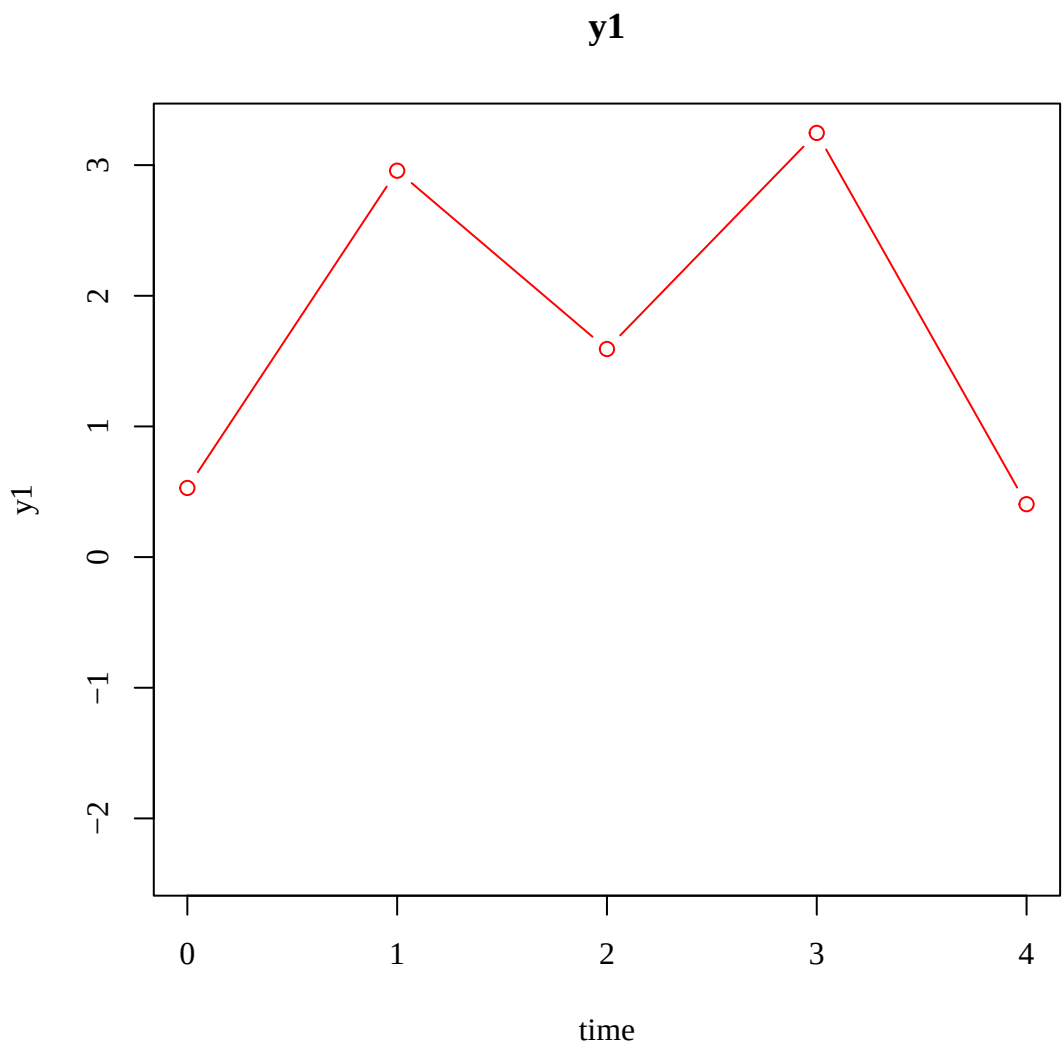

y3
2
1
0
−1
y3
0
1
2
3
4
time

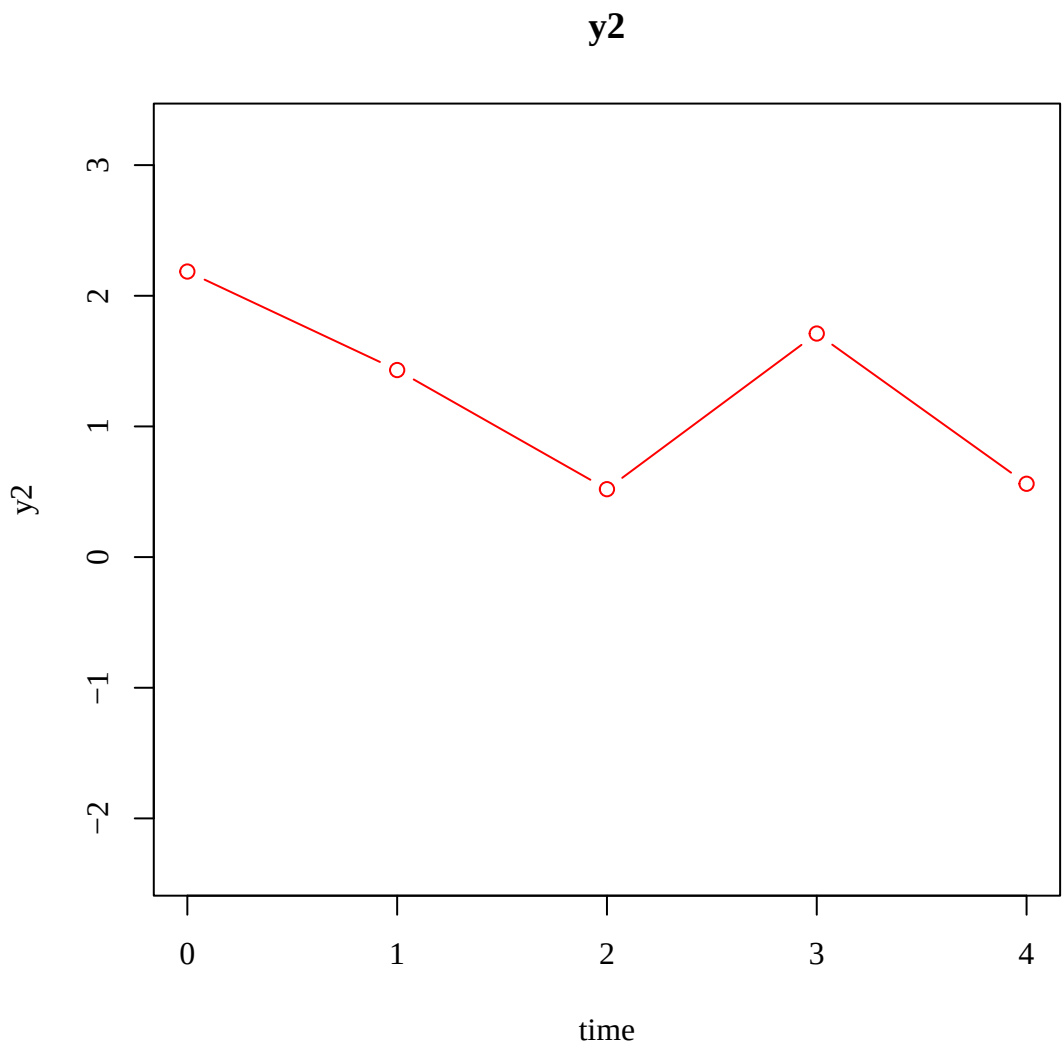

eta1
eta1
2
1
0
−1
−2
−3
0
10
20
30
40
50
time

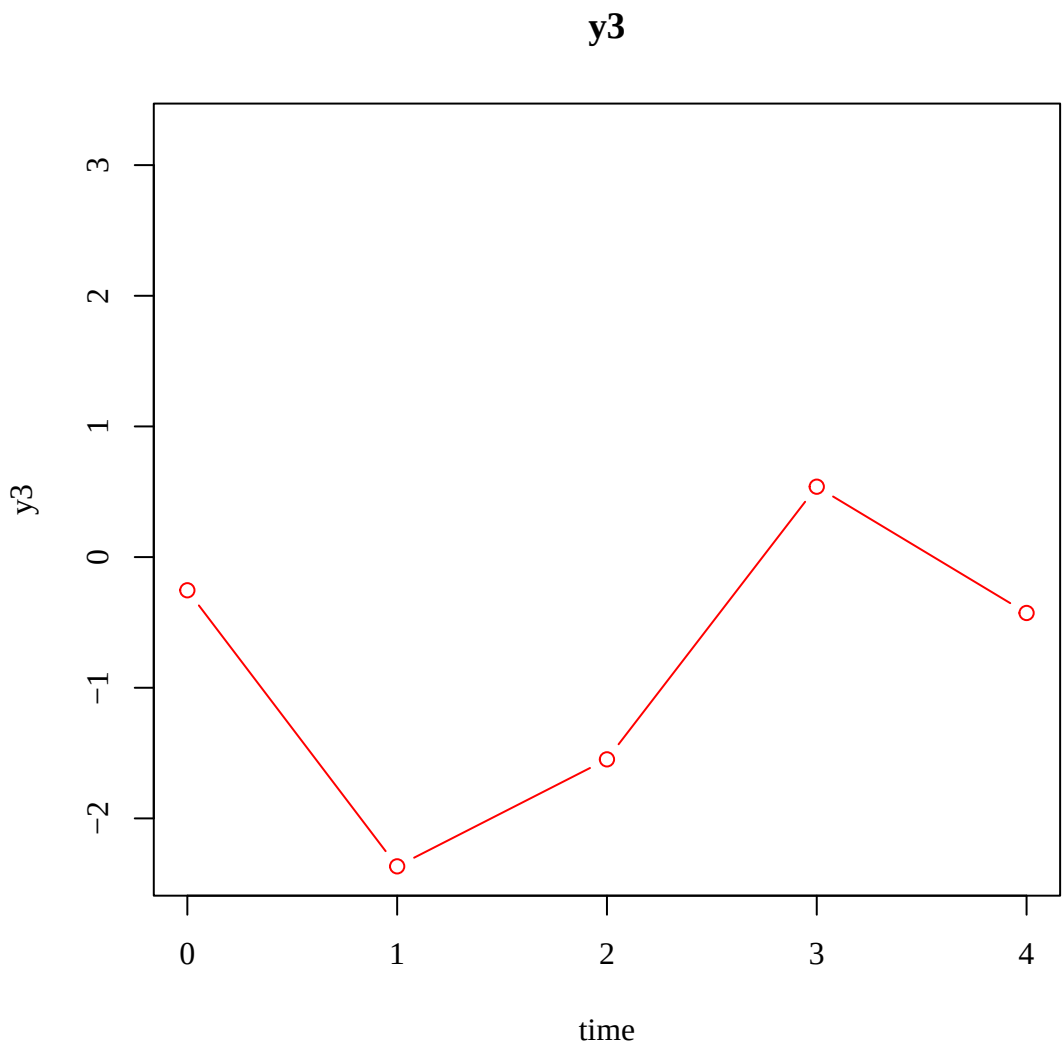

eta2
eta2
2
1
0
−1
−2
−3
0
10
20
30
40
50
time

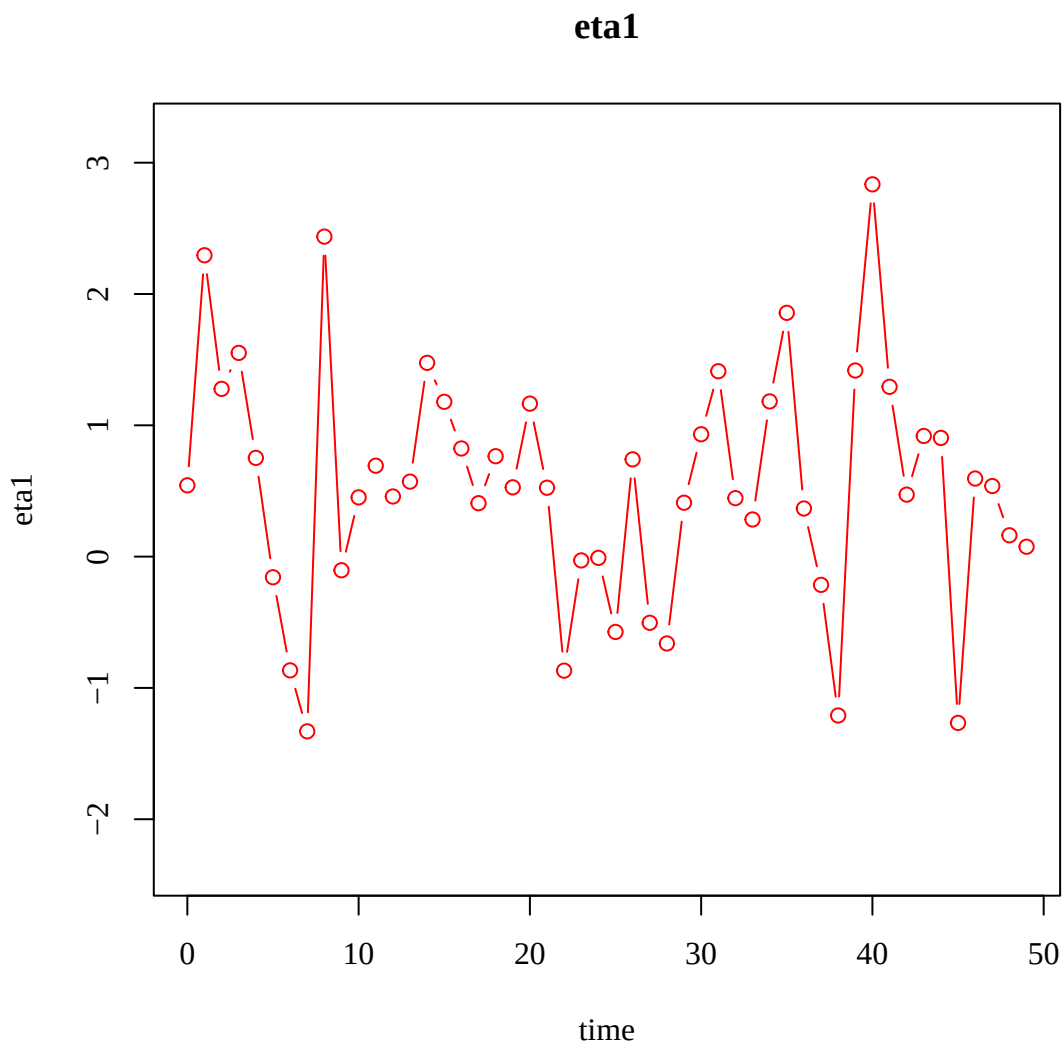


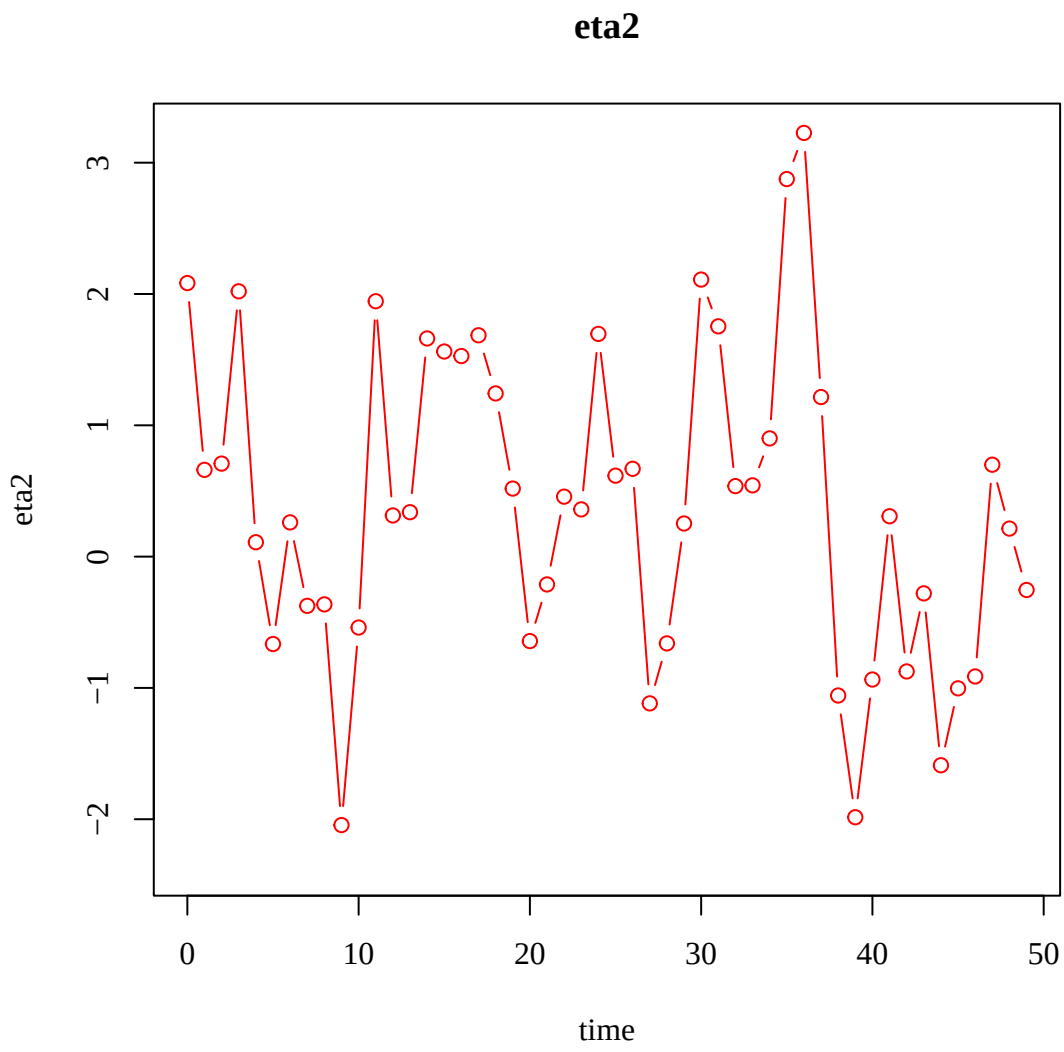

```
#> Call:
#> simStateSpace::SimSSM(mu0 = mu0, sigma0 = sigma0, alpha = alpha,
#>     beta = beta, psi = psi, nu = nu, lambda = lambda, theta = theta,
#>     gamma_y = gamma_y, gamma_eta = gamma_eta, x = x, type = 2,
#>     time = time, burn_in = burn_in)
#> Use `as.data.frame` or `as.matrix` on the output of `SimSSM`
#> to convert it to a data frame or a matrix.
```


y1
y1
3
2
1
0
−1
−2
0
1
2
3
4
time

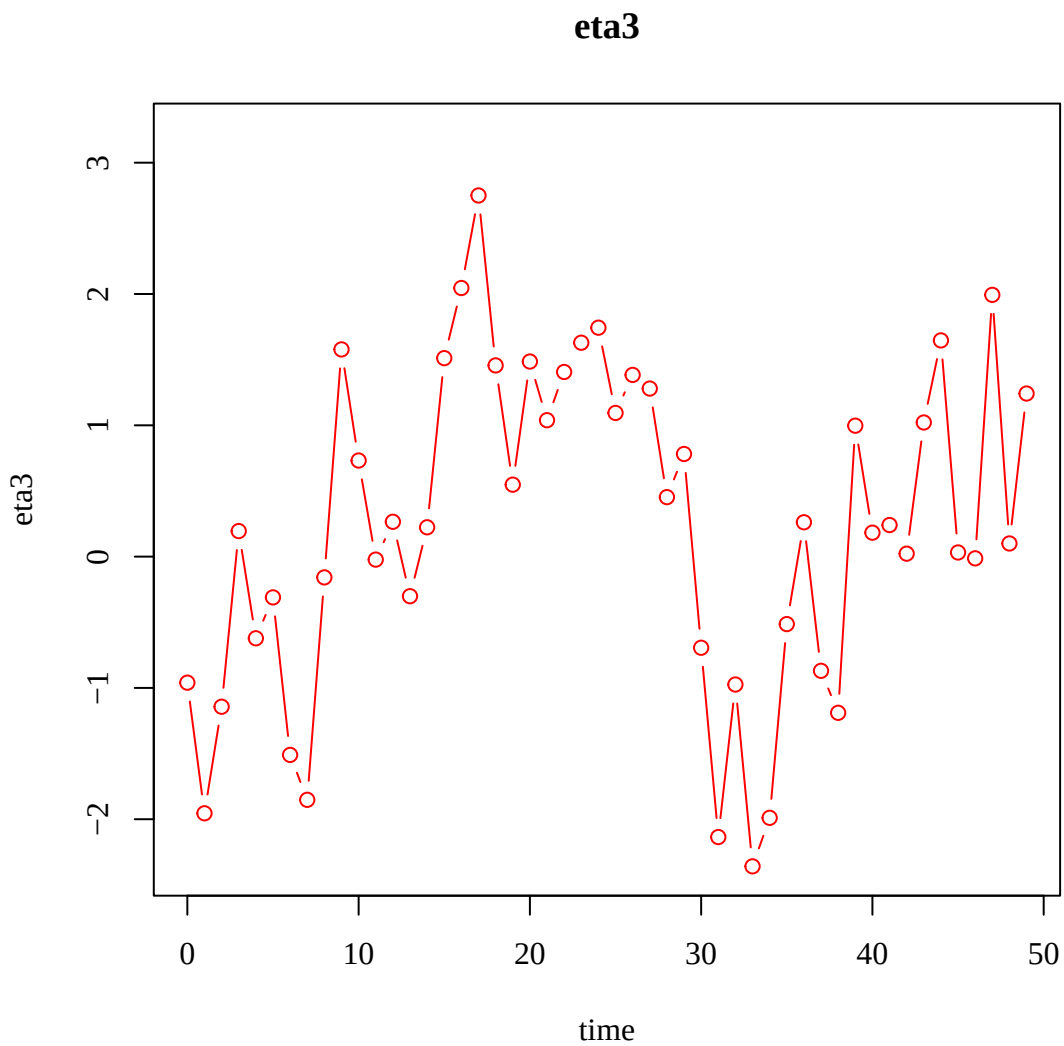

y2
y2
3
2
1
0
−1
−2
0
1
2
3
4
time


y3
y3
3
2
1
0
−1
−2
0
1
2
3
4
time


eta1
eta1
3
2
1
0
−1
−2
0
10
20
30
40
50
time




```
#> Test passed
#> Test passed
#> Test passed

#> test-simStateSpace-tests
```




```
#> Test passed
#> Test passed
#> Test passed
#> Test passed
#> [[1]]
#> [[1]][[1]]
#> [[1]][[1]]$value
#> [[1]][[1]]$value[[1]]
#> [1] TRUE
```

```
#>
#>
#> [[1]][[1]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[2]]
#> [[1]][[2]]$value
#> [[1]][[2]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[2]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[3]]
#> [[1]][[3]]$value
#> [[1]][[3]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[5]]$visible
#> [1] TRUE
#>
#>
```

```
#> [[1]][[6]]
#> [[1]][[6]]$value
#> [[1]][[6]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[6]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[7]]
#> [[1]][[7]]$value
#> [[1]][[7]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[7]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[8]]
#> [[1]][[8]]$value
#> [[1]][[8]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[8]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[9]]
#> [[1]][[9]]$value
#> [[1]][[9]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[9]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[10]]
#> [[1]][[10]]$value
#> [[1]][[10]]$value[[1]]
#> NULL
#>
```

```
#>
#> [[1]][[10]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[11]]
#> [[1]][[11]]$value
#> [[1]][[11]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[11]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[12]]
#> [[1]][[12]]$value
#> [[1]][[12]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[12]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[13]]
#> [[1]][[13]]$value
#> [[1]][[13]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[13]]$visible
#> [1] TRUE
```

Environment

```r
ls()

#> [1] "i"        "root"     "tex_file"
```

Class

```
#> [[1]]
#> [1] "character"
#>
#> [[2]]
#> [1] "root_criterion"
#>
#> [[3]]
#> [1] "character"
```

References

R Core Team. (2023). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/